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Diagnosis**

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2013



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Chapter 1

Bovine Tuberculosis: Diagnosis, Prevention and Control

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Bovine Tuberculosis (BTB) is an infectious and chronic disease of domesticated animals especially cattle. BTB has also been well documented in other domesticated and wild species such as buffaloes, bison, sheep, goats, equines, camels, pigs, wild boars, deer, antelopes, dogs, cats, foxes, mink etc. It is caused by *Mycobacterium bovis* (*M. bovis*), characterized by the progressive development of caseous granulomatous lesions forming tubercle nodule in affected tissues. Bovine tuberculosis is primarily considered as an airborne infection. Lung and lymph nodes are considered as most favored predilection site of *M. bovis* organisms. Usually, first it affects head and thorax lymph node followed by lung tissue. Distribution of pathological lesion in body is variable which depend upon, route of infection, dose inoculum, host species and environmental conditions. Infectious agent is usually transmitted through aerosol route followed by oral route. Initially innate immunity (macrophage, neutrophil, natural killer cell and gamma delta T-cell) try to contain and destroy mycobacterium organism, then Th1 mediated cell immune response destroy mycobacteria laden macrophage either by CD4+ mediated activation of macrophage or CD8+ mediated lysis of infected macrophages. When this cell mediated immune response starts to wane, antibody mediated immune response start to progress which is usually correlated with disease establishment and disease progression.

Bovine tuberculosis is usually diagnosed by tuberculin skin testing based on delayed type hypersensitivity in which minimum 2000 IU bovine PPD antigen is injected intradermally in mid third of neck skin into infected animal. Specificity of tuberculin skin testing is low due to sharing of mycobacterial antigen among pathogenic and nonpathogenic environmental mycobacteria causing false positive results. Lymphocyte proliferation test and interferon gamma test offer higher sensitivity to detect CMI and these test are able to detect infection at an early stage of disease. Serological based assay are not so sensitive and specific for bovine tuberculosis diagnosis but their simplicity, inexpensiveness and ability to develop some pan side diagnostic test made them an alternative of CMI based test. Diagnostic test specificity may be increased by using more specific and refined antigen such as ESAT-6, CFP-10 and MPB-70.

Molecular diagnostic test has opened a new era in bovine tuberculosis diagnosis. These are rapid, precise with high sensitivity and specificity but these tests require sophisticated equipment and costly molecular reagent.

Diagnosis: BTB is considered as a chronic debilitating disease. Animal usually do not exhibit overt clinical sign unless it has reached in advanced stage of disease. In many cases, the course of the infection is chronic and signs may be lacking, even in advanced cases when many organs may be involved. When present, clinical signs vary, lung involvement may be manifested by a cough, dyspnoea and low-grade pneumonia.

1. **Isolation and Identification of *Mycobacterium bovis* (*M. bovis*) from Clinical Specimens:** Isolation of *Mycobacterium bovis* from infected animal is considered as gold standard in diagnosis but it is very tedious, time consuming, high skill demanding procedure. The presence of *M. bovis* in clinical and post-mortem specimens may be demonstrated by examination of stained smears or tissue sections and confirmed by cultivation of the organism on primary isolation medium. *M. bovis* is very fastidious and slow growing organism, usually contaminated by rapidly growing environmental mycobacteria. Therefore, extra precaution should be ensured during collection and shipment of samples to laboratory. Prompt delivery of specimens to the laboratory greatly enhances the chances of cultural recovery of *M. bovis*. If delays in delivery are anticipated, specimens should be refrigerated or frozen to retard the growth of contaminants and to preserve Mycobacteria. In warm ambient conditions, when refrigeration is not possible, boric acid may be added (0.5% [w/v] final concentration) as a bacteriostatic agent, but only for limited periods, no longer than 1 week.

M. bovis can be demonstrated microscopically on direct smears from clinical samples and on prepared tissue materials. The acid fastness of *M. bovis* is normally demonstrated with the classic Ziehl–Neelsen stain, but a fluorescent acid-fast stain may also be used. Immunoperoxidase techniques may also give satisfactory results. The presumptive diagnosis of mycobacteriosis can be made if the tissue has characteristic histological lesions (caseous necrosis, mineralization, epithelioid cells, multinucleated giant cells and macrophages). As lesions are often paucibacillary, the presence of acid-fast organisms in histological sections may not be detected.

2. **Cell-Mediated Immunity Based Diagnosis:** Immune response to *M. bovis* is mainly based on T-helper type-1 response as evidenced by IFN- γ , interleukin (IL)-12, and tumor necrosis factor (TNF)- α production to pathogen-associated antigen and exacerbation of disease when these cytokine combinations are disrupted. TB diagnosis is mainly based on evaluation of these cell-mediated responses either by skin test or IFN- γ -based assay. When cell mediated immune response starts to wane, host induce humoral immune response against mycobacterium antigens but humoral response develop at the very late stage of infection and mostly, till that time, disease would have already established. In general, specific antibody responses are most easily detected in late stage of disease and correlate with severe progression of disease.

(a).Tuberculin Skin Testing (The Prescribed Test for International Trade): The standard method for detection of bovine tuberculosis is the tuberculin test, which involves the intradermal injection of bovine tuberculin purified protein derivative (PPD) (2000IU) and the subsequent detection of swelling (delayed hypersensitivity) at the site of injection 72 hours later. This may be performed using bovine tuberculin alone or as a comparative test using avian and bovine tuberculin. The tuberculin test is usually performed on the mid-neck, but the test can also be performed in the caudal fold of the tail. The skin of the neck is more sensitive to tuberculin than the skin of the caudal fold. To compensate for this difference, higher doses of tuberculin may be used in the caudal fold.

The interpretation is based on observation and the recorded increases in skin-fold thickness. In the single intradermal test, the reaction is commonly considered to be negative if only limited swelling is observed, with an increase of no more than 2 mm and

without clinical signs, such as diffuse or extensive oedema, exudation, necrosis, pain or inflammation of the lymphatic ducts in that region or of the lymph nodes. The reaction is considered to be inconclusive if none of these clinical signs is observed and if the increase in skin-fold thickness is more than 2 mm and less than 4 mm. The reaction is considered to be positive if clinical signs, as mentioned above, are observed or if there is an increase of 4 mm or more in skin-fold thickness. Moreover, in *M. bovis*-infected herds, any palpable or visible swelling should be considered to be positive. Sometimes a more stringent interpretation is used, particularly in a high risk population or in-contact animals. Animals that are inconclusive by the single intradermal test should be subjected to another test after an interval of 42 days to allow desensitization to wane (in some areas 60 days for cattle and 120 days for deer are used). Animals that are not negative to this second test should be deemed to be positive to the test. Animals that are positive to the single intradermal test may be subjected to a comparative intradermal test or blood test.

Delayed hypersensitivity may not develop for a period of 3–6 weeks following infection. Thus, if a herd/animal is suspected to have been in contact very recently with infected animals, delaying testing should be considered in order to reduce the probability of false-negatives. As the sensitivity of the test is less than 100%, it is unlikely that eradication of tuberculosis from a herd will be achieved with only a single tuberculin test. It should be recognized that when used in chronically infected animals with severe pathology, the tuberculin test may be unresponsive. However, animals which have been sensitized by non-pathogenic environmental strains of mycobacteria may react positively to PPD-B, due to sharing of antigens among virulent and non-virulent environmental mycobacterial strains. To overcome this problem, comparative intradermal tuberculin skin testing may be done in which host responses to PPD-A and PPD-B is compared to discriminate cattle infected with *M. bovis* and those exposed to environmental strains.

The complexity of tuberculin and the presence of cross-reactive components shared by different nonpathogenic environmental mycobacteria and other organism led to develop diagnostic test based on specific antigen conserved among pathogenic mycobacterium species only. One such antigen, ESAT-6 has been assessed as a diagnostic reagent in different test formats. The specificity of the delayed-type hypersensitivity reaction in cattle has been improved by the introduction of ESAT-6.

(b).Gamma-Interferon Assay (The Alternative Test for International Trade):

In this test, lymphoblastogenic activity is indirectly expressed in terms of interferon gamma production by peripheral blood lymphocytes from infected animals following antigenic stimulation. The assay is based on the release of IFN- γ from sensitized lymphocytes during a 16–24-hour incubation period with specific antigen (PPD-tuberculin). The test makes use of the comparison of IFN- γ production following stimulation with avian and bovine PPD. The detection of bovine IFN- γ is carried out with a sandwich ELISA that uses two monoclonal antibodies to bovine gamma-interferon. It is recommended that the blood samples be transported to the laboratory and the assay set up as soon as practical, but not later than the day after blood collection. IFN- γ test is capable in detecting early infections; the use of both tests in parallel allows detection of a greater number of infected animals before they become a source of infection for other animals as well as a source of contamination of the environment. The

use of defined mycobacterial antigens such as ESAT 6 and CFP-10 enhance specificity. These antigens have been used in many developed countries including United Kingdom and New Zealand for serial testing. The use of such antigens may also offer the ability to differentiate BCG-vaccinated from unvaccinated animals. The other advantage of this test is that animals that are difficult or dangerous to handle, such as excitable cattle or other bovidae, need to be captured only once. The IFN- γ test may be used for serial testing (to enhance specificity) and parallel testing (to enhance sensitivity). The test is available as commercial kits for bovine species and primates, however it has been validated in only a few species.

(c). Lymphocyte Proliferation Test: In this method, a lymphoblastogenic response by antigen stimulation is measured. This type of in-vitro assay compares the reactivity of peripheral blood lymphocytes to tuberculin PPD (PPD-B) and a PPD from *M. avium* (PPD-A). The assay can be performed on whole blood purified lymphocytes from peripheral blood samples. These tests endeavor to increase the specificity of the assay by removing the response of lymphocytes to 'nonspecific' or cross-reactive antigens associated with non-pathogenic species of mycobacteria to which the animal may have been exposed. Results are usually analyzed as the value obtained in response to PPD-B minus the value obtained in response to PPD-A. The B-A value must then be above a cut-off point that can be altered in order to maximize either specificity or sensitivity of the diagnosis. The assay has scientific value, but is not used for routine diagnosis because the test is time-consuming and the logistics and laboratory execution are complicated. It requires long incubation times and the use of radio-active nucleotides. Lymphocyte proliferation assay should be performed shortly after blood is collected. The test may be useful in wildlife and zoo animals. The test is relatively expensive and has not yet been subject to inter-laboratory comparisons.

3. Serological Based Diagnosis: Antibody based diagnostic test for bovine tuberculosis are simple, easy to perform, easy to interpret and it requires capturing of animal once only but their sensitivity and specificity are questionable due to inherent nature of *M. bovis* immune response. It is well established that cattle infected by *M. bovis* first develop cell-mediated immune response so serology is less efficient to identify cattle in the early stages of infection, when antibodies titers are low. Antibodies are often considered to be unimportant in protective immunity against intracellular mycobacteria, but a recent review has indicated that, under certain circumstances, antibodies may have a role in protection. Serological test using crude mycobacterial preparations gives generally satisfactory test sensitivity but having poor specificity because of broad cross-reactivity with non-TB mycobacteria, such as *M. avium*. Sensitivity and specificity of serological assay can be successfully increased by using *M. bovis* specific antigen. Antigen discovery efforts have unveiled numerous sero-reactive targets. MPB70 and MPB83 proteins of *M. bovis* appear to be the major serodominant antigens for detection of tuberculous cattle. Various mycobacterial antigens such as ESAT-6, 14 kDa, MPT63, MPT70, MPT51, MPT32 and MPB83 have been shown to contain B-cell epitopes that can be exploited for the sero-diagnosis of tuberculosis infections. Recent advances in both antigen discovery and immunoassay technology have facilitated progress in developing novel antibody-based tests for bovine tuberculosis. Nowadays cocktail antigen (using more than one

antigen) or multi-epitope fusion proteins is being used to improve sensitivity and specificity of these tests.

4. **Molecular Diagnosis:** Molecular diagnostic test for bovine tuberculosis diagnosis are highly sensitive with adequate specificity, rapid, convenient and alternative of conventional diagnosis. Conventional diagnosis relies on isolation and identification of causative organism from clinical specimen. It takes atleast 8 week to grow and 4 week for further biochemical identification. This time gap permit infected animal to excrete organism which infect other susceptible animals and contaminate environment. Non recovery of mycobacterial growth from clinical specimen cannot be considered as tuberculosis free animal because it may be related with early stage of infection (non-secretory stage), very low no. of bacilli in clinical specimen, destruction of mycobacterium bacilli during decontamination procedure, superseding of environmental mycobacteria over slow growing fastidious pathogenic mycobacteria, latency disease status, etc.

Molecular diagnosis relies on amplification of organism specific sequence (DNA, RNA). Molecular diagnostic test give result in a single day, it obviate need of isolating Mycobacterium, and it is very safe to laboratory personnel. PCR is universal choice to identify bacterial pathogen and this technique has been used successfully targeting different gene or insertion sequence of mycobacterium tuberculosis complex for diagnosis and epidemiology both. Various mycobacterium tuberculosis complex specific PCR have been developed targeting 16s rRNA, 16S–23S rRNA, the insertion sequences IS6110 and IS1081, and genes coding for *M. tuberculosis*-complex-specific proteins such as MPB70 and the 38 kDa antigen. Specific identification of an isolate as *M. bovis* can be made using PCR targeting single nucleotide polymorphism in oxyR gene, pncA gene, gyrB gene and presence/absence of RDs (Regions of difference). Molecular assays have been extensively used to analyze the epidemiology of bovine tuberculosis. Diversity of mycobacterial strains, identification of point source of infection, transmission dynamics of disease and geographical and temporal distribution of bovine tuberculosis.

Prevention and Control- Bovine tuberculosis can be controlled effectively by test-and-slaughter policy but in India, where TB is endemic and a significant proportion of animals are infected. Test-and-segregation methods may be followed in which infected animal may be separated from healthy animals. Affected animals should be handled and managed by separate personnel and those personnel should not be allowed to handle healthy non infected animal. Affected herds should be re-tested periodically to eliminate cattle that may shed the organism.

Managerental practices can reduce spread of infection from infected to other susceptible animals such as animal handlers must be healthy (TB free), avoiding nursing of calves on tuberculous dam, disinfection of infected fomites, providing balanced feed to animals etc. Sanitation and disinfection may reduce the spread of the agent within the herd. *M. bovis* is relatively resistant to disinfectants and requires long contact times for inactivation. Effective disinfectants include 5% phenol, iodine solutions with a high concentration of available iodine, glutaraldehyde and formaldehyde. In environments with low concentrations of organic material, 1% sodium hypochlorite with

a long contact time is also effective. In addition, biosecurity measures on farms decrease interactions between wildlife and domesticated animals. BCG vaccine is only available vaccine against tuberculosis disease but we cannot use this vaccine in our cattle herd in India because (1) it does not give full protection against tuberculosis disease. (2) Vaccination will hinder animal testing with tuberculin skin testing. In near future, we will have to develop new vaccine with better efficacy and simultaneously we will have to develop new diagnostic tests (DIVA) which can be used to differentiate infected animal and vaccinated animals.

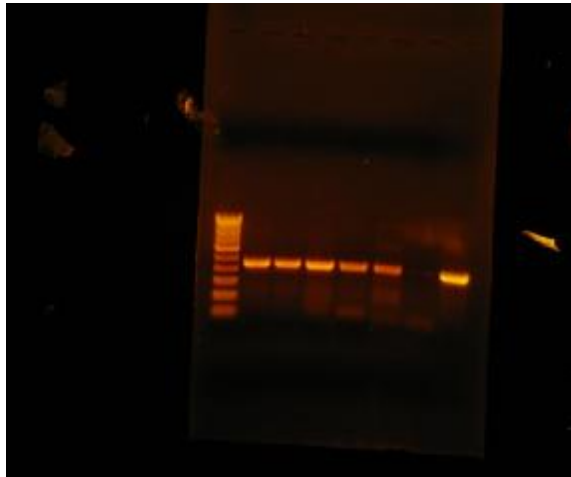


Fig.1 PCR targeting MTB complex specific insertion sequence IS6110 producing 445 bp amplicon in all MTB complex Mycobacterium

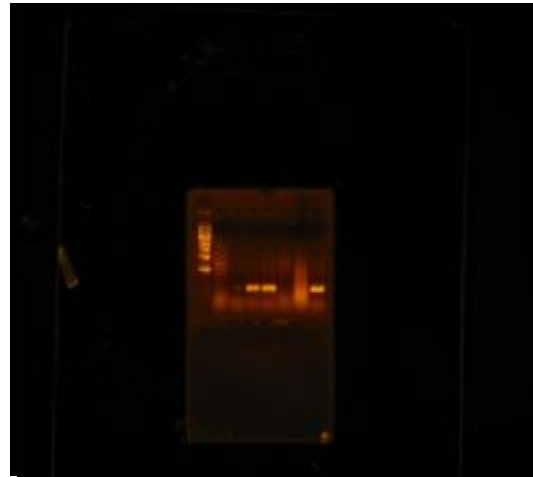


Fig. 2 PCR targeting *M. tuberculosis* specific SNP in pncA gene producing 185 bp amplicon in *M. tuberculosis* only

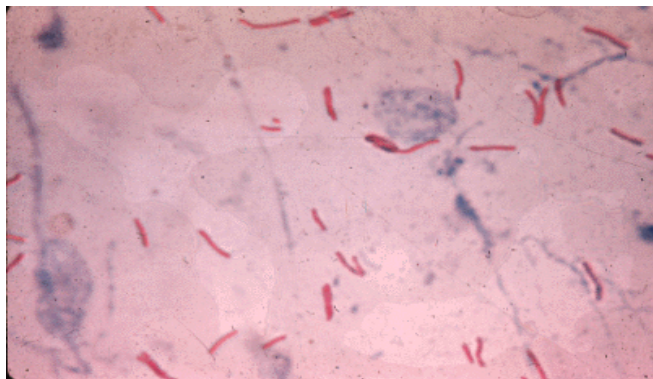


Fig. 3 Acid fast bacilli in Ziehl-Nielsen staining

Chapter 2

Antimicrobial Sensitivity Assay and Antimicrobial Chemotherapy in Animals: A Practical Approach

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Antibiosis was first described in 1877 in bacteria when Louis Pasteur and Robert Koch observed that an airborne bacillus could inhibit the growth of *Bacillus anthracis*. Salvarsan discovered by Paul Ehrlich was produced in 1932 at the Bayer Laboratories and was used for treatment of syphilis. Alexander Fleming: Discovered penicillin in 1928, could not purify. In 1939, gramicidin from *B. brevis* was one of the first commercially manufactured antibiotics. Ernst Chain and Howard Florey successfully purified penicillin, and in 1941 tested on human subjects. The term antibiosis was coined in 1889 by Louis Pasteur's pupil Paul Vuillemin however; the term antibiotic was given by Selman Waksman in 1942. After discovery of antimicrobial agents, clinicians felt relieved as these wonder drugs substantially reduced burden of infectious diseases. Over the years, antibiotics have saved lives, eased the suffering of millions of people, contributed to the major gains in life expectancy. However, these wonder drugs have started to lose ground rapidly. With each application of antibiotic to kill bacteria a new micro environment is created where the sensitive microbes get killed but the resistant organisms start to flourish. New selection pressure each time leads to rapid evolution in bacteria. As a result, now almost all important infection causing bacteria are armoured to survive in antibiotic loaded environment with much deadlier infective power.

Worldwide, infectious diseases are the number one cause of death accounting for approximately one-half of all the deaths in tropical countries. Perhaps it is not surprising to see these statistics in developing nations, but what may be remarkable is that infectious disease mortality rates are actually increasing in developed countries, such as in the United States, deaths from infectious disease ranked 5th in 1981, has become the 3rd leading cause of death in 1992, an increase of 58%. It is estimated that infectious diseases are the underlying cause of death in 8% of the deaths occurring in the US. This is alarming given that it was once believed that we would eliminate infectious disease by the end of the millenium through the use of antimicrobials. The increase in deaths attributed to increases in respiratory tract infections and HIV/AIDS. Other contributing factors are an increase in antibiotic resistance in nosocomial and community acquired infections. Furthermore, the most dramatic increments in infectious disease deaths are in 25-44 year old age group. These negative health trends call for a renewed interest in infectious disease in the medical and public health communities and renewed strategies on treatment and prevention. Proposed solutions are outlined by the CDC as a multi-pronged approach that includes: prevention, (such as vaccination); improved monitoring; and the development of new treatments. It is this last solution that would encompass the development of new antimicrobials.

The control of microorganisms is critical for the prevention and treatment of diseases, however, the increasing number and variety of drug-resistant pathogens is a

serious public health problem. When microbes began resisting penicillin, medical researchers fought back with chemical cousins, such as methicillin and oxacillin. By 1953, the antibiotic armamentarium included chloramphenicol, neomycin, tetracycline, and cephalosporins. But today, researchers fear that we may be nearing an end to the seemingly endless flow of antimicrobial drugs. Resistance to antimicrobial agents has resulted in morbidity and mortality from treatment failures and increased health care costs. Although defining the precise public health risk and estimating the increase in costs is not a simple undertaking, there is little doubt that emergent antibiotic resistance is a serious global problem.

A detailed evaluation of all infectious pathogens revealed that out of the 1415 species of microbes pathogenic for humans 868 (~61%) are zoonotic, i.e., affect animals too. Most of the emerging and re-emerging pathogens are also zoonotic. Emergence refers to the appearance of a newly recognized or newly evolved pathogen or the appearance of a known pathogen in a geographical area where it has never been recognized before. Re-emergence similarly refers to known pathogens whose incidence in a given geographical area (ranging from a county to the world in the case of pandemics) is significantly increasing or whose ecology is newly altered, enhancing its potential to cause human disease (e.g. by using novel hosts, jumping from wild animals to domestic ones, exhibiting a wider vector range, and so on).

The bacterial infections that contribute most to human and animal diseases are usually due to MDR strains such as penicillin-resistant *Streptococcus pneumoniae*, vancomycin-resistant enterococci, methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin resistant *S. aureus* (VRSA), glycopeptide intermediate *S. aureus* (GISA), vancomycin insensitive *S. aureus* (VISA), vancomycin-resistant *Enterococcus* (VRE), linezolid-resistant *Enterococcus* (LRE), extended spectrum beta-lactamase (ESBL) producing *Klebsiella* spp., *Pseudomonas aeruginosa*, *E. coli*, *Serratia* spp, *Citrobacter* spp., metallo-beta-lactamase producing *Pseudomonas aeruginosa*, MDR-*Acinetobacter baumannii*, MDR-*Stenotrophomonas maltophilia*, MDR-salmonellae, and MDR-*Mycobacterium tuberculosis*, *Pseudomonas* and *E. coli*.

There is no place on the earth surface where potentially dangerous drug resistant bacteria have not reached. They have infiltrated even into wilderness of virgin and barren islands including the Arctic region. After discovery of antimicrobial agents in the first half of 20th century, clinicians felt relieved as these wonder drugs substantially reduced the threat of infectious diseases. Over the years, antimicrobials have saved lives, eased the suffering of millions of people. And have contributed to the major gains in life expectancy. However, these wonder drugs have started to lose ground rapidly. With each application of antibiotic to kill bacteria a new micro environment is created where the sensitive microbes get killed but the resistant organisms start to flourish. New selection pressure each time leads to rapid evolution in bacteria. As a result, now almost all important infection causing bacteria are armoured to survive in antibiotic loaded environment with much deadlier infective power.

Although a range of more than 150 antibiotics (some naturally produced and others synthesized) is available, bacteria resistant to almost all drugs are prevalent in the environment. The emergence of antibiotic drug resistance (ADR) in the bacteria of human and animal health significance and those present in the environment constitute a global problem. However, it is of much concern in the developing countries where the

poor patients cannot afford too many or newer antibiotics. The problem is much more severe with livestock the owners because most of them are fighting with poverty and have no resource other than few animal heads as resource. Monitoring of ADR is essential to assess and control the burden of multiple drug resistant (MDR) pathogens that are responsible for serious respiratory, uro-genital tract or gastrointestinal tract infections and febrile illness claiming many lives in early age. However, the antibiotic sensitivity testing is expansive making it less practicable in the developing countries. Moreover, the reliability of ADR testing carried out in ill-equipped laboratories is also doubtful.

Antibiotic Drug Resistance (ADR)- ADR is the ability of bacteria to withstand the effects of antibiotic(s). It may evolve naturally via natural selection under antibiotic pressure through random mutation or just induction or through genetic engineering by applying an evolutionary stress on a population. Once ADR is generated, bacteria can transfer the genetic information in a horizontal fashion (between individuals of same or other groups in close vicinity) and vertically to their progeny. If a bacterium carries several resistance genes, it is called multi-resistant or, informally, a superbug or epidemic clone. The epidemic clones of bacteria may be resistant to just one common antibiotic of choice for treatment or a few commonly used antibiotics (multiple drug resistant) or to all the available drugs. These spread rapidly and are difficult to control.

ADR may be of two major types, community acquired (CA) and hospital acquired (HA). Hospital acquired ADR is usually more dangerous and often the root of most of the epidemic clones. The infection by such pathogens is not easy to treat and control. Although both CA as well as HA- ADR may lead to the emergence of epidemic infective clones, CA-ADR clones are much faster in spreading and cause rapidly progressive fatal diseases.

Antimicrobial Drug Use- Appropriate antimicrobial drug use is unquestionably useful in therapeutics, but veterinarians use these wonder drugs most of the time inappropriately. Examples of inappropriate drug use includes prescribing antimicrobial drugs to treat non-bacterial infections, using inadequate criteria for diagnosis of infections, unnecessarily prescribing expensive, broad-spectrum agents, new in market and drugs reserved for human use and not following proper dose and schedule. Antibiotics are used in animals for three major reasons:

- A. In high doses for short periods to treat sick animals.
- B. In high doses for short periods to prevent diseases (after weaning, or during transport). This use “usually involves treating a whole herd or flock, which increases the likelihood of selecting for organisms that are resistant to the antibiotic.”
- C. Antibiotics are commonly given in the feed at low doses for long periods to promote the growth of cattle, poultry, and swine (How? No one knows exactly, 1-15% improvement in production, poorer the hygiene better the effect).

A=17%, B+C=83% (about ten million Kg alone in USA, i.e., 70% of total antibiotic production).

A+B+C=87% of total antimicrobials used on earth (human, agriculture, aquaculture, veterinary, research etc.), 40% of it is added in to feed.

Problems Associated With Antimicrobial Drug Use- The common problems associated with antimicrobial drug use are associated with **a.** Reactions due to toxic properties of antibiotics, **b.** Hypersensitivity reactions and **c.** Super-infection (or also called Supra-infection). These side effects may lead to:

1. Change in microbial education and fine tuning of the immune system.
2. With increase in use of antibiotics the following disorders have shown an increasing trend:
 - a. Asthma, autoimmune disorders and allergies (penicillins, carbapenems, cephalosporins and sulpha drugs),
 - b. Mutagenicity and carcinogenicity (sulphamethazine, oxytetracycline, furazolidone)
 - c. Nephrotoxicity (aminoglycosides, penicillins, sulphonamides)
 - d. Bone-marrow toxicity (chloramphenicol, tetracyclines, linezolid, sulphonamides)
 - e. Hepatotoxicity (macrolides, telithromycin, penicillins)
 - f. Brain and Nerve damage (carbapenems, penicillins, polypeptides, quinolones, fluoroquinolones)
 - g. Hearing impairment –ototoxicity (aminoglycosides)
 - h. Visual disturbances (telithromycin)
 - i. Foetotoxic (tetracyclines)
 - j. Tendinosis (quinolones), kills your athletic carrier.
 - k. Photosensitivity (sulphonamides, tetracyclines)
 - l. Change in Taste (metallic by metronidazole, bitter by tinidazole)
 - m. Multiple sclerosis
 - n. Soaring obesity rates around the globe
 - o. Increased cases of mental illness
 - p. Insulin-dependent diabetes mellitus.
3. Reproductive disorders: Wassenaar (2011) explained how the antibiotic use alters intestinal bacterial microflora of a fruit fly which in-turn drives mating preference. If this applies to Humans! Animals! What may be the fate of different species and biodiversity on earth?
4. Gastro-intestinal problems
 - a. IBD (Krohn's disease)
 - b. Peptic ulcers
 - c. Antibiotic associated pseudo-membranous colitis (AAPMC) due to *Clostridium difficile* (common in horses).
 - d. Antibiotic-associated diarrhea (AAD); although all antibiotics may be associated with AAD, cephalosporins, extended spectrum penicillins (β -lactams), and clindamycin are the main.

Inappropriate, indiscriminate and widespread antibiotic use establishes a selective pressure, a driving force in evolution of antibiotic drug resistance (ADR). Therefore, for better therapeutic affectivity some guidelines developed on the basis of routine monitoring of drug resistance pattern in clinical and environmental strains must be followed. The selection of antibiotic panel for therapeutic use should be based on commonly observed susceptibility patterns, and should be revised periodically.

Methods of Susceptibility Testing- There are several methods of testing susceptibility of bacteria to antibiotic but Disk Diffusion Assay developed by Kirby Bauer is one the most acceptable and practiced. We must remember while testing bacteria that there are several factors which may affect the interpretation of the data viz., media, growth conditions, divalent ions in media, pH of media etc.

Problems with Antibiotic Sensitivity Testing Include-

1. **Sulphamethoxazole, trimethoprim and combinations of both-** Media containing thymidine allow bacteria to grow even when they are sensitive. Combination discs containing drugs in 19:1 ratio (19 sulpha: 1 trimethoprim) may give erroneous results. Both drugs must be tested separately particularly for UTI isolates, so that Trimethoprim can be utilized as the option with less toxicity in UTI.
2. **Polymyxins and colistin-** These drugs diffuse poorly so interpretation following general rule with disc diffusion method may yield wrong results. On using *Escherichia coli* NCTC 10418 test are interpreted always with caution. Inhibition zone not more than 3 mm smaller in diameter than for *Escherichia coli* NCTC 10418 or zones bigger or equal to reference strain is considered sensitive otherwise classified as resistant.
3. **Ciprofloxacin and related drugs-** 1 µg discs are preferred. For sensitive strains with inhibition zone not more than 7 mm smaller in diameter than for *Pseudomonas aeruginosa* NCTC 10662, or *Staphylococcus aureus* (Oxford) NCTC 6571 or zones bigger or equal to reference strain is considered sensitive otherwise classified as resistant.
4. Disc diffusion method is not reliable for- Fungi, Mycobacteria and other slow growing pathogen. Preferably antibiotics are incorporated into solid media and MIC is calculated using radiometric measurement of liberation of C¹⁴ labeled CO₂ released during growth.
5. Testing strains that fail to grow satisfactorily- Certain fastidious bacteria such as *Haemophilus* spp., *Neisseria gonorrhoeae*, *Brucella* spp, *Streptobacillus moniliformis*, *Streptococcus pneumoniae*, and viridans and β-haemolytic streptococci do not grow sufficiently on unsupplemented MHA. These organisms require supplements or different media to grow, and they should be tested on the media suitable (Chocolate MHA or Brain heart infusion agar or *Brucella* base agar) for their growth.

Antimicrobial Discs- The readymade antimicrobial discs can be obtained from several manufacturers (Hi-Media, Oxoid, BBL diffco etc) or can be prepared using Whatman filter paper no. 1. To prepare discs approximately 6 mm in diameter discs are cut, placed in a Petri dish and sterilized in a hot air oven. The loop used for delivering the antibiotics is made of 20 gauge wire and has a diameter of 2 mm. This delivers 0.005 ml of antibiotics to each disc.

Storage of Commercial Antimicrobial Discs- Discs should be stored as follows: Refrigerate the containers at 8°C or below, or freeze at -14°C or below, in a nonfrost-free freezer until needed. Sealed packages of disks that contain drugs from the β-lactam class should be stored frozen, except for a small working supply, which may be refrigerated for at most one week. Some

labile agents (e.g., imipenem, cefaclor, and clavulanic acid combinations) may retain greater stability if stored frozen until the day of use.

Turbidity standard for inoculum preparation: To standardize the inoculum density for a susceptibility test, a BaSO₄ turbidity standard, equivalent to a 0.5 McFarland standard or its optical equivalent (e.g., latex particle suspension), should be used.

Disc Diffusion Methods- *The Kirby-Bauer and Stokes' methods are usually used for antimicrobial susceptibility testing, with the Kirby-Bauer method being recommended by the CLSI.*

E-Test- E-test can be used to determine minimum inhibitory concentration (MIC). It is done after determining the sensitivity through disc diffusion method. E-strips are filter paper strips adsorbed with gradient of antimicrobial, and available from different producers. The test is done in the same way as the disc diffusion method using the desired strip.

Reference strains for quality control

Escherichia coli ATCC 25922 (beta-lactamase negative)

Escherichia coli ATCC 35218 (beta-lactamase positive)

Staphylococcus aureus ATCC 25923 (beta-lactamase negative, oxacillin susceptible)

Staphylococcus aureus ATCC 38591 (beta-lactamase positive)

Pseudomonas aeruginosa ATCC 27853 (for aminoglycosides)

Enterococcus faecalis ATCC 29212 (for checking of thymidine or thymine level of MHA)

Haemophilus influenzae ATCC 49766 (for cephalosporins)

Haemophilus influenzae ATCC 10211 (for medium control)

Neisseria gonorrhoeae ATCC 49226

Stock cultures should be kept at -70°C in Brucella broth with 10% glycerol for up to 3 years. Before use as a QC strain, the strain should be subcultured at least twice and retested for characteristic features. Working cultures are maintained on TSA slants at 2-8°C for up to 2 weeks.

Therapeutic Use of Antimicrobial Drugs- Outcome of antibiotic treatment depends on:

1. Susceptibility of the disease-causing microbe to the drug used and its attained tissue concentration during treatment at the site of predilection.
2. State of host defense, bodily conditions of the host
3. Nature and severity of the infection.
4. Duration of treatment
5. Physical removal of the necrotic tissue, pus or foreign bodies.

Suggested Battery of Antibiotics for Susceptibility Testing

Gram positive bacteria

Gram negative bacilli

Penicillin	Ampicillin
Oxacillin	Piperacillin
Cephalothin	Cephalothin
Gentamicin	Cefotaxime
Netilmicin	Ceftazidime
Amikacin	Gentamicin
Chloramphenicol	Netilmicin
Tetracycline	Amikacin
Erythromycin	Chloramphenicol
Co-trimoxazole	Tetracycline
Clindamycin	Co-trimoxazole
Ofloxacin	Nalidixic Acid
Rifampicin	Ciprofloxacin
Vancomycin	Ofloxacin
Teicoplanin	Nitrofurantoin
Amoxiclav	Imipenem
Ampicillin+sulbactam	Meropenem

How to Decide the Dose of Antibiotic Drug?- Susceptibility of the disease-causing microbe, highly susceptible bacteria at nonspecific sites can easily be cured with low doses given at long intervals too, but for less susceptible bacteria large and more frequent doses are needed.

1. Site of infection, when infection site is such where drugs cannot penetrate easily large doses are required.
2. Host toxicity of the drugs. Some drugs as Aminoglycosides and polymyxins have tissue toxicity and their toxicity level decides their dose.
3. Half-life of the drug in host, it depends on host's excretory system and care should be exercised according to patients' liver and kidney functions.

For How Long One should Administer Antibiotic Therapy-

1. It depends on kind of infection and site of infection.
2. In general, one should wait for 2 days for effect of therapy, if effective continue for 2-3 days otherwise reconsider diagnosis and treatment.
3. Where humoral immunity is important treatment of standard duration is sufficient but the infections associated with intracellular parasitism and cell mediated immunity require prolonged treatment (7-10 days).

Why Antibiotic Therapy Fails?

1. Selection of inappropriate antibiotic due to
 - a. Misdiagnosis
 - b. Failure to culture microbe
 - c. Inaccurate laboratory results of testing

- d. Sampling errors
2. Resistance of pathogen
3. Intracellular parasitism by bacteria
4. Metabolic state of pathogen
5. Inadequate doses or low bio-availability at the site of infection
6. Persistence of foreign body, necrosed tissue etc.
7. Impaired host defenses due to a. Neoplasia, b. Immunodeficiency diseases, c. Radiotherapy, d. Corticosteroids therapy etc.

Should we Use Corticosteroids with Antibiotics?- It is a controversial issue and use of corticosteroids should be carefully decided as corticosteroids affect both specific and nonspecific host defenses by:

1. Suppressing inflammation
2. Impairing phagocytosis
3. Delaying healing
4. Reducing fever
5. Impairing immune response

Therefore, Use of Corticosteroids is Justifiable only when-

1. Severe, life threatening bacteremia, septicemia, endotoxemia, septic shock
2. Severe, acute and extensive local infection so that lysosomal enzyme release from neutrophils can be prevented to inhibit further tissue destruction as in HS.
3. Cerebral edema complicating meningitis.
4. Short acting steroids should not be used for >3 to 5 days.

Spectrum of Activity of Common Antibacterial Drugs Against Microbes

Antibacterial drug	Bacteria	Mycoplasma	Rickettsia	Chlamydia	Protozoa
Aminoglycosides	1	1	0	0	0
Chloramphenicol	1	0	0	0	0
Beta-lactams	1	0	1	1	0
Lincosamides	1	1	0	0	1
Macrolides	1	1	0	1	0
Pleuromutilins	1	1	0	1	0
Tetracyclines	1	1	1	1	0
Quinolones	1	1	1	1	0
Sulphonamides	1	1	0	1	1
Trimethoprim	1	0	0	0	1

1= effective, 0= not effective

Spectrum of Activity of Common Antibacterial Drugs Against Bacteria

Name of antibiotics	Aerobic Bacteria		Anaerobic bacteria	
	Gram +ve	Gram -ve	Gram +ve	Gram -ve
Azlocillin, cefoxitin, chloramphenicol, fluoroquinolones, imipenem, moxalactam, tetracyclines	1	1	1	1
Carbenicillin, cefoperazone, cefotaxime, ceftraxone, cephalosporins, clavulanic acid, subactam potentiated penicillins	1	1	1	2
Ampicillin, amoxycillin	1	2	1	2
Aztreonam, mecillinam, polymyxins, cefsulodin	0	1	0	0
Benzyl penicillin	1	2	1	2
Aminoglycosides, spectinomycin, sulphonamides, trimethoprim	1	1	0	0
Lincosamide, macrolides, pleuromutilins, spiramycin, vancomycin	1	0	1	1
Bacitracin	1	0	1	0
Nitroimidazole	0	0	1	1

1= effective, 0= not effective, 2= variable results

Clinically Useful Rational Combinations of Antibacterial Drugs

Drug combinations	Indications	Interaction of drugs
Ampicillin + clavulanic acid Amoxycillin + clavulanic acid	Cystitis caused by <i>E. coli</i> and <i>Proteus</i> spp.; <i>S. aureus</i> mastitis in bovines	Synergistic combination
Ampicillin + subactam Amoxycillin + subactam	Bovine-pasteurellosis (shipping fever)	Synergistic combination
Amoxycillin + gentamicin	Severe infection of unknown origin	Synergistic combination
AmphotericinB + flucytosine	Cryptococcal meningitis	Synergistic, decreased toxicity
Erythromycin + rifampin	<i>Rhodococcus equi</i> pneumonia in foals	Synergistic combination
Gentamicin + clindamycin	Peritonitis after intestinal perforation	Broad spectrum activity
Minocycline + streptomycin	<i>Brucella canis</i> infection in dogs	Synergistic combination
Suphamethox + trimethoprim	Coliform meningitis	Synergism and good CSF penetration

Adverse Interaction of Antibiotics and Other Drugs *in vivo*

Antimicrobial drug	Interacting drug	Adverse effect
Aminoglycosides	Cephaloridine, cephalothin, polymyxins, furosemide, amphotericin B	Nephrotoxicity
	Polymyxins, curare-like drugs, anesthetics	Neuromuscular blockade
Chloramphenicol	Dicoumarol, barbiturates	Prolonged anesthesia and anticoagulation effect
Griseofulvin	Dicoumarol, barbiturates	Reduced anticoagulant effect
Lincosamin	Kaoline-pectate	Decreased absorption of lincosamin
Monensin	Tiamulin	Neurotoxicity
Rifampin	Theophylline	Enhanced theophylline clearance
Sulphonamides	Oral anticoagulants	Prolonged anticoagulation effect
Tetracyclines	Barbiturates, oral iron, Calcium, magnesium	Decreased tetracycline absorption

Incompatibility of antibiotics with other medications

Antibiotic	Incompatibility
Benzyl penicillin G	Normal saline, dextrose saline
Ampicillin	Normal saline, dextrose saline
Tetracyclines	Any solution containing calcium or magnesium

Bactericidal drug should be preferred to bacteriostatic drugs by Clinician:

1. When host defenses are weak or impaired,
2. In case of serious infections as meningitis, endocarditis, osteomyelitis
3. In immunodeficient or immunosuppressed animals
4. When corticosteroids are necessary to be used.

Contraindicated Antibiotics in Animals

Animal	For oral use
Horses	Tetracyclines, Chloramphenicol, Lincomycin, Clindamycin, Tylosin, Tiamulin, Spiramycin, Ciprofloxacin, Norfloxacin, Enrofloxacin, Tilmicosin, Erythromycin (in adults), cefotaxime, ceftriaxone, ceftiofur, cefoxitin, latamoxef (moxalactam), Imipenem, neomycin, gentamicin in foals (by injection also), amphotericin B (should not be used locally)

Dogs & cats	Tobramycin, spectinomycin, Gentamicin (neuromuscular & renal toxicity), phenylbutazone, chloramphenicol and griseofulvin, cefachlor/ cefadroxil/ cephalixin/ cepharadine (cause diarrhoea and vomiting), eneroxacin/ ciprofloxacin/ norfloxacin (may induce diarrhoea and vomiting, nephroticity, must be avoided in pregnant and neonates), erythromycin/ clindamycin/ lincomycin (causes vomiting, diarrhoea and hepatotoxicity), metronidazole (avoid in early pregnancy, anorexia, vomiting, neuro problems), nitrofurans and rifampin (hepatotoxic, discolour tears and urine, avoid during pregnancy), ketoconazole, itraconazole (anorexia, depression, diarrhoea, vomiting, coat change, in cats fever, anemia)
Ruminants	Antibiotics should not be used orally and those antibiotics have hepatobiliary excretion should be avoided by injections too. Tetracyclines, sulfonamides and coccidiostats can be used orally. Monensin, a coccidiostat, has often problem with uniform mixing in feed thus poisoning may result.
Swines	Sulfamethazine (due to residue in meat), monensin, salinomycin, arsanilic acid (causes muscle tremors, listlessness)
Poultry birds	Furazolidones, Nitrofurans, Apramycin, dimetridazole, ronidazole (Not approved for food animals)
Guinea pigs Rabbit Hamsters	Penicillins, tetracyclines, clindamycin or lincomycin causes Enteritis, profuse diarrhoea and death in 4-5 days, enterotoxemia due to Clostridial-infection. In rabbits diarrhoea due to <i>Clostridium spiroformae</i> .
Mice & gerbil	Dihydro-Streptomycin (Cardiovascular effect death within minutes)
Pet birds	Tetracyclines during egg production and breeding season results in to soft shelled eggs and rickets in chicks

Recommended Antibiotics in Horses

Diagnosis/ disease	Drugs recommended
Endometritis	Amikacin 2 gm in 200 ml saline intrauterine (iu) Ampicillin 1-3 g in 200 ml saline iu (may be irritating) Gentamicin sulfate 0.5- 3 g in in 200 ml carbonate buffer iu Kanamycin sulfate 1-3 g in 200 ml saline iu (kills sperms) Neomycin 3-4 g in 200 ml saline iu Clotrimazole 400-600 mg in 200 ml saline iu for fungal endometritis.
Retained placenta and placentitis	Streptopenicillin, gentamicin, amikacin, ceftiofur, cefotaxime (iu) Ampicillin. Trimethoprim sulfamethoxazole by injection
Mastitis	Penicillin G and gentamic/ kanamycin for bacterial intramammary Tetracycline for mycoplasmal intramammary

Balanoposthitis or penis infections	Polymyxin B 1 mg/ ml of cream, gentamicin dissolved in carbonate buffer or amikacin 0.1 mg/ ml in cream (these drugs should also be used in semen extenders to use semen of such stallions.
Orchitis/ epididymitis	Strepto-penicillin (injection), Trimethoprim-Sulfa (oral/ injection)
Upper respiratory infections (cough, sinusitis, strangles, guttural pouch infections)	Procain penicillin G (injection), Trimethoprim-Sulfa (oral/ injection), Metronidazole (injection as well as local), Povidone-iodine for local application, lavage etc.
Pneumonia and pleuropneumonia	Streptopenicillin (inj), Rifampin and erythromycin (in foals only in Rhodococcal infection, gentamicin/ amikacin+ metronidazole, ticarcillin+ clavulenic acid, ceftiofur/ cefotaxime (iv)+ metronidazole, chloramphenicol (only as last resort)
Fungal infections (candidiasis, ring worms)	Amphotericin B (150 mg iv in 1 lt dextrose with 50 mg increment every 2 nd day till 400 mg is given up to 30 days) or Ketoconazole, fluconazole Captan (50 g of 50% powder/ in 5 lt water) or lime sulfur, topical hypochlorite or povidone iodine, apply for 3 days daily then weekly.
Diarrhoea/ gastroenteritis	Trimethoprim-sulfa (oral/ inj)Amikacin/ gentamicin with ceftiofur/ cefotaxime
Rhodococcus equi infection	Erythromycin+ rifampin
Local or systemic mycosis/ candidiasis	Fluconazole/ ketokonazole with rifampin
Fistulous withers	Oxytetracycline+ streptomycin or trimethoprim-sulfonamide, tetracycline with gentamicin or rifampin
Omphalophlebitis	Ampicillin+kanamycin/ gentamicin , ticarcillin+clavulenic acid, or ceftiofur/ cefotaxime
Clostridial infection	Systemic penicillin G and metronidazole
Abscess	Streptopenicillin with metronidazole or rifampin, chloramphenicol, trimethoprim+ sulfonamide
Potomac horse fever (<i>Ehrlichia risticii</i>)	Tetracycline, rifampin
Tyzzar's disease (<i>B. piliformis</i>)	Penicillin G, tetracycline, erythromycin or streptomycin
Burns	Topical: silver sulfadiazine, ticarcillin+ clavulenic acid
Septicemia	Ampicillin+ gentamicin/ kanamycin
Septic arthritis/ polyarthritis	Ampicillin+gentmicin/ kanamycin, with or without metronidazole, ticarcillin+ clavulenic acid, Ceftiofur/ cefotaxime+ metronidazole or chloramphenicol.

Rain Dermatitis/ streptotrichosis/ folliculitis, Furunculosis,	rot/	Procaine penicillin G, ampicillin or cephalexin/ cephalothin with gentamicin
Sporotrichosis (S. schenckii)	(S.	Sodium iodide 40mg/kg for 2-5 days then potassium iodide 2mg/ kg OD orally for several weeks or Ethylene diamine dihydro-iodide 1mg/kg BID, for 7 days
Phycomycosis (swamp cancer, bursatte, horse leech, coast fungus		Amphotericin B, 150 mg OD iv in 1 lt 5% dextrose with an increment of 50 mg every 2 days until a dose of 400 mg is achieved for 30 days, or potassium iodide treatment as above, or Etisazole with dimethyl sulfoxide applied topically
Bacterial meningitis		Ceftiofur, cefotaxime, orally trimethoprim+sulfa
Mycotic meningitis		Amphotericin B+ flucytosine or ketoconazole +flucytosine
Bacterial keratitis		Subconjunctival gentamicin, tobramycin, chloramphenicol, cephalexin
Fungal keratitis		Amphotericin B+ miconazole for topical application
Eye injury/ perforation with foreign object		Topical gentamicin/ tobramycin, cefazolin/ chloramphenicol with injection of chloramphenicol, trimethoprim+sulfa

Antibiotics Recommended for Use in Canines

Disease/ syndrome	Recommended drugs
Pyoderma (deep/ superficial)	Oxacillin, cloxacillin, dicloxacillin, flucloxacillin, lincomycin, clindamycin, tylosin, erythromycin, Amoxy-clav, chloramphenicol, trimethoprim-sulfa
Ringworms/ dermatophytosis	Griseofulvin, ketoconazole
Wounds and trauma, bite wounds	Oxacillin, cloxacillin, dicloxacillin, flucloxacillin, penicillin G, amoxy-clav
Anal sac inflammation/ abscess	Amoxy-clav, chloramphenicol
Otitis externa	Topical: Neomycin/ framycin, polymyxin B, chloramphenicol, nystatin, miconazole or cuprimycin for Malassezia.
Otitis media	Amoxy-clav, chloramphenicol, ticarcillin, cipro/enero/ norfloxacin, ketoconazole
Penetrating-eye-wounds, infection	Topical: Neomycin, polymyxin, bacitracin, gentamicin, Episcleral: Chloramphenicol, amoxy-clav, gentamicin
Bacterial rhinitis	Amoxycillin, trimethoprim+ sulfa
Fungal rhinitis	Topical: Enilconazole, itraconazole, ketoconazole
Infectious Tracheobronchitis	Tetracycline, chloramphenicol, trimethoprim-sulfa
Bacterial pneumonia	Amoxycilav, trimethoprim-sulfa, penicillins, cephalosporin+ gentamicin, amikacin
Peumocystis pneumonia	Trimethoprim-sulfa, pentamidine
Purulent pleuritis/ pyothorax	PenicillinG, amoxycillin/ ampicillin/ hetacillin, amoxy-clav, chloramphenicol

Periodontitis/ gingivitis, molar abscess or carnassial abscess, ulcerative stomatitis, tonsillitis	Amoxycillin/ ampicillin/ hetacillin, clindamycin, lincomycin
Enterocolitis, enteritis	Chloramphenicol, trimethoprim-sulfa, erythromycin, tylosin, tetra/oxytetra/doxy/ minocycline, amoxyclav, cephalosporin + gentamicin, amikacin, tobramycin
Giardiasis	Metronidazole, tinidazole, furazolidone, quinacrine
Coccidiosis	Trimethoprim+sulfa, pyrimethamine, furazolidone, amprolium
Parvoviral enteritis	Amoxycillin/ ampicillin/ hetacillin ±gentamicin, amikacin, tobramycin amoxyclav
Colitis	Metronidazole, erythromycin, tylocin, chloramphenicol, amoxycillin, hetacillin
Cholecystitis/ cholangitis, peritonitis, abdominal sepsis	Amoxyclav, amoxycillin/ ampicillin/ hetacillin ±gentamicin, amikacin, tobramycin, clindamycin
Cystitis, pyelonephritis,	Amoxyclav, amoxycillin/ ampicillin/ hetacillin, trimethoprim+sulfa, enrofloxacin, norfloxacin, ciprofloxacin, cephalexin
Prostatitis, orchitis, epididymitis	Trimethoprim-sulfa, chloramphenicol, amoxyclav, erythromycin, lincomycin, tylosin, clindamycin, doxycycline+gentamicin (for brucellosis)
Metritis, pyometra	Amoxyclav, ampicillin/amoxycillin/ hetacillin+gentamicin/amikacin/ tobramycin
Mastitis	Chloramphenicol, gentamicin/amikacin/ tobramycin, erythromycin, tylosin amoxyclav (if pH is >7.4), trimethoprim-sulfa (pH <7.3)
Osteomyelitis discopondylitis, septic arthritis	Amoxyclav, Oxacillin, cloxacillin, dicloxacillin, flucloxacillin ±gentamicin, amikacin, tobramycin, clindamycin, doxycycline+gentamicin (for brucellosis)
Bacterial meningitis	Amoxyclav, trimethoprim-sulfa
Cryptococcal meningoencephalitis, hepatic encephalopathy	Amphotericin B +flucytosine/ fluconazole, itraconazole, metronidazole, tetracycline+metronidazole, ampicillin/amoxycillin/ hetacillin
Actinomycosis	Penicillin G, chloramphenicol, erythromycin
Brucellosis	Doxycycline/ minocycline +gentamicin, dihydrostreptomycin
Leptospirosis	Dihydrostreptomycin, ampicillin/amoxycillin/ hetacillin, tetracycline, minocycline
Nocardiosis	Trimethoprim-sulfa, erythromycin
Tuberculosis	Rifampin+isoniazid, isoniazid+ dihydrostreptomycin/ ethambutol
Septicemia	Amoxyclav, Oxacillin, cloxacillin, dicloxacillin, flucloxacillin ±gentamicin, amikacin or gentamicin+cefazolin/ cephapirin/ cephradine

Babesiosis & hepatozoonosis	Imidocarb, dimiazene, phenamidine, primaquine
Leishmaniasis	Meglumine antimonite, sodiumstibogluconate
Neosporosis, toxoplasmosis	Sulfa drugs + pyrimethamine, clindamycin
Trypanosomiasis	Diminazene+difluoromethylornithine, nifurtimox
Bartonella, Rickettsia, chlamydia, mycoplasma infections	Tetracycline, oxytetracycline, doxycycline, minocycline, chloramphenicol, thiacetarsamide, erythromycin, tylosin, lincomycin, clindamycin
Aspergillosis, blastomycosis	Itraconazole, fluconazole, erythromycin, tylosin, lincomycin, clindamycin
Cryptococcosis, histoplasmosis	Itraconazole/ fluconazole±amphotericin B
Coccidiomycosis	Ketoconazole, Itraconazole/ fluconazole±amphotericin B

Drugs Recommended in Cats/ Felines

Pustular dermatitis, Furunculosis, cellulites, folliculitis, cat fight absces	Penicillin G/ V (oral), lincomycin, clindamycin, amoxicillin/ amoxiclav, chloramphenicol
Dermatophytosis/ ringworms	Griseofulvin/ ketoconazole
Leprosy (M. lepraemurium)	Clofazimine, rifampin & dapsone
Atypical mycobacterial granuloma	Amikacin, chloramphenicol, erythromycin, clofazimine, rifampin & dapsone
Otitis (Malassezia, Staphylococcus, Streptococcus, Pasteurella)	Topical application of neomycin, framycin, gentamicin, polymyxins, chloramphenicol, nystatin/ miconazole/ cuprimyxin/ thibendazole (Malassezia)
Conjunctivitis (Chlamydia, Mycoplasma)	Topical tetra/ oxytetra/ chlortetracycline, chloramphenicol, Idoxuridine for early herpes
Penetrating eye wounds	Amoxyclav with topical or episcleral gentamicin, chloramphenicol by all route
Bronchitis, rhinitis, simple/ aspiration pneumonia, pyothorax	Amoxyclav, amoxicillin, trimethoprim-sulfa, chloramphenicol, metronidazole, clindamycin long acting penicillin.
Cryptococcal rhinitis/ pneumonia	Fluconazole, itraconazole or ketoconazole + flucytosine
Periodontitis, ulcerative stomatitis	Penicillin G, amoxyclav ± metronidazole
Enteritis, cholecystitis, cholangiohepatitis, abdominal sepsis	Amoxyclav, erythromycin, chloramphenicol, amikacin, tobramycin, streptomycin, trimethoprim+sulfa
Giardiasis	Metronidazole, tinidazole, furazolidone, quinacrine
Coccidiosis	Sulfonamides, trimethoprim+sulfa, furazolidone

Meningitis	Amoxiclav, cephalosporins, fluconazole/ amphotericin B+flucytosine (for cryptococcal meningitis)
Hepatic encephalopathy	Metronidazole, neomycin, amikacin, tobramycin, streptomycin, amoxyclav
Babessiosis (Babesia felis)	Primaquin
Cytauxzoonosis (C. felis)	Parvoquine and fluid therapy
Toxoplasmosis	Clindamycin, pyimethamine + sulfadimidine/ sulfadiazine/ sulfadimethoxine
Feline infectious anemia (haemobartonellosis)	Tetracycline, oxytetracycline, doxycycline, minocycline, chloramphenicol
Mycoplasmosis, Chlamydiosis (C. psittaci), ocular infection, rhinitis, pneumonia	Topical oxy/chlor tetracycline for conjunctivitis and systemic for pneumonia, chloramphenicol. Erythromycin, clindamycine

Doses of Drugs Used Orally in Dogs and Cats

Drugs	Doses	Comments
Penicillin V, oxacillin, cloxacillin	25-50 mg/kg, TID	Reduce food consumption
Aoxicillin, amoxiclav, ampicillin, carbenicillin, hetacillin	12.5-30 mg/kg, TID	Reduce food consumption
Cefachlor, cefadroxil, cephalixin, cephadrine	15-30 mg/kg, TID	Cause vomiting, diarrhoea
Dihydrostreptomycin, kanamycin, neomycin	10-12 mg/ kg, TID, QID	Renal, vestibular & auditory toxicity, avoid in preg. & infants
Ciprofloxacin, enrofloxacin, norfloxacin	5-15 mg/kg, BID	Cause vomiting, diarrhoea
Chloratetra, doxycycline, minocycline, oxytetra, tetracycl	5-20 mg/kg TID/ BID	Anorexia, emesis, diarrhoea
Clindamycin, erythromycin, tylosin	10 mg/ kg, TID/ BID	Anorexia, emesis, diarrhoea
Sulfadiazine, sulfadimethoxine, tripplesulfa	50-100mg/kg, BID/ OD	Anorexia, emesis, diarrhoea
Sulfadiazine-trimethoprim, sulfamethoxazole-trimethopri	30 mg/kg, BID/ OD	Anorexia, emesis, diarrhoea
Chloramphenicol, dapsone, quincrine, fluconazole	50-60 mg total for cats and 50 mg/ kg for dogs	Anemia
Amprolium	Total 100-400 mg to dogs, 60-100 mg to cats	Anorexia, dpression, CNS signs, thiamine deficiency
Furazolidone	8-10 mg/kg BID for 7 d	

Metronidazole	15-30 mg/kg BID/ OD	
Fluconazole, itraconazole, ketoconazole, thiabendazole	5-10 mg/kg BID	Anorexia, emesis, diarrhoea
Flucytosine, griseofulvin	50-60 mg/kg, BID	Anorexia, anemia, diarrhoea

Doses of Drugs Used Through IV/ IM/ SC Injections in Dogs and Cats

Drugs	Doses mg/kg,	Comments
Penicillin G (QID), procaine (OD)	20000-40000	Reduce food consumption
Cloxacillin, oxacillin, dicloxacillin, methicillin	25-50, TID/ QID	Reduce food consumption
Amoxycillin, ampicillin, carbenicillin	10-20 TID	Reduce food consumption
Ticarcillin	40-75, QID	Reduce food consumption
Cefazolin, cephapirin, cephradine, cefotaxime, cefamandole, cefoxitin,	15-30 TID/ QID	
Moxalactam	50, TID	
Amikacin, getamicin, kanamycin, streptomycin	10, TID/ QID	
Netimicin, tobramycin	1-2, TID	
Doxycycline, oxytetra, tetracycline	10, BID	
Clindamycin, lincomycin, tylosin	10, TID/ BID	
Chloramphenicol	50, TID/ BID	
Metronidazole	10, TID	

Antibiotics Recommended for Use in Cattle and Buffaloes

Problem	Cause	Recommended drugs
Enzootic calf pneumonia & HS, pneumonic pasteurellosis	Pasteurella	Trimethoprim+sulfadoxime (3 ml/45 kg, im, OD), ceftiofur (1 mg/kg, im OD) , sublactam-ampicillin, tilmicosin (10 mg/kg SC once in 3 d), long acting oxytetracycline ((10-12 mg/kg OD or long acting @ 20 mg/kg, im once in 3d)
CBPP & acute undifferentiated respiratory disease (AUBRD)	Mycoplasma, + Pasteurella	Oxytetracycline Trimethoprim+sulfadoxime, ceftiofur, sublactam-ampicillin
Chronic pneumonia, necrotic laryngitis (<i>F. necrophorum</i>)	Actinobacillus, Haemophilus	Tilmicosin, erythromycin, penicillin G (~45 000 IU/kg im,daily), Oxytetracycline

Acute peritonitis, TRP	<i>F. necrophorum</i> , <i>A. pyogenes</i>	Trimethoprim+sulfadoxime, sublactam-ampicillin, tilmicosin, long acting oxytetracycline
Necrotic-stomatitis, oral necrobacillosis	<i>F. necrophorum</i>	Tilmicosin, penicillin G, Oxytetracycline
Neonatal colibacillosis, salmonellosis	<i>E. coli</i> , <i>Salmonella</i>	Trimethoprim+sulfadoxime, ceftiofur, sublactam-ampicillin
Enterotoxemia	<i>Clostridium spp.</i>	Trimethoprim+sulfadoxime, tilmicosin, long acting tetracycline
Footrot	<i>Bacteroides spp.</i>	Topical copper sulphate solution (2-5%) & trimming of hooves
Interdigital necrobacillosis	<i>F. necrophorum</i>	Trimethoprim+sulfa, sulfamethazine, penicillinG Oxytetracycline
Infectious keratoconjunctivitis	<i>Moraxella bovis</i>	Subconjunctival penicillin + cortisone, oxytetracycline
Omphalitis, omphalophlebitis	<i>Many</i>	Lance and flush the abscess, trimethoprim+sulfa, ampicillin
Ringworms	<i>Trichophyton spp.</i>	4% thibandazole or 2.5% tincture iodine topically, griseofulvin (5 mg/ kg OD for 7 days)
Dermatophilosis	<i>D. congolensis</i>	Penicillin G, Oxytetracycline
Acute septic metritis, endometritis	<i>Many</i>	Trimethoprim+sulfadoxime, sulfamethazine, penicillin G, Oxytetracycline
Polyarthritis & neonatal meningitis, listeriosis	<i>Actinobacillus</i> , <i>E. coli</i> , <i>Mycoplasma</i>	Tilmicocin, Trimethoprim+sulfadoxime, sublactam-ampicillin, oxytetracycline, ceftiofur
Blackleg, malignant edema	<i>Clostridium spp.</i>	Penicillin G, Oxytetracycline
Actinomycosis (lumpy-jaw, wooden tongue)	<i>A. bovis</i>	20% sodium iodide 1 g/ 15 kg, iv once weekly, isoniazid (10 mg/kg od, 28 d)
Cystitis, pyelonephritis, leptospirosis	<i>Coryne. Renale</i> , <i>Leptospira</i>	Penicillin G, Oxytetracycline, Trimethoprim+sulfadoxime
Endocarditis & myocarditis	<i>A. pyogenes</i> , <i>Strep.</i> , <i>H. somnus</i>	Penicillin G, Oxytetracycline, Trimethoprim+ sulfadoxime
Septicemia, haemoglobinuria	<i>Many</i> , <i>Clostridium haemolyticum</i>	Trimethoprim+sulfadoxime, Penicillin G, Oxytetracycline
Anaplasmosis, babesiosis	<i>Babesia</i> , <i>A. marginale</i>	Imidocarb +tetracycline, primaquin, berenil

Antibiotics Recommended for Use in Sheep and Goats

Problem	Cause	Recommended drugs
Enzootic, Listeria, Leptospira, Coxiella, Campylobacter abortions	<i>Chlamydia psittaci</i> , and respective bacteria	Long acting oxytetracycline (20 mg/kg im once in 3d) and 500 mg tetracycline/day/sheep till lambing finishes, vaccination. Streptopenicillin at 10-15 day interval, penicillin G 40000 iu/kg, QID
Salmonellosis and abortions	<i>Salmonella spp.</i>	Ampicillin-sublactam (10 mg/kg OD, im), ceftiofur (1-2.2 mg/kg, im, OD), fluoroquinolones (5-8 mg/kg/day, im), trimethoprim-sulfamehazine (25-30 mg/kg BID, OD)
Metritis, posthitis	<i>A. pyogenes</i> , <i>E. coli</i> , <i>C. renale</i>	Ampicillin-sublactam, ceftiofur, fluoroquinolones, trimethoprim-sulfamehazine, penicillin G
Lab epididymitis	<i>H. somnus</i> , <i>Actino.</i>	Tetracycline
Enterotoxemia/pulpy kidney	<i>C. perfringens</i>	Give no antibiotic but withdraw grains from ration and inject C & D antitoxin, vaccination to risked animals
Omphalophlebitis, polyarthritis	<i>A. pyogenes</i> , <i>E. coli</i> , <i>Erysipelothrix</i>	Penicillin G, Ampicillin-sublactam, ceftiofur, fluoroquinolones, trimethoprim-sulfamehazine
Watery moth in sheep, metabolic-acidosis goats	<i>E. coli</i>	Oral amoxicillin (10-20 mg/kg, im, TID), Ampicillin-sublactam, ceftiofur, fluoroquinolones, trimethoprim-sulfamehazine
Colibacillosis, salmonellosis	<i>E.coli</i> , <i>Salmonella</i>	Ampicillin-sublactam, ceftiofur, fluoroquinolones, trimethoprim-sulfamehazine
Cryptosporidiosis	<i>Cryptosporidium</i>	Sulfonamides oral 50 mg/kg, OD, 100 mg/kg loading dose
Coccidiosis	<i>Eimeria spp.</i>	Monensin (11-22 g/ tonne of feed, lasalocid (30g/ tonne feed), salicinomycin 11-16 g/tonne feed, 6% decoquinate @ 1 kg/tonne feed
Pasteurellosis	<i>P. haemolytica</i>	Long acting (LA) tetracyclines
CCPP, mycoplasmosis	<i>Mycoplasma spp.</i>	Tetracycline, lincomycin or tylocin 20 mg/kg, im, BID,
Pink eye	<i>C. psittaci</i>	Tetracyclines
Lumpy-wool, dermatophytosis	<i>D. congolensis</i>	LA tetracyclines, dusting powdered alum to prevent reinfection

Foot-rot, foot-scald	<i>B. nodosus, F. necrophorum</i>	Zinc-lauryl sulphate (10-20% ZnSO ₄ with 2% Lauryl sulphate as foot bath, penicillin+streptomycin, lincomycin, LA tetracycline
Strawberry foot rot	<i>D. congolensis</i>	LA tetracyclines, dusting powdered alum to prevent reinfection
Gangrenous mastitis	<i>S. aureus, P. haemolytica</i>	Ampicillin-sublactam, ceftiofur, fluoroquinolones, trimethoprim-sulfamethazine
Contagious agalactia	<i>M. agalactiae</i>	Tetracycline, tylocin
Sub-clinical mastitis	<i>S. aureus</i>	Na-cloxacillin, cephalixin (5-10 mg/kg/day, im)

Withholding Times and Milk Discard Times for Intramammary Antimicrobial Drugs Used for Mastitis in Lactating Animals

Drug	Lactating	Dry animal	Milk should be discarded for
Amoxicillin	12 days		60 hr
Cephalexin (Benzethine)		42 days	72 hrs postcalving
Cephalexin (sodium)	4 days		96 hrs
Cloxacillin (Benzethine)		30 days	72 hrs postcalving
Cloxacillin sodium	10 days		48 hrs
Dihydrostreptomycin		60 days	96 hrs postcalving
Erythromycin	14 days		36 hrs
Hetacillin potassium	10		72 hrs
Novobiocin	15 days	30 days	72 hrs
Oxytetracycline	4 days		4 days
Procaine penicillin G	4 days	14 days	60-96 hrs

Withholding Time and Milk Discard Time for Antimicrobial Drugs

Drug	Cattle	Sheep	Swine	Chicken	Turkeys	Milk should be discarded for
Amoxicillin parenteral	25 days					96 hrs
Amoxicillin oral	20 days		15 days			
Ampicillin parenteral	9 days		15 days			48 hrs
Ampicillin oral	20 days		15 days			
Ceftiofur sodium parenteral	0 day					0 hr

Dihydrosterptomycin parenteral	30 days	30 days	30 days			72 hrs
Erythromycin parenteral	14 days	3days	7 days			72 hrs
Gentamicin parenteral			40 days	60 days		
Gentamicin oral			14 days			
Lincomycin parenteral			2 days			
Oxytetracycline parenteral	28 days		28 days	5 days	5 days	
Oxytetracycline oral	7 days					
Procaine penicillin G parenteral	10 days	9 days	7 days			48 hrs
Benzethine penicillin G parenteral	30 days					
Spectinomycin parenteral				0 days		
Spectinomycin oral			21 days			
Sulfachlorpyridazine parenteral	5 days					
Sulfachlorpyridazine oral	7 days		4 days			
Sulfabromomethazine oral	18 days					96 hrs
Sulfadimethoxine oral	12 days					60 hrs
Sulfadimethoxine parenteral	5 days					60 hrs
Sulfaethoxypyridazine parenteral	16 days					72 hrs
Sulfaethoxypyridazine oral	16 days		10 days			72 hrs
Sulfamethazine oral	28 days					
Sulfamethazine parenteral	10 days					
Tetracycline oral	14 days					
Tylosin parenteral	21 days		14 days	3 days	5 days	

Antibiotics Recommended for Use in Swines

Problem	Cause	Recommended drugs
Exudative dermatitis, greasy pig disease	<i>Staphylococcus hyicus</i>	Procain penicillin G (45 000 iu/kg/day, im), lincomycin (11 mg/kg/day, im or 12.5 mg/L water), ampicillin (10-20 mg/kg, TID, im), trimethoprim-sulfa (24-30 mg/kg. Im, BID)
Erysipelas, diamond skin disease	<i>E. rhusiopathiae</i>	Vaccinate, procain penicillin G, Tylosin (20-30 mg/kg, BID/TID, im or 100 g/ton feed or 80 mg/L water), tetracycline (10-20 mg/kg/day, im or 200-800 g/ton feed), lincomycin
Mange and lice	<i>Sarcoptes scabiei</i> , <i>Haematopinus suis</i>	Avermectin (0.3 mg/kg sc injection), organophosphate, chlorinated hydrocarbons
Arthritis, polyarthritis	Many bacteria	Procaine, trimethoprim-sulfa, lincomycin, tylosin, gentamicin (2-4 mg/kg, BID/TID, im), chloramphenicol (10-25 mg/kg, BID, im)
Foot-rot	<i>F. necrophorum</i>	Procaine penicillin G
Meningitis	<i>S. suis</i> , <i>H. parasuis</i>	Procaine (im), penicillin V (oral)
Edema disease	<i>E. coli</i>	Apramycin (10-20 mg/kg, OD/BID, oral or 150 g/ton feed, or 100 mg/ L water), carbadox (250 g/ ton feed), furazolidone (100-500 g/ton feed or 100 mg/L water), trimethoprim sulfa
Otitis media	Many bacteria	Trimethoprim-sulfa, chloramphenicol
Tetanus	<i>Clostridium tetani</i>	Antitoxin with relaxant and procaine penicillin G
Enzootic pneumonia, mycoplasmosis	<i>M. hyopneumoniae</i>	Tiamulin (10-15 mg/kg/day, im or 200g/ton feed or 60 mg/L water), tetracycline, enrofloxacin (2.5-5.0 mg/kg/day, im), lincomycin, tylosin
Pleuropneumonia	<i>Actinobacillus pleuropneumoniae</i>	Chloramphenicol, trimethoprim-sulfa, procaine, tetracyclines, ampicillin (10-20 mg/kg TID, im), spectinomycin ((22 mg/kg/day, im), tiamulin
Atrophic rhinitis	<i>Pasteurella</i>	Sulfamethazine (240 mg/kg/day, 0.4-2kg/ton in feed, 100 mg/L in water), trimethoprim-sulfa, tetracycline, lincomycin
Glassers disease	<i>H. parasuis</i>	Procaine, trimethoprim-sulfa, ampicillin, chloramphenicol, tetracycline

Salmonellosis	<i>S. chleraeusis</i>	All Oral Nitrofurans, chloramphenicol, trimethoprim-sulf, ampicillin, neomycin ¹ (10 mg/kg/day, 40g/ton feed, 100 mg/L water), spectinomycin, gentamicin, apramycin
Leptospirosis	<i>Leptospira spp.</i>	Streptomycin, tiamulin, tetracycline
Brucellosis	<i>B. suis</i>	Tetracyclines, streptomycin, sulfonamides
Pyelonephritis	<i>E. suis</i>	Tetracycline, ampicillin
Mastitis	<i>Many bacteria</i>	Trimethoprim-sulfa, ampicillin, chloramphenicol
Metritis	<i>Many bacteria</i>	Tetracyclines, ampicillin
Colibacillosis	<i>E. coli</i>	Gentamicin, trimethoprim-sulfa, apramycin, neomycin, carbadox
Coccidiosis	<i>I. suis</i>	Sulfonamides, decoquinat, amprolium, tetraturil
Dysentery	<i>S. hyodysenteriae</i>	Carbadox, tiamulin, dimetridazole (100-135 g/ton feed, 250 mg/L water), ronidazole (55g/ton feed, 60 mg/L water), lincomycin
Necrotic enteritis	<i>Cl. perfringens C</i>	Ampicillin/ penicillin V, Bacitracin
Proliferative intestinal adenomatosis	<i>Campylobacter</i> like organism	Tetracycline, dimetridazole, tylosin, carbadox, tiamulin
Eperythrozoonosis	<i>E. suis</i>	Tetracycline, arsenicals

Antibiotics Recommended for Use in Poultry Birds

Problem	Cause	Recommended drugs
Chronic respiratory disease (CRD) & other Mycoplasmosis	<i>Mycoplasma gallisepticum</i> & others	Oral erythromycin (100-185 g/ ton feed, 116 mg/L water), enrofloxacin/ danofloxacin (8 mg/ kg BW), lincomycin (2g/ ton feed or 17 mg/ L water)-spectinomycin (264-532 mg/L water or 10 mg /chick), Tetracyclines (oxy/chlor tetracycline, tera/doxycycline, 25-50 mg/bird, oral/ SC), tylosin (1 kg/ton feed, 530mg/L water or 15-25 mg/Bird), spectinomycin
Complicated CRD	CRD with colibacillosis	Enero/danofloxacin, gentamicin (0.2-1 mg/ chick, SC), lincomycin+spectinomycin (530-850 mg/ L water), tetracyclines, ceftiofur
Fowl cholera	<i>P. multocida</i>	Tetracyclines, novobiocin (200-350 g/ ton feed), ormetoprim-sulfamethoxazole (136.2+227 to 272.4+454 g/ ton feed), spectinomycin, sulfonamides (sulfadimethoxine, sulfamethazine, sulfaquinoxaline, sulfathiazole, 250 mg- 1 g/ L of water)

Fowl typhoid/ paratyphoid/ Pullorum disease, sinovitis	<i>S. Gallinarum</i>	Enerofloxacin/danofloxacin, gentamicin, spectinomycin, sulfonamides
Vibronic hepatitis	<i>Campylobacter sp.</i>	Erythromycin, oxytetracycline, neomycin (35-140 g/ton feed, 10mg/kg BW)
Crop mycosis	<i>Candida albicans</i>	Nystatin (50-100 g/ ton feed)
Hexamitiasis, ulcerative enteritis, histomoniasis	Respective pathogens	Furazolidone (100-200 g/ ton feed), dimetridazole (450g/ ton feed), , nitrofurazone (50 g/ ton feed)
Leucocytozoonosis	<i>L. smithi</i>	Clopidol (225 g/ ton feed as preventive)
Necrotic enteritis	<i>C. perfringens</i>	Bacitracin (100-200 g/ton feed), lincomycin

Antibiotics Recommended for Use in Pet (Pigeons, Parrots, Quails) Birds

Problem	Cause	Recommended drugs
Psittacosis	<i>Ch. psittaci</i>	Injectable doxycycline 100 mg/kg sc, im every 5 th day, long acting tetracyclines 100 mg/kg, sc every alternate day, enterofloxacin 10-20 mg/kg oral, im, sc every day, for prevention and treatment in feed doxycycline hyclate 200-400 ppm feed, chlortetracycline 500-1000 ppm feed, enterofloxacin 100-200 ppm in water.
Colibacillosis	<i>E. coli</i> infection	Licomycin+spectinomycin 50 mg/ kg in medicated water is preferred drug
Other infections	Same as for poultry	Same as for poultry

Antibiotics Recommended for Use in Laboratory Animals

Drug	Mice	Hamster	Gerbil	Rats	Guinea pigs	Chinchillas	Rabbits
Ampicillin	20-100 po, sc, tid	XX	20-100 po, sc, tid	20-100 po, sc, tid	XX	XX	XX
Chloramphenicol palmitate	50-200, po, tid	50-200, po, tid	50-200, po, tid	50-200, po, tid	50 po, tid	50, po, tid	50, po, tid
Chloramphenicol succinate	30-50, bid, im, sc	30-50, bid, im, sc	30-50, bid, im, sc	30-50, bid, im, sc	30-50, bid, im, sc	30-50, bid, im, sc	30-50, bid, im, sc
Enerofloxacin	2.5, bid, po			2.5, bid, po	2.5, bid, po	2.5, bid, po	5, bid, po
Furazolidone		30, sid, po					5, sid, po

Gentamicin	5, sid, im, sc	5, sid, im, sc	5, sid, im, sc	5, sid, im, sc	5, sid, im, sc	5, sid, im, sc	4, sid, im, sc
Metronidazole	2.5mg/ml in water	80, tid, po		200-400 sid, po			40, sid, po, 3 days
Neomycin	50, sid, sc	100, sid, po	100, sid, po	50, sid, sc	8, sid, po	15, sid, po	30, bid, po
Oxytetracycline	20, tid, po	16, sid, sc	10, tid, po	20, tid, po		50, bid, po	50, bid, po
Trimethoprim+ sulfamethoxazole		5, bid, po	15, bid, po		15, bid, po	15, bid, po	15, bid, po
Trimethoprim+ sulfadiazine		30, sid, sc	30, sid, sc		30, sid, sc	30, sid, sc	30, sid, sc
Tylosin	10, sid, im/sc/po	2-8, bid, im/sc/po	10, bid, im/sc/po	10, sid, im/sc/po			10, sid, im/sc/po

NB: Doses are given as mg per Kg body weight, doses are given in divided doses (sc= subcutaneous, po= per oral, iv= intravenous, im= intramuscular, sid=once daily, bid= twice daily, tid= thrice daily, XX= not used)]

Antibiotics Recommended for Laboratory Animals in Water/ Feed mg/ ml or g

Drug	Mice	Hamsters	Gerbils	Rats	Guinea pigs	Chinchillas	Rabbits
Chloramphenicol	0.5		0.83		1		1.3
Dimetridazole	1	0.5	0.5	1		0.8	1.3
Enrofloxacin					0.05-0.1	0.05-0.1	0.05-0.1
Furazolidone					5.5		5.5
Neomycin	2	0.44		2			
Oxytetracycline	0.4	0.5-1.0	0.8	0.4	1		1
Sulfamerazine	0.2			0.2			
Sulfamethazine	1	1	1	1	1	1	1
Sulfaquinoxaline		1	1	1	1		1
Tetracycline	2-5	0.4	2-5	2-5	0.70	0.5-2	0.7
Tylosin	0.5	0.5		0.5			

Antibiotics Recommended for Infections in Laboratory Animals

Infection of	Mice	Hamster	Gerbil	Rats	Guinea pigs	Chinchillas	Rabbits
Skin and subcutis	Chlor, ampicillin	Chlorl, tetra	Chlor, sulfa	Chlor, tetra, amp	Chlorl, tetra	Chlor, tetra, sulfa, genta	Chlor, tetra, genta, sulfa, enero
Respiratory tract, pneumonia	Tylo, Tetra, enero, genta, kana, chlor	Chlor, tetra	Chlor, sulfa	Tylo, chlor enero, genta, tetra, sulfa	Chloral, tetra	Tetra, genta	Chlor, tetra, genta, kana, sulfa
Gastroenteritis	Chlor, tetra	Tetra, genta, neo, sulfa	Tetra, Chlorl, sulfa		Chlor, tetra, genta	Genta, sulfa, neo	Chlor, metro, furazoli
Urogenital, Infertility, abortions	Tylo, tetra. Enero			Tylo, enero	Tetra, chlor, genta	Genta, tetra, sulfa	Sulfa, tetra, genta, chlor
Torticollis, head-tilt circling, eye-rubbing, corneal ulceration, squinting	Tylo, enero chlor	Chlor, tetra		Tylo, chlor, enero, genta	Tetra	Tetra, sulfa	Chlor, tetra, genta
Abscess				Chlor, tetra, ampi			Genta, chlor, tetra
Mastitis/metritis				Chloral, Genta			Chlor, tetra, genta

Guidelines to Good Antimicrobial Stewardship-

1. Veterinary surgeons should arm themselves with a working knowledge of commonly used antimicrobials.
2. If possible, a narrow spectrum should be chosen. In reality the majority of authorized veterinary antimicrobials are broad-spectrum, increasing the challenge.
3. Prophylactic use should generally be avoided except in high-risk immune-suppressed patients and surgical prophylaxis.
4. If use of an antimicrobial is justified, the correct dose, dose frequency and duration of treatment should be used.
 - § Too much or too little antimicrobial is equally bad in terms of resistance development.
 - § Sub-therapeutic dosing is more of a risk for in-feed or in-water medication.
 - § Environmental contamination with antimicrobials should be avoided.

- § Optimal dosing regimens maximize bacterial killing and minimize the window for resistance development to occur.
 - § Time and concentration dependent killing of pathogen should be considered.
 - § The client should be educated in terms of the correct use of the antimicrobials prescribed.
 - § Topical or local use should be considered (if appropriate) as this will, for example, reduce selective pressure on gastrointestinal flora.
5. If combinations of antimicrobials are considered necessary, care should be taken to choose rational combinations.
 6. Routine use of antimicrobials considered being important in treating resistant infections in human and veterinary medicine (e.g. fluoroquinolones, third- and fourth-generation cephalosporins and amikacin) should be absolutely avoided.
 - § In all species fluoroquinolones and third- and fourth-generation cephalosporins should be used judiciously and never considered as first-choice options.
 - § There is also a strong argument that 'last resort' antimicrobials, such as imipenem and vancomycin, should not be used for veterinary patients.
 7. Taking time to institute practice-based guidelines for therapeutic antimicrobial use:
 - § For example, it is feasible to work out appropriate first option antimicrobials for uncomplicated urinary tract infections, pyoderma and surgical prophylaxis, which should then be used by all practice members.
 - § Creating a table of first-, second- and third-choice antimicrobials to be considered.
 - First-choice antimicrobials would comprise agents appropriate for initial treatment, not necessarily based on culture and sensitivity information.
 - Second-choice antimicrobials should be prescribed based on culture and sensitivity data.
 - Third-choice antimicrobials should only be prescribed for serious and life-threatening infections, based on culture and sensitivity data, and only where no first- or second-choice agents are appropriate.
 - § The key considerations are that all guidelines should be reviewed regularly and be flexible so that change could easily be adopted should inadequacies arise or new information become available.
 8. Report treatment failures. This is a vital part of monitoring for developing and emerging resistance. Situations where this would be appropriate include:
 - Treatment failure despite culture and sensitivity results indicating that an appropriate antimicrobial class had been used
 - Treatment failure where a particular antimicrobial product is authorized for the specific condition and species, and where the clinician's experience would suggest that a positive response should have occurred.

Our Observations on Sensitivity Patterns of Important Animal Pathogens:

Disease/ Pathogen	Sensitive (>90%), Use	Resistant (>50%), Not use
Brucellosis	Tetracyclines, chloramphenicol, gentamicin	Fluoroquinolones (enroflox), azithromycin, aztreonam, doxycycline, ampicillin, amoxycillin, amoxyclav, ampicillin+sulbactam
Edwardsiellosis	Azithromycin, aztreonam, ceftriaxone, fluoroquinolones, chloramphenicol	Tetracyclines, gentamicin, nitrofurantoin, ampicillin, amoxycillin, amoxyclav, ampicillin+sulbactam
<i>Escherichia coli</i>	Nitrofurantoin, gentamicin, ceftriaxone+tazatobactam, chloramphenicol, cefoperazone+sulbactam	Tetracyclines, gentamicin, nitrofurantoin, ampicillin, amoxycillin, amoxyclav, ampicillin+sulbactam
<i>Klebsiella pneumoniae</i>	Fluoroquinolones, gentamicin, imipenem, meropenem, cefoperazone+sulbactam, ceftriaxone+tazatobactam, aztreonam	Azithromycin, tetracycline, chloramphenicol, co-trimoxazole, nitrofurantoin, ampicillin, amoxycillin, amoxyclav, ampicillin+sulbactam
<i>Moraxella bovis</i>	Strepto-penicillin, and most of the common antibiotics	Aztreonam, ceftazidime, cotrimoxazole
Pasturellosis	Tetracycline, fluoroquinolones, amoxycillin, gentamicin, ceftriaxone	Chloramphenicol, co-trimoxazole, ampicillin, amoxycillin, amoxyclav, ampicillin+sulbactam
Salmonellosis	Chloramphenicol, gentamicin, ceftriaxone+tazatobactam, cotrimoxazole, imipenem	Tetracycline, nitrofurantoin, cefoperazone+sulbactam, ampicillin, amoxycillin, ampicillin+sulbactam
Staphylococcal infections	Tetracycline, nitrofurantoin, chloramphenicol, amoxyclav	Ampiclox, aztreonam, ceftazidime+clavulanic acid, amoxy+sulbactam, vancomycin,
Streptococcal infections	Clindamycin, gentamicin, amoxyclav, vancomycin,	Streptopenicillin, ampicillin, fluoroquinolones, nitrofurantoin
<i>Pseudomonas</i> infections	No, use only after testing	Ampicillin, Streptopenicillin tetracycline, amoxy+sulbactam

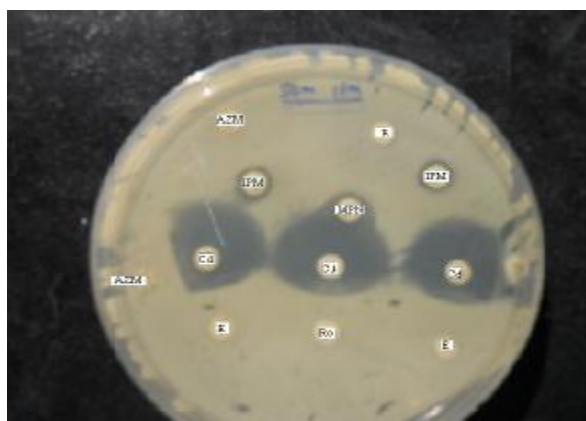


Fig. 1. D-Effect of Macrolides on Clindamycin sensitivity

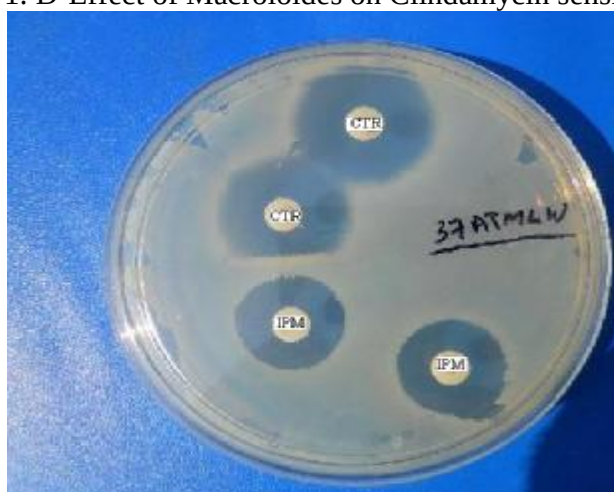


Fig. 2. D Effect of Imipenem (IPM) on Ceftriaxone (CTR) sensitivity

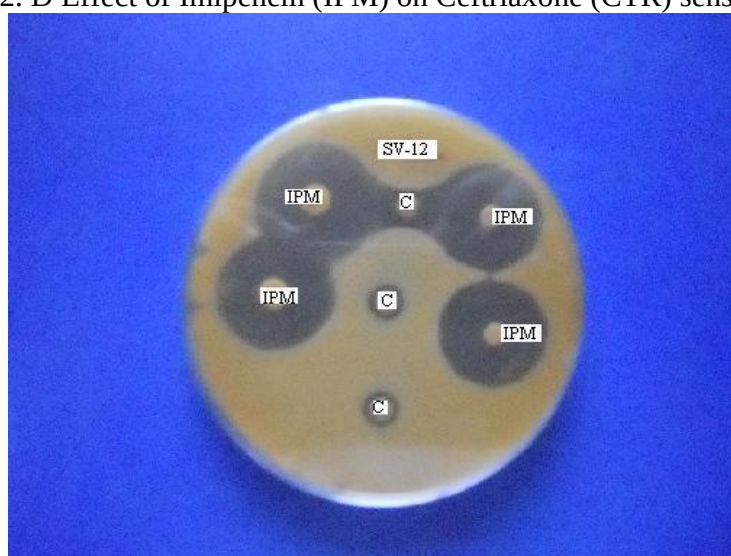


Fig. 3. Synergistic antibacterial effect of Imipenem (IPM) and an herbal drug (C)

Chapter 3

Diagnosis of Commonly Occurring Viral Diseases of Animals

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Livestock animals are vulnerable to a variety of infectious and non-infectious diseases. There are a number of bacterial, viral, rickettsial and mycoplasmal diseases which are prevalent in India and have the potential to cause enormous economic losses. Out of many diseases, viral diseases namely FMD, PPR, sheep pox, goat pox, rabies, Canine Parvovirus, Canine distemper, IBR, BT and swine fever are very important in context to Indian subcontinent. All the diseases caused by the pathogens except the viral diseases can be treated with antimicrobials. As there is no specific treatment available for viral diseases or treatment is expensive, establishing a viral cause would definitely help in protecting the valuable animals through implementation of immunization programme. Diagnosis at an early stage of the onset of a disease lays the basis of its successful therapy. To have wider applicability of the diagnostic method, there should be cost-effective, simple and easy to perform by semiskilled para-auxillary personnel, have requisite sensitivity, specificity and unambiguous readability. Technological developments based on hybridoma technology, molecular biology, enzymology and amplification techniques have made possible to study the development of diseases and analysis of molecules of importance.

Apart from giving valuable information in respect of disease diagnosis and occurrence it also helps to a great extent in providing rational treatment and prophylactic measures. Quick and reliable diagnosis through laboratory aids not only curtails the total cost of treatment but also cuts short the course of disease and period of convalescence, thus resulting in quick recovery and earlier initiation of normal phase of activity and productive life once again. Although the diagnosis can be made on the basis of observation and clinical symptoms manifested by the sick animals, for precise and accurate diagnosis the use of laboratory tests is invariably a must, especially now-a-days. The immunodiagnostic and molecular approaches have significant role in clinical diagnosis of diseases, for epidemiological studies and also for developing strategies for disease control programme. Such method of diagnosis is based on the techniques that are rapid, simple to perform and inexpensive. Commonly occurring viral diseases of animals will be discussed here.

Foot and Mouth Disease- Foot and mouth disease (FMD) is an acute, highly communicable viral disease affecting cloven hoofed animals, both domesticated and wild. It is the most contagious disease known in the animal kingdom. It is characterized by vesicular eruptions in the epithelium of the buccal cavity, tongue, nares, muzzle, feet, teats and udder.

FMD virus is under the genus *Aphthovirus* within the family *Picornaviridae*. There are seven antigenically distinct types of FMD virus; O, A, C, Asia-1, SAT-1, SAT-2 and SAT-3. Strains designated as SAT-1, SAT-2 and SAT-3 were recognized from South African territories during 1934 to 1948. Later, a strain could be isolated from Pakistan (South East Asia) in 1954 and was named as Asia-1. Cattle, buffalo, sheep, goats and pigs are the main domesticated species infected. The water buffalo (*Bubalis bubalis*) can become infected and may also transmit infection to other species. Of all the species cattle

are the most susceptible. The disease is spread at an extremely high speed through direct contact with the infected animal or through infected fomites i.e. food stuff, feeding utensil, water, cloth, harness, manger, beddings, straws, hay etc. The virus may occur in all the secretions and excretions of acutely infected animals. Airborne transmission is also an important mode of spreading the disease.

The carrier state of FMD virus is an inapparent persistent infection in which intermittent recovery of virus from the oropharynx can be detected. The persistent infection in cattle and buffalo and to a lesser extent in sheep and goats is a common sequel to both clinical and subclinical FMD. It is also established in vaccinated animals following contact with FMD virus, where virus replication is restricted to the oropharynx by protective level of circulating antibody. The duration of the carrier state is species related i.e. cattle continue to excrete virus for up to 2.5 years P.I., sheep and goat for up to 9 months.

Clinical signs- Infection with FMD virus can result in a variety of disease manifestations, depending on the species on animal, the strain of the virus, the route of infection and general health of the host. The disease is not usually fatal. Morbidity is 100% but mortality is less in indigenous cattle and comparatively more in exotic and cross-bred cattle. The severity of infection depends on the strain, subtype, serotype of virus and immune status of the susceptible hosts. The incubation period of the disease is 2-8 days under natural condition but it may extend to as long as 2-3 weeks. The disease is characterized by elevated temperature of 40-41.1°C and appearance of vesicles or erosions in the mucosa of the mouth, including tongue, lips, gums, pharynx and palate. Vesicles may appear in the teat leading to mastitis and occlusion of teat canal. Abortion and subsequent infertility are common sequels. In rare instances, lesions may be found on other portions of the skin such as vulva or scrotum. Young animals die without exhibiting clinical manifestations. Some animals show signs of gastro-enteritis. Young calves are rather more susceptible than adults and die without showing any clinical manifestations. Lesions in the myocardium are most common in the fatal disease in young calves, lambs, piglets and kids. The disease is usually mild in sheep, goat and pigs but is important mainly because of the danger of transmission of the disease to cattle. Sheep rarely develop acute form of the disease but there is high mortality in lambs due to myocardial and skeletal muscle necrosis.

Diagnosis- A range of laboratory tests is available. Although conventional tests like complement fixation test, agar gel precipitation test, fluorescent antibody test are still in use in small laboratories, well equipped central laboratories prefer to perform the highly sensitive latest techniques such as enzyme linked immunosorbent assay (ELISA) using monoclonal and polyclonal antibodies and radio-immunono assay (RIA). Besides virus isolation in cell culture, plaque reduction neutralization, radial immunodiffusion and serum neutralization tests are also available. Indirect ELISA and liquid phase blocking ELISA can be used to monitor the immune response in the vaccinated animals. Further recombinant 3ABC protein based indirect ELISA can be employed to detect the antibody and to differentiate the infected from vaccinated animals. The RT-PCR is more sensitive than standard virus isolation techniques and may be used for the rapid detection of FMD virus in specimens obtained during acute state of FMD as well as persistently infected animals. Further, molecular epidemiology based on the sequencing of the nucleotides of several isolates of FMD virus would help in the understanding of their source of origin

and their relationship with isolates of same serotype of FMD and definitely will have impact in the prevention and control programme.

Immunoprophylaxis- The BHK-21 cell culture based inactivated FMD vaccines are still popular around the world and are being used widely as immuno-prophylactic agents against FMD. The trivalent vaccine contains O, A, and Asia-1 serotypes of FMD virus, inactivated with AEI or BEI and adjuvanted with Al(OH)₃ gel or mineral oil or saponin. Al(OH)₃ gel can elicit immune response only in ruminants but not in pigs however mineral oil based vaccine can be used for both ruminants and pigs. Al(OH)₃ gel based vaccine gives protection against the FMD only for 6 months whereas mineral oil gives protection for 1 year. In India, control programme on FMD is now going on covering most of the states and BEI inactivated and oil adjuvanted trivalent vaccine containing O, A and Asia-1 serotypes of FMD is being used at an interval of 6 months. All the susceptible animals should be vaccinated with a potent and effective vaccine besides strict legislation on animal movement and animal import from disease prone countries. While vaccinating the animals wild and captive susceptible animals should also be brought under the immunization umbrella. The carrier animals which harbour the FMD virus without showing any clinical symptoms should not be neglected as they are the potential source of virus to the susceptible animals and responsible for disease outbreak.

Canine Parvovirus Infections- Canine parvovirus-2 (CPV-2), the causative agent of acute haemorrhagic enteritis and myocarditis in dogs, is one of the most important pathogenic viruses. It is a highly contagious and often fatal disease. CPV-2 was first recognized in 1978 and since then it has been well established as an enteric pathogen of dogs throughout the world. CPV-2 is thought to be evolved most probably from feline panleucopenia virus (FPV) by mutation and adapted to infect members of family *Canidae*. CPV-2 is genetically and antigenically unrelated to canine minute virus (CnMV), formerly known as canine parvovirus type 1 (CPV-1). The virions have icosahedral symmetry, are non-enveloped and have a diameter of 20 nm. The CPV-2 genome is a single stranded negative sense DNA having size of 5.2 kb which comprises two open reading frames, one coding in a nested fashion for nucleocapsid proteins (VP1 and VP2) and the other coding for nonstructural proteins NS1 and NS2 involved in replication of the virus. In the 1980s, two antigenic variants, distinguishable using monoclonal antibodies (MAbs), emerged within few years and were termed as CPV-2a and CPV-2b. In 2000 CPV-2c has been reported in Italy, with the amino acid substitution aspartic acid to glutamic acid (Asp-Glu) at residue 426. Currently, these antigenic variants have completely replaced the original type 2, which is still used in most commercial vaccines and are variously distributed in the canine population worldwide. CPV- 2a differs from CPV-2 at six different positions *i.e.*, Met87Leu, Ile101Thr, Ala300Gly, Asp305Tyr, Asn375Asp and Val555Ile. CPV-2b differs from CPV-2a at two positions *i.e.*, Asn426Asp and Ile555Val caused by two single nucleotide polymorphism (SNPs) at positions 4062 and 4449 of gene sequence coding for VP2 protein. Subsequently New CPV-2a, New CPV-2b and Asp-300(2a/2b) have been emerged in canine population.

CPV-2 infection occurs worldwide in domestic dogs and other members of the dog family. Incidence is higher in animal shelters, pet stores and breeding kennels. CPV can affect dogs at any age. Severe infection is most common in puppies between 6 weeks and

4 months old. All breeds of dogs are susceptible. The crossbreds are less susceptible in comparison to pure breeds like Rottweilers, Doberman Pinchers, English Springer Spaniels and German Shepherd, the exception to this being Toy Poodles and Cocker Spaniels. CPV-2 affects only canines and cannot be transmitted to humans or other species. If a dog survives the first four days, they will usually recover rapidly and become immune to the virus for life. Most puppies die without medical treatment. The CPV-2 infection is more severe in young puppies especially those younger than three months of age. All infected dogs including stray dogs may not necessarily exhibit clinical manifestations but they may shed the virus in feces during the acute phase of enteric fever and become potential source of infections to other susceptible dogs in spite of significant rise in the serum antibody titers.

Clinical signs- In parvovirus enteritis the most commonly encountered clinical signs include depression, vomiting, diarrhoea, anorexia and fever. There is slight rise of temperature in the initial stage of the disease but gradually turn to subnormal level with advancement of vomiting and diarrhoea. There is no consistent character of the stool, it may be watery, yellow in color or tinged with frank blood. The second form of CPV is cardiac syndrome, or myocarditis, which can affect puppies under three months of age. The collapsed dying pup may have cold extremities, pale mucosae and show gasping respiration or terminal convulsions. Acute heart failure with respiratory distress occurs in pups between 4 to 8 weeks of age. Subacute heart failure occurs in older pups usually aged 2 months or more.

Diagnosis- The laboratory diagnosis of CPV-2 infection is performed by demonstration of the virus in feces or in tissues after post mortem examination. Haemagglutination (HA) assay followed by HA inhibition (HI) with a CPV-2 specific antiserum or virus isolation (VI) in cell cultures were generally used for diagnosis. Immuno-electron microscopy has been shown to be a sensitive and reliable method for CPV-2 detection, but is not practical for routine use. Although virus neutralization test, fluorescent antibody test (FAT), counter immunoelectrophoresis (CIE) test, nucleic acid hybridization or dot blot, *in situ* hybridization, ELISA etc. are employed for diagnosis of CPV-2, they have varying degree of sensitivity and specificity. Recently, the polymerase chain reaction (PCR) and real time PCR (RT-PCR) have been increasingly used as a tool for the diagnosis of CPV infections. The detection of canine parvovirus in paraffin embedded tissues by PCR has also been described. Nested PCR was developed for more sensitive detection of genomic DNA of CPV-2. Another recently described novel diagnostic method for the sensitive and accurate detection of CPV-2 is the loop-mediated isothermal amplification (LAMP) test.

Immunoprophylaxis- Effective vaccines are available for the prevention of CPV -2 infections. Both modified live and inactivated parvovirus vaccines have been used to fully susceptible sero-negative pups. Attenuated strains of CPV have been derived by repeated passage of the virus in cell culture. The vaccine virus is shed at much lower titres in the faeces suggesting that the absence of enteritis results from decreased viral replication in the intestine. Experimentally live virus vaccines have been shown to protect dogs for at least 3 years or longer. Inactivated vaccine however, provides only a limited duration of immunity to infection and dogs are protected against disease for several months. For parvoviral prophylaxis, modified live virus (MLV) vaccines have proved to be much more effective than inactivated vaccines. MLV vaccines have been

shown to be safer and neither vaccine induced diseases, reversion of virulence or the involvement of vaccine viruses in the generation of new viruses have been confirmed.

Control- Due to nonenveloped feature of the canine parvovirus, it is stable in the environment for a long time. It is not killed by common disinfectants. Virus is shed in the stool of infected dogs in gigantic amounts during the 2 weeks following exposure. A typical/average infectious dose for an unvaccinated dog is 1000 viral particles. An infected dog sheds 35 million viral particles per ounce of stool. Virus loses its infectivity within one month; therefore, it should be safe to introduce a new puppy indoors one month after the active infection has ended. Shaded areas should be considered contaminated for seven months. Areas with good sunlight exposure should be considered contaminated for five months. Chlorine bleach is quite effective in the ratio of 1 part bleach and 30 parts water. There is no way to completely disinfect contaminated dirt and grass, although sunlight and drying has some effect. Potassium peroxymonosulfate can be sprayed on contaminated areas using an applicator as it has relatively good activity in the face of organic matter.

Canine Distemper- Canine distemper (CD) is one of the most severe infectious diseases affecting wild and domestic canidae as well as many other species of carnivores. It is caused by canine distemper virus (CDV) which is an enveloped negative strand RNA virus classified under family *Paramyxoviridae* and genus *Morbillivirus*. The virus mainly affects respiratory system, gastrointestinal tract, lymphoid organs, urinary bladder and the central nervous system. This leads to either subclinical infection or a combination of respiratory, ocular, gastrointestinal, neurologic and cutaneous signs or lesions that appear simultaneously or sequentially. The contagious nature and the high mortality rates of the canine distemper make it important to provide the prompt, accurate and specific diagnosis to detect even small amounts of virus during early stage of the infection. Although vaccination against canine distemper has been widely used, this infection remains an important disease of canines. Generally young, unvaccinated or not fully-vaccinated puppies, under 6 months of age, are most susceptible to canine distemper virus. Very young puppies (under 3 months old unable to get their mother's colostrum (maternal immunity) are also at high risk. Older, unvaccinated dogs can also contract the disease if they are unlucky enough to encounter a sick dog, which is shedding distemper, or encounter the rare virus in their local environment. Clinical signs of canine distemper virus infection depend upon the strain of the virus, the host age and immune status and levels of environmental stress. A significant proportion (estimated to be 50%) of infections are subclinical or so mild as not to require veterinary care. Dogs with mild disease exhibit fever, signs of upper respiratory tract infection, and become listless and inappetent. Bilateral serous ocular discharges can become mucopurulent with coughing and labored breathing, signs that are often indistinguishable from those of "kennel cough" (acute respiratory disease of canines). In severe generalized distemper, infected dogs first develop a fever after an incubation period of 3–6 days, but a second febrile response ushers in the more serious phase of the infection that coincides with systemic spread of the virus and accompanying profound leukopenia. Signs occurring at this time include anorexia, inflammation of the upper respiratory tract with serous or mucopurulent nasal discharge, conjunctivitis and depression. Some dogs show primarily respiratory signs, whereas others develop gastrointestinal signs; respiratory signs reflect

inflammation and injury to the upper respiratory tract and large airways, causing a productive cough, bronchitis and interstitial pneumonia follow.

Diagnosis- Clinical diagnosis of canine distemper can be complicated by the use of modified live vaccines. Cases of canine distemper can occur in recently vaccinated puppies, raising the obvious question of whether the signs are caused by the vaccine virus or a field strain. This question is not satisfactorily resolved with standard serological, virus isolation, or antigen detection tests. RT-PCR is now becoming a standard method of testing, but the distinction of field and vaccine viruses also requires very specialized RT-PCR assays that are not routinely available. Laboratory diagnosis is necessary to exclude other diseases with similar clinical manifestations. Virus isolation can be achieved by co-cultivation of lymphocytes from suspect animals with cell lines expressing the CD150 (SLAM) molecule, which has eliminated the need to use activated mononuclear cells for isolation of field strains of canine distemper virus. After initial isolation, the virus can then be adapted to grow in primary dog lung cells or conventional cell lines, including Madin–Darby canine kidney or Vero. Immunohistochemical or fluorescent antibody staining methods are useful for demonstrating the presence of viral antigen in impression smears of the conjunctiva and skin biopsies (antemortem) or sections of lung, intestine, stomach, kidney, brain and bladder tissue collected at necropsy. RT-PCR tests can be done on conjunctival swabs, blood mononuclear cells, any tissue sample that includes epithelium, and urine. The serological status of dogs can be assessed with virus neutralization assays, ELISA or indirect fluorescent antibody tests.

Prevention and control- Control of canine distemper virus infection is based on adequate diagnosis, quarantine, sanitation and vaccination. The virus is very fragile, and susceptible to standard disinfectants. Thorough disinfection of premises, however, can be very challenging. Successful immunization of pups with attenuated canine distemper virus vaccines depends on the absence of interfering maternal antibody. Alternatively, pups can be vaccinated with modified live-virus vaccine at 6 weeks of age and then at 2- to 4-week intervals until 16 weeks of age, which is often now the standard practice. For treatment, hyperimmune serum or immunoglobulin can be used prophylactically immediately after exposure. Antibiotic therapy generally has a beneficial effect by lessening the effect of secondary opportunistic bacterial infections. Standard modified live vaccines should not be used in species other than canids. Adverse reactions (disease) have occurred in other species, including red pandas and foxes. Inactivated (killed) vaccines previously were used to immunize zoo animals; however, these vaccines were often of marginal efficacy.

Classical Swine Fever- It is an acute highly infectious viral septicaemia of swine of all ages characterized by rapid and sudden onset, high morbidity and mortality and generalized haemorrhages. A chronic course followed by recovery after treatment has been reported in older pigs and appearance of new born pigs with congenital defects. The most virulent virus strains produce clinical disease of pigs of all ages. The less virulent strains may produce little clinical disease primarily to fetal or new born pigs. This variation has occurred in the field strain of the virus but use of inadequately attenuated vaccine virus is also a contributory factor. The pigs are the natural host for the virus and disease occurs naturally only in this species. All breeds, sex and ages of pigs are susceptible to this infection. It is a highly contagious disease and the infection is usually

acquired by direct contact with infected pigs, ingestion or inhalation. Due to resistance and high infectivity, the virus may get access to the body through inert materials especially uncooked meat. All the secretions and excretions of the pigs infected with virulent swine fever virus contain virus including oronasal and lachrymal secretions, urine, feces and semen and tissue. Virus shedding can begin before the onset of clinical signs and occurs throughout the course of acute or subclinical disease. Chronically or persistently infected pigs can shed virus continuously or intermittently for months. The virus continues to excrete in the urine for some days before clinical illness appears and for 2-3 weeks after clinical recovery.

Clinical signs- The hog cholera virus is highly invasive and virulent. It apparently enters into the body through upper digestive tract or respiratory tract followed by septicaemia and invasion of vascular endothelium. The pathogen usually enters by oral route and the tonsils act as a primary site for viral replication beginning within several hours after infection. The virus then moves through lymphatic vessels and enters blood capillaries resulting in an initial viremia at around 24 hours. The virus multiplies rapidly in the blood and gives rise to various clinico-pathological manifestations viz., fever, hyperemia, inflammation, oedema of various organs and mucous membranes. The incubation period usually varies from 5-10 days although longer incubation period of 35 days or more are recorded. The disease may appear in three different forms viz.:

- (a) Peracute
- (b) Acute and
- (c) Chronic.

(a) Peracute form: It is most commonly noticed in young pigs. Young pigs die in about 24 hours without showing any clinical signs except high rise of temperature (106-107°F).

(b) Acute form: There is high rise of body temperature (107°F). The peak of temperature rise usually occurs between 4th and 8th day of illness. Animals show dullness, depression, anorexia, vomiting and constipation followed by diarrhea, dehydration and loss of body weight.

(c) Chronic form: In chronic form mortality is 60-70% and morbidity is 90%. It occurs in the field outbreaks due to vaccination with serum virus mixture and characterized by prolonged incubation period, emaciation, alopecia, dermatitis, blotching of ears and abdominal skin. Recovery is usual after a brief period of illness but animals subsequently develop clinical disease and die in stress condition. There are signs of chronic diarrhea and chronic pneumonia.

Prompt and quick diagnosis of swine fever is extremely important to avoid enormous economic losses due to high mortality in the herd. A confirmatory diagnosis is often difficult to make without laboratory examination as the signs and lesions of this disease are similar to that of swine erysipelas, septicaemic salmonellosis, pasteurellosis and streptococcosis. It would have been more complicated if African swine fever is present in that country since marked similarity of its lesions to those of hog cholera.

Diagnosis- The diversity of clinical signs in CSF in field condition as well as experimental condition makes it difficult to diagnose the disease. Furthermore, BVD virus and BDV infections can seriously interfere with the clinical and laboratory diagnosis of the disease. In most CSF epizootics, symptoms observed are fever, dullness, decrease food intake, coughing, ataxia, conjunctivitis and diarrhoea. Increased mortality

and perinatal death or still birth or abortions were also observed in fattening and breeding farms respectively.

Detection of antigen- The rapid and sensitive detection of viral antigen can be achieved by FAT of tonsil as early as 2 days after infection and together with lymph node, spleen and ileum; it is the most appropriate tissue. To rule out the BDV and BDV, it is to be carried out using a panel of MABs. It can also be tested using MABs based avidin-biotin complex (ABC) immunoperoxidase test and AGPT. ELISAs (enzyme linked immunosorbent assays) especially antigen capture are used for rapid herd screening. It is also an OIE recommended test for international trade. Various ELISAs have been described. They are double antibody sandwich using polyclonal or monoclonal antibody against various viral protein antigen in serum, blood leucocytes fraction or anti-coagulated whole blood. The sensitivity of ELISA is lower than that of virus isolation and PCR.

Detection of infectious virus- Virus can be isolated from tissue or blood and is the most sensitive *in vitro* method for detection of CSF virus. It has been shown that CSF virus from whole blood or plasma is more sensitive in the early phase of infection when compared to leukocyte fraction. The most frequently used cells for virus isolation are porcine kidney cell lines PK-15, SK-6 or swine testicular (STE) cells. Cultures are examined by FAT after 24-72 hours or 4-5 days for immunoperoxidase staining. As a diagnostic tool, virus isolation is more sensitive than the direct immunofluorescence test (IFT) on frozen tissue sections.

Detection of antibodies- Several diagnostic techniques for detection of antibodies to CSF virus has been developed and VNT is most commonly used in Western Europe. In some Asian countries exaltation of ND (END) virus neutralization method is used to detect antibodies to CSF virus in serum samples. The immunoperoxidase monolayer assay (IPMA) is often used in Northern America and Some Latin America for detection of antibodies to CSF virus. To avoid the false positive results, parallel assays with BDV virus /BDV and analysis by differences in antibody titre is taken into consideration. Several CSF viruses specific ELISAs are available for detection of antibodies to E2, E^{ms} or NS53 protein. The sensitivity of CSF virus E2 protein based ELISA varies from 90-99% and specificity is about 99% when compared to VNT. Samples for antibody detection are collected in ordinary (nonheparinised) tubes from convalescent pigs and from contact herds after 21 days have passed since confirmed outbreak occurred. ELISAs are also used for herd testing of the disease.

Reverse transcription polymerase chain reaction: RT-PCR is a rapid and sensitive test than ELISA and virus isolation. After RNA extraction and reverse transcription step, the DNA is amplified and can then be detected and analysed. It is useful in preclinical diagnosis and several conventional and real-time PCR assays have been performed. Due to its rapidity and sensitivity it is used worldwide. Care should be taken of the false positive due to contamination during processing and false negative due to inhibitors in the sample. This is helpful in differentiating other pestivirus infections and enables to group the isolates on the basis of sequence homology and important in tracing various outbreaks during epidemics. Molecular epidemiology can be performed by testing the samples and then further sequencing of the positive samples to compare various virus strains. The nested PCR, an RT-PCR followed by a second PCR, is generally considered most sensitive *in vitro* method for the detection of CSF virus. A

method based on a pan-pestivirus specific single tube nested PCR and a CSF virus specific fluorescent probe has been reported. It allows detection of CSF virus in the blood of piglets infected with moderately to highly virulent CSF with greater sensitivity than virus isolation method.

Control-

In hog cholera free areas- The slaughter of all in contact and infected pigs is generally practiced followed by burning or burial.

- All the vehicles, pens, premises and utensils must be disinfected with strong chemical disinfectants.
- Movement of the pigs should be restricted and in contact animals must not be sold in the market.
- Entry to and departure from infected premises must be controlled to avoid the spread of disease through footwear, garments etc.
- Boiling of all the garbage before offering as food to pigs should be practiced.
- Different persons should handle the affected and healthy pigs.

In enzootic areas: Farmers should be educated about the highly infectious nature of the disease and the ease with which it can be spread by feeding uncooked garbage and purchase and sale of infected or in contact pigs.

Rabies- Rabies is an acute highly fatal viral disease of all warm blooded animals including man characterized by signs of abnormal behaviour, nervous symptoms (increased excitability and irritability), impairment of consciousness, ascending paralysis and death. Rabies is primarily a disease of terrestrial and airborne mammals, including dogs, wolves, foxes, coyotes, jackals, cats, bobcats, lions, mongooses, skunks, badgers, bats, monkeys and humans. The dog has been, and still is, the main reservoir of rabies in India. Other animals, such as monkeys, jackals, horses, cattle and rodents, seem to bite incidentally on provocation. Rabies virus (RABV) is the type species of the genus *Lyssavirus* within the family *Rhabdoviridae*. There is variation in susceptibility in different species to rabies. Foxes, cotton rats and coyotes are extremely susceptible, cattle, rabbit and cats are highly susceptible and dogs, sheep and goats are moderately susceptible. Cattle and equidae may be considered as dead end hosts. The rabies virus is maintained by two different but interrelated cycles; a. urban cycle, where disease transmission takes place through dogs and b. sylvatic cycle, where the disease is transmitted through wildlife. Transmission of rabies under natural conditions is commonly by bite of rabid animals usually carnivorous animals. The 90% of human cases occur due to the bite of rabid dog. In India, besides the problem in domestic animals like dog, cattle, cat etc, a limited survey reported the evidence of rabies in various species of rodents. Rabies infection also prevails in bandicoots, mongooses and rats. Death reports of human beings infected by other animals like dog, cat, fox, jackal, wolf, mongoose and rat bite were recorded in India. Many clinically infected animals excrete virus in the saliva, particularly if infected with a relatively low dose of a homologous strain. The species difference exists in the duration of viral excretion before the onset of clinical signs. The period is short in cats (up to 24 hrs), longer in dogs (up to 13 days) and longest in foxes (up to 29 days) following experimental infection. Apparently healthy carriers and excretors of rabies virus in saliva have been reported.

Clinical signs- Two types of clinical manifestations are noticed in dogs. They are furious form and dumb form. But this differentiation is not always perfect and one symptom may overlap with other.

(A) **Furious form-** In this form excitation is predominant changes and can be divided into stage of melancholy and stage of excitation.

Stage of melancholy- There is change of behaviour of the affected animal which does not obey its master. It often shows unusual violence and frenzy behaviour. The restrained dog may show the tendency to bite either nearby inanimate objects but set free dogs show the tendency to bite other animals and human beings which come across their ways till death. Rabid dog may move from one village to another in a circular manner (circling disease) thus spread the disease over wide areas.

Stages of excitement- In this stage, dogs become aggressive due to increased excitability and irritability. Initially the animal will hide in dark place due to photophobia and will have the urge to bite. There is change in the bark which is very characteristic and due to paralysis of the vocal cord. There is champing of jaws and dribbling of saliva. The affected dogs will wander aimlessly and bite animals and persons who will come across its way.

(B) **Dumb form-** It is also known as paralytic form. In this form there is paralysis of the lower jaw, tongue, larynx and hind quarters. The dogs can not bite, saliva remains infective and a peculiar voice is produced known as howl. There is hanging of jaws due to paralysis of jaw muscles. The dog used to seek solitude and appear sluggish and morosed. In the terminal stage of the disease dogs show progressive weakness and paralysis which cause them to stagger. There is coma and finally death. The episode lasts for 1-7 days.

Diagnosis- The tentative diagnosis is made on the basis of clinical signs. However, the familiarity with the clinical signs is important for the diagnosis of rabies in affected human beings, dogs, cats and other animals. In dogs and other animals, the clinical disease can be divided into three phases. However, these phases may not be distinct, as some of the early phases are not always apparent. In some dogs, the furious stage may be quite pronounced whereas in others it is not at all apparent. In some dogs, the furious stage may be quite pronounced whereas in others it is not at all apparent. In other animals, the clinical signs of rabies are not always as distinct as in dogs, but there is progressive dysfunction of CNS. However, the clinical signs can be mistaken with other diseases or conditions –i.e. canine distemper in dogs, tetanus, pseudorabies, other CNS infections and toxicoses with neurological manifestations.

In the laboratory, Seller's staining is usually carried out on unfixed impression smears of hippocampus (Ammon's horn), cerebral cortex, cerebellum, medulla oblongata and spinal cords to detect the typical I/C acidophilic 2-8 μ in size inclusion bodies (Negri bodies) in the neurons. The test is very inexpensive and results are obtained in under 1 hour. These bodies appear magenta or heliotropic to bright red in colour and have black basophilic inner granules only in fresh tissues. This test has been replaced by fluorescent antibody test (FAT) over the last 30 years. When conducted by a trained person using quality reagents this technique is most accurate and quick to perform (within 2-4 hours). The direct FAT is employed as a routine rabies diagnostic test by all the laboratories on fresh, frozen or glycerolized materials. When further characterization is necessary the virus can be isolated in albino mice (3-4 weeks old and

a litter of two days old new born mice) or in cell lines (mouse neuroblastoma cell line – NAC1300), Vero, BHK21, McCoy cell. The mouse neuroblastoma cell line is inoculation test can be performed by making 10% suspension of the suspected brain materials and inoculating into the mice of 4-6 weeks age by intracranial route @ 0.03 ml. The inoculated mice are observed for 21 days and dullness, rough coat and paralysis of the hind limbs are the commonly observed symptoms in positive cases.

Detection of nucleic acid- Several nucleic acid based test (NA hybridization, PCR, genome sequencing etc) have been developed and are in use for rabies diagnosis. These methods detect viral RNA in suspected brain materials. The nucleic hybridization technique provides the high degree of sensitivity and specificity than any other tests. *In situ* hybridization (ISH) can be done on formalin fixed paraffin embedded tissue. The ISH involves interaction of specific rabies probes with complementary target sequence of rabies RNS to form hybrids. The probes used in ISH involve interaction of specific rabies probes with complementary target sequence of rabies RNA to form hybrids. The probes used in ISH are either radio-labelled (^3H or ^{32}P) or non-radioactive label (digoxigenin). PCR based tests also be used for studying viral pathogenesis and epidemiological work besides diagnosis. PCR can provide diagnosis in the sample which contains less number of viral particles. The PCR coupled with sequencing of amplified product can further provide accurate genotyping of rabies virus.

Prophylaxis- A galaxy of anti-rabies vaccines is available to be used in animals and human beings. Different types of vaccines are nervous tissue rabies vaccine (mouse brain, rat brain, goat brain and sheep brain), tissue culture, oral vaccine and recombinant vaccines. Nervous tissue vaccines are not used now-a-days because of neuromuscular reactions in the vaccinated individuals.

(1) Tissue culture rabies vaccines: Modern tissue culture anti-rabies vaccines are safe, potent and efficacious and imported into India for several years. Few can afford these vaccines owing to its higher cost compared to nervous tissue vaccine.

(a) Human diploid cell vaccine: The human diploid cell vaccine (HDCV) is grown in human diploid cell strain MRC-5 or WI-38. The vaccine is marketed in India either by Serum Institute as Human diploid cell vaccine or by Serum International as Rabivax. The 5 doses of a conventional post exposure course of human diploid cell vaccine consists of 1 ml on days 0, 3, 7, 14 and 30.

(b) Purified chick embryo cell (PCEC) (Rabipur, Sanofi Aventis): 5 doses are administered. This vaccine is equally effective in inducing the immune response in the vaccinated individuals compared to HDCV without any side effects.

(c) Purified Vero cell rabies vaccine (PVRV; Verorab, Ranbaxy): Five doses are administered. The vaccines are imported. However, indigenous production has been started at Pasteur Institute, Coonoor and the vaccine is now available. Intramuscular dose of the latter is 0.5 ml, whereas that of the others is 1.0 ml. The minimum acceptable potency for these is 2.5 I.U./dose, but in practice the potency normally exceeds 5 I.U./dose. Other vaccines of Vero cell origin are Verovax-R marketed by Aventis Pasteur and Abhayrab marketed by Human Biological Institute.

2. Rabies immune globulin (RIG): Rabies deaths despite optimal vaccine treatment have been attributed to lack of RIG. RIG is expensive and not generally available in large areas of Nigeria, Togo, Cameroon, Bangladesh, Pakistan, India and Nepal. When presented with severe bite wounds with a high risk of rabies infection, the clinician in

developing countries should make the best use of the available resources for post exposure cases.

Peste des Petits Ruminants (PPR)- It is severe rinderpest like disease of small ruminants' viz., sheep, goat and wild ruminants characterized by pyrexia, catarrhal inflammation of ocular and nasal mucous membrane, erosive stomatitis, enteritis and pneumonia. The disease is also called goat plaque, kata, and pseudorinderpest and stomatitis pneumo-enteritis complex. PPR is rinderpest like disease of sheep and goats. Clinical signs and pathological lesions are confusingly similar to those of RP. Both sheep and goats are susceptible to PPR but goats are more susceptible than sheep in field outbreaks. The young, adult and both the sexes of sheep and goats are susceptible. PPR virus does not attack cattle, buffaloes and pigs under natural conditions. Experimentally, cattle have been infected without clinical signs and resisted experimental challenge inoculation with virulent RP virus suggesting PPR virus conferring cross protection against RP virus in animals. Natural infection occurs mainly through direct contact with infected sheep or goat. Secretions, excretions and faeces contain high concentration of virus and spread of the disease take place through inanimate objects contaminated with it like RP. There is no carrier state in animal but disease can be spread through animals with subclinical infection. The disease is also transmitted through ingestion of infected materials or inhalation.

Clinical signs- The clinical signs of acute PPR in goats and sheep are fever (39-41°C) after an incubation period of 6 days. There is profuse serous nasal and ocular discharge which may turn from mucopurulent to purulent. Some animals may show signs of conjunctivitis. There is increased respiration rate (120/min) with extended head and mouth breathing. Signs of pneumonia become prominent and animal may die due to respiratory distress.

The lesions are found in alimentary tract mucosa. Necrotic lesions are evident in lip, buccal mucosa (stomatitis), gum (gingivitis), dental pad, palate and tongue accompanied with oedematous lips. Diarrhoea is most commonly observed after 3-4 days and faeces contain mucous and blood. Due to diarrhea, emaciation and dehydration set in, temperature goes down, animals become exhausted and finally death occurs. Subacute form is mostly seen in sheep with signs of acute form in lower grade. The most prominent pathological lesions such as erosions in the lips, dental pad, tongue, soft and hard palate and cheeks, esophagus, pillars of the rumen, abomasums and intestine, conjunctivitis and gastroenteritis. There is pronounced necrosis of lymphocytes in lymph nodes, tonsil, Payer's patches and spleen. The distinguishing pathological features between RP and PPR are the frequent occurrence of primary broncho-interstitial pneumonia in the latter disease. This is due to the affinity of the virus to bronchial epithelial cells and alveolar macrophages which are infected and destroyed.

Diagnosis- Laboratory diagnosis of the PPR is done either by serological test or recently developed non serological tests. Various techniques have been applied in the serological diagnosis of PPR in sheep and goats. These include AGPT, CIE, VNT, ELISA (indirect, competition and immunocapture). Differential electrophoretic profile in the 'N' protein, nucleic acid hybridization using either radio-labelled or biotynated c-DNA probes and RT-PCR using P and F gene specific primers have been used for precise, correct and reliable diagnosis of PPR from RP. Tissue culture rinderpest vaccine has been proved to

be effective against PPR as it confers satisfactory level of cross immunity. Vaccine should be administered at 3-4 months of age when there is little or no maternal antibodies. The vaccine can be used in an actual outbreak to protect the animals after keeping them segregated from the infected ones for a week or two because there is a risk that some animals already may be incubating the disease may succumb.

Vaccines- Tissue culture rinderpest vaccine has been proved to be effective against PPR as it confers satisfactory level of cross immunity. Vaccine should be administered at 3-4 months of age when there is little or no maternal antibodies. The vaccine can be used in an actual outbreak to protect the animals after keeping them segregated from the infected ones for a week or two because there is a risk that some animals already may be incubating the disease may succumb. Furthermore, some researchers do not suggest vaccinating the animals against PPR with TCRP vaccine as there is a chance that PPR virus may mutate and result in evolution of a new variant of PPR virus. So, both the inactivated and live attenuated vaccines against PPR have been developed. Inactivated PPR vaccine is not suitable as immunity following its use is short lived. Live attenuated Vero cell adapted PPR vaccine developed by IVRI and TNUVAS have been proved to be effective against PPR in sheep and goat. Vaccinia virus recombinants expressing the fusion (vRF) or the haemagglutinin (vRH) and a double recombinant expressing both the genes (vRFH) of RP virus provided complete protection to goats against PPR. The recombinant vaccine using capripox viruses for expression of the fusion and HA protein genes of RP virus also exerted satisfactory level of immunity in goats to be protected against PPR.

Goat Pox- It is a malignant disease of goats characterized by high rise of body temperature and generalized pock lesions. Goat pox has been reported from all parts of the world and may attain epidemic proportion with mortality rate of 80% in kid and ~50% in adults. The pox virus is one of the largest and most complex of all viruses. The causative agent of the goat pox is double stranded DNA containing virus belongs to the genus *Capripox virus*, subfamily *Chordopoxvirinae* and family *Poxviridae*. The other members of the genus *Capripox virus* are lumpy skin disease virus and sheep pox virus. Capripox viruses are antigenically and genetically very close to each other. It has been reported that it is possible to infect goats with sheep pox virus and sheep with goat pox virus.

The capripox virus is peculiarly host adapted and is pathogenic only for sheep or goats. Although the Kenyan sheep and goat pox virus strain is the mildest producing only a local lesions in both sheep and goats and local necrotic lesions in cattle. Many strains of Mediterranean regions and from India appear highly adapted to either sheep or goats. Virus strains will produce local reactions when inoculated into the other species and there is no natural cross infection between the two, appears to occur under field condition. Sheep pox and goat pox viruses isolated from Scandinavia and India differs in some aspects from the usual forms. A unique property of most of the sheep pox and goat pox virus strains is the high level of host adaptation and inability to infect other animals except man. The Swedish strain produced lesions in rein deer. None of the sheep pox strains including Kenya sheep and goat pox virus strain were found to have the zoonotic properties.

The usual mode of transmission is either through contact with infected animal or inhalation. The virus may gain entry through wound and abrasion. The greatest

concentrations occur in the skin from 10-14 days and the titre then falls but virus can be detected up to 28 days in necrotic tissue or scabs. The virus may also be transported to other places by dog, cat etc. The disease may be transmitted through food and water mechanically.

Clinical signs- The clinical signs of goat pox virus infection vary with different host and in different geographical areas. Goats of all ages can be affected. The disease is more severe in young animals. The first signs of disease are rhinitis, conjunctivitis and pyrexia (40-42°C). Animals stand with an arched back, have poor body coat and have inappetence. One to two days later, cutaneous lesions develop (0.5-1.5 cm in diameter) often accompanied with lesions on the external nares, lips and within the mouth. The lesions on the skins are largely confined to the areas where the wool or hair is shortest such as on the head, neck, ears, axilla, groin, and peritoneum and under the tail.

Lesions present on the tongue and gums tend to ulcerate and high mortality rates occur when lesions develop in the respiratory and alimentary tract. Post-mortem examination shows multiple consolidated areas (0.5-2.0 cm in diameter) in the lungs. Secondary bacterial pneumonia is the common cause of death. Similar lesions are also visible in the liver, kidneys, abomasums and other organs. The appearance of the skin lesions varies and in some outbreaks the lesions are obviously vesicular, they coalesce and involve large areas of the body, especially the lower abdomen.

A nodular form of goat pox has been described in many parts of the world and has been called 'stone pox'. The clinical appearance of the skin lesions is similar to lumpy skin disease. While lesions may be restricted to predilection sites, they often involve the whole of the body in both young and old animals. Some lesions heal *in situ*, but the majority form necrotic plugs, which are shed leaving a shallow 'punched out' ulcer. This heals leaving a depressed permanent scar.

When an epidemic occurs in a completely susceptible flock with no previous contact with the virus, the morbidity may be as high as 75% and mortality 50%. Mortality in young animals may approach to 100%. A sheep and goat pox virus isolate from Kenya is often associated with low morbidity and high level of background immunity (90%). However, 60% morbidity may be found in areas where there was no previously detectable antibody. Such situation is thought to exist elsewhere and cause very variable figures for morbidity and mortality obtained in different outbreaks.

Clinical Management and Supportive Therapy- Infected goat should be placed in clean, well ventilated pens and fed on a high plane of nutrition. Animals that are reluctant to feed should be given 10% glucose saline. All the diseased animals should be under the antibiotic coverage (Terramycin injectable solution I/M @ 5 mg/kg body weight) for 5 successive days. The eyes of the affected animals should be washed with 2% boric acid solution and nostrils cleaned and washed with a weak solution of potassium permanganate (1:10,000). Respiration should be stimulated with oleum eucalyptus inhalation or coramine. Terramycin ointment and powder should be applied topically to dry scabby and ulcerated skin lesions.

Diagnosis- The disease is diagnosed on the basis of

- a) **Clinical signs-** Pock lesions on the skin throughout the body.
- b) **Histological examination-** I/C inclusion bodies in epidermal cells.
- c) **Isolation of virus** in lamb or kid kidney, testicle cell culture, BHK₂₁ cell line, Vero cell line or chorio-allantoic membrane.

- d) **AGPT**- It gives precipitation bands in 24-48 hours.
- e) **CIE**- It gives precipitation band in 2-3 hours.
- f) **FAT**- Direct FAT is used to detect the presence of pox virus in the oedema fluid. Indirect FAT is employed to determine the antibody status of a flock.
- g) **ELISA**- Immunocapture ELISA can be used to detect the goat pox virus in scab suspension. Indirect ELISA is employed to screen the serum samples for the presence of antibodies against goat pox virus in a flock.
- h) **Polymerase chain reaction (PCR)** - Recently PCR using primers based on the attachment gene has been successfully adopted to detect the capripox virus genomic DNA in skin scab samples. The amplified product could be visualized in phenol-chloroform extracted DNA samples as well as samples which were boiled for 10 minutes only in water bath.

Control-

- a) Import of animals from disease prone areas are to be prohibited.
- b) The destruction of affected flocks or the quarantine of the sick animals from healthy ones.
- c) Strict sanitary measures are to be adopted.
- d) A killed or BPL inactivated immunoprophylactic agent along with non-specific immunostimulant from *M. phlei* has shown great promise exerting one year protection when administered S/C @ 1 ml/animal.
- e) Live attenuated goat pox vaccine adapted and attenuated in cell culture (lamb/kid kidney, testicle, BHK21, Vero etc) are also available in the market giving solid lifelong protection when administered S/C or I/D @ 1 ml/animal.

Sheep Pox- Sheep pox causes a severe and highly contagious disease in sheep which is listed in group A disease of the OIE. It is defined as a malignant pox of sheep characterized by fever and generalized development of pock lesions. Lesions first appear as vesicles which later turn into pustules on the exposed parts of the body. There is high mortality rate in lambs. The sheep are naturally susceptible. The susceptibility also depends on the breed and age of animals. Younger sheep are more susceptible over older ones. In young lambs, the disease may flare up in epidemic proportion. Merino breed is more susceptible over indigenous one.

The disease spreads by direct or indirect contact with contaminated objects. In infected pens, the scab virus can remain viable for as long as 6 months. Infection can occur through either the respiratory or the cutaneous route. Mechanical transmission by arthropods may occur. Affected sheep shed the virus at every stage of the disease up to 8 weeks after the lesions have healed. Merino sheep are most susceptible whereas Algerian sheep are comparatively resistant. On inhalation, the virus enters the lungs where from it enters the circulation by lymphatic channels and is then carried to skin and mucous membranes. The virus may pass from the infected ewe to the foetus through placental blood and the lambs may be born with pox lesions.

Clinical signs- The clinical signs of sheep pox depend on the breed of sheep, the virulence of the virus strain, the nature of secondary infection and organ involved. The disease may appear usually in three different clinical forms. (a) Malignant form (b) Mild (benign) form and (c) Abortive form.

(a) Malignant form-This is the most common form. It is characterized first by appearance of marked general distress with elevated high temperature (106-107°F),

rough staring coat, and loss of appetite and numerous areas of erythema all over the body surface. The eruptions are more prominent on the cheeks, nostrils, lips, ears, eyelids, vulva, armpits, thighs, chest, mammary glands, and testicles and under surface of the tail. Affected animals may also die of disease due to pyaemia or septicaemia caused by secondary bacterial complications.

(b) Mild (benign) form- Adult Algerian and indigenous sheep are usually affected with this form and eruptions are confined around eyes, lips and nose.

(c) Abortive form- Generalization is rare and mortality is low (5%). Ewe may abort foetus with pock lesions. Lactating ewe may develop the signs of mastitis due to lesions in the udder. The petechial haemorrhages are found on epicardium and endocardium of the heart. The liver is found enlarged and acutely congested. The kidneys are usually affected with white infarcts and subscapular haemorrhages and showed intense congestion of both medulla and cortex. Adrenals are also congested.

Diagnosis- A number of serological tests, viz., complement fixation test, immunodiffusion test, counterimmunoelectrophoresis test, spot agglutination test, reverse phase passive haemagglutination test, single radial haemolysis test, enzyme-linked immunosorbent assay and dot- enzyme linked immunosorbent assay are available for the detection of sheep pox virus antigen and antibody. Besides, the sheep pox can be diagnosed by inoculating the lymph or skin materials of the affected sheep on the scarified skin of the suspected animal which will show rise of temperature, characteristic papules, vesicles and pustules. Secondary pox lesions will appear on abdomen, thighs, eyelids and ears. Recently Polymerase chain reaction (PCR) using primers based on the attachment gene has been successfully adopted to detect the capripox virus genomic DNA in skin scab samples. The amplified product could be visualized in phenol-chloroform extracted DNA sample as well as samples which were boiled for 10 minutes only in water bath.

Control- Vaccines being the most economical and efficient way to control and eradicate a disease has been proved in several other viral diseases. A variety of vaccines both inactivated and live attenuated have been developed since the beginning of the century by a number of researchers employing aluminium hydroxide, alum and saponin as adjuvant and formalin, merthiolate and BPL as inactivating agent. The route of administration of most of the vaccines is either I/D, S/C or I/M and elicited an immune response lasted for 6 months to 1 year. However, in India now-a-days live attenuated tissue-culture sheep pox vaccines are mainly produced by two manufacturers, viz., IVRI, Izatnagar and Biomed, U.P. It is a live attenuated freeze dried vaccine which confers immunity in all breeds of sheep against sheep pox. The vaccine evokes thermal and/or local cutaneous reaction following vaccination without generalization. The content of each freeze dried ampoule should be reconstituted in cold sterile normal saline as per the doses indicated on the label. It should be used within 2 hours of reconstitution. Dose is 0.1 ml for all age groups, I/D in the caudal fold. It can also be inoculated on the tip of the ear. The vaccine is expected to provide immunity for about 1 year. The vaccine can be stored at 2-4°C for 3 months.

Vaccination schedule- (i) All the sheep in the enzootic area should be regularly vaccinated every year. In case of breakdown of immunity the reason must be investigated.

(ii) Lambs or sheep not showing 'take' following vaccination should be revaccinated.

(iii) In few cases, where local reaction progresses to ulceration application of boroglycerine is suggested to alleviate the condition.

Bluetongue- It is an infectious, noncontagious arthropod borne viral disease of domestic and wild ruminants namely sheep, goat, cattle, camels, llamas, deer and antelopes. This is predominantly a disease of sheep but occasionally cattle and goats are subclinically affected. The disease is characterized by high fever, catarrhal stomatitis, rhinitis, enteritis and lameness due to inflammation of the coronary bands and sensitive laminae of the feet. Although all the domesticated and wild ruminants are susceptible to bluetongue disease, it is basically a disease of sheep. Young sheep under one year are more prone to infection. British and Merino sheep are more susceptible than African sheep. The disease remains in subclinical form in cattle but cattle may show overt clinical signs with high mortality rate. The degree of susceptibility of goats, although variable, is markedly less than that of sheep. Bluetongue virus causes serious disease in sheep but little or subclinical illness in cattle, buffalo and goat. Under natural condition the viral infection is common in sheep, cattle, elk, white tailed deer, pronghorn antelope, camels and other ruminants. It is interesting to note that deer can be infected by bluetongue or epizootic haemorrhagic disease of deer or both. The diseases are very similar and the viruses are also similar but distinguishable serologically and in culture in chick embryos. Both the viruses have also been isolated from cattle. Recovered animals are considered to be reservoir of infection and thus act as carrier of the virus from one season to other. Virus has been isolated from recovered sheep 4 months and from cattle 49 days after infection. The virus has been observed in the semen of the infected bulls and infection is transmitted from infected to healthy cows by insemination. Evidence of vertical transmission from infected dam to offspring has been described but does not generally occur.

Bluetongue virus is transmitted by *Culicoides species* and is only enzootic in areas where adults of competent vector species are present for most of the time of the year, thus maintaining a continuous series of virus. The species of culicoides which are known to act as vectors for bluetongue virus are most active in the temperature range of 18-29°C and almost inactive below 10°C or above 30°C. The disease is divided into 3 forms (a) acute, (b) subacute and (c) chronic. Naturally occurring acute bluetongue disease in sheep has an incubation period of less than a week (2-4 days in experimentally infected sheep). There is high rise of temperature (105-106°C), nasal discharge, salivation and lacrymation. The nasal discharge is mucopurulent and often blood stained and the saliva is frothy. Swelling and edema of the lips, gums, dental pad and tongue may be evident. There may be ulceration of the lips, dental pad, gum and tongue. Necrotic ulcers develop on the lateral aspects of the tongue which is swollen and purple in colour hence the disease is called bluetongue. Foot lesions including laminitis and coronitis and manifested by lameness and recumbency appear when the mouth lesions begins to heal. The appearance of a dark red to purple band in the skin just above the coronet is considered to be an important diagnostic sign. Subacute or subclinical form is noticed in cattle and goats. There is a mild to moderate fever, hyperemia of the mucosa and conjunctiva. Some cattle may develop syndrome like that seen in severely affected sheep. This form of the disease in cattle appears to be a hypersensitivity reaction in a previously affected animal. The same hypersensitivity does not occur in sheep.

Cattle and sheep infected during early pregnancy may cause abortion of congenital deformities which include hydranencephaly, microcephaly, curvature of the limbs, blindness and deformity of the jaw. Under field conditions, the mortality rate usually varies between 2-30%. Death occurs mostly from 1-8 days after the appearance of visible signs. Bronchopneumonia often resulting from aspiration of rumen ingesta is probably the most common cause of death.

Diagnosis- The serological tests which are generally used for confirmatory diagnosis of bluetongue consist of agar gel diffusion test, plaque inhibition test, neutralization test, fluorescent antibody test, complement fixation test, ELISA and dot-ELISA. Besides, isolation of bluetongue virus in suitable system viz., sheep, chicken embryo or cell culture inoculation offer the alternative method of confirmatory diagnosis. Recently molecular biological techniques viz., RT-PCR based on the structural protein and nonstructural protein gene specific primers, SYBR green or Taqman based real time PCR, nucleic acid hybridization and nucleic acid sequencing offer the most sensitive ways to detect the viral genome in the clinical samples.

Control-

1. Grazing of the animals should be avoided in areas where there is lot of culicoides vectors.
2. Attempts should be made to curb down the vector population by spraying insecticides and good water management.
3. Good hygienic and sanitary conditions should be adopted to control the spread of the disease.
4. Movement of all the animals susceptible to bluetongue should be restricted.
5. Contact with wild ruminants should be avoided.
6. Strict quarantine of the affected and sick animals should be made.
7. Animals' importation from bluetongue disease prone countries should be strictly avoided.
8. Susceptible animals can be vaccinated with pentavalent BEI inactivated and Montanide adjuvanted (Serotype 1, 2, 10, 18 and 23) vaccine which can provide protection against the specific serotypes for about 1 year.

Infectious Bovine Rhinotracheitis- It is a highly infectious disease of bovine characterized by high fever, rhinotracheitis, conjunctivitis, with a short course and high recovery rate. Encephalitis, the systemic form of the disease in new born calves, infectious pustular vulvovaginitis/balanoposthitis is other syndromes caused by the same virus. Cattle of all ages are susceptible to experimental challenge but the disease occurs naturally mostly in animals over 6 months of age. Dairy and beef cattle are equally susceptible. The main sources of infection are the nasal exudates and droplets, genital secretions, semen and fetal fluids and tissues. Aerosol is the important mode of transmission of respiratory disease; venereal transmission is the method of spread of the genital disease. The IBR virus may survive for up to 1 year in semen frozen at -196°C. Introduction of new carrier/infected animal may spread the disease. One of the important features of the BHV-1 is its ability to become latent following primary infection with a mild strain or following vaccination with a live attenuated strain. The virus may remain latent indefinitely and recrudescence, reactivation and shedding of the virus may be possible by the use of large doses of corticosteroids which mimics the effects of stress.

The incubation period varies from 3-7 days following experimental infection and 10-20 days under natural condition.

Clinical signs- The disease occurs in several forms including, respiratory form, genital form, ocular form, encephalomyelitis form and abortive form.

1. **Respiratory form-** IBR occurs as a subclinical, mild or severe disease. Morbidity approaches to 100% and mortality may reach 10%. This form is characterized by fever, inappetance, acceleration of respiration rate and dyspnoea. There is profuse nasal discharge, initially serous and later mucopurulent. The nasal mucosa is hyperemic and lesions progress from pustular necrosis to large haemorrhagic and ulcerated areas covered by a cream coloured diphtheric membrane. Foul breath, mouth breathing, salivation and a deep bronchial cough are common. The mortality rate is less unless there is secondary bacterial or viral infection. Animal may show signs of bronchitis and pneumonitis. The disease is called 'red nose' as there is red appearance of nasal mucosa. Acute uncomplicated cases last for 5-10 days and remain as carrier to shed virus for a considerable period.
2. **Genital form-** IPV is most commonly noticed in dairy cows. Affected animals develop fever, depression, anorexia and stand apart avoiding tail from contact with the vulva. Animals micturate at frequent intervals. Vulva may remain swollen and there may be mucopurulent discharge from vulva and vagina. Vestibular mucosa is reddened with many small pustules which coalesce to form fibrinous pseudomembrane that covers an ulcerated mucosa. The acute case of the disease lasts 4-5 days.
3. **Ocular form-** There is inflammation of the conjunctiva (conjunctivitis) and serous to purulent ocular discharges. Petechial haemorrhages and corneal opacity may be evident. This form may appear along with respiratory form.
4. **Encephalomyelitic form-** The signs include high rise of temperature, incoordination, tremor, cycling, falling, coma and death of calves within 4 days from onset of neurological disorders.
5. **Abortive form-** There may be 'abortion storm' as cows may abort foetus died at 4 months of gestation. Foetus is autolysed and expelled within 1-7 days of death.

Diagnosis- The confirmatory diagnosis may be provided through laboratory tests e.g. virus isolation in MDBK cells and FAT using liver and spleen of the foetus. SNT is the routine serological test to evaluate antibody titre of current or recent infection. ELISA and dot-ELISA may provide alternative and sensitive method for diagnosis of the disease. Molecular biological techniques namely PCR, RE mapping, nucleic acid sequencing may be adopted for detection and characterization of viral genome/isolate not only in clinical but also in semen samples.

Control-

- 1) Affected animals should be isolated from the healthy animals.
- 2) A variety of live modified virus vaccines are available in some part of the world. In India, recently an inactivated IBR virus vaccine (IBRIVAX) grown in cell culture and emulsified with fortified special oil adjuvant has been introduced in the market by Intervet. Primary vaccination in young and adult animals should be carried out at 4-6 weeks of age and at any age respectively. All the animals should be given booster dose 3 months later followed by annual revaccination. The vaccine should be given @ 2 ml subcutaneously or intramuscularly.

Chapter 4

Diagnosis of Important Parasitic Diseases by Faecal and Blood Smear Examination

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Parasitic diseases of animals are important from mortality point of view they cause along with very high morbidity and loss of production. Accurate diagnosis of these diseases is crucial not only in sick animals but a routine checkup of apparently healthy animals through examination of faeces and blood is also required to maintain optimum production and also to prevent losses due to morbidity and mortality. For accurate diagnosis it is necessary to collect, preserve and dispatch the diagnostic material properly.

1. Faecal Examination: Parasites inhabiting in digestive canal and in biliary system produce eggs/larva, which leave host body through faeces. Parasitic eggs or larva from lower respiratory tract are coughed in pharynx and swallowed. These also appear in the faeces. Gross examination of faeces will reveal presence of blood, mucous and fragments of parasites without eggs. Where motile organisms i.e. protozoa are to be searched, materials are examined as soon as possible since the organisms may lose their motility in cold conditions. The preparation should be kept warm using warm saline until examination is completed. The use of phase contrast/dark field microscope greatly helps this.

(A) Direct Smear: Small portion of faeces mixed with normal saline or water and a drop of it is placed on glass slide. After removing coarse fibers a cover slip is put over it and examined under microscope. First, preparations are examined under low power objective. If any suspicious thing is visible, it should be focused under high-power objective for detailed study. In case, a few numbers of eggs / larva / cyst is there, it indicates that infection is probably absent. However, some parasites produce relatively low number of eggs hence no conclusive diagnosis can be given by this method.

(B) Concentration Method: Concentration method gives good result even in low infection. Two methods are employed. These are (1) sedimentation and (2) floatation.

(i) Sedimentation Method: In light infection it is necessary to concentrate eggs and larvae. This can be done by several methods. Take 10gm of faeces. Add 10-20 times water and suspend by sacking. Strain the mixture through a sieve to remove coarse fiber particles. Pour in conical cylinder and allow the sediment to settle down. After about 30-40 minutes $\frac{2}{3}$ rd top of supernatant is poured off without disturbing the sediment in bottom. Add more water and repeat it until supernatant become clear from the suspended particles. Finally pour off supernatant and transfer into 10ml sedimentation cone and allow standing for 30-40 minutes. Remove 2-3 drops of sediment with pipette, transfer on slide cover with cover slip and examine under microscope. Sedimentation method is

good for trematode eggs. For schistosome eggs, the addition of 0.5% glycerol to the water results better recovery. Sedimentation is recommended for eggs of trematodes and operculated ova of cestodes, which have high specific gravity and cannot lift on top by floatation.

(ii) Floatation Method: Floatation of eggs is achieved by using aqueous solution of chemical of higher specific gravity. The specific gravity of eggs and larva of helminthes and protozoan cysts varies from 1.05 to 1.15. It is 1.20 in case of infertile Ascarid eggs. The specific gravity of solution ranges from 1.18 to 1.20 or at the times certain solutions have a specific gravity ranging from 1.20 to 1.40. This method is good & efficient for majority of nematodes eggs. It is, however, not suitable for large and operculated eggs such as *Fasciola*, *Fasciolopsis*, *Diphylobothrium* etc. Chemicals like sugar (sucrose), NaCl, MgSO₄, sodium nitrate, sodium acetate, zinc sulphate and calcium chloride can be used for making saturated solution in floatation method. A gram of faeces is mixed with saturated solution in test tube/centrifuge tube. Tube is first filled half and after thoroughly mixing it is filled up to the top with solution. A clean cover slip is placed over the top of the filled tube so that it should come in contact with solution. Care must be taken that there should not be air bubbles between the cover slip and solution. After about 10 minutes the cover slip is quickly and carefully removed and placed on glass slide. The slide is examined under microscope for eggs/ cysts. Best results are obtained with centrifugation at 1000RPM for 3-5 minutes.

(C) Faecal Culture Method: Specific diagnosis cannot be made on the basis of eggs as eggs of many helminthes parasites having similarity in their morphology. Therefore, it is necessary to culture the hatched eggs. Infective third stage larvae of nematodes have some characteristics and on basis of this specific identification of nematodes can be done. For this purpose faeces is cultured at 27°C for 7-10 days. Fungal contamination is checked daily for fungal growth. Fungal contamination can be controlled by spraying fine mist of tap water and mixing and rolling the faeces to break the fungal growth. Most of the GI nematodes eggs will hatch in seven days except *Nematodirus* spp. which requires 10-14 days. Third stage larvae are harvested and identified under microscope. It will be better to kill the larva by heating with precaution over a small flame until movement of it ceases and lies fully extended.

2. Blood and Lymph for Detection of Parasite: Blood constitutes an important specimen for the diagnosis of infections caused by blood parasites. The examination of blood is useful for the diagnosis of theileriosis, anaplasmosis, babesiosis, trypanosomosis and filarial infection.

(A) Examination of Direct Wet Film of Fresh Blood: Moderate to heavy infections of trypanosomes and filarial worms can be detected by direct examination of fresh blood. A drop of fresh blood is placed on a glass slide; a coverslip is put on it and examined by using a high power objective (40×) of the microscope. The trypanosomes and microfilaria are identified by their size, shape and motility. The correct identification of parasites, however, is made only after permanent staining of the blood smear.

(B) Examination of Permanent Stained Blood Smears: Examination of permanent stained blood smears is essential for accurate identification of blood parasites.

Two types of blood films generally are used for the diagnosis. There are: thin blood smears and thick blood smears.

(C) Thin Blood Smear: Alcohol cleaned and grease free slide is taken. The old slides if used are cleaned first with detergent and then with 70% ethyl alcohol. A one small drop of fresh blood is placed on a glass slide, about 1.5cm from the end of the slide. The drop of blood is touched with the edge of another slide and blood is allowed to spread along the edge. The second slide is quickly pushed across about a 30 angle, along the surface of the horizontal slide to the far end. This helps in distribution of blood in a thin film. Then the film is allowed to air dry at room temperature.

The thin blood smear is used primarily for specific identification of parasites. The disadvantage of the smear is that less amount of blood is examined; the number of parasites per field is much less compared with thick blood smear. Therefore, examination of thin blood smear is not a sensitive method.

(D) Thick Blood Smears: Two to 3 small drops of fresh blood are placed on a grease-free slide. The drop of blood is spread and mixed with the corner of another slide over an area of about 2cm in diameter. The mixing is continued for about 30 seconds to prevent formation of any fibrin strands which may conceal the parasites in a stained smear. If the blood smear is too thick it may fall off during staining. The blood smear is then allowed to air dry at room temperature. Thick blood smear is primarily used as a screening procedure. It is a sensitive procedure for detection of a larger amount of blood. Disadvantages of the thick blood smear are that species identification of parasites can not be made.

Staining Blood Film

1. One type of stain has both fixatives and staining reagents. e.g., Wright, Leishman stains.
2. The other type of stain has only staining reagents e.g., Giemsa stain. Hence the thin films should be fixed before staining with any this stain.

Leishman's Stain: Leishman's stain is prepared by dissolving 0.15gm of Leishman's dry powder in 100ml of absolute methyl alcohol in a bottle. The bottle is shaken until the powder is dissolved and allowed to stand for 48 hrs with frequent shaking in between.

Method:

1. Thin blood smear is flooded with 5-10 drops of stain. After 2 minutes, the stain is diluted by adding twice as many drops of buffered distilled water. The solution is allowed to mix.
2. The solution is allowed for 15-20 minutes for staining.
3. The slide is washed with buffered distilled water, rinsed dry and examined under the microscope.

Staining Buffer:

Na ₂ HPO ₄	0.22g
KH ₂ PO ₄	0.74g
Distilled H ₂ O	1000 ml

Adjust pH between 6.4-6.8

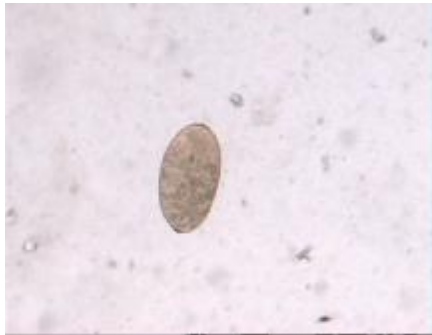
Giemsa Stain:

The stain can be purchased as a ready-made solution. The method of staining is as follows:

1. The film is first fixed with pure methyl alcohol for 3 to 5 minutes and allowed to dry.
2. Giemsa's stain is diluted by adding either 1 drop to each ml of buffered water (pH 6.4 - 6.8) or 1 part stain: 9 parts buffered water.
3. The diluted stain is poured over the film (about 5ml per film is required) and kept for 30 to 45 minutes.
4. The slide is then flushed in a gentle flow of tap water, after which it is placed in an upright position with the film-side inwards to drain and dry.
5. The stained film is examined under oil-immersion lens.

Concentration of Blood: Several concentration methods are used for blood parasites. The methods followed to concentrate protozoan parasites such as trypanosomes and microfilaria. These consist of taking a large quantity of blood (5 to 10 ml) and centrifuging it at a speed of 2,000 revolutions per minute for 5 minutes. The supernatant fluid is decanted and the sediment is examined for microfilariae. The sediment is drawn into a film which is then air dried, fixed and permanently stained or the sediment as a whole may be vitally stained. The various concentration methods vary as to the use of dehaemoglobinising agents which may be distilled water, acetic acid (2 per cent), formalin (2 to 5 per cent) or saponin (1 to 10 per cent). The following are some of the methods:

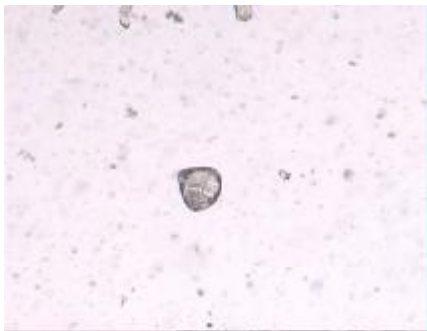
- (a) One to two ml of blood is taken in a test tube containing 5 to 10 ml of a 2 per cent acetic acid solution. The blood is centrifuged next morning and a smear made from the deposit is stained and examined for microfilaria.
- (b) About 5 ml of blood is taken from a vein and is placed in a centrifuge tube containing 10 ml distilled water. The mixture is shaken thoroughly until the blood is haemolysed. It is then centrifuged and the deposit is examined for microfilaria.
- (c) About 20 drops of blood are placed in 10 ml normal saline. To this a few drops of 10 percent saponin solution is added to haemolyse the red blood cells. The specimen is then centrifuged and the living motile larva may be found in the sediment.
- (d) About 5ml of blood is taken in 10ml of citrated saline (2.5 per cent). Next morning the blood is dehaemoglobinised by adding 1.0% saponin solution in normal saline, drop by drop, till the haemolysis is completed. A drop of heparin is then added and mixture is centrifuged. The sediment is examined for microfilaria.
- (e) One milliliter of blood is mixed with 9 ml of a 2 per cent solution of formalin. The mixture is centrifuged at a speed of 2,000 rpm for 5 minutes; the microfilaria is to be found in the sediment.



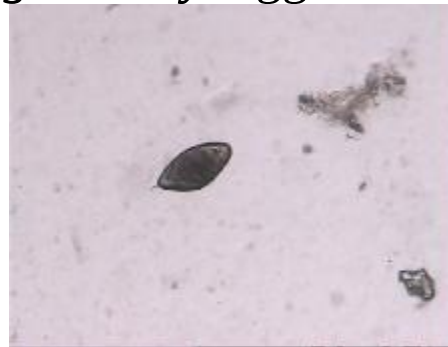
Fasciola egg



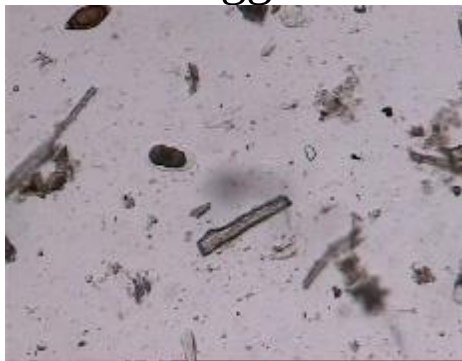
Gigentocotyl egg



Monezia egg



Schistosoma egg



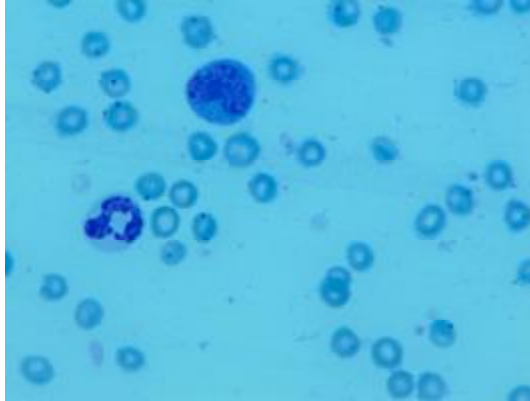
Strongyle egg



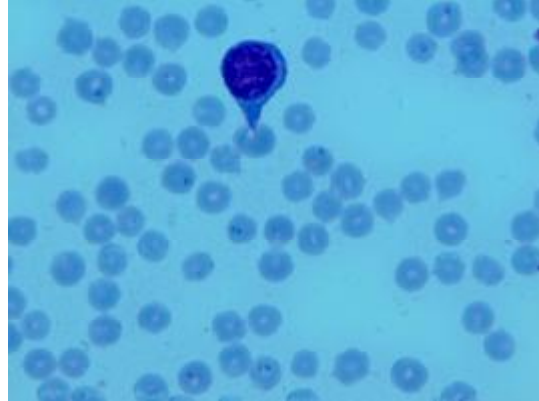
Toxocara egg

Parasitic eggs seen in faecal samples

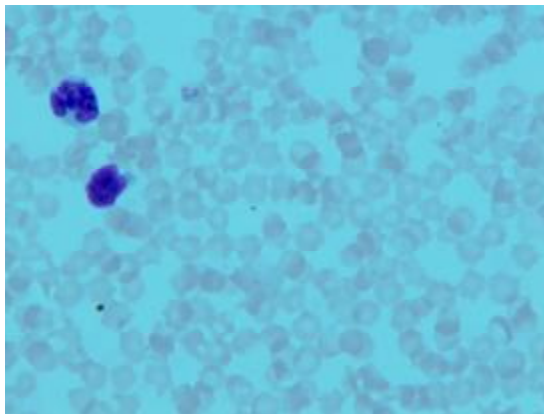
Parasites seen in blood smear of sick animals



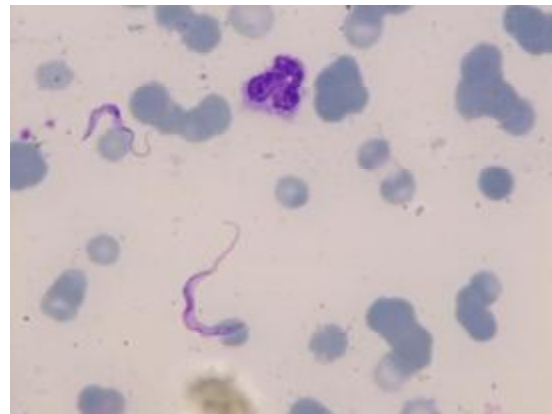
Anaplasma



Babesia



Theileria



Trypanosomes

Chapter 5

Diagnosis and Management of Reproductive Problems in Cattle and Buffaloes

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During the last two decades, the dairy sector in India went through major development programmes, which call in most cases for increased milk production to ensure the country's self-sufficiency in milk and dairy products. Cattle and buffaloes rearing is an important subsidiary to agriculture in India and it has been playing a significant role in India's rural economy. The buffaloes and cows are the multipurpose animals and occupy an important place among the domestic animals as a provider of dairy produce, beef and valuable dung for compost and draught power. India is the largest milk producer in the world, producing 7.2% and 66.3% cow and buffalo milk of the world, respectively. It is well known that the buffaloes are known for its feed conversion ability and are labour intensive and cost-effective. All buffalo breeds have a strong milk/meat trait.

Reproduction efficiency is one of the most important factors for productivity and profitability of dairy animals and it's the primary factor affecting productivity in female buffalo, but is greatly hampered by late attainment of puberty, seasonality of calving, long postpartum anoestrus and subsequent calving interval. In the absence of regular breeding and calving at the appropriate time cattle rearing will not be profitable. In order to improve them, systematic breeding is necessary. A healthy calf each year is the usual goal. So far it has been a neglected aspect world wide. But now a lot of studies related to breeding are in progress. The reproductive efficiency is a complex phenomenon controlled by both genetic and non-genetic factors, the non-genetic factors being climate, nutrition, and level of management. It varies not only between species and breeds but also among the animals within the same breed. Even the best feeding and management can not coax performance beyond the genetic limit of an inferior animal. Improving the genetic merits of livestock populations is important at all levels of management.

Buffalo breeding; like any other branch of animal husbandry, is an entrepreneur in nature, and its success depends to a great extent on the understanding of the whole process of its reproduction and the various factors involved in it. The efficiency of the reproductive process in the water buffalo is linked up with a number of factors controlled by heredity and environment. Among these, its slow growth rate, delayed maturity, seasonality in breeding, higher age at first calving, longer calving intervals etc. are some of the major problems in buffalo breeding. A sound knowledge of the physiology of reproduction of the buffalo is essential for the farmer to have better control over these factors.

The sustainability of dairy farming requires that the dairy sector productivity growth needs to be fostered because only efficient farmers are likely to stand the competitive pressure and to remain competitive in the ever-changing world economy. The objectives of this paper are to address current managerial practices and environmental factors that affecting reproductive indices and to suggest alternative measures that have potential

to improve reproductive efficiency of dairy cows and buffaloes. Here, the reproductive indices of some Indian cattle and buffalo breeds have been listed below:

The major causes of conception failure in individual cows and buffaloes are given below-

Age of Pregnancy- The cows breeding efficiency can be greatly enhanced by lowering the interval between successive pregnancies. The wise general policy is to breed for the first time at an early age and to rebreed at almost the earliest opportunity after each pregnancy. In this way the lifetime efficiency is increased. Cows can be rebred in 10-12 weeks after parturition. The age of the female can influence first service and overall conception rates. The first service and overall conception rates may tend to be higher in yearling heifers than in cows provided the heifers have reached puberty and are cycling. When heifers calve as two year olds, conception rates can be considerably lower, compared to mature cows. Conception rate is generally lower in older cows.

Indian buffaloes, the age at first calving and interval between first and second calving traits showed low estimates of heritability, indicating that those traits should not have a good response for selection; long calving intervals and a large number of days open are characteristics typical of buffalo. First estrus in young animals is generally not prominent than other stages of life. Different experiments show that buffalo heifers should be bred for the first time at their age of 30 months.

Weight of Animals- Weight changes near breeding time affect pregnancy rate. Sixty-seven percent of the cows that held their weight from calving to breeding conceived on first service as compared with 43% in cows losing weight during this period. The pregnancy rate after 21 and 90 days of breeding was also higher in cows holding their weight as compared with cows losing weight.

Body weight of buffalo also influences the estrous cycle. Different experiments show that buffalo heifers should be bred for the first time either at about 360 kg body weight or at the age of 30 months whichever is earlier. During the day, onset of oestrus behaviour was expressed during the cooler periods of the morning and the evening, but the peak was in the morning, while the lower percentage was at noon. In the hot season, buffaloes display the most remarkable sexual activity in the morning and midnight and the lowest activity at noon time. In buffaloes, the average estrous cycle length is 21 days and oestrus period could be for 24 hours. It has been observed that the onset of heat is not very pronounced and often it is difficult to detect. This is the main reason for the highest percentage of the conception failures during the first service.

Season and Environmental Affects- Extensive records on dairy cattle have shown that season of the year influences conception rates. Among all environmental stressors, the temperature and the relative humidity are the major factors, which affect the reproductive performance of female and male. The lower conception rates are associated with both high and low temperatures. The effect is usually observed in August, but sometimes in July or September. To minimize the environmental effects on fertility, producers should reduce heat stress during summer by providing shade if cows are outside.

High environmental temperature above 40°C upsets the normal physiological functions of the buffaloes. High ambient temperature also adversely affects spermatogenesis in the buffalo bulls and ovarian activities in the buffalo. In a study conducted on murrah buffaloes it was found that the conception rate was highly correlated with day length and rainfall. The rainfall may be indirectly correlated with the conception rate through its effects on the growth of vegetation.

Health and Reproductive Disorders- Body condition is important in determining when a cow shows estrus following calving. Ninety-one percent of the cows in good body condition at calving had shown estrus by 60 days post-calving, as compared with 61 % of the cows in moderate condition and 46% of the cows in thin body condition. Calving and post-calving reproductive disorders seriously affect conception rates. The reproductive diseases (*Vibriosis* and *Brucellosis*) will cause lower conception rates. Cows will breed, but then return to heat fairly soon afterwards. A number of diseases (*Leptospirosis*, *Vibriosis* and *Brucellosis*) will cause abortion to occur and leave the animals infertile. Some evidences also indicate cows suffering from metabolic disorders, like milk fever, may have a higher incidence of reproductive disorders and lower conception rates. It generally takes from 30-100 days for cows to cycle following calving. Nutrition prior to calving or even nutrition after calving will influence the post-partum interval. First calf heifers will often require an additional 20-30 days to reach first estrus following calving. Long post-partum intervals can affect conception rates two ways:

1. Lack of estrual activity prevents any chance of conception occurring and
2. Cows will be bred on their first post-partum estrus which often slightly reduces conception rate.

Poor reproductive performance of the animals leads to economic losses due to reduced production and additional cost on management. The incidence of reproductive disorders in buffaloes is increasing over years, ranging from 24.36% to 32.66% and comparatively higher incidences were reported in murrah buffalo as compared to surti buffalo. Among the peri-parturient disorders (pre-parturient disorders, parturition associated disorders and post parturient complications) and general disorders (not associated with parturition), the most important factors affecting fertility is peri-parturient disorders.

Time of Calving- The factors that influence the percent of cows cycling at the start of the breeding season or conception rates early in the breeding season, is when the cow calved relative to the start of the breeding season. Cows bred the first estrus following calving, or soon thereafter, are not as fertile as cows that have had an opportunity to cycle a number of times prior to the start of the breeding season. Cows calving late in the calving season generally have a lower pregnancy rate because they do not have time to show estrus early in the breeding season. Conception rate is higher in cows bred 60 days or more after calving.

The breeding and corresponding calving seasons are almost same throughout India, the breeding season from September to February and the calving season from July to November. During this breeding period, the bulls have been found to be very active sexually and the quality and quantity of semen is very high particularly during winter

(November to February). The buffalo cow show the maximum of ovarian activity and largest percentage of them conceive during this period. The breeding frequency in buffaloes was highest during the winter, decreased in autumn and spring, and was lowest in the summer.

Calving Conditions- A cow at calving contributes to retained placentas and reproductive tract disorders slow the reproductive tract repair processes and can lower conception rates, especially if they do not receive timely and effective treatment. With increased emphasis on growth rate, unfortunately, the selection of sires that can transmit genetic growth through their progeny also causes a fairly dramatic increase in the birth weight of their calves and, subsequently, an increase in calving difficulty.

Interval from Calving to Conception- The conception rates are reduced in cows bred before 50 days after calving. Conception rates increase slightly, after 50 days. Calving interval was affected significantly by season of birth in Indian buffaloes. The lowest length of calving interval was during summer and autumn. Such changes result in depressions in live body weight, growth rate and total body solids and daily body solids gain weight averages and thus impairment of reproduction.

Level of Milk Production- As milk production of dairy cows increases, postpartum nutrition play a larger role in reproductive performance and thus the profitability of dairy herds. The effect of milk production is probably mainly due to the suckling stimulus affecting post-partum intervals by reducing early conception. This improved conception rate will generally relate back to the condition of the animal and the ability to survive more efficiently under the nutritional regime being provided them. However, timing and magnitude of negative energy balance affect luteinizing hormone secretion and, therefore, secretion of progesterone, which affects expression of heat and support of the uterus during the early pregnancy. Early postpartum negative energy balance may cause low fertility by negatively affecting the quality of follicles destined to ovulate during the breeding period.

Three general theories have been proposed to describe how excess dietary protein may negatively influence fertility:

1. Toxic by-products of nitrogen metabolism from the rumen (ammonia) and liver (urea) may impair sperm, ova, or early embryo survival.
2. Imbalances in protein and energy supply may affect efficiency of metabolism.
3. Nitrogen by-products or efficiency of energy utilization may alter gonadotropin and (or) progesterone secretion.

In buffalo, milk yield and fertility are the main factors that affect the profitability of milk herds. As the milk yield is related to the variations in the reproductive activity, then the shorter calving intervals can be associated to bigger milk production during the animal's productive life, besides the possible increase in the number of calves per year. Thus, the genetic importance of the fertility in these herds must be evaluated according to the reproductive performance of the buffalo and its relations to the milk yield.

Semen Quality- Semen quality at the time of breeding is dependent on the quality of semen at the time it was purchased, storage conditions on the farm, thawing procedures

and handling techniques. To maximize conception rate, high quality semen must be placed in the cow's reproductive tract. Short-cuts or bad habits can damage semen and lower its fertility.

Insemination Technique- Faulty insemination technique is a major factor causing low conception rate in many herds. Various studies have indicated that the major difference between artificial insemination technicians with high conception rates and those with low rates was semen placement. The spermatozoa of the buffalo bull are morphologically different from those of the cattle bull which are shorter and narrower.

Timing of Service- Standing behavior is the most reliable sign for predicting when cows will ovulate and, therefore, when they should be inseminated. This is based on the fact that ovulation occurs 24 to 30 hours after the animal first stands to be mounted. The first limitation on the breeding efficiency of fertility of an animal is the number of functional ova released during each cycle of ovulation. Ovulation is the process of shedding of ovum from the Graffian follicle. In the case of cow, usually a single ovum is capable of undergoing fertilization only for a period of 5-10 hours.

Estrus detection errors are usually occurs in herds that breed a large number of cows on the basis of secondary signs of estrus. Progesterone levels are always low for 5-6 days around the day they are in estrus (heat). Cows are never in true estrus when progesterone is high. Estrous detection errors are an individual herd problem and should always be considered as a possible cause of low conception rates. Thus, estrus detection becomes the single most critical factor in timing the service to maximize conception rate.

1. Sperm must be in the cow's reproductive tract for about 6 hours before they acquire the ability to fertilize the egg.
2. Fertilization fails to occur when cows are bred too early or too late.
3. Sperm cells remain alive in the cow's reproductive tract for 18-24 hours. The fertile life of the egg is 10-12 hours, but the most fertile period is the first few hours after ovulation.

Embryonic Mortality- From the time of fertilization till birth, embryonic mortality may occur due to a variety of reasons. Hormone deficiency or imbalance may cause failure of implantation of fertilized ova which die subsequently. Death may occur as a result of lethal genes for which the embryos are homozygous. Other causes may be accidents in development, over-crowding in the uterus, insufficient nutrition or infections in tile uterus.

Management Practices to Improve Breeding Efficiency

Some of the management suggestions which will tend to improve breeding efficiency of cattle are listed below:

1. Keep accurate breeding records of dates of heat, service and parturition. Use records in predicting the dates of heat and observe the females carefully for heat.
2. Breed cows near the end of heat period.
3. Call a veterinarian to examine females not settled after three services and have females with abnormal discharges examined and treated by veterinarian.

4. Get the females checked for pregnancy at the proper time after breeding, preferably after 60 days post AI.
5. Buy replacements only from healthy herds and test them before putting them in your herd.
6. Have the females give birth in isolation, preferably in a parturition room and clean up and sterilize the area once parturition is over.
7. Follow a programmed of disease prevention, test and vaccination for diseases affecting reproduction and vaccinate the animals against such diseases.
8. Practice a general sanitation programme.
9. Supply adequate nutrition, (provide additional energy through challenge feeding during the last 50 days of gestation).
10. Artificially inseminate heifers with semen from calving ease proven sires.
11. Provide early calving assistance when intervention is needed.
12. Provide young cows/buffalo with the best feed resources available after calving.
13. Induce/synchronize estrous cycles in young cows even with natural service.
14. Consider early weaning during drought and cheap feed availability.
15. Provide suitable shelter management.
16. Detect silent or mild heat, by using a teaser bull.

Chapter 6

Recent Approaches in Diagnostics and Management of Bovine Mastitis

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Mastitis, inflammation of the mammary gland is a multifactorial disease which attributes to one of the most difficult mammary pathologies to control. Mastitis is the single most costly disease of dairy cattle resulting in the reduction of milk yield and quality. Inflammatory reaction accompanying mastitis is caused by multifactorial aetiopathological factors. Severe inflammation damages the mammary secretory epithelium and subsequently reduces milk production. The success in control of mastitis is difficult due to the complex aetiopathology and multitude of microbial involvements. The National Mastitis Council estimates an overall loss to animal agriculture of \$2 billion (approximately \$180.00 per cow) due to mastitis (National Mastitis Council, 2005). These costs include reduced production, discarded milk, drug therapy, premature culling and increased labor. Mastitis causes annual losses of over Rs. 6000 crores to India's Dairy Sector. Of this Rs. 1700 crores are lost due to clinical mastitis and Rs. 4400 crores due to sub clinical mastitis (Financial Daily, 2002). Presently antibiotics are used for the treatment of mastitis. However, therapeutic success rate is poor due to indiscriminate use of antibiotics leading to development of multiple drug resistant mastitogen, besides these, antibiotic residues, drug resistant bacteria and their products present in the milk causes threat to consumer's health. These facts highlight the need for completely newer moieties for treatment of mastitis. In the recent years researchers throughout the world are investigating the role of alternative methods for prevention and control of bovine mastitis. These ameliorative measures are bacterial enzymes, antibacterial peptides, bioresponse modifiers, corticosteroids, cytokines, micronutrients, vitamins and traditional medicine. Vaccination against the disease using the recombinant DNA technology has an important role in future as it boosts the animal's immunity and reduces the dependability on antibiotics. Bovine lactoferrin or lactoferrin in combination with a lactam antibiotic can increase the antibacterial activity of these antibiotics against *Staphylococcus aureus*. Lacticin 3147 (broad spectrum bacteriocin) produced by *Lactococcus lactis* is bactericidal against a range of mastitis causing *Streptococci* and *Staphylococci*. The other approach towards the control of mastitis is genetic manipulation by incorporating protective genes against mastitis into the genetic code of dairy cows. A revolution was made with the development of the first transgenic cow clone "Annie" in the year 2000, having mastitis disease resistance. Bovine mastitis can be tackled either by minimizing the pathogens or by stimulating the host immunity. Elimination of pathogen requires both the effectiveness of antimicrobial drugs and optimum functioning of the defense system. One possible approach to control mastitis involves manipulation of the host defense mechanisms. Hence recent strategies aimed at improving the immune cells of diseased udder during immunosuppressive stages would greatly impact the ability of the animal to resist pathogenic infection.

Major Pathogens- Intramammary infection caused by major mastitis causing bacteria is of great importance in terms of treatment and control. These major pathogens are either animal bound and referred as contagious pathogens or the microorganism surrounding the dairy premises and termed as environmental pathogens. Contagious mastitis pathogens are *Staphylococcus aureus*, *Streptococcus agalactiae* are spread from infected dairy animals to healthy animals during the milking process. Streptococci such as *Str. uberis* and *Str. dysgalactiae* and coliforms such as *Escherichia coli* and *Klebsiella* are environmental pathogens. Coliform infections are usually associated with an unsanitary environment. Environmental pathogens are often responsible for most of the clinical cases. About 50% of environmental streptococci infections display clinical symptoms.

Types of Mastitis-

1.Sub clinical mastitis (SCM) - It is estimated that economic losses due to sub clinical mastitis are many fold more than clinical mastitis. Sub clinical mastitis leads to poor quality milk production and it contains higher bacterial load. Such milk is unsuitable for preparation of quality milk products. In sub clinical mastitis there is no visible signs of the disease as such, like there are no changes in the milk or in the udder. SCM is more harmful for the dairy owners than clinical mastitis, because it often goes undiagnosed. SCM attributes to increased Somatic cell count (SCC), higher bacterial load in the milk. It is estimated that there are 20- 40 SCM cases on every cases of clinical mastitis.

2.Clinical mastitis- There is changes in milk and udder. The visible signs which include flakes, clots, blood, pus in the milk or the wateriness of the milk. The udder may be swollen, painful and hardness may be felt on palpation in milder cases. In acute clinical mastitis the animal may have fever loss of appetite, dehydration and depression, if not treated the animal may die.

Diagnosis-

1.California mastitis test (CMT)- It is conducted during milking of cows. The CMT paddle is used to collect about three to four ml of fore milk from each quarter of cows after discarding few streams to which an equal volume of CMT reagent is mixed immediately by Swirling/circular motion. The reaction is graded by intensity of gel formation and colour changes as follows:

CMT Grade	Description	Point Score
Negative (N)	No change	0
Trace (T)	Slime formation which disappeared with continuous movement of paddle	1
Weak (+)	Distinct slime, but no gel formation slight purplish color	2
Distinct positive (++)	Viscous with gel formation which adherent to the margin of the cup with colour changes	3
Strong positive (+++)	The gel form convex projection, the gel does not dislodge after swirling movement of paddle	4 & 5

California mastitis test (CMT) is routinely used for rapid diagnosis of mastitis in dairy. CMT reaction is due to the presence of polymerized DNA originating from inflammatory cells of mammary tissue leading to gel formation. This test is based on increased leukocyte count and high alkalinity of milk due to inflammatory exudate as well as increased level of basic salt.

CMT Reagent

Bromocresol purple	5 mg
Sodium hydroxide (NaOH)	15 g
Teepol (A.G.)	15 ml
Distilled water up to	1000 ml

2.Somatic cell count (SCC)- The SCC in milk is done by taking the milk samples thoroughly mixed by shaking the vials and 10 µl of milk was taken on the predrawn one sq. cm. area over a grease free clean glass microslide which is uniformly smeared with a fine sterile glass rod. The smears are dried and examined after staining them with modified Newman's Lampert stain . The counting of cells in 10 different fields was carried out under oil immersion lens (100X) and the counting was repeated thrice per smear to assess average number of somatic cells in 30 fields. Somatic cell count (SCC) basically comprises leukocytes e.g. Polymorphonuclear neutrophils (PMNs), lymphocytes, macrophages and epithelial cells (0-7%). SCC can be used as a guidance tool for predicting mastitis. A SCC of <1, 00,000 cells/ml is often considered to be normal, reflecting a healthy mammary gland, whereas an SCC of >2, 00,000 cells/ml is suggestive of bacterial infection

Modified Newman's Lampert Stain

Methylene blue	1 g
Ethyl alcohol	54 ml
1,1'2,2' tetrachloroethane	40 ml
Glacial acetic acid	6 ml

3.Isolation of pathogenic bacterial from milk samples- The bacteriological isolation procedures are carried out under strict sterile environment. The identification of pathogenic causative organism in milk samples is carried by inoculating 10 µl of milk which is spreaded over 5% bovine blood agar plates with a sterile 'L' shape glass rod (Griffin et al., 1977).

Procedure- With the help of a sterile tip 10 µl of milk is pipetted with the help of micropipette and put over the agar plate. For spreading a sterile 'L' shape glass rod is used and the milk is spread all over the agar plate evenly. The plating is done in sterile environment. The inoculated plates are incubated at 37°C for 24 hrs. The causative organism of the milk samples can be identified initially on the basis of colony morphology and smell on 5% blood agar as per Cruikshank (1962).

Blood agar- Blood agar base is generally used for making of the blood agar. The blood agar base is weighed and transferred into a Erlenmeyer flask .The distilled water is poured over the agar powder to dissolve the content and sterilized at 15 pound pressure for 15 minutes. 5% cow or sheep fresh blood is added when the temperature of the blood agar base comes to 45 ° C and the mixture is poured in sterile petri dishes to make blood agar plates. The plates are then incubated at 37 ° C for 18 hours and after that only those plates are used for bacterial isolation, which have no bacterial growth.

Preparation of Blood agar media

Nutrient agar	6.8 g
Distilled water	200 ml

4.Determination of milk pH- The milk pH is determined with the help of bromocresol purple indicator dye.

Interpretation- The color change to dove grey with the dye- normal milk; the color change is purple with the indicator dye- mastitic milk. The pH of the normal milk ranges from 6.5 to 6.8. The pH of the colostrums' is 6.4 or lesser. The milk secretion during the midlactation generally varies from 6.6 to 6.7 and near to the end of lactation is 6.8 or more.

Bromothymol blue (BTB) test- For BTB card test, Whatman filter paper No. 1. is prepared by adding one drop of BTB test solution (Bromothymol blue-1.6 g in 100 ml ethanol) at 4 different spots on the paper like , left fore (LF), left hind (LH), right fore (RF) and right hind (RH). One drop of suspected milk is directly poured on the indicator spot and observed for the change in color and scored as follows:

Pale green indicates normal quarter and +,

Green ++

Dark Blue Green +++

The disadvantage of this test is that cow in later stages of lactation may give false positive reaction.

5.Total bacterial count-

- a. **Pour plate method-** One ml of milk is diluted with sterile normal saline solution at the ratio of 1:20, 1:40 and 1:100.1 ml of the diluted milk is placed into a sterile Petri disc and melted blood agar id poured into the plate and rotated to mix with the milk. After that the plates are allow to solidify and transferred to the incubator at 37⁰ C for 20 hrs. The number of colonies are counted under the colony counter and multiplied with the dilution factor, which gives the total bacterial count/ ml of milk.
- b. Numerous immunoassays have been developed for the detection of pathogens in milk and are used for monitoring milk quality. Immunoassays, such as enzyme-linked immunosorbent assay (ELISA), can provide a reliable and inexpensive approach provided that suitable antibodies are available against specific inflammation-related biomarkers or the causative microorganisms.

6. Immunoassays

- i. **Staphylococcus aureus antibody test kit (SAATK)-** More than one hundred known organisms can be responsible for causing mastitis but ELISAs have only been developed for some of the most prevalent pathogens, such as *S. aureus*, *Escherichia coli* and *Listeria monocytogenes*. For example, an *S. aureus* antibody test kit (SAATK) was assessed as a primary screen for cows suspected of having an *S.aureus* infection. Immunoassays can also be used to detect inflammation- related biomarkers present in the milk at different stages of sub-clinical mastitis. For example, Hp (heptaglobin) concentrations have been reported to increase significantly in plasma,

as well as in milk, during mastitis and thus Hp was suggested as a potential marker for diagnosis.

- ii. **A magnetic-bead-based ELISA** was developed for the detection of staphylococci using beads coated with an anti- *S. aureus* monoclonal antibody .Flow cytometry was also used to detect antibodies to *S. aureus* in milk.

7. Nucleic acid detection assays- The genome sequences of many of the major mastitis causing pathogens are now available and can be utilized to develop nucleic acid-based testing methods, such as PCR. Such tests are generally more expensive than, for example, immunoassays. However, they are highly sensitive and specific, can be performed rapidly (e.g. 'real-time' PCR) and can overcome the sensitivity and time-constraints sometimes encountered with culture-based tests and thus could complement or replace them in the long-term. PCRs allow the identification of closely related organisms within a few hours.

- i. **Multiplex PCR and 'real-time'**- PCR assays that can simultaneously detect different mastitis- causing organisms in milk samples .The most recently developed assay is capable of detecting 11 of the major mastitis-associated pathogens, including *E. coli*, *S. aureus*, *Str. agalactiae* and *Streptococcus uberis*.
- ii. **Nucleic acid sequence based amplification (NASBA)**, is used for quantification of RNA, has an advantage over PCR methods in that it is capable of discriminating between dead and living organisms, and real-time NASBA for the detection of *Bacillus cereus* in milk has been developed.

8. Biosensors have also been developed to detect mastitis. The instrument use a biological receptor molecule (e.g. antibody, enzyme, nucleic acid) in combination with a transducer to produce an associated signal, allowing observation of a specific biological event (e.g. an antibody–antigen interaction). Biosensor assay using surface plasmon resonance to monitor the interaction between Hp, which was immobilized onto the chip surface, and haemoglobin (Hb) to discriminate between sub-clinical mastitic and non-mastitic milk. Hp binds strongly to Hb. Therefore, by mixing the milk sample with Hb, any Hp present in the milk sample will bind to the Hb, thus preventing the binding of the Hb to the immobilized Hp. In the absence of Hp (i.e. in an uninfected milk sample), Hb will bind to the immobilized Hp, resulting in a positive signal.

Conventional Antibiotic Therapy- The most common bacterial infection in dairy cows is mastitis. It is generally treated by conventional antibiotic therapy, however, antibiotic therapy of established mammary infection during lactation are only moderately efficacious. The conventional antibiotic therapy generally fails due to lactation drainage, inflammation of the mammary parenchyma, poor activity of mammary cellular defense and infections around the dairy farms. Most common antibiotics which are used to treat mastitis are generally marketed in the form of intramammary tubes or injectable antibiotics for parenteral administration. These antibiotics are effective in such cases which are diagnosed immediately and rushed for treatment by the veterinarians. Proper dosing and duration of the treatment should be strictly followed by optimal results. Along with antibiotics, adjunct therapy can also be advised such as Vitamin E +

selenium, vitamin C and mineral mixture for minimizing the duration of antibiotic and for fast recovery.

Prevention and Control - Pre and post milking teat antisepsis is regarded as the single most effective mastitis control practice in lactating dairy animals. The key control point in minimizing mastitis organism spread involves teat dipping, immediately after milking with an effective teat dip solution. Teat dip, applied properly, kills these bacteria and the risk is minimized. Teat dip not only clears up existing infections or shortens their duration but it also reduces the spread of bacteria from infected cows to uninfected cows and it therefore limits the spread of the infection. Teat dipping is simple, effective and economical means to reduce bacterial population on teat skin. An effective teat dip, correctly used reduces the incidence of new udder infections by 50-90%. Though the teat dip with the conventional chemical sanitizers are very effective in reducing the incidence of new intramammary infection but the major concern is the potential of increased chemical residues in milk. Most of the pre and post milking teat dips are chemical in nature, such as iodine, chlorhexidine based, hydrogen peroxide, glutaraldehyde and various acid based compounds against the major mastitic pathogens, like *Staphylococcus aureus*, *Streptococcus agalactiae*, CNS, *Corynebacterium* and *Escherichia coli* and were found to be very effective against the major pathogens. The chemical based teat dips are often associated with very poor teat skin condition such as cracking of the teat skin, hardening of the skin, roughness of the skin, and trans epidermal water loss, the demerits of chemical teat dip warrants the need of alternative teat dip for prevention of bovine mastitis. Preparation of polyherbal post milking teat dipping powder for the prevention of bovine sub clinical mastitis may be of great value. A poly herbal teat dip powder is prepared containing *Phyllanthus emblica*, *Azadirachta indica*, *Ocimum sanctum* and *Lawsonia innermis*. Teat dip application in lactating cows is done by taking teat dip powder soaked in water for over night, strained and used as teat sanitizer by immersing the teat in the teat dip solution up to 2/3 length of the teat of the individual cows, teat dipping was done in diagonal fashion.

Chapter 7

Diagnosis and Management of Sexually Transmitted Infections

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The Indian livestock is in the state of great revolution with the objective of increased production by proper health care, management and breeding programme. There are number of diseases transmitted sexually through natural mating, AI, embryo transfer etc. One good bull can utilized for breeding many cows, either by natural service or AI; if one bull or cow is infected with any of the sexually transmissible disease it can infect a large number of animals. The important sexually transmissible diseases that are of great concern in our country are Brucellosis, IBR, Campylobacteriosis, Leptospirosis, Trichomoniasis etc.

Brucellosis- rucellosis is an acute and chronic contagious disease of domestic animals causes placentitis and abortion. It is widely prevalent throughout the country among bovine population. It causes huge economic loss to livestock industries through abortion, delayed conception, temporary or permanent infertility.

Etiology and epidemiology- Brucellosis is an infectious disease of animals caused by *Brucella* spp. The disease in cattle, water buffalo, and bison is caused almost exclusively by *B. abortus* and occasionally by *B. suis* or *B. melitensis*. Infection spreads rapidly and causes abortions in unvaccinated cattle herd. In a herd in which disease is endemic, an infected cow typically aborts only once after exposure; subsequent gestations and lactations appear normal. Organisms are shed in milk and uterine discharges, and the cow may become temporarily infertile. Bacteria may be found in the uterus during pregnancy and following parturition shedding from the vagina largely disappears with the decrease of fluids. Some infected cows that previously aborted shed brucellae from the uterus at subsequent normal parturitions. Organisms are shed in milk in most cattle for life.

Transmission occurs by ingestion of contaminated feed and water with aborted fetuses, fetal membranes, and uterine discharges. Cattle may lick contaminated genitals of other animals. Venereal transmission by infected bulls to susceptible cows appears to be rare. Transmission may occur by artificial insemination when *Brucella* -contaminated semen is deposited in the uterus but, reportedly, not when deposited in the midcervix. *Brucellae* may enter the body through mucous membranes, conjunctivae, wounds, or intact skin.

Clinical signs- Abortion in last trimester of pregnancy (6 months onwards) is the most obvious manifestation. Infections may also cause stillborn or weak calves, retained placentas, and reduced milk yield. Organisms are localized in the supra-mammary lymph nodes, iliac lymph nodes and retropharyngeal lymph nodes. Animal may develop hygroma (bursitis). Usually, general health is not impaired in uncomplicated abortions. In bulls, seminal vesicles, ampullae, testicles, and epididymides may be infected leading to epididymitis and orchitis; therefore, organisms are present in the semen. Testicular abscesses may occur.

Diagnosis- Diagnosis is based on clinical sign, bacteriology, serology and molecular detection. *B. abortus* can be isolated from the placenta, stomach and lungs of an aborted

fetus. Most cows cease shedding organisms from the genital tract when uterine involution is complete. Foci of infection remain in some parts of the reticuloendothelial system, especially supramammary lymph nodes, and in the udder. Serum agglutination tests have been the standard diagnostic method. Agglutination tests may also detect antibodies in milk, whey, semen, and plasma. An ELISA has been developed to detect antibodies in milk and serum. Other tests that may be used are complement fixation, rivanol precipitation, and acidified antigen procedures.

Serological tests	Immunoglobulin class	Remarks
Herd test- Brucella milk ring test	IgM ₁ , IgM, IgG ₁	Pooled milk from herd used
Agglutination test, Rose Bengal plate test (RBPT)	IgG ₁ , IgM	Antigen buffered to 3.6 - 4.0 & IgG agglutinate
Tube agglutination test, standard serum agglutination test	IgM, IgG	Widely used IgG fail to agglutinate and possibility of false negative
Complement fixation test (CFT)	IgG ₁ , IgM	
ELISA	Detect all IgGs	
Coombs antiglobulin test	IgM, IgG ₁ , IgG ₂	Sensitive and can detected incomplete antibody
Heat Inactivation test	IgG	Used for differentiation of non specific
Rivanol precipitation test	IgG	Reaction by destroying IgM
Mercaptoethanol treatment test	IgG	Antibody produced post vaccination

Screening Tests-

- 1) **Brucella milk ring test (BRT), Milk ring test (MRT) or Abortus Bang ring test (ABRT)**- In official control and eradication programs on an area basis, the BRT has been effective in locating infected dairy herds, but there is a high percentage of false positive tests. The brucellosis status of dairy herds in any area can be monitored by implementing the BRT at 3 to 4 month intervals. Cows in herds with a positive BRT are individually blood tested, and reactors are slaughtered.
Two drops of antigen is added in 2 ml thoroughly mixed suspected pooled milk and gently shake to mix. Incubate 1 hr at 37°C followed by room temp for 2 hrs and

positive result indicates by cherry colour ring at creamy layer in the milk. Whole milk remains pink is considered as negative.

- 2) **Rose Bengal plate test (RBPT)**- It is performed for the rapid screening of brucellosis in nondairy and dairy herd animals or big groups of animals. Place a drop of suspected serum on a slide and adjacent to it add one drop of *B. abortus* colored antigen. Mix with matchstick. Check the antigen for auto agglutination replacing the serum with drop of saline. The presence of antibodies in the serum against *B. abortus* has shown by distinct clumps of colored antigen within 1-3 minutes. If serum does not contain antibodies, then the mixture will remain smooth and homogenous.
- 3) **Standard tube agglutination test (STAT)**- The quantitative estimation of exact amount of antibodies in the serum has determined by this test as well as indicates opinion about the infected cows or buffalo. **This test is most widely used for the purpose of international trades.** In this test bacterial cells forms clumps at the bottom and supernatant will remains clear and transparent is indicative of positive test and no clumps are formed in negative test and supernatant will look turbid just like antigen control. The highest dilution of serum in which there is agglutination is called titre of serum. A titre of 1:40 was considered positive for cattle and buffaloes while a titre of 1:20 was considered positive for sheep, goat and breeding bulls.

Brucellosis-free areas can be achieved and maintained, effectively and economically, by using the BRT on dairy herds and through agglutination test. Supplemental tests using sensitive screening methods may be used in cattle in which the brucellosis status is unclear. Use of a battery of these tests improves the probability of detecting infected cattle that have remained in some herds as possible reservoirs of infection. Supplemental tests are also used to clarify the results of plate or tube tests, especially in serum samples from vaccinated cattle.

Molecular detection: Polymerase chain reaction for omp gene or 16S rRNA are also useful for confirmation of *Brucella* infection.

Control- Efforts are directed at detection and prevention because no practical treatment is available. Eventual eradication depends on testing and eliminating reactors. The disease has been eradicated from many individual herds and areas by this method. Herds must be tested at regular intervals until 2 or 3 successive tests are negative.

Non infected herds must be protected. The greatest danger is from replacement animals. If pregnant or fresh cows are added, they should originate from brucellosis-free areas or herds and be sero negative. Replacements should be isolated for ~30 days and retested before being added to the herd.

Vaccination of calves with *B. abortus* Strain 19 or RB51 increases resistance to infection. *B. abortus* killed 45/20 vaccine commonly used in tropical countries to vaccinate adult animals. Vaccination as the sole means of disease control has been effective. Reduction in the number of reactors in a herd is directly related to the percentage of vaccinated animals. However, when proceeding from a control to an eradication program, a test and slaughter program is necessary. The low prevalence of brucellosis in cattle in the USA has resulted in reduced use of vaccines and current emphasis on depopulation of infected herds.

Bovine Genital Campylobacteriosis- Bovine genital campylobacteriosis is a venereal disease of cattle characterized primarily by early embryonic death, infertility, a protracted calving season, and occasionally abortion. Distribution is probably worldwide.

Etiology and epidemiology: The causative agent is a motile, gram-negative, curved or spiral, polar flagellated microaerophilic bacteria *Campylobacter fetus* subsp. *venerealis* (Cfv) or *C. fetus* subsp. *fetus* (Cff). Cfv is associated with BGC, causing fertility problems with considerable economic losses, particularly in endemic regions. Cff present in intestine and are associated with sporadic abortion (4-7 months gestation) after getting entry to the reproductive system. In bulls, Cfv is confined to the glans penis, prepuce and distal portion of the urethra. *Campylobacter* spp are very labile and are destroyed quickly by heating, drying, and exposure to the atmosphere. Unless cultured quickly after collection from the animal and grown under microaerophilic or anaerobic conditions, campylobacters will not grow.

Cfv is transmitted venereally and also by contaminated instruments, bedding, or by artificial insemination using contaminated semen. Individual bulls vary in their susceptibility to infection; some become permanent carriers, while others appear to be resistant to infection. The primary factor associated with this variability seems to be the age-related depth of the preputial and penile epithelial crypts. In young bulls (<3-4 year of age), in which the crypts have not yet developed, infection tends to be transient. Spontaneous clearance in these younger bulls does not seem to be related to any immune response, so reinfection can readily occur. In bulls >3-4 year old, the deeper crypts may provide the proper microaerophilic environment required for the establishment of chronic infections.

In cows, the duration of the carrier state is also variable; some clear the infection rapidly, while others can carry Cfv for ≥ 2 year. IgA antibodies are shed in cervical mucus in significant amounts in ~50% of cows for several months after infection and are useful diagnostically. Although most of the genital tract may be free of infection when a cow eventually conceives, the vagina may remain chronically infected through pregnancy.

Clinical signs- Cows are systemically normal, but vaginal mucus may be red with mild vaginitis. The mucopurulent endometritis can occur after early embryonic death leading to prolonged luteal phases, irregular estrous cycles, repeat breeding. Observed abortions are not common. Old cows and heifers are the most susceptible animals due to low levels of immunity. Infected bulls typically do not show overt signs of disease and could remain as carriers for extended periods of time and produce normal semen.

Diagnosis-

1. Clinical observations such as increased rate of infertility, irregular estrus, repeat breeding and increased calving interval may be useful in diagnosis.
2. **Isolation of organism** – prescribed test for international trade:
 - a. **Collection of samples**
 - i) **Preputial smegma and semen-** In bulls, smegma is commonly collected by scraping into a tube with approximately 5 ml of phosphate buffered saline (PBS) with 1% of formalin for immunofluorescence (IFAT) diagnosis. Smegma can also be collected from the artificial vagina after semen collection, by washing the artificial vagina with 20–30 ml of PBS. Preputial washing can be done with 20–30 ml of PBS introduced into the preputial sac. After vigorous massage for 15–20

seconds, the infused liquid is collected. Collected semen samples diluted with PBS and are sown directly onto culture medium or transport and enrichment medium.

- ii) **Cervico vaginal mucus (CVM)**- Collected by aspiration, or washing the vaginal cavity. An artificial insemination (AI) pipette is inserted into the vaginal cavity so that the anterior reaches the cervix (3). Gentle suctioning is applied while moving the pipette gently backwards and forwards.
- iii) **Aborted fetuses, placentas**- The placenta as well as the liver, lungs and stomach contents of the fetus collected for isolation of the causative bacteria.
- b. **Transport of samples**- The samples should be transported in transport and enrichment media (e.g. Clark's, Lander's, SBL, Foley's and Clark's, Weybridge's, Cary-Blair's).
- 3. **Cervicovaginal mucus agglutination test (CMAT) and ELISA (detection of IgA antibodies)**- Systemic antibody responses are not helpful because they are often due to nonpathogenic *Campylobacter* spp. A CMAT is useful, but due to variability in individual responses, at least 10% of the herd or at least 10 cows should be sampled. An ELISA test has been developed for use on vaginal mucus and is said to be more sensitive and able to detect a wider range of antibody responses than the CMAT.
- 4. **Molecular detection by PCR**- Recent molecular techniques for the identification of *C. fetus* subspecies have been described, including 16S sequencing, PFGE, AFLP, and MLST.

Treatment and control- Infected animals must be kept isolated and their services must be avoided. The bulls used for natural service as well as semen used for AI must be free from campylobacter infection. Newly purchased bulls and cows must be tested before introducing to the herd. Aborted fetus, placenta, fetal membranes, discharges etc must be disposed off hygienically. Entry of dogs and wild birds should be prohibited to prevent spread of infection. Resting of two heat periods before breeding can be done. Treatment of semen used for AI with broad spectrum antibiotics. Treatment with antibiotics may be useful in cows but difficult in bulls. Vaccination with vaccine prepared against S layer protein of *C. fetus* should start as soon as genital campylobacteriosis is diagnosed. Both infected cows and cows at risk should be vaccinated. Vaccination of infected cows hastens the elimination of *C. fetus*. In routine use, the vaccine should be given once, thirty days before breeding starts. Bulls are vaccinated for the same reason as cows (i.e., for treatment as well as for prophylaxis) but are given twice the dose used for cows, 3 week apart. The infection can also be eliminated in bulls by treatment with streptomycin (20 mg / kg, s.c., 1-2 treatments) together with 5 g of streptomycin in an oil-based suspension applied to the penis for 3 consecutive days.

Infectious Bovine Rhinotracheitis (IBR)- Infectious Bovine Rhinotracheitis (IBR) is a highly contagious, infectious respiratory disease that is caused by Bovine Herpesvirus-1 (BHV-1). It can affect young and older cattle. In addition to causing respiratory disease, this virus can cause conjunctivitis, abortions, encephalitis, and generalized systemic infections.

Etiology and epidemiology- Bovine herpesvirus 1 (BHV-1) is associated with several diseases in cattle: infectious bovine rhinotracheitis (IBR), infectious pustular vulvovaginitis (IPV), balanoposthitis, conjunctivitis, abortion, encephalomyelitis, and mastitis. Only a single serotype of BHV-1 is recognized; however, three subtypes of

BHV-1 have been described on the basis of endonuclease cleavage patterns of viral DNA—BHV-1.1 (respiratory subtype), BHV-1.2 (genital subtype), and BHV-1.3 (encephalitic subtype). BHV-1.3 has been reclassified as a distinct herpesvirus designated BHV-5. BHV-1 infections are widespread in the cattle population. The viral infection alone is not life-threatening but predisposes to secondary bacterial pneumonia, which may result in death. In breeding cattle, abortion or genital infections are more common. Genital infections can occur in bulls (infectious pustular balanoposthitis) and cows (IPV) within 1-3 days of mating or close contact with an infected animal. Transmission can occur in the absence of visible lesions and through artificial insemination with semen from subclinically infected bulls. Cattle with latent BHV-1 infections generally show no clinical signs when the virus is reactivated, but they serve as a source of infection for other susceptible animals.

Clinical signs- Clinical signs in respiratory form ranged from mild to severe, depending on the secondary bacterial pneumonia. High fever, anorexia, coughing, excessive salivation, nasal discharge which progresses from serous to mucopurulent, conjunctivitis with lacrimal discharge, inflamed nares (hence the common name “red nose”), and dyspnea are the clinical signs. Conjunctivitis with corneal opacity may occur as an important manifestation of BHV-1 infection. Abortions may occur concurrently with respiratory disease but may be seen up to 100 days after infection. Abortions generally occur during the second half of pregnancy, but early embryonic death is possible. Frequent urination, elevation of the tail, and a mild vaginal discharge are the first signs of infections. The vulva is swollen, and small papules, then erosions and ulcers, are present on the mucosal surface.

Diagnosis-

- Identification of characteristic lesions
- Virus isolation-Samples should be taken early in the disease
- Demonstration of intra-nuclear inclusion bodies in infected cells.
- Demonstration of virus in fetal tissues by immunoperoxidase, or fluorescent antibody staining.
- Haemagglutination test
- Virus Neutralization test : prescribed test for international trades
- ELISA: prescribed test for international trades (indirect or blocking ELISA)
- Detection of nucleic acid by PCR

Treatment and control- Antimicrobial therapy is indicated to treat secondary bacterial pneumonia. Immunization with modified live or inactivated virus vaccines generally provides adequate protection against clinical disease. Both i/m and intranasal modified live vaccines are available, but the i/m types may cause abortion in pregnant cattle. The intranasal vaccines can be used in pregnant cattle. Calves should be vaccinated two to three weeks before weaning at which time they start to be at risk of infection. The marker vaccines should also be used.

Eradication of the virus is possible by serologic testing and either culling reactors. To aid in eradication, deletion mutant vaccines have been developed that permit discrimination between antibodies produced in response to the vaccine and antibody produce in response to natural exposure. Carrier cattle should be identified and removed from the herd.

Chapter 8

Important Poultry Diseases and their Diagnosis

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The poultry industry in India is considered as the most dynamic industry in the agribusiness sector, with a contribution of 1% to the national GDP, however, the constant risk of disease is the one which causes great economic losses to the industry apart from adversely affecting the export of poultry and poultry products in the WTO regime due to public health implications. Among the diseases which are more commonly encountered in layer and broiler chicks are colibacillosis, pullorum, fowl typhoid, fowl cholera, brooder pneumonia, coccidiosis, chronic respiratory disease (CRD), hydropericardium syndrome (HPS), avian influenza, infectious bursal disease, chicken infectious anemia (CIA), infectious laryngotracheitis (ILT), mycotoxicosis etc.

Colibacillosis- Colibacillosis in chickens occurs as an acute fatal septicemia or sub-acute pericarditis and airsacculitis and is caused by pathogenic *Escherichia coli*. In poultry, it is a common systemic disease of economic importance and is seen worldwide. The disease mainly affects young growing chicks especially broilers and is characterized by decreased growth rate and feed conversion efficiency, increased mortality in the flock, and down grading and condemnation of carcasses in processing plants. It causes serious threats if favored by stress in growing broilers, unhygienic management of farms and in immunosuppressive conditions caused by chronic respiratory disease and other respiratory viral infections. Diets containing aflatoxins can lead to immunosuppression leading to increased susceptibility to *E. coli* infection. Similarly flocks infected with vertically transmitted or acquired mycoplasmosis are extremely susceptible to *E. coli* airsacculitis. The most common age group for this disease is chickens in the age group of 5 to 10 weeks and this age is higher than those for paratyphoid infection. Broilers are most susceptible to this disease. The symptoms of colibacillosis depends on presence of concurrent disease condition, age of the chickens and organs involved. The main symptoms in young chicks include diarrhea, soiled vent feathers and the faeces are pasty. Symptoms in broilers and growers are drop in food consumption, depression, difficult breathing, sneezing, huddling in a corner, ruffled feathers standing dejectedly with heads down. The symptoms resemble to fowl typhoid. Affected birds show poor weight gain. In layers decreased egg production. Mortality may vary from 5 to 10% and in some cases it can be upto 50%. Young chicks dying due to this disease have acute septicaemia, enlarged hyperemic liver and spleen and presence of fluid in body cavities. Most common lesion of this disease is presence of milky fluid in the pericardium due to fibrinous pericarditis. Liver, spleen, lungs and kidneys become dark and congested. Airsacs are thickened and show cloudy appearance. Birds surviving acute air sacculitis show stunted growth and develop caseous exudates in the air sacs accompanied by peritonitis resulting in down grading of carcass at processing. There may be fibrinous layer on the liver. Chickens those survive septicemia develop subacute fibrinopurulent airsacculitis, pericarditis, perihepatitis, and there is lymphocytic depletion of the bursa and thymus. Airsacculitis is a classic lesion of colibacillosis.

Affected birds can be treated by administration of water soluble furazolidone, sulfonamides or the birds can be given antibiotics such as enrofloxacin, pefloxacin, tetracycline for 3 to 5 days. It is necessary to perform the culture sensitivity to ensure that selected drugs are effective. During outbreak of this disease addition of vitamin C at the rate of 330 mg per kg of feed reduce mortality to a great extent. Vitamin A and vitamin E increases cell mediated and humoral immunity and may be given during outbreak of this disease. Supply of clean water with chlorination at 2 ppm level is effective in prevention of this disease. Dirty and open drains, ditches or ponds should be avoided as these serve source of *E. coli*. Penetration of *E. coli* in hatching eggs should be avoided by washing of eggs in hot water (43.5 to 46 °C) and having chlorine as disinfectant.

Pullorum Disease- Pullorum disease or also known as bacillary white diarrhoea is a bacterial disease which world-wide in distribution. It is highly contagious and egg transmitted disease and is transmitted from infected hen to the egg. This disease is caused by *Salmonella pullorum*. This organism can survive outside the body for many months. This disease occurs in poultry, ducks, turkey, quails and pigeons. As it is the disease of chicks predominantly below 3 weeks of age and the first indication of this disease is excessive number of dead-in-shell chicks and deaths shortly after hatching. The newly hatched chicks usually die without showing any prominent signs. Chicks which are infected with this infection show loss of appetite, huddling near source of heat, depressed and white diarrhoea with accumulation of whitish fecal pasting around the vent. In birds above 14 days of age main symptoms include poor growth rate and feathering, frequently lameness due to arthritis. In adult birds lower egg production and symptoms are very much similar to fowl typhoid. Also there may be depression, paleness and shrinkage of comb with ruffled feathers. The newly hatched chicks die from this disease without showing any signs but some show classic gray nodules in the liver, spleen, lungs, heart, gizzard, and intestine. Consistent findings are presence of firm and cheesy material in the caeca, peritonitis, and inflamed, coagulated and unabsorbed egg yolk sac. Liver congested with grayish necrotic spots of 1 to 2 mm in size. Lungs may be congested with liver dark and swollen. Distinct, small, white, necrotic foci are usually found in the liver, lungs, heart and gizzard wall. Affected birds may be given antibiotics such as ampicillin and other drugs such as nitrofurantoin and sulphamizole but treatment does not prevent the birds from becoming carriers. More effective method of disease eradication of this disease is testing of the birds and sacrificing the positive and doubtful reactors among the breeding flock.

Fowl Typhoid- This is an acute or chronic septicaemic disease which affects birds more commonly at 3-6 weeks of age. This disease also affects the newly hatched chicks and mortality in this disease continues up to the age of maturity. Mortality in flock may reach up to 50%. This disease occur in chickens more commonly but other species such as quails, turkeys, guinea fowl and pheasants are also known to be affected with this disease. Mortality due to fowl typhoid in young birds is similar to *S. pullorum* infection but may be higher in older birds. Clinical signs and lesions in young birds are similar to those of infection with *S. pullorum*. The older bird may be pale, dehydrated, and have

diarrhea. Lesions in the older bird may include a swollen, friable, and often bile-stained liver, with or without necrotic foci, enlarged spleen and kidneys, anemia, and enteritis.

Usually this disease occurs in growing and adult birds. The symptoms are similar to those seen in pullorum in young birds with drop in food consumption. Affected birds show depression, loss of appetite and the most characteristic sign is a watery to mucoid yellowish diarrhoea. Birds show yellow, pasty droppings which adhere to the feathers around the vent. There is decrease in weight gain and ruffled feathers. The carcass of the birds died in the acute phase of the disease show septicaemic and jaundice appearance. Pathologically gross enlargement of liver and spleen with characteristic copper colour is pathognomonic. Necrotic spots on liver, lungs and gizzard as seen in pullorum are also evident. Therapy with number of antibacterial agents will reduce clinical signs and mortality in a flock affected with *S. gallinarum*. Furazolidone continuously in feed at a level of 0.04 % is effective in treatment of this infection. Treatment should also be accompanied with culling of chronic carrier of this infection by a blood test and prompt removal and incineration of dead birds.

Pasteurellosis (Fowl Cholera)- Pasteurellosis which is also known as fowl cholera, is a septicaemic caused by bacteria called *Pasteurella multocida*. This disease is an acute, highly contagious, which is more commonly affects adult birds. It is distributed worldwide and was one of the first infectious diseases to be recognized, by Louis Pasteur in 1880.

The disease can range from acute septaemic form to chronic and localized form and the morbidity and mortality may be heavy and may range up to 100%. This organism can easily be destroyed by environmental factors (sunlight, drying, and heat) and disinfectants, but it may persist for prolonged time in soil. Reservoirs of infection may be present in other species such as rodents and cats and can act as source of infection to domestic poultry birds. Predisposing factors for this disease include high density of birds and concurrent infections such as respiratory viruses. Although *P. multocida* may infect a wide variety of animals, strains isolated from nonavian hosts mostly do not produce fowl cholera. The susceptibility to this infection vary with age as well as breeds of chicken with older chickens are more susceptible to this disease than young ones, and some breeds of chickens are more susceptible than others.

This disease is seen more often in poultry, geese and ducks. This can also affect other birds such as turkeys, guinea fowl and pigeons. In poultry this disease occur more commonly in birds more than 16 weeks of age means it affects the laying birds more severely than the young birds, however, the disease can occur in broilers between 3 to 6 weeks of age. *Pasteurella multocida* is the causative organism of this disease. It is a Gram-negative, bipolar, non-motile, rod shaped bacteria which may exhibit pleomorphism after repeated sub-culturing. The bacterium has a bipolar appearance when stained with Wright's stain especially in freshly isolated cultures. Based on capsular antigens *P. multocida* of poultry have been classified in to serogroups A, B, D and F. This organism is susceptible to wide rang of disinfectants and it can be killed by 1 % formalin or 1% caustic soda or phenol. Outside the body of the birds the organisms are very susceptible to drying and disinfectants. Drying, sunlight or heat can kill the bacteria within 15 minutes at 56°C and 10 minutes at 60°C.

Transmission occurs by direct contact with clinically affected birds or carrier birds or carcasses of birds which have died from the infection. Recovered birds become chronic carriers. The organism enters through the respiratory (nasal) or digestive tract (oral). The organism is excreted through nasal exude and faeces. This disease can also be transmitted through contaminated feed bags, water, equipments and clothing of personnel. Environmental contamination, rodents and wild birds also serve as source of infection. Airborne spread of infection does occur between pens. Handling of infected birds for vaccination and other purposes may also transmit this disease.

The incubation period is usually 5-8 days. Morbidity and mortality depends on pathogenicity of strain and the susceptibility of the flock and according to the severity of the disease it has been characterized as peracute, acute, chronic and localized. In acute form, symptoms are not seen and large numbers of birds are found dead. The symptoms in this form include marked depression, loss of appetite, mucus discharges from the orifices (nasal, ocular and oral discharge) ruffled feathers, cyanosis and swelling in comb and wattles, increased respiratory rate and foul-smelling diarrhoea. Diarrhoea at first watery with whitish in colour which later becomes greenish and may contain mucus.

The chronic form is seen in birds which survive the more acute form and the symptoms include depression, difficult breathing, some times later lameness, torticollis and swelling of wattles in male birds. Swelling of joints, wattles, tendon sheaths, and footpads are seen because of fibrinosuppurative exudate accumulation. Mortality in affected birds occurs due to dehydration and septicaemia. Mortality in affected cases may vary from 10 to 90% depending upon the strain involved.

Post-mortem lesions in acute cases are enlargement and dark colorations of liver of liver, kidney and spleen. Petichial haemorrhages on heart are characteristic lesions. In most of the cases enteritis is seen. Lungs show congestion in most of the cases. The liver may be swollen and often develops multiple, small, and necrotic foci. Fibrinous covering on heart and air sacs is evident. In subacute cases there may be gray granulomatous foci in the liver. In chronic cases arthritis is the main lesion besides increased amounts of peritoneal and pericardial fluids.

Diagnosis in acute form is done through observation of sudden onset of disease like Newcastle disease and spirochaetosis followed by high mortality. Blood smears prepared from heart blood, liver and spleen stained with Giemsa stain in septicaemic form shows Gram-negative, bipolar organisms. Presence of petichial haemorrhages on the heart and other serosal surfaces are also diagnostic features. Subcutaneous or intraperitoneal inoculation of suspension prepared from tissue homogenate from affected birds into mice or pigeon at a dose of 0.2 ml cause death within 24 hours. Again this organism can be demonstrated in the blood of inoculated bird or animal.

Treatment of this disease must be done based on drug sensitivity of antimicrobial drugs. In addition, early treatment and adequate dosage are important. The following treatment is effective in this disease. For treatment tetracycline can be given in feed or drinking water or by injection. High levels of tetracycline antibiotics in the feed (0.04%), drinking water, or even parental administration may be useful in treatment of this disease. The disease can also be treated by antibiotics such as penicillin, oxytetracycline, chlortetracycline and erythromycin which can be given in drinking water. Moreover,

sulfaquinoxaline sodium in feed or water usually controls mortality, as is seen with sulfamethazine and sulfadimethoxine; however, sulfas should be used with care especially in breeders because of potential toxicity. In severe outbreak antibiotics such as gentamycin, erythromycin can also be injected intramuscularly. Along with the medication, sodium salicylate in the drinking water for 3 days appears to help in recovery.

Chronic Respiratory Disease (CRD)- Chronic respiratory disease which is more commonly known as CRD is one which occurs more commonly in birds reared under bad management. The disease is caused by bacteria called *Mycoplasma gallisepticum*. This condition is triggered by respiratory viruses such as Newcastle and infectious bronchitis and subsequently complicated by bacterial invasion. It is an important disease affecting both layers and broilers. The disease is responsible for extensive losses in broilers especially when other viral respiratory diseases occur simultaneously. Depressed growth rate and poor feed conversion efficiency are the manifestation of this disease in broilers. In layers there is decreased egg production. There may be reduced hatchability of chicks. The main symptoms associated with *M. gallisepticum* infection include chronic respiratory symptoms such as abnormal respiratory sounds, ocular discharge, coughing, sneezing, tracheal rales (gurgling and snicking), and respiratory distress with breathing through open beak. There may be nasal discharge and frothiness about the eyes. Marked reduced growth rate, feed efficiency is greatly reduced and increased susceptibility to other respiratory diseases. This bacteria is sensitive to many antibiotics such as streptomycin, oxytetracycline, chlortetracycline, tiamulin, tylosin, lincomycin; however, sensitivity to these antibiotics varies in certain isolates.

Aspergillosis (Brooder Pneumonia)- Aspergillosis is also known as brooder pneumonia, is a fungal infectious disease, caused by *Aspergillus fumigatus*, in which the typical sign is gasping for breath, especially in young chicks. Sometimes this organism causes eye lesions or chronic lesions in older birds. The fungus can infect many species of animals including birds and man. The disease may be observed as endemic in nature on some farms. The disease occurs in two forms; acute outbreaks with high morbidity and high mortality in young chicks, and a chronic form which mostly affects adult birds. Sometimes similar lesions are produced by other species of *Aspergillus* or even other fungi such as *Penicillium* and *Absidia*. The infection has an incubation period of 2-5 days. Mortality among young chicks may go up to 5-50%. These organisms are present in the environment of all poultry houses. Rapid and difficult breathing especially those chicks which are infected in the hatchery and some time there is breathing with open mouth. This happens due to obstruction of the air passages.

If aspergillosis is associated with other respiratory diseases such as infectious bronchitis and infectious laryngotracheitis, then the symptoms are gurgling and rattling noises. Other symptoms include loss of appetite, increased thrust, and drowsiness; develop eye swelling or blindness, weakness, show torticollis and other nervous signs. In chronic form there may be ocular discharge and wasting. Postmortem lesions include yellow to grey nodules which (hard in consistency) or plaques in lungs, air sacs, trachea, nasal passages and peritoneal cavity. They may have greenish surface. Some times lungs

are diffusely grayish yellow. Conjunctivitis or keratitis may be present. Usually no treatment is there, however, environmental spraying with effective antifungal spray will reduce challenge. Amphotericin B and Nystatin have been used in high-valued birds. Good hatchery sanitation is very effective in reducing aspergillosis outbreaks. The eggs which are grossly contaminated should not be set for incubation because they may have large population of *Aspergillus* spores and may disseminate spores throughout the hatching machine. Clean eggs should be preferred for setting. In hatchery, the contaminated setters and hachers should be fumigated with formaldehyde.

Avian Influenza- Avian influenza also known as ‘bird flu’ is a highly contagious viral disease by type A strains of the influenza virus. The disease affects mainly chickens, turkeys, ducks and other birds. It was first noticed in Italy in the year 1878 and named as fowl plague. The disease is characterized by variety of syndromes, ranging from generalized fatal to mild upper respiratory disease.

Avian influenza viruses are member of the family Orthomyxoviridae. The viruses that constitute this family are classified into type A, B and C, based on differences between their nucleoprotein and matrix protein antigens. Avian influenza viruses belong to type A, which can be further classified into subtypes, according to the antigens of haemagglutinin (H) and neuraminidase (N). At present, 16 H subtypes (H1-H16) and 9 N subtypes (N1-N9) are recognized, so that there are $16 \times 9 = 144$ subtypes of avian influenza viruses. Out of these 144 subtypes, more than 90% are less pathogenic called low pathogenic avian influenza (LPAI), few are moderately pathogenic called as moderately pathogenic avian influenza (MPAI) and only few are pathogenic avian influenza like H5 and H7 with N1, N2, N3, N7 and N9 combinations, especially H5N1 type is highly pathogenic avian influenza (HPAI) both for birds and it also affects humans.

The characteristic feature of avian influenza virus is constant mutation, results in development of new, highly virulent strains. The avian influenza viruses lack mechanisms for the proofreading and repair of errors that occur during replication. As a result of these uncorrected errors, the genetic composition of the viruses change as they replicate in humans and animals, and the existing strain is replaced with a new antigenic variant. These constant, permanent and usually small changes in the antigenic composition of influenza viruses are known as antigenic drift.

The influenza viruses have a second characteristic to swap or reassort genetic materials and merge. The reassortment process is known as antigenic shift, results in evolution of novel subtype, different from both of the parent viruses. Of the 16 avian influenza virus subtypes, H5N1 is of particular concern for several reasons. H5N1 mutates rapidly and has propensity to acquire genes from viruses infecting other animal species. In addition, the isolates of H5N1 have a high pathogenicity and can cause severe disease in avian and humans. Knowledge about the survivability of this virus in the environment is necessary. It has been reported that the highly pathogenic H5N1 virus can survive in bird faeces for at least 35 days at low temperature (4°C) and at a much higher temperature (37°C), H5N1 viruses have been shown to survive, in faecal samples, for six days.

In the mild form, signs of illness may be expressed only as ruffled feathers, reduced egg production, or mild effects on the respiratory system. Highly pathogenic avian

influenza is characterized by sudden onset of severe disease, rapid contagion, and a mortality rate that can approach 100% within 48 hours. In this form of the disease, the virus not only affects the respiratory tract, as in the mild form, but also invades multiple organs and tissues and various symptoms include:

- i) Severe depression, in appetite, difficulty in breathing and ruffled feathers
- ii) Drastic decline in egg production.
- iii) Facial oedema (head and neck)
- iv) Bluish colouring and swelling of cyanotic combs and wattles.
- v) Swollen sinuses with nasal discharge can be seen with respiratory involvement.
- vi) Petechial haemorrhages on internal membrane surfaces
- vii) Signs of blood in nose discharge
- viii) Small haemorrhages (most visible on feet and shanks)
- ix) watery diarrhoea
- x) Mortality rate is very high especially in highly pathogenic form that can approach 100% within 48 hours

Postmortem Lesions:

- i) Lesions may be absent in cases of sudden death
- ii) Severe congestion of the musculature
- iii) Dehydration
- v) Subcutaneous oedema of the head and neck area
- vi) Nasal and oral cavity discharge
- vii) Severe congestion of conjunctiva
- viii) Excessive mucous exudate in the lumen of the trachea, or severe haemorrhagic tracheitis
- ix) Petechiae on the inside of the sternum, on the serosa and abdominal fat, serosal surfaces and in the body cavity
- x) Severe kidney congestion, sometimes with urates deposits in the tubules
 - xi) Haemorrhages and degeneration of the ovary.
- xii) Haemorrhages on the mucosal surface of the proventriculus, particularly at the juncture with the gizzard.
- xiii) Haemorrhages and erosions of the gizzard lining
- xiv) Haemorrhagic foci on the lymphoid tissues in the intestinal mucosa. The lesions in turkeys are similar to those in chickens, but may not be as marked. Ducks infected with HPAI and excreting the virus may not show any clinical signs or lesions.

This disease has to be differentiated from acute fowl cholera, velogenic Newcastle disease, fowl cholera, Mycoplasma infection, Staphylococcus, and *Escherichia coli* and respiratory diseases like infectious laryngotracheitis.

Laboratory Diagnostic Procedures:

- (a) Identification of the agent by inoculation of 9-11-day-old embryonated chicken eggs followed
 - i) Demonstration of haemagglutination.
 - ii) Immunodiffusion test to confirm the presence of influenza A virus.
 - iii) Subtype determination with monospecific antisera.

- iv) Strain virulence evaluation: evaluation of the intravenous pathogenicity index (IVPI) in 4-8-week-old chickens.

Samples to be collected for detection of avian influenza virus include tracheal and cloacal swabs or faeces) from live birds or from pools of organs and faeces from dead birds

- (b) Serological tests include haemagglutination and haemagglutination inhibition tests and agar gel immunodiffusion and the sample required for this are clotted blood samples or serum.
- (c) Polymerase chain reaction (PCR) based molecular tests.
- (d) ELISA has also been used in detection of avian influenza.

Specimens sent to the laboratory should be accompanied by a history of clinical and gross lesions, including any information on recent additions to the flock. Specimens should be collected from several birds. It is not unusual for many of the submitted specimens to fail to yield virus. Swabs are the most convenient way to transfer AI virus from tissues or secretions of the suspect bird to brain and heart infusion broth or other cell culture maintenance medium containing high levels of antibiotics. Dry swabs should be inserted deeply to ensure obtaining ample epithelial tissue. Trachea, lung, spleen, cloaca and brain should be sampled. If large numbers of dead or live birds are to be sampled, cloacal swabs from up to five birds can be pooled in the same tube of broth. An alternative technique is to place 0.5 cm³ of each tissue into the broth. Blood for serum should be collected from several birds. If the specimens can be delivered to a laboratory within 24 hours, they should be placed on ice. If delivery will take longer, quick freeze the specimens and do not allow them to thaw during transit. There is no treatment for avian influenza. Antibiotics will help to prevent secondary bacterial infections.

Hydropericardium Syndrome (HPS)- HPS is a disease of young chickens characterized by acute mortality, often with severe anaemia and hepatitis. This disease is usually seen in broiler chickens in young chickens in the age group of 4 to 9 weeks of age. However, the disease has also been observed in growers and in older birds. Immuno-suppression produced by some other diseases such as infectious bursal disease helps in producing adenoviruses to produce inclusion body hepatitis. Severity of this disease increases when the birds are immuno-suppressed with diseases such as chicken infectious anaemia and infectious bursal disease. The main symptoms in this disease are sudden mortality in the young chickens especially which are below 6 weeks of age and become maximum after 3-4 days and thereafter declines on 5th day and in rare instances mortality continues for 2-3 weeks. Mortality may reach 10% but sometimes as high as 30%, however, morbidity in this disease is low. Antibiotics administration either through feed or water may help prevent secondary bacterial infections. Use of sulfonamides is contraindicated if there is evidence of hematologic disease or immunosuppression is seen. Multivitamins are considered as useful tools in early the recovery of this disease.

The main symptoms in this disease are sudden mortality in the young chickens especially which are below 6 weeks of age and become maximum after 3-4 days and thereafter declines on 5th day and in rare instances mortality continues for 2-3 weeks. Mortality may reach 10% but sometimes as high as 30%, however, morbidity in this disease is low. Mortality rates vary depending on the pathogenicity of the virus IBH/HP

virus and infection of birds with other viral or bacterial agents. The birds in the flock usually look normal but depression of birds may be there. Sick birds may have ruffled feathers and may die within 48 hour. There is decrease in feed conversion and weight gain. Broiler chickens with age group of 3 to 5 weeks, there may not be specific clinical symptoms. However, there appear abrupt onset of mortality, birds appear lethargy, droppings appear mucoid in consistency and birds may huddle with ruffled feathers.

Gross lesions include up to 10 mL of a straw-colored transudate in the pericardial sac, generalized congestion, and an enlarged, pale, friable liver. Histopathologic lesions include myocardial edema in the heart with degeneration, necrosis, and mild mononuclear cell infiltration. Basophilic intranuclear inclusion bodies may be present in the liver. A tentative diagnosis is based on typical microscopic findings and confirmed by isolating adenoviruses from the liver. Serology, restriction enzyme analysis and PCR are used to classify adenoviruses isolated from clinical cases. This information is used for epidemiologic studies.

The main post-mortem lesions in this disease include swelling of liver with pale, friable and mottled appearance. Liver may be congested also and there may be presence of small and large haemorrhages in the liver and skeletal muscles. Kidneys are swollen and pale in colour. Other lesions include anaemia and jaundice of the skin over legs, breast and subcutaneous tissues. The bone marrow is thin and watery and blood is thin in consistency. Microscopically, there are basophilic intranuclear inclusions bodies in the liver. A presumptive diagnosis may be made on history (signs) and lesions. Diagnosis is confirmed through isolation and identification of adenoviruses from the liver. The main disadvantage with isolation of adenoviruses is that these are present in healthy birds also. Antibody against adenoviruses can be detected by agar-gel precipitation test, indirect fluorescent test and ELISA. Since antibodies against adenoviruses are present in healthy birds also and therefore there are difficulties in interpretation of results.

Another technique for the detection of this virus is the PCR based technique. By this technique, genomic DNA sequences of various sizes using specific primers can be prepared and can be used for amplification of the desired genome and this can be used to classify adenoviruses isolated from clinical cases. In addition, restriction fragment length polymorphism (RFLP) of PCR products with restriction enzymes can be used to differentiate strain of this virus. Differential diagnosis is must from diseases such as chicken anaemia syndrome, sulphonamide intoxication, Infectious Bursal Disease, vibronic hepatitis, fatty liver syndrome, and deficiency of vitamin B₁₂.

Infectious Bursal Disease (Gumboro Disease)- Infectious bursal disease is an acute, highly contagious viral disease of young chickens. The disease is characterized by sudden onset, short course and extensive destruction of lymphocytes particularly in the bursa of fabricius and also in other lymphoid organs. Chicks in the age of 2 to 7 weeks are the most susceptible. It is responsible for lower productivity and profitability in poultry. This disease leads to extensive immuno-suppression and the affected chickens have reduced antibody response to vaccinations, strong post-vaccinal reactions, and increased susceptibility to concurrent or secondary infections such as *E. coli*, gangrenous dermatitis and inclusion body hepatitis. The disease in chicks follows two courses,

depending on the age at which the birds are affected. The subclinical form of the disease occurs in chicks below 3 weeks of age. In this form of disease, chicks exhibit no clinical signs of disease, but the affected chicks experience permanent and severe immunosuppression due to damage to the BF. In field conditions, most of the outbreaks of IBD are due to this form and this form is important from economic point of view. IBD related diseases such as inclusion body hepatitis are more frequent in these birds. In broiler chicks this form of the disease results in bad performance with lower weight gains and higher feed conversion ratios. The clinical form of disease usually occurs from 3 to 6 weeks of age. This form has a sudden onset and the mortality rate in the flock increases very rapidly. Main clinical signs include dehydration, ruffled feathers, inappetance, and unsteady gait, vent picking, diarrhea with urates in mucus, closed eyes, lying down in exhaustion, depression and inflamed vents. Occasionally the birds show trembling or shivering. Affected birds also appear listless, pale and huddling. Body temperature significantly increases at 48 hours after infection. Mortality reaches its peak after 3-5 days of infection.

Post-mortem examination of carcasses of dead birds are dehydrated, hemorrhages are present in the leg (thigh) and breast muscles and under the skin. Hemorrhages are also seen on the mucosa of the proventriculus. Kidneys may appear swollen with presence of urates that die or those are in the advanced stage of disease. Appearance of kidneys looks whitish in colour due to dilatation of tubules with ureters. In intestines, increased mucus (a sticky fluid) on the lining is evident. Liver of affected birds may be swollen and there may be areas of dead tissue near the margin. Enlargement of bursa of Fabricius initially within 3-5 days of disease and has a gelatinous exudate covering the serosal surface followed by atrophy of bursa after 5 days and after 8th day post-infection it is one-third its normal weight. Hemorrhage and areas of necrosis may be present in more severe cases. Some times hemorrhages are also seen through the entire bursa and this is the reason that birds sometimes pass blood in the droppings. Hemorrhages are also seen at the junction of the proventriculus and gizzard.

No specific treatment is available for IBD. Use of a multivitamin supplement for 3-5 days to increase the immunity in affected birds and facilitating access to water may help. To avoid any secondary bacterial infection along with IBD and then antibiotics like amoxycillin, pefloxacin and enrofloxacin etc. can be given in drinking water. Levamisole may be given for 2-3 days to boost the immunity. Vitamin E and vitamin C has also been found to be good for recovery and production. Immunization of breeders is an important part of the IBD control programme. Three categories of vaccines, based on their pathogenicity, have been described: (1) mild, (2) intermediate, and (3) virulent. The intermediate type IBD vaccines are most commonly used as these vaccines can stimulate the broiler to produce antibodies earlier than the mild-type vaccines, without significant damage to the BF as may occur with the virulent type vaccines. Chicks are vaccinated with live vaccine at 14 days and then at 35 days by intermediate strain by putting one drop in eye and nose.

Chicken Infectious Anaemia- Chicken infectious anaemia (CIA) is a disease which affects young chicks mostly below 3 weeks of age and the day-old bird are most susceptible. CIA infects chickens, however, antibodies to CIA has been detected in

Japanese quail also. Disease due to chicken infectious anaemia virus (CIA) is characterized by aplastic anaemia, immuno-suppression and generalized lymphoid atrophy and may be responsible for stunting and for secondary viral such as inclusion body hepatitis, fungal like aspergillosis and bacterial infections including dermatitis which results in downgrading. The immuno-suppression induced by CIA may reduce the vaccinal immunity of the affected birds. Mortality in affected birds may go up to 60%. The infection is widespread in broiler and replacement parent and laying-strain pullets. The disease is also known by other names such as blue-wing disease, anemia dermatitis syndrome. The disease due to chicken infectious anemia has been observed in most countries where chickens are raised for commercially. Symptoms of this disease appear at about 10-14 days of age in broiler chicks, however some time symptoms may appear between 2 to 4 weeks of age also. Major symptoms of CIA include depression, paleness, anorexia, weakness, sub-cutaneous hemorrhages and watery blood. Anemic changes appear on non-feathered parts such as comb, eye lids, wattles and legs. Skin lesions in CIA appear on wings which may lead to secondary infections which ultimately result in necrotizing dermatitis. Haemorrhages are on skin and in the muscle of the CIA affected birds. The most important changes are anemia and atrophy of the thymus, bursa of Fabricius and spleen. The surviving chicks recover from anemia by 20-28 days after infection with disease. Main disadvantages after recovery from this disease are that birds are susceptible to other viral and bacterial infections. On postmortem major lesions include paleness of organs, atrophy of thymus, mild atrophy of bursa and spleen, petechial hemorrhages of breast muscles, leg muscles, heart and proventriculus. Haemorrhages are also seen in sub-cutaneous tissues and muscles. Bone marrow is pale or yellow. Haemorrhages in the proventriculus, under the skin and in the other organs associated with severe anemia. The most characteristic lesion of CIA is the yellowish bone marrow. Enlargement mottled appearance of liver and hemorrhages are seen on the surface of heart. Swelling of kidneys is also seen. Histopathologically, appearance of yellow or white color of bone marrow and intense atrophy of lymphoid organs are important. There is no effective treatment is known, however, the affected birds should be given antibiotics to avoid secondary bacterial infection along with multivitamins and B-complex vitamins are also given.

Infectious Laryngo-Tracheitis -Infectious Laryngo-Tracheitis is an acute, highly contagious disease of respiratory disease of chickens resulting in severe production losses due to mortality and decreased egg production. In India, this virus causes mild pathogenicity in poultry, however, in some cases high mortality have been recorded. Disease is seen in chickens of all ages; however, young birds are most susceptible. The most common age is 3- 9 months of age. This disease was first described in 1925. This disease results in respiratory depression, gasping and expectoration of body exudates. The disease caused by ILT becomes serious when other disease producing agent, such as the viruses of Ranikhet disease, infectious bronchitis, fowl pox, *Haemophilus paragallinarum* and *Mycoplasma gallisepticum* occur at the same time. This virus affects mainly poultry but can cause disease in turkeys and pheasants. The disease is severe in adult birds but mild in chicks below 3 weeks of age. Embryonated eggs of turkey and chicken are susceptible to ILT virus. Eggs of guinea fowl and pigeons are not

susceptible. Even vaccine of the virus has been shown to cause disease due to its reversion to parental virus type virulence.

ILT is caused by a DNA herpes virus which is a double stranded virus. Only one serotype is known. This virus does not seem to be very invasive in nature and it affects the respiratory tract and conjunctiva. The multiplication of the virus is limited to respiratory tract and there is no viremia. The virus can survive for longer period in farm conditions especially under cold environment. This virus gets killed at 55°C in 10 to 15 minutes. In the body of dead birds it survives for about 2 days. Symptomless bird carries the virus for as long as 16 months which may be excreted through the faeces of such carrier birds.

The virus in infected birds is present in discharges from mouth, nostrils, trachea and the conjunctiva. The virus spread by air and enters into the body through the upper respiratory tract and conjunctiva. It can also enter through ingestion of contaminated feed or litter. Equipments can also be source of virus for transmission through mouth. Transmission occurs more readily from actually infected birds than those through contact with clinically recovered carrier birds. The recovered birds are carriers of this virus and when there is stress of movement of birds, handling etc. then such birds shed this virus for long period. Because of longer survival of the virus outside the body of the host, substances such as infected vehicles, equipments, crates and mechanical carriers such as wild birds, people, vermin (rodents, insects etc.) , dogs and cats can be important transmitters of the virus of this disease. Infection of ILT also may be spread mechanically. Several epidemics of this disease have been traced to the transport of birds in contaminated crates having presence of this virus.

In the acute form of this disease the birds may die without showing any symptoms. After 6-12 days of natural exposure, the main symptoms include respiratory distress, marked coughing, gasping, rattling, extension of the neck while inspiration and discharge of mucus and blood stained material blood clots from the mouth. In this form of disease, young birds (upto 3 weeks of age) show only conjunctivitis or watery fluid between the eye lids. In adults it is characterized by difficult respiration, nasal discharge, moist rales followed by coughing and gasping. Obstruction of trachea with exudates (inflamed material) causes the birds to breathe with long gasps and is the main factors in mortality. During inhalation of air birds lower their heads and during exhalation they extend the beak with wide open and with a loud harsh cry. Marked dyspnoea and expectoration of blood stained mucous are the characteristics of severe form of disease. Mortality in acute form can reach upto 70%. Signs normally subside after 2 wk, although birds may cough for a month.

In sub-acute form of the disease, the adult birds show dullness, wet eyes, conjunctivitis and discharge of exudates from nostrils. Effect on eye lids can cause blindness, starvation and mortality. Less pathogenic strain can cause failure to gain weight. Mortality in this form ranges from 10-20% in adult birds while in young birds it may be around 5-10%. Conjunctivitis and respiratory sounds (wheezing) can also be seen, with little or no mortality in this form of disease. After recovery from this disease, some birds remain carriers for longer periods and become a source of infection for susceptible birds. The latent virus can be reactivated under stressful conditions.

Lesions vary with the form of the disease and most of the lesions are restricted to upper respiratory tract. In peracute form, haemorrhagic tracheitis is present; trachea is inflamed and contains blood clots or filled with blood stained mucus. The primary bronchi may be affected. Lesions in acute form include Inflammation of larynx and trachea with accumulation of blood tinged exudates in them. At some occasions, the mucosa of larynx and trachea may be necrosed and may contain cheesy exudates. Oedema, congestion of epithelium of conjunctiva and infraorbital sinuses are lesions associated with subacute form.

Diagnosis of this disease is based on the clinical signs and by presence of blood, mucus and yellow caseous exudate in the trachea. Microscopic changes include desquamation, necrotizing tracheitis, presence of intranuclear inclusion bodies in the epithelium of trachea especially in early course of the disease. Diagnosis is confirmed through isolating and identifying the virus. For isolation of virus, chicken embryos (9-12 days old) are preferred and specimen is inoculated in chorioallantoic membrane of developing chicken embryos in which chorioallantoic membrane lesion shows intranuclear inclusions by microscopically. This disease must be differentiated from the diphtheritic form of fowlpox (with tracheal lesions) the main differentiating feature between these two diseases is the produces intra-cytoplasmic inclusions in fowlpox virus. This disease can also be differentiated using nucleic acid probes which can be prepared from cloned genomic fragments of ILT virus. By nucleic acid probes method, disease can be differentiated from the diphtheritic form of fowlpox especially tracheal lesions. Another technique for the detection of this virus is the PCR based technique. By this technique, genomic DNA sequences of various sizes using specific primers, can be prepared and can be used for amplification of the desired genome. This method is very useful in samples having low levels of virus. In addition, restriction fragment length polymorphism (RFLP) of PCR products with restriction enzymes can be used to differentiate strain of this virus. The best way to control this disease is slaughtering of the affected birds. The slaughtered birds should be properly disposed off. The affected birds can be treated with antibiotics like Terramycin or Aureomycin to avoid secondary infections.

Coccidiosis- Coccidiosis is one of the most commonly reported parasitic diseases of poultry and is a major cause of mortality, suboptimal growth and poor feed conversion efficiency especially in rainy season leading to heavy economic losses to poultry industry. Coccidiosis in poultry is characterized by heavy mortality and bloody diarrhea. Predisposing factor is the high moisture content of litter exceeding 30% due to leakage of rain water or leaking waterers. Since high level of moisture helps in the quick sporulation in coccidial oocysts within 24 to 48 hour which otherwise take longer time period. Apart from other factors, immune-suppression due to Marek's, infectious bursal disease (IBD) or mycotoxins may also aggravate the disease. Bloody diarrhoea is not a common feature in this form. Other signs include poor egg production, paleness of mucous membranes, loss of yellowish colour of shanks and the main post-mortem lesion include presence of blood tinged fluid in caeca and there may be presence of localized or diffuse haemorrhages. Treatment includes amprolium, sulfonamides (sulfamethazine or sulfaquinoxaline), diclazuril and toltrazuril.

Mycotoxycosis- Mycotoxycosis refers to all of those diseases caused by the effects of toxins produced by moulds. Disease is mostly subclinical and may be difficult to diagnose. Problems occur worldwide, but more problems are in climates with high temperature and humidity and where grain is harvested with high water content. Mortality is variable but all are detrimental to bird health and are resistant to heat inactivation. Among the important mycotoxins that affect chickens include aflatoxin, T2 fusariotoxins, ochratoxins and zearalenone. The main adverse effect of mycotoxin in poultry birds include. Reduction of feed intake, alteration in contents of feed interns of nutrient absorption and metabolism and most important is the suppression of immune system. Among the measures to prevent the toxin production include keeping the moisture content of feed ingredients below 12%, storage of feed ingredients below 20°C an RH <60%. Mycotoxin adsorbents and binders should be added in poultry feed. There are some measures which the poultry farmers must follow during rainy season to prevent the poultry birds from diseases. These include:

Prevention and Control

1. Application of sound principles of biosecurity.
2. Hygienic measures should be adopted in daily farm practices
3. Always keep the litter in the farm in dry condition.
4. Prevent ammonia production at low level as ammonia levels.
5. Proper ventilation.
6. Following proper schedule of vaccination
7. Providing balanced nutrition.
8. Following sound principles of management.
9. Disease profiling as a routine practice.
10. Regular monitoring of feed and water quality.

Chapter 9

Diagnosis of Central Nervous System with Special Reference to Bovine Spongiform Encephalopathy and Rabies

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Nervous system (central and peripheral) plays a pivotal role in co-ordination of all bodily functions through complicated but well organized process, involving various kinds of receptor and effector organs. The receptor organs receive stimulus from appropriate environmental change, which travels the length of several neurons and or interneuron and then via motor neurons, it ends on an effector organs (glands and muscles). In this process the neurons act as main player for rapid communications of impulses, while accessory cells like astrocytes, oligodendrocytes, microglial, ependymal and vascular endothelial cells maintain the functionality of the neurons. All these cell types together are called brain parenchyma. The brain is bestowed with effective blood brain barrier, an interface between the brain parenchyma and the circulatory system, the membranous coverings, the serous fluid within sub-dural spaces, the cerebrospinal fluid within sub-arachnoids space and the bony encasement etc, for protection. Despite the protective barriers, brain may suffer with various kinds of insults during the life of the living beings. The brain, which consisted of forebrain (olfactory bulb and tract, cerebral hemisphere and diencephalon), midbrain (corpora quadrigemina, crura cerebri) and the hindbrain (pons, cerebellum, medulla oblongata), together with spinal cord, peripheral nerves and ganglia regulate sensory, motor, cognitive, memory, and autonomic functions through complex network of neuronal cells connections, located in different anatomical sites of the brain. The functional properties of the nervous system are topographically localized. Further, the disturbance in one anatomical part of the brain may affect the other parts; thereby it becomes difficult to pinpoint the exact location of the lesion based on the clinical signs. Similarly, the neurological disorders/ diseases are also regionally distributed due to varying degree of vulnerability of different cell types and regions of the brain to various etiological agents. Therefore, systematic examination of the brain is recommended to find out the pattern and extent of lesions in knowing the known and documentation of new diseases. The brain cells, spanning within the white and the gray matter, exhibit limited cytopathological and inflammatory responses and their susceptibility to the stimuli vary. The neurons are being more susceptible followed by neuroglial cells. In the event of neuronal damage, the accessory cells (neuroglial cells-astrocytes, oligodendroglia, microglial, ependymal cells, endothelial cells) show the reactive changes, which help in distinguishing the artifacts, resulted due to native and technical variability from real pathological lesions. The brain may be affected either directly or indirectly by various kinds of disease causing agents like virus, prion protein, bacteria, rickettsia, chlamydia, fungus, protozoa, parasite, toxins, nutritional and metabolic disorders, etc. These agents break open the brain barriers and reach the brain by way of the blood stream, or may pass along the axis cylinders of motor or sensory nerves,

establish the infection and produce pathological changes. These changes are limited, diffuse in nature, and produce clinical signs more or less similar and overlapping. The inflammatory process at the blood brain barrier results into generation of various inflammatory cytokines/ mediators, which contribute to brain pathology. The structural changes in brain like hemorrhage, tumour, parasitic cysts, abscesses and malacic lesions may be visible grossly without any difficulty but alterations involving the neurons, glial cells; endothelial cells, ependymal cells and inflammatory cells require microscopic examination. Among the various affections of the brain, malformations occur disproportionately high in frequency and variety in domestic animals than in other tissues. The malformations are caused by number of teratogens (toxins, infectious agents and genetic causes), which affect specific cell population of the brain at particular developmental stage and result into morphological, microscopic or functional disturbances. Many of the agents like BVD, Bluetongue, Hog cholera, Akabane virus, fungal and plant toxins, drugs, etc are known to cause developmental disturbances like microencephaly, hydrocephalous, cyclopia, dysraphic states, etc in domesticated animals. The cytopathology of various cell types of the brain is also special which is reflected by the presence of specific changes within the neurons, the glial cells and the vascular endothelium. The brain affections in animals have not been thoroughly investigated as compared with human beings. In many surveillance studies, workers have documented brain affections including rabies, besides BSE, however, in similar kind of surveillance studies in our country no BSE cases were detected so far.

The diagnosis with brain and or spinal dysfunction includes a minimum data base such as clinical signs, hemogram, serum chemistry and urine analysis and plain and contrast radiograph. The additional techniques for imaging are computerized tomography (CT), magnetic resonance imaging (MRI) and positron emission tomography (PET) (Sapierzynski, 2007;). PET is used to probe biochemical imaging in vivo using physiological tracers (C11, N13, O15 and F18) to quantify regional blood flow, metabolic rates and various neurotransmission functions in the brain.

Diagnosis of BSE- Bovine spongiform encephalopathy (BSE) or mad cow disease is a neurodegenerative disease of cattle, which is 100% fatal. It was first reported in Great Britain in 1986 and then subsequently in the native cattle in many of the European countries. The disease was also reported in Canada, USA and Oman in cattle imported from United Kingdom. Between 1986 and 2002, approximately 1.82 lakhs cases of BSE were confirmed in the UK and 4.4 million cattle were slaughtered during the eradication program. The outbreak of BSE in cattle was resulted because of feeding of scrapie contaminated meat-bone meal (MBM) of ruminant origin, which escaped inactivation due to change in the rendering process from dry rendering to steam and hydrocarbon solvent extraction in 1970, as a result of fall in cost of tallow. The ban on MBM and mass slaughter of cattle in UK showed decline in BSE cases in cattle and humans. The onset of the disease is insidious and progressive in nature with long incubation period. The disease is usually seen in cattle over 2 year of age (average age at onset of clinical signs about 4 - 5 years, earliest case recorded in 20 months and the oldest in a 19 year old animal). In BSE, there occurs brain degeneration leading to considerably variable clinical signs and nearly 66% of the BSE cases do not exhibit clinical signs. The clinical signs exhibited like disturbances in behavior (apprehension to enter the milking parlor or

may kick vigorously during milking), disturbance in locomotion (dry cows show pelvic limb in-coordination and weakness, abnormal posture and movement when the animal is led by the owner, swaying of the pelvic limbs and pelvic limb hypermetria at pasture, with the advancing severity, there is generalized weakness, resulting in falling and recumbency and as the disease progresses, there is loss of bodily condition, decreasing body weight, and reduction in milk, besides nervous signs, the protracted clinical course, extending over a period of weeks (6-8) or months, would eventually require slaughter on welfare ground or animal dies) and disturbance in sensitivity (hypersensitive to touch, sound, light), which are sufficiently distinctive to lead to suspicion of disease. The disease should be differentiated with other infectious (rabies, MCF, salmonella, encephalic listeriosis, aspergillosis, parasitic cysts, etc.), toxic syndromes (tetanus, botulism, plant toxins, mycotoxicoses, urea poisoning, ammonia toxicity, insecticides, lead poisoning, salt poisoning), nutritional deficiency/excess (thiamine/vitamin B1 deficiency, Cu deficiency, cerebrocortical necrosis), metabolic disorders (ketosis, hypomagnesaemia, hypocalcaemia, hepatic encephalopathy), neoplasms, degenerative and developmental disorders, etc.

Bovine spongiform encephalopathy comes under prion group of diseases (earlier known as transmissible spongiform encephalopathies). The disease has possible link to the human BSE (named as new variant CJD, 1996). The other prion group of diseases includes **scrapie** in sheep and rarely goats (endemic for over 250 years in UK), **transmissible mink encephalopathy** (TME, Hartsough, 1947) of mink, **chronic wasting disease** (CWD) of deer, **feline spongiform encephalopathy of cats** (FSE) and the exotic ungulate encephalopathy of greater Kudu, nyala and oryx. Human prion diseases include, **Kuru** (known for 200 years ago in Papua New Guinea due to ritual endo-cannibalism), **Creutzfeldt Jacob Disease** (CJD) with three phenotypic variants viz., sporadic CJD (accounts for 85% of the cases in 50-70 years age group), familial CJD (10-15% of the cases due to gene mutation) and iatrogenic CJD (<5% of the cases due to medical and surgical intervention), Gertsman-strauss syndrome (GSS, due to germline mutation in gene) and fatal familial insomnia (FFI). The diseases are caused by abnormal isoform (PrP^{Sc}) of normal cellular prion protein (PrP^C) (Prusiner (1997)). The diseases are characterized by insidious onset, long incubation period and progressive development with invariably fatal outcome but all have common histopathological lesions. The agent manifests as infectious, genetic and sporadic and involves modification of normal cellular prion protein (PrP^C) into infective prion protein (Pr P^{Sc}).

Pathogenesis of BSE involves ingestion of ruminant (sheep and cattle) derived contaminated MBM. The infectious prion (PrP^{Sc}), being mostly resistant to proteolytic enzymes, escapes digestion in the gastro-intestinal tract and enters through M cells lining the mucosa of Peyer's patches. After amplification in the Peyer's patches, it travels to the brain through nerve endings. Mechanisms in the pathogenesis of the disease involve conversion of PrP^C in the CNS, slowly and progressively, to PrP^{Sc} (acts as template) which is capable of inducing other normal PrP^C molecules to undergo conformational change to the PrP^{Sc} form, resulting in the generation of extremely large numbers of abnormal molecules. When this happens, the proteins, which are normally in the liquid form, begin to solidify within the brain cells. The abnormal prion accumulates in the neurons (vacuolation) of the nervous system, causing nervous symptoms and finally death. This is a prolonged process, and symptoms of the disease may not appear for

years. In BSE, the histological changes are found monotonously uniform in the most caudal level of the medulla oblongata at the obex. At this level spongiform changes occur mainly in the nucleus of the solitary tract and in the spinal tract nucleus of the trigeminal nerve.

The clinical manifestation in prion disease are varying owing to species specific PrP^{Sc} in different conformation. BSE affected adult cattle (3-6 yrs old) represents clinical signs of apprehensive behaviour, locomotor deficits, lowered head, hyperesthesia and weight loss within 6-8 weeks after the onset. In human prions, there are symptoms of decline in cognition, and loss in locomotion, besides observation, etc. The history and neurological signs provide some clue to the neurodegenerative disorders. But in majority of the cases, these are confused with other infectious and nutritional disorders. There are no valid preclinical tests available. However, the erythroid precursor protein detection in lymphoreticular tissue, plasminogen in peripheral blood in preclinical stage, PrP gene analysis from DNA of blood leukocyte, MRI scan and CT scan and detection of higher levels of protein 14-3-3 in CSF of BSE affected bovine are some of the tests under trial.

Since, there are no authenticated clinical tests available to diagnose BSE in live animal and also during incubation period; it is diagnosed by histo-pathological lesions and by detection of modified prion protein (PrP^{Sc}) by immune-histochemistry (IHC) and by rapid ELISA on brain samples. The deposition of PrP^{Sc} in the brain causes triad of histopathological changes: i) single or multiple vacuolation in neurons and dendrites (neuropil) scattered either focally or diffusely in gray matter ii) neuronal degeneration and loss of neurons which appear dark, shrunken and angular and iii) astrocytosis that includes hypertrophy and hyperplasia of astrocytes in gray matter.

BSE surveillance and diagnosis-Examine the targeted cattle population (native born, crossbred and high risk cattle (imported from BSE infective countries or offspring of BSE affected cows or animals fed with BSE contained MBM of ruminant origin) over 24 months of age displaying nervous signs compatible with BSE.

1. Examine the clinical cases showing any nervous clinical signs particularly change in locomotion (wobbling, inability to stand), behavior (apprehension) and sensitivity (hypersensitive to touch, sound, light) and if all three are present the possibility of BSE is >90%.
2. Collect brain or brain stem from cattle carcasses preferably from fallen stock / emergency slaughter and also from slaughter house, preferably cattle lying in the lairage for more than 15 days. In slaughter house study of brains, the chances of getting a case of BSE are very remote.
3. Sensitize farmers, fellow veterinarians, officials of State Animal Husbandry Departments and State Agriculture Universities, butchers and workers engaged in transportation, marketing and slaughter of animals about BSE through personal contacts, the literature, etc but without causing any panic. They should be encouraged for reporting all cases of neurological disease in adult animals.
4. Carry out risk analysis of all other potential factors for BSE viz., look for rendering units and their functioning in the area, source and production of MBM or greaves of ruminant origin, other TSEs in animals, importation of animals and animal by-products from countries with BSE, etc.

How to collect brain or brain stem for BSE surveillance?-

1. Remove the head from atlanto-occipital articulation.
2. De-skin the head by cutting skin around head, ear, and horns.
3. Remove the muscles, lymph and salivary glands.
4. Remove the skull cap by severing with saw the ethmoid, the frontal bone, the foramen magnum and then laterally up to eye orbits.
5. Remove meninges by cutting dura mater over the cerebellum and medulla oblongata.
6. Slightly tilt the skull on either side, then with support of hand take out brain by tilting the skull backward and cutting the optic chiasma and cranial nerves.
7. Clear the meninges and make cuts over the cerebellum till ventricles are visible for better fixation of the brain.
8. Cut the brain into two halves and preserve half of the brain in 10% formalin and half over the ice and send for examination to CADRAD.
9. Care should be taken not to damage the parts of brainstem like medulla at the Obex, caudal cerebellar peduncle and rostral colliculi.
10. Enclose detailed history sheet of cattle like whether indigenous, exotic, crossbred, dead carcass, emergency slaughtered carcass, abattoir slaughtered cattle, clinical BSE suspect cattle, other neurological signs, age, origin of specimen, owners name, with the sample as well as separately.
11. If there is any problem regarding collection, preservation and dispatch of brain or brain stem of cattle, please contact CDDL at CADRAD, IVRI, Izatnagar or nearby RDDDL'S.
12. Risk analyses of potential risk factors should be carried out continuously.

(Note: while collecting the brain, persons should wear protective clothing like thick gloves, goggles, plastic apron, and gum boots etc. as the samples may be infected with communicable viral and bacterial infections. Always use 1 N NaOH or 2 N NaOH for decontamination in case of BSE.

Laboratory examination-

1. After proper fixation of brain, the coronal sections at the level of the Obex, caudal cerebellar peduncle, and rostral colliculi are cut (specific for BSE surveillance).
2. Thin tissue pieces from these sites are processed for paraffin embedding technique to get 4-5 μ thick sections, stained with H&E and examined microscopically for Hallmark spongiform lesions.
3. In the absence of histopathological spongiform changes or in autolysed case, detection of PrP^{sc} (infective prion protein) in different specific sites of brain is done by immunohistochemistry (IHC) on formalin fixed duplicate paraffin sections taken on poly-L-lysine treated glass slides or by dot blot assay in fresh samples.

Biosafety Guidelines Related for BSE (Bio safety level 2 or 3)-Following precautions are suggested in case of handling BSE suspected case-

1. Necropsies and processing of formalin fixed contaminated tissue required bio-safety level precautions.

2. Use waterproof gown, cut resistant gloves underneath two pairs of surgical gloves and surgical mask with a wraparound splash guard transparent visor while conducting the necropsy. Avoid contamination from aerosol during necropsy and tissue or fluid manipulation.
3. Avoid puncture of skin. If it is there swab with 1N NaOH for 5 minutes and then wash with plenty of water.
4. Place instruments in large stainless steel dish soaked in 2N NaOH for 1 hr or 1N NaOH for 2 h or autoclave at 134°C for 1 hr. Decontaminate the table and other surfaces with repeated wetting over 1 h. with 2N NaOH.
5. Formalin fixed tissues are post fixed in 95-100% formic acid for 1 h followed by fresh 10% buffered formalin for at least 48 h before histopathological processing.
6. Examine formalin fixed brain (at least 10-14 day) on a table covered with an absorbent pad with an impermeable backing.
7. All absorbent cotton, covers, pads, disposable clothing, gloves, imbedding moulds, section waste and other handling materials are disposed in bio-safety receptacles for incineration.
8. Liquid waste is collected in a 4 lit. waste bottle containing 600 ml 6N NaOH and diluted to a final volume of 4l to maintain the optimum concentration for disinfection.
9. Use disposable specimen cups or slide mailers for reagents, disposable Petri dishes for IHC.
10. During tissue preparation wear gloves, apron, eye protection.
11. Use disposable specimen cups for manual staining.

Decontamination of PrP^{sc}-

1. 1N NaOH or 2N NaOH or 4Mol/lit. guanadine hydrochloride, sodium hypochloride ($\geq 2\%$ free Cl_2).
2. Steam autoclaving at 132°C for 4.5 h or 1 h.
3. Dry waste at 132°C for 4.5 h or incinerated.
4. Liquid waste treated with 1N NaOH followed by autoclaving at 132°C for 4.5 h.
5. Boiling in SDS or prolonged protease digestion. High infected brain by autoclaving at 132°C for 4.5 h

Rabies Diagnosis- Rabies is an enzootic and a serious public health and economic problem in India. Rabies continues to be an endemic in Asia due to the large population of stray dogs, together with a lack of effective control strategies. A recent national survey by the Association of the Prevention and Control of Rabies in India estimated that in India a total of 20,000 human deaths occur as a result of rabies each year (Sudarshan *et al.*, 2006). In India, rabies occurs mainly in the urban form, although the existence of a sylvatic cycle cannot be ruled out. In the urban form, dogs play an important role as the reservoir and transmitter of the disease to humans and domestic animals, while jackals and foxes maintain the virus in sylvatic form. Worldwide, transmission from dogs accounts for more than 90% of human cases. In developed countries, bats, foxes, coyotes, raccoons, and skunks are major reservoirs. A child is at least 4 times more likely to be bitten than an adult in developed and developing countries. Boys are affected more often than girls, and younger children are at increased risk for facial injury (Mani and

Muarry, 2006). Phylogenetic analysis supports the evolution of lyssaviruses in bat vectors with occasional but regular spill over and host switching to carnivore vectors to extend the virus host range. Rabies is caused by classical rabies virus (CRV) belong to genotype-1, since the other member of genus lyssavirus (RRVs) have been found to cause illness indistinguishable from classical rabies. These facts indicate a great need to strengthen accurate and rapid laboratory diagnostic capabilities, which involves virus detection and genotyping. For suspected human cases, rapid antemortem diagnosis ensures appropriate patient management. Postmortem diagnosis enables the rapid administration of postexposure treatment to all human and animal contacts.

Rabies virus possesses a single-stranded, linear non-segmented, negative-sense RNA approximately 12 kb in length with helical symmetry. The principal genotype of lyssaviruses is the classical RV, genotype-1, which is present throughout the world. The remaining six members of the genus form a group of viruses termed as rabies related viruses (RRV) that show a more restricted geographical distribution and are more commonly associated with the specific host. RRV includes, Lagos bat virus (genotype 2), Mokola virus (genotype 3), Duvenhage virus (genotype 4), European bat Lyssavirus 1 and 2 (genotypes 5 and 6 respectively) and Australian bat Lyssavirus (ABLV) (genotype 7). The RRVs have been found to cause illness indistinguishable from classical rabies (Smith, 1996). The viral genome encodes five structural proteins: nucleoprotein (N), phosphoprotein (P), matrix protein (M), glycoprotein (G), and RNA-dependent RNA polymerase (L). The highly conserved and abundant nucleoprotein (N), a key structural component of the viral ribonucleoprotein core essential to viral propagation, constitutes the main target for rabies diagnosis and virus identification (Dean *et al.*, 1996). The G protein, which forms spike projections of the virion, is composed of homotrimers of single type which controls major aspects of host cell interaction, such as receptor recognition and receptor binding, host range, immunogenicity, neurovirulence, spread from the postsynaptic site to the presynaptic site and host adaptation.

Pathogenesis of rabies- The virus is usually transmitted by infected saliva through the bite of a rabid animal. Less commonly, the virus can be transmitted by contamination of an open wound, scratch, abrasion or mucous membrane with fresh saliva from a rabid animal. Rarely, the virus can be transmitted through aerosol inhalation or by corneal or organ transplants. After entry through a skin break, the virus may remain latent near the inoculation site. Subsequently, the virus replicates slowly within muscle cells. The neuromuscular junction is the major site of virus entry into the neurons. G protein plays an important role in viral entry and interaction with various cell receptors such as nicotinic acetylcholine receptor (nAChR) at the neuromuscular junction, the neural cell adhesion molecule, and the p75 neurotropic receptors on neural cell membranes. The RV then migrates along peripheral nerves to the central nervous system (CNS) via retrograde fast axonal transport at a rate of 12 to 100 mm/d. The P protein may be an important determinant in the retrograde transport of the virus within the axons. When the CNS is reached, massive viral replication and spread occurs, causing widespread inflammation and degeneration. From the CNS, the virus spreads centrifugally to the rest of the body, especially the salivary glands, via the peripheral nerves. It is at this stage that productive viral replication occurs with budding, particularly in the salivary glands, resulting in high viral concentrations in the saliva. The ability of RV to produce neurological disease is not well understood.

Diagnostic techniques and their efficacy-

History and clinical findings- The clinical signs of rabies are confused with other neurological signs caused by some other etiological agents. In 75% of dogs, the early signs (2 to 5 days duration) progress to dumb or paralytic rabies and to the furious form in the remainder, this is very risky while attending the dumb form of cases. Paralysis and death occurs in both clinical forms 4 to 8 days after the onset of symptoms. Based on some typical clinical signs in dogs and other animal species, a tentative diagnosis can be made. The animals showing abnormal behavior should be kept in isolation where they can not bite others for 10 days. In the **prodromal phase of rabies**, marked changes in behavior are seen which vary with the species like irritable, increased sensitivity to noise and light, more alert, restless and friendly, aggressive (more in cat) and attack without provocation, or become depressed, hiding in dark places, slight pyrexia and impaired corneal reflex, self-mutilation at the site of the bite. In excitative phase, nervous signs like irritable, vicious, bite and attack, muscle tremors, flaccidity or incoordination, pica, spasm and paralysis of deglutination, change in voice, difficulty in swallowing drooling and frothing of saliva, dropping of jaw, paralysis, coma and death.

Grossly, no specific lesions are observed in the brain. However, wound due to the bite and presence of foreign bodies in stomach due to pica may be suspected for rabies. The disease should be differentiated from Canine distemper, canine encephalitis, equine encephalitis, bovine encephalitis, hepatic encephalopathy, thiamine deficiency (cats), lead poisoning, organochloride compounds, benzoic acid, strychnine poisoning, Pseudorabies, spongiform encephalopathy (BSE/Scrapie/CJD) and listeriosis. In human, progression of the disease occurs in both forms: the encephalitic (“furious”) form in about 80% of patients and the paralytic (“dumb”) form in about 20% of patients.

Laboratory procedures for rabies diagnosis- Suspected animals should be captured carefully and confined for 10 days observation. Premature killing and poisoning of suspected animal is avoided, and if necessary, the animal may be shot through heart.

1. Wear protective clothing like thick gloves, plastic apron, gumboots, spectacles and facemask before opening/decapitating the skull of suspected rabies animal carcass. Get post-mortem kit, 50% buffered glycerine, 10% formalin, chilled acetone, Sellers stain, filter paper, wooden spatula, sterile vials, disinfectant (1% mercuric chloride, 4% lysol, 10% washing soda, etc.) in advance.
2. If the person is not skilled, the head of the carcass should be decapitated and kept cool in thick plastic bag (0.01cm thick) measuring 45cm x 100cm. Tightly secure the knot and place it in a larger double-walled outer bag (0.008 cm thick, 60 cm wide x 120 cm deep) filled with enough crushed ice. Place the knot in outer bag and keep entire package in corrugated cardboard carton (25 cm² x 25 cm² and 45 cm deep). Seal carefully with adhesive plastic tape. Label caution on the carton and send it immediately, to avoid inactivation of virus if the cooling is not maintained, through a person /courier to the nearby lab along with full history as per the WHO (2005).
3. The rabies viruses though present in all parts of the brain, its abundant presence has been seen by FAT in hippocampus (Ammon's horn) and the medulla oblongata, and hence these parts must be included. To reach these locations in cranium in field conditions, occipital foramen route or retro orbital routes may be followed.

4. For taking out the whole brain from the suspected animal, the skull is opened by a trained person in necropsy room with all precautions. If possible, salivary glands and trigeminal ganglion should also be collected. Half of the brain in 50% buffered glycerine over ice in well secured and leak proof container and half in 10% formalin (10-15 times volume of tissue) in wide mouthed plastic container be sent to the reference laboratory (Kaplan and Koprowski, 1973).

Safety measures while handling rabies virus in the laboratory-

1. Persons who run a high risk of repeated exposure, such as laboratory staff working with rabies virus must be protected by pre-exposure immunization which consists of a short course of three injections of a potent Anti-rabies vaccine at 0, 7, 21/28 days. Protective antibodies to be assessed before start handling virus.
2. Rubber or strong plastic gowns/protective clothing, face mask, goggles and thick gloves should be worn when dealing with virus.
3. For pipetting, rubber teat/automatic suction devised should be used, and mouth pipetting should be forbidden. The pipetting should be carried out under a negative draught hood.
4. Needles and sharp objects must be used very carefully and placed in boxes/containers with imperforate and not readily penetrable walls which can be autoclaved and incinerated.
5. Hands and workbenches must be cleaned and disinfected with 1:500 dilution of Quaternary ammonium compound (Cetrimide, Benzalkonium chloride etc.) or 45-70% alcohol before starting work and after completion of the work.
6. The waste materials including specimen containers must be decontaminated before their disposal.
7. The wetting of caps/cotton plugs with the culture must be avoided.
8. Eating, drinking, smoking and applying cosmetics must be forbidden in laboratory area.
9. Entry of virus into the effluent from laboratory must be checked.
10. High speed mixing and centrifugation procedures should be carried out in closed containers.
11. A 1% concentration of soapy water/detergent or a standard disinfectant solution must be used for pipette, glassware, and plastic ware and instrument receptacles. The solution should be autoclaved and discarded after every use.
12. The solution containing virus should never be poured from a container being held above eye level.
13. Chipped, cracked or broken glassware must not be used.
14. While working in a Laminar Air Flow, care must be taken not to splash the specimen containing virus onto the filter.
15. The work bench should be kept tidy and uncluttered to prevent accidents occurring due to spillage or breakage.
16. Aerosol formation (as occurs due to forceful pipetting, frothing, centrifugation, etc.) should be reduced up to a possible minimum level during handling the virus.
17. All the materials and cultures maintained in the laboratory should be properly labeled along with date.

18. The person must avoid working with rabies virus, if having cuts, open sores or scratches on skin or must take proper care to cover it with water proof adhesive material.

Action taken if any accident occurs while handling rabies virus-

1. All cuts and wounds should be made to bleed freely and then wash immediately and thoroughly with plenty of water and soap for several minutes, before applying a standard chemical disinfectant (1:500 dilution of Quaternary ammonium compound (Cetrimide, Benzalkonium chloride etc., 45-70% alcohol, 1% soap solution, or 5-7% iodine solution).
 2. Splashes in the eyes must be washed out thoroughly with saline or antiseptic eye wash.
 3. Splashes into mouth must be washed thoroughly with repeated application of water (Mouth wash).
 4. If infective or potentially infective material spilled on the laboratory protective clothing, this should be removed and immediately autoclaved/disinfected.
 5. If an accident has involved breakage of glassware, the piece must be picked out using forceps or dustpan and brush. They must never be picked up by hands even when protective gloves are being worn.
 6. If a fluid specimen containing virus is spilled/dropped, the area should be rendered ventilated for least 19 minutes to allow the aerosols and droplets to disperse. After this the potentially infective material should be covered in a cloth, soaked in an appropriate disinfectant for 30-60 minutes and then the spillage can be cleaned away with swabs and dustpan, which can be placed in a disinfectant for an hour or autoclaved. The area then can be cleaned with water. All swabs and clothes must then be disposed off as hazardous waste in appropriate containers.
 7. As a precautionary measure, after exposure to rabies virus, a post -exposure immunization course of Anti-rabies vaccine can be used:
 - ▼ In case of a person, previously immunized with a potent cell culture vaccine, and within the immunization period of previous vaccination, two injections at day 0 and 3 are given.
 - ▼ In case of a non-immunized person, or previously immunized more than 3 years before the exposure, 5 injections at day 0, 3, 7, 14 and 28 are given.
- *Usually a panel of laboratory tests is used to rule out the rabies. As per OIE Manual, 1996 following tests are required to be conducted for rabies diagnosis: Among all the tests, dFA is the most rapid, reliable and sensitive of all the tests available.*

1. Rapid Seller's (1972) staining: The test is inexpensive and results are obtained in under 1 h. Seller's stain reveals well differentiated intracytoplasmic, acidophilic, 2-20µ in size bodies (Negri bodies) in the cytoplasm of neurons. These bodies appear magenta or heliotrope to bright red in colour and have black basophilic inner granules only in fresh tissues. In putrefied samples, this differentiation is lost and identification becomes more difficult unless examined by trained person. In 15% or more cases, false negative results are encountered.

Procedure-

- Ø Make a direct smear by impression on glass slide by touching on cut surfaces of tissue placed on clean wooden spoon/ blotting paper. 2-3mm thick unfixed brain

tissues viz., hippocampus (Ammon's horn), cerebral cortex, cerebellum, medulla oblongata and spinal cord are used for making the slides (Ammon's horn and brain stem are preferred).

- Ø Stain immediately the wet smear slides by immersing in Seller's staining solution for 10 -30 sec. depending upon thickness of smear. No fixation is required as the tissue is fixed and stained immediately.
- Ø Rinse the slides in running water and air- dry. Do not blot. See the slide under the oil immersion of a light microscope for the presence of well-differentiated Negri bodies.
- Ø Care should be taken to differentiate these bodies from nonspecific inclusions, like Lafora bodies, etc. which are normally found in aged animals.

2. Histopathology- Grossly, except for mild congestion of meninges occasionally, no other changes may be seen during post-mortem examination. Microscopic lesions in animals are variable from sparse to occasionally severe. In about 30% or more suspected cases, Negri bodies are not developed due to many reasons. Changes of non-suppurative encephalomyelitis in cerebral cortex, hippocampus, pons, medulla, cerebellum and spinal cord are characterized by perivascular cuffing with mononuclear cells and ring haemorrhage, neuronophagia, satellitosis and diffuse or nodular aggregation of microglial cells (Babe's nodule) surrounding the degenerating neurons. The presence of specific Negri bodies (2-20 μ m), single or multiple, in cytoplasm of neurons is pathognomonic for rabies. In positive cases, Negri bodies are seen most frequently in hippocampus in dogs and Purkinje cells of the cerebellum in cattle. Lesions of rabies are seen earlier and more constant in Gasserian ganglia. Sometimes, infected wild and domestic animals show vacuolation in the neuropil of the grey matter, thalamus and inner layers of the cerebral cortex. Special stains like Mann's, giemsa, or Seller's stains can permit differentiation of rabies inclusions from other intracellular inclusions. The Negri inclusions appear magenta in color with dark-blue interior basophilic granules. This test is neither as sensitive nor as specific as other tests. So, it is not considered diagnostic for rabies. In contrast, the dFA test shows rabies antigen in nearly 100% of the samples.

3. Fluorescent antibody test (FAT)- When conducted by a trained person using quality reagents, this technique is most accurate and quick to perform (within half-hour to 2 - 4 h). The **direct FAT** is employed as a routine rabies diagnostic test by all the laboratories on fresh, frozen or glycerolised materials. However, FAT does not work on highly autolysed and putrefied material.

Procedure-

- Ø Use positive (CVS-24 challenged mice brain impressions) and negative control mice brain impression smears during staining.
- Ø The optimal working solution of anti- rabies fluorescent (FITC) conjugate (either polyclonal conjugate specific to whole rabies virus or specific to rabies nucleoprotein or mix of different mAbs) should be titrated before carrying out FAT.
- Ø Prepare impression smears from the suspected brain portions (Ammons horn, cerebral cortex, cerebellum, medulla oblongata and, if possible from salivary glands) or from skin, corneal smears, CSF, etc. from clinical cases.

- Ø Air dry the slides and fix in chilled acetone (-20°C) for 30 min. to 2 h. The slides may be used immediately or may be stored at -70°C after wrapping in paper for later use.
- Ø Encircle smears (1 positive, 1 negative and 1 test) with quick fix to retain the conjugated drop within the smear area.
- Ø To avoid autofluorescence treat the slide with freshly prepared 1mg/ml sodium borohydride in PBS (Apply this solution immediately while fizzing) for 2 changes of 5 min each or 2.5mg/ml of trypan blue in water or buffer for 1 min
- Ø Wash with PBS
- Ø Add 1 drop of working solution of anti- rabies fluorescent (FITC) conjugate.
- Ø Put the slides in a moist dark petridish and incubate at 35 - 37°C for 30-60 min.
- Ø Rinse the slides individually in PBS (7.6 pH) with squirt bottle and then put the slides separately in couplin jar containing PBS (7.6 pH). Put the jars over magnetic stirrer and wash under agitation for 10 min, repeat the process thrice.
- Ø Take out the slides and add 1 drop of 50% buffered glycerine and place a coverslip, excluding the air bubble. The fading can be extended by adding 2.5% of 1, 4 diazabicyclooctone or by using commercially available antifade mountants (Vector Laboratories).
- Ø Examine smears using 40x objective of the fluorescence microscope and evaluate the viral antigen that appears as fine particles (dust) of yellow green fluorescence.
- Ø Check the results of positive and negative control slides.

If the results of Seller's staining and FAT are doubtful, undertake biological test immediately.

4. Mouse inoculation test (MIT)- The rabies virus is detected in clinical samples by isolation in suckling mouse or in neuroblastoma cell line (CCL 131, ATCC). MIT is quite expensive and does not give rapid results but allows easy identification of rabies strains. In cell line, which is sensitive to street virus, the results are obtained only after 18 h. This test is much cheaper than MIT and gives more rapid results.

Procedure-

- Ø Pool the portions of cerebral cortex, cerebellum and Ammon's horn of both the sides of the brain, and medulla oblongata and, if possible salivary glands.
- Ø Ground the tissue to smooth paste using 10% of chilled PBS (pH 7.6) in a sterile and cold, mortar and pestle.
- Ø Add 200 units of penicillin and 200µg of streptomycin/ml. Do not exceed the concentration of streptomycin beyond 200µg otherwise mice may die showing convulsions.
- Ø Centrifuge at 1000-rpm (165g). Supernatant is used for injection in the mice. Take at least 10 healthy new-born mice (2 - 3 days old) or 3 weeks old.
- Ø Inoculate the suspension (0.03ml) intracerebrally in suckling mice without anaesthesia or in adult after anaesthesia (ether) using 26-gauge needle.
- Ø Keep the control and healthy mice separately in iron cages.
- Ø Observe adult mice for 28 days. Every dead mouse from the 5th day onward is examined for rabies using FAT on its brain impression. Clinical signs of ruffled feather, tremor, in-coordination, paralysis, prostration and death appear after nine

days post inoculation. For faster results out of new-born mice 1 baby mouse may be checked by FAT on days 5, 7, 9 and 11 post inoculations.

Use of mAbs, nucleic acid probes or PCR may support the above tests.

5. Reverse Transcription - Polymerase Chain Reaction (RT-PCR)- The PCR based tests may allow rabies diagnosis even in the situation that precludes virus isolation (MIT). In this method, RNA is first converted to c-DNA by the enzyme reverse transcriptase before its *in vitro* amplification by PCR. The test requires only small piece of brain obtained via the retro orbital or occipital foramen route using a disposable pipette. **Even in highly decomposed brain tissue, it compares better than dFAT.**

6. Real-time PCR- This technique works based on the amount of florescence, which is directly proportional to the amount of accumulated amplicon during the reaction by using different fluorescent dyes (Syber green, TaqMan probe, ROX etc.). Real-Time chemistries allow for the detection of PCR amplification during the early phases of the reaction. Measuring the kinetics of the reaction in the early phases of PCR provides a distinct advantage over traditional PCR detection. Advantages of using Real-Time PCR: Traditional PCR is measured at End-Point (plateau), while Real-Time PCR collects data in the exponential growth phase, An increase in Reporter fluorescent signal is directly proportional to the number of amplicons generated, The cleaved probe provides a permanent record amplification of an Amplicon, Increase dynamic range of detection, No-post PCR processing and Detection is capable down to a 2-fold change.

Chapter 10

Animal Disease Outbreak Investigation

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When a disease affects a large number of animals than one would normally expect, that constitutes an outbreak/epidemic. Different combinations of infectious or toxic agents, individual and herd immunity, population age and movement, nutrition and environmental factors may contribute to an outbreak of disease. Treatment of the affected animals can usually be initiated immediately, before the cause of the outbreak is fully understood. The occurrence of a disease in epidemic form is quite common and as such there is no strict rule to follow while investigating the outbreak. It depends upon the objective of the investigation, availability of the resources, facilities and tools of investigation. In general the most common objectives of any disease investigation are:

1. To identify causal factor(s) of the disease
2. The origin of these factor(s), and
3. Mode of disease transmission

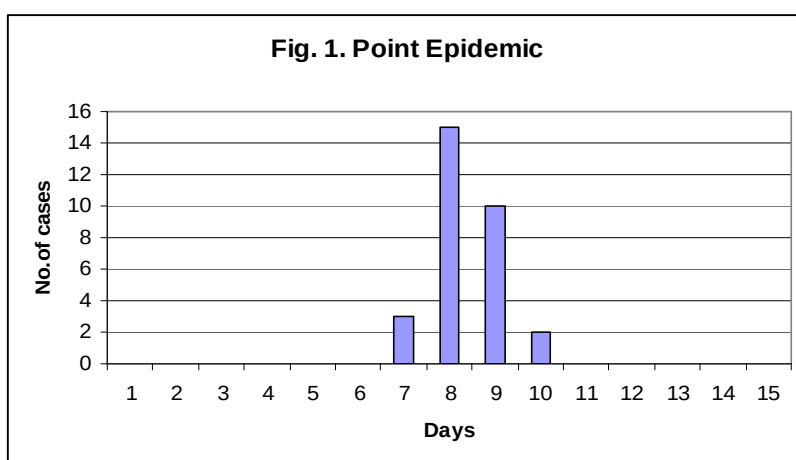
The disease outbreak investigation is a team approach. Contribution and expertise from specialists from different disciplines are required to reach at a conclusion. Specialist from microbiology, parasitology, pathology, nutrition, toxicology, medicine, epidemiology and others may be required. The utility of the information obtained after the investigation will be utilized in development of suitable and appropriate managemental measures to prevent and control the occurrence of the disease in future. The disease investigation includes descriptive epidemiology as the first step followed by analytical and experimental epidemiology. The entire exercise can be described step-wise as follows:

- a. First step is to define the disease i.e. the characteristic sign and symptoms seen in the diseased animals. Obtain individual case histories of the affected animals, identify cases and controls (not affected), and look for the source (factors) of exposure. A good case definition includes the animals that have the primary disease under investigation and excludes those that are healthy or may be sick but affected by an unrelated disease. A tentative diagnosis based on common clinical signs can be made, before a confirmatory diagnosis has not been reached. Reviewing physical findings and laboratory test results from each case can be used to confirm the clinical diagnosis and rule-in or rule-out alternative differential diagnoses. Collect clinical and morbid samples from diseased as well as healthy animals for laboratory tests to further confirm the diagnosis.
- b. Find out, if indeed it is an epidemic i.e. the number of animals affected are more than the expected level. This is done by comparing the current frequency of disease with what would be expected in a similar group of animals under similar conditions. First, a count of the affected animals and the total number of animals at risk (affected and unaffected) is needed. Then an overall attack rate (AR) can be calculated.

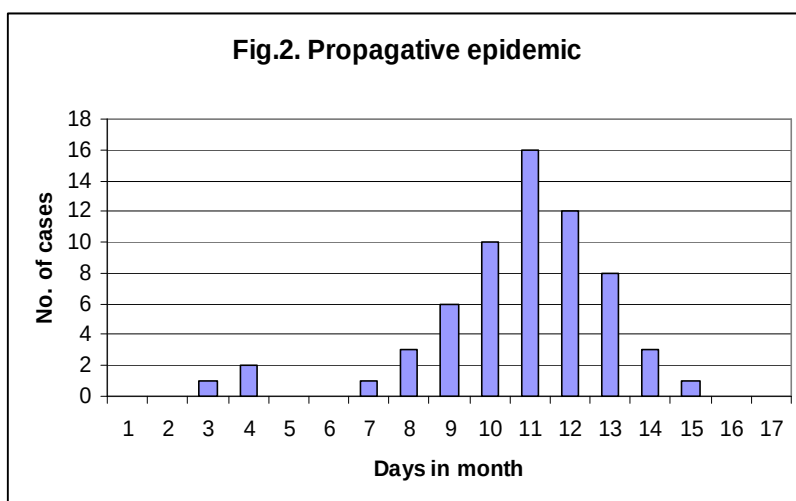
$$AR = \frac{\text{No. of new cases of a disease since onset of outbreak}}{\text{Total Number of animals at risk at onset of outbreak}} \times 100$$

The attack rate for the population under investigation is then compared to the attack rate in other similar population which is not affected i.e. not exposed to the causative factor.

If AR or incidence is really more than the expected level, then look for type of epidemic i.e, either point or propagative? For example a rapid increase in the number of cases of a disease over a very short period of time, suggestive of point source epidemic (Fig. 1) i.e. large number of animals exposed simultaneously to a common source like food or water or a highly virulent infectious agent.



A gradual increase in frequency of disease is suggestive of a propagative epidemic (Fig. 2) in which transmission of causative agent from animal to animal occur directly (physical contact) or indirectly through vehicle or vectors. Describe the disease pattern on the basis of information collected.



- c. Relate distribution of disease to time (temporal distribution) i.e. days, weeks, months, seasons, lactation number, stage of lactation, and relationship with calving or weaning time or time of change of pasture, feed, shed, vaccination, and therapy etc.
- d. Relate distribution of disease to space (spatial distribution) i.e. herd, flock, pen, and shelter, village, district, country etc. The role of farm management in causation of disease can not be ignored.
- e. Relate distribution of disease to animals or groups of animals (demographic distribution) as per the age, sex, species, and breed. The addition of new animal or temporary / permanent withdrawal of animal from herd or vaccination, dipping, therapy of animals play significant role in causation of disease.
- f. Examine the role of collateral factors like water, feed, vectors, air (ventilation), biological and drugs. Collect feed and water samples as well as vector specimens to confirm their role in disease causation.
- g. Based on the information, identifies factors associated with the outbreak. If the attack rate is high in the exposed group and low in the non-exposed group, this factor is more likely to be the cause of the outbreak. Formulate causal hypothesis on the basis of information obtained. The lesser the number of hypothesis, the better would be the judgment.
- h. Evaluation of hypotheses must be carried out. Depending on the nature of the data, there are two approaches: i) comparison of the hypotheses with the established facts or ii) analytic epidemiology.
 - (i) The first method should be used when the evidence is so strong that the hypothesis does not need to be tested, as in the outbreak of aflatoxicosis. All of the cases ate GN cake infested with *Aspergillus* spp. The large difference in AR of aflatoxicosis in exposed (i.e. consumed GN cake infested with *Aspergillus* spp.) and non-exposed (i.e. not consumed GN cake infested with *Aspergillus* spp.) population will suggestive of the source of the outbreak. In this case it is easier to hypothesized that the GN cake was the source. When GN cake was tested, found to have toxin levels higher than the recommended dose were inadvertently added to the animal feed.
 - (ii) The second method, analytic epidemiology, is used when the cause is less clear. With this method, the hypothesis is tested by using a comparison group to quantify relationships between various exposures and the disease. There are two types of analytic studies: **cohort studies** and **case-control studies**. A cohort study compares a group of animals who share a similar feature, such as a demographic characteristic or a particular exposure to a group of animals who do not share the particular feature to compare the rates of illness. A case-control study compares people with a disease (cases/patients) with a group of people without the disease (controls) to ascertain differences in exposures. The nature of the outbreak, time and resources will help determine which of these studies will be used.

- i. Analyze the data, estimate statistical measures of association of cause (factor) and effect (disease). Depending upon the result, accept or reject the hypothesis. If rejected, explore another hypothesis by re-examining the collected data.
- j. Authenticate the findings on hypothesis (causal association) by conducting experiment. Draw suitable inferences.
- k. Prepare a detail report including the data analysis, photographs, laboratory testing, maps, charts and experimental findings and finally make appropriate recommendations for preventing new cases and future outbreaks. The report is a key part of the veterinary record and may be a useful educational tool for owners, practitioners, or students and communicate the same to the concerned authorities for implementation.

Chapter 11

Postmortem Examination of Animals and Collection of Materials for Disease Diagnosis

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Necropsy examination of animal helps in diagnosis of diseases and ultimately their control. It is said “Necropsy is the message of wisdom from dead to living”. It includes systemic examination of dead animal, recording of gross pathological lesions, and their correlation with history to make diagnosis of diseases. Sometimes it is difficult to make any conclusion merely on the basis of post mortem. In this situation material is to be collected for further laboratory analysis such as histopathology, microbiology and toxicology for confirmation of the etiological agents. Necropsy examination is an integral part of disease investigation. Therefore veterinarian must have the basic knowledge of post mortem technique, recording of lesions, collection of proper material for laboratory examination. Necropsy is perhaps the sole diagnostic technique available to veterinarian to find out the cause of death of the animals and as it is well known that all the control measures for their effectiveness depend upon a correct diagnosis.

Necropsy Techniques in Different Species-

Bovine- Before dissecting the carcass, possibility of anthrax is to be ruled out first. Blood smears, prepared by pricking the tips of ears are examined for this purpose. The carcass is deskinning leaving the genital organs and udder attached in the normal position. The skin is examined on both sides for any marks and lesions. The body is observed for nutritional status, cadaverous changes and possible discoloration or anaemia. To detect anaemia the visible mucous membranes are seen, especially the third eye lid (membrane nictitans) by pulling it out with help of forceps.

The cadaver is supported on its back and inclined towards the left side. The hind legs are abducted by cutting through the medial thigh muscles anteroposteriorly close to the pelvic symphysis, opening the hip joints and severing the ligaments. Incise around the udder in female and remove it from its attachments. In the case of male the penis along with the prepuce is detached and drawn backward up to the ischial arch. The forelimbs at this stage may also be separated by incising in the axilla close to the chest wall. Do not sever the dorsal attachment and leave the forelimbs partly attached with the body.

The scrotum is cut to expose the testicles. Incise the tunica vaginalis, draw out the testicles and separate the spermatic cords and leave as such hanging on either side of the abdomen. Make a superficial incision in the linea alba, taking due care for not puncturing the viscera. In cases of excessively tense wall due to bloat etc., a small puncture on the left abdominal wall with the point of knife would allow the escape of gas and relax the wall. Incise from the opening in the linea alba, guided by two fingers upto the xyphoid cartilage and pubic symphysis posteriorly.

Reflect the abdominal wall along the costal arch on either side and in female along the border of the pubic bones as well. Observe and measure if there is any increased peritoneal fluid or abnormality. Separate if not already done the testicles and spermatic

cord from the lateral abdominal wall by incising along the inguinal canal. Examine for the normal position of the viscera and note the changes. Locate and examine the epiploic foramen for herniation of the loop of intestine. This is done by inserting two fingers along the caudate lobe of liver, a clear space allowing the two fingers in adult is the epiploic foramen. Remove the omentum from the longitudinal grooves and examine.

Separate the duodenum between two ligatures at the pylorus. Detach rumen possibly by hand from diaphragm and expose the oesophagus. Ligate at two places and cut the oesophagus between ligations. Pull out the rumen with spleen and abomasum on the left side. While doing PM care is taken to separate the pancreas manually and is to be taken out with liver. Divide duodenum between two ligations at the junction of its second turn with the jejunum. Free the rectum, ligate at two places and incise between these. Remove the whole intestine by cutting the mesenteric roots. Give a small incision in the vena cava near its entrance in diaphragm and with the help of one finger tear the vena cava the whole length of the liver. Observe here for any thrombi or abscess in the vena cava. Now remove the liver from its attachment along with pancreas and duodenum in one piece.

The kidneys and adrenals are freed and separated from its lateral and anterior side. Cut the renal vessels along the medial border and carefully pull both the kidneys backward, thus freeing the ureters from the abdominal wall. In female the broad ligament of uterus is separated from its insertions on the abdominal wall. In male separate the already drawn penis from the ischial arch. Now the urinary and genital organs are all free. The floor of the pelvic is first drawn out from anterior to posterior borders through the obturator foramen. Then all the urinogenital organs are gathered in one hand and backward removing simultaneously the attachments by knife. Make an incision around the anus and in the female include the vulva, thus freeing these organs which are safely pulled out.

The diaphragm is taken out by cutting at its insertion at xyphoid cartilage, following on either side along the costal attachment and finally the sub lumbar insertion. Damage to the pericardial sac is avoided. The pericardial sac is separated from the sternum. Any fluid or abnormal contents in the pericardial sac is observed and collected. This can also be done after removing the heart along with the lungs. Separate the trachea and oesophagus in neck region and divide them in the middle. The free end of trachea along with oesophagus is pulled towards the thoracic aperture. This is afforded by making a short transverse incision between the two cartilage rings and inserting the fingers for tight grip. After freeing the trachea up to thoracic inlet, it can be now pushed in the cavity. It is then pulled backward and thoracic organs are separated by cutting through the mediastinum and brachial vessels. The thoracic aorta is removed with these organs and cut through the lumbar region, leaving behind the abdominal aorta.

The structure of oral cavity and neck are then removed. Give an incision on each side of the intermandibular space along the medial border of the mandible as far as the symphysis. The tongue is now drawn out and pushed backward, the incision on either side are also extended backward. The soft palate is cut by incision from each side, forward and medially to meet in the middle line forming a triangular area. The hyoid bones are divided on either side by placing the knife with its edge upward between the thyroid branch of the hyoid bone and larynx. A single jerk is sufficient to separate hyoid bones provided the knife is in correct position. Now pull the tongue with pharynx, oesophagus and trachea backward along the neck where previous incision was made.

The head with the salivary glands intact is severed through the atlanto occipital joint. The joints are examined by opening from the medial side commencing distally on the limbs. The spinal cord is exposed by sawing through the arch of the vertebrae. It can also be exposed by cutting through edge of the spinal canal slightly off the midline.

Small ruminants- The procedure adopted is the same as for large ruminants expect the following:

1. The sternum is removed by cutting the ribs at costochondral joints.
2. The neck and thoracic organs are then removed together without cutting the trachea and oesophagus in middle.

Equines- The body of the horse is inclined toward the right side. The rest of the technique is the same as in ruminants except for the abdominal organs. The pelvic flexure of great colon and caecum are reflected ventrally away from the body cavity. Spleen, left kidney, left adrenal and small floating colon are removed in order, duodenum is then severed behind the root of mesentery where it passes medially from the right side and is separated from the mesentery till the ileocaecal valve is reached.

Another method of locating the duodenum is following the duodenocolic fold mesentery from its colonic attachment near the place where the small and large colons divide, to its duodenal attachment few centimeters posteriorly. Stomach is then removed. Anterior mesenteric artery and its branches supplying the great colon and caecum are examined first by removing the abdominal aorta before separation of the great colon and caecum. Liver, right kidney and right adrenal are subsequently removed. All the thoracic organs are removed together by cutting the trachea and oesophagus at the thoracic inlet.

Canines and felines- The dog or cat is placed on its back and incision in the skin is given and legs abducted. Abdominal cavity is exposed by a midline incision along the linea alba from xyphoid cartilage to the pubis and transverse incisions from xyphoid cartilage to dorsal extent of the body cavity along the last rib. Sternal cartilage or ribs are cut through and sternum removed completely to expose the thoracic cavity. The viscera may be removed in piecemeal but *enmasse* removal of all organs from tongue to rectum is equally good if not better, because anatomic relationships are not disturbed and the procedure is quicker. For a hasty examination, Rokitansky procedure, i.e. examination of all organs *in situ* without detaching them by lifting them up, cutting, examining and returning them back to body cavities, is followed, the piecemeal method intestines from rectum to duodenum, spleen with omentum and liver with pancreas, stomach and duodenum are removed separately.

Swine- The technique is the same as applied for canines except the removal of intestinal tract. The removal of the intestinal tract in this case should begin at the tip of colonic spiral which is made free along with the intestine from the mesentery. Another way is to loosen both forelegs and hind legs and a strip of ventral body wall to expose trachea esophagus, thorax and abdomen for convenient removal. After post mortem examination the viscera is returned to the body cavity and ventral flap replaced so that the organs are removed together without the tongue.

Laboratory animals- Laboratory animals are necropsied either to know the cause of their spontaneous death or to evaluate the results of experimentation by performing complete or line necropsy. There is no single standard procedure. The carcass is invariably soaked in a disinfectant solution and fixed on the board for easy manipulation opened ventrally. At times a direct approach to the visceral organs is made through a midline incision. Brain of laboratory animals often require examination and can easily be removed either sharp scissors or a pair of bone-forceps.

Collection and Dispatch of Material- All pathological processes lead to morphological, physiological, chemical and other alterations. These alterations in certain infectious diseases are pathognomonic of the disease e.g. infectious canine hepatitis (intranuclear inclusions in hepatocytes), canine distemper (intranuclear and intracytoplasmic inclusions in different organs), rabies (Negri bodies), malignant catarrhal fever (intense lymphocytic reaction and vasculitis), equine viral arteritis (generalized arteritis with oedema and cellular infiltrations). In many cases the changes are suggestive of probable etiological agent and in others indicate the ill effects of the disease and are rarely of little consequence. It is, therefore, obligatory in each and every case that material for histopathological examination be sent for laboratory diagnosis.

In order to get the optimal histopathological interpretations of lesions in tissue samples for disease diagnosis, it is obligatory on the part of veterinarians/technicians to pay due attention for proper method of collection, preservation and its dispatch to the laboratory. On many occasions, it is experienced that the valuable material is gone waste when received in laboratory in a state of either putrefaction or incomplete preservation due to ignorance and not following the proper methods. The following methods would work as a guide for the same. Samples from all vital organs, tissues showing lesions and lymph nodes are necessarily required. Tissues from putrefied carcasses should also be collected. Tissue samples are collected from dead or live animals for laboratory examination to confirm the tentative diagnosis.

Purpose of Collection of Samples

- ▼ Diagnosis of disease or for identification of new disease.
- ▼ Confirmation of tentative diagnosis.
- ▼ Prognosis.
- ▼ To observe the effect of treatment and give direction for future therapy.

Precautions

- ▼ The tissue sample should be fully representative of the organ and the lesions.
- ▼ Collect the tissues as early as possible after death of animal.
- ▼ Representative tissue/sample should be collected.
- ▼ Sharp knife should be used for cutting.
- ▼ Collect the tissues directly in fixative.
- ▼ Size of tissue should not be more than 1cm for histopathology in 10% formalin.
- ▼ Hollow organs should be taken on paper to avoid shrinkage.
- ▼ Hard organs like liver, kidneys etc. should be collected along with capsule.
- ▼ Wide mouth glass or plastic bottle of varying capacity should always be used.

- ▼ 10% formalin solution is the best fixative for routine histopathological diagnosis.
- ▼ Zenker's fluid may be used in viral suspected cases for demonstration of inclusions.
- ▼ For virological examination, small pieces of spleen, lymph node and from lesion sites may be sent either in 50% buffered glycerine or on ice.
- ▼ For bacteriological isolations heart blood, tissue pieces and swab may be sent on ice.
- ▼ Proper labeling of the bottle/specimen samples is very essential.
- ▼ A piece of absorbent cotton should be placed on the surface of the sample in formalin to keep the tissue moist in case the bottle is broken during transit.
- ▼ The mouth of the specimen bottle or bags may be sealed carefully by paraffin was or adhesive tape to avoid the leakage during transit and make it watertight.

Chapter 12

Autolytic Changes

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Post mortem changes result from degradation of tissues associated with the release of proteolytic lysosomal enzymes from the cells. These processes (called post mortem autolysis) occur automatically after death of the animal. It must be remembered that autolysis also occur following death of a limited portion of an organ or tissues (necrosis). The term post mortem autolysis appropriately distinguishes the same processes that occur after death of the whole animal. Autolysis means self digestion by the tissue enzymes that are present in or release into the cytoplasm of the cell. Autolysis is one of the most important and most striking post-mortem changes observed in most carcasses, particularly in hot weather. These changes are evident on both gross and microscopic examination of tissue and must be differentiated from ante-mortem lesions. Considerable experience is required to make a clear cut distinction and to avoid generally occurring common confusions arise in interpretation of findings until some confidence is gained. Therefore, it is important to learn specifically the range of post-mortem changes with special emphases to “autolysis changes” that may occur after death in each of the tissues and organs.

All post-mortem should be conducted on freshly dead animals. Autolytic changes occurs very rapidly in organs such as the brain, liver and kidneys, especially in warm weather or in animals with heavy coats which keep the body temperature high for some time after death. The rumen because of its bulk, also acts as a heat reservoir and enhances a rate of autolytic changes. If it is not possible to do a post-mortem soon after death the carcass should be chilled. This is possible with small animals but with cattle and horses rarely feasible unless the post-mortem house has the chilling facilities. When chilling is out of question the animal should be put in a cool place and abdomen slightly open so that heat loss from the carcass is accelerated. Cloths soaked in normal saline solution should cover the opening to prevent undue evaporation. If this is not done abdominal contents, although cool, will dry out, become crinkled and hard. This type of changes is about as bad as autolysis. It makes subsequent examination difficult and masks some type of pathological changes. The recognition of autolytic changes is extremely important and something with which any person conducting post-mortem should become familiar.

Autolysis plays an important role in producing the picture of necrosis. Cells undergo enormous swelling due to the enzyme breaking down of the large molecules into a greater number of small molecules with the increase of osmotic pressure and inhibition of fluid. For this reason dead cell floating in a fluid medium become very swollen. In the dying cells respiration ceases, glycolysis proceeds for a while and there results a drop of pH due to production of lactic acid. The synthetic activities of the cell stop but the lytic destructive enzymes continue their work. These destructive enzymes are mostly active of a low pH and include wide range of proteases, lipases, esterases, deoxyribonuclease, ribonuclease etc. The cell thus undergoes a process of self-digestion or autolysis. The enzyme may be derived from lysosomes and it has been suggested that the disruption of these bodies is an important factor in the causation of cell damage as well as in the

autolysis, which follows cell death. Electron microscopic study of cells undergoing autolysis-*in vitro* study shows that the changes (mitochondrial swelling etc.) are diffuse and not more advanced near the lysosomes. Such evidence would suggest that lysosomal enzymes do not initiate the change of necrosis. All of the nuclear and cytoplasmic changes seen in coagulative necrosis occur in autolysis, however the changes will probably be uniform in the tissue with no reaction such as occurs in necrosis.

Biochemical Mechanisms of Autolysis-

1. ATP depletion
2. Anoxia
3. Membrane damage by bacteria
4. Loss of Ca²⁺ homeostasis

Cathepsins: Hydrolases, such as catheptic enzymes (cathepsin D, cathepsin B and cathepsin L), as well as calpain are considered to be the main cause of post-mortem muscle softening. Cathepsin L- are mostly responsible for muscle tissue degradation. Cathepsin L from muscles, hydrolyses myosin, troponin T and I, and tropomyosin. Cathepsins are liberated from lysosomes during ageing.

Putrefaction: Decomposition of tissues brought about by the protein splitting anaerobic saprophytic organisms, results in the formation of gas & variety of foul smelling substances- ammonia, hydrogen sulphide, indol, skatol & putrescent amines-like “putrescine & cadaverine”. The tissue turns black or dark-green as a result of formation of iron sulphide from break down haemoglobin. The common putrefactive organisms are *Clostridium spp.* normally present in faeces, leads to pronounced postmortem changes in the body like gaseous distention, softening etc. bacterial flora -Present in GIT and Respiratory tract bring about the post-mortem changes rapidly in favourable condition. Post mortem autolysis is influenced by a host factors. It is particularly enhanced by the action of bacteria. Bacteria that form part of the microbial flora of surface mucosa particularly in the gut proliferate soon after death. This possibly occurs due to alterations in the environment, where the usual defense mechanism of the animal body ceases to operate. Also, the bacterial proliferation may be due to the availability of a large amount of growth promoting substrates. Following their proliferation, invasion of organs and tissue occurs possibly through the vessels and lymphatics. Animals that died of bacterial diseases usually show post mortem changes rapidly than those that died of other causes.

Factors Influencing the Rate of Post Mortem Autolysis- The factors that influence the rate of post mortem changes include the following:

1. **Environmental Temperature:** The mechanism of enzyme action is temperature-dependent. Ambient environmental and body temperature of the carcass accelerates post mortem autolysis. Therefore, in summer rapid onset occurs but in winter-delayed onset.
2. **Size of Animal:** In large animals rapid onset occurs while in smaller animals delayed onset is reported.
3. **External Insulation-** Like thick fur, feather, wool-rapid onset.
4. **Fat Content:** Fatter the animal-rapid onset.
5. **Species of animal:** Pig-soft and moist muscle-rapid onset. Horse muscles are dry and firm and onset is slow.

- 6. Cause of Death:** Generalised infection such as septicaemic bacterial diseases hasten the appearance of post mortem changes. In tetanus, glycogen is less and muscles are stiff and rapid onset occurs.
- 7. State of Body at the Time of Death-** The condition of the animal before death, particularly its nutritional status influences the rate of post mortem autolysis. Obese animals tend to show an accelerated appearance of post mortem changes. This condition is probably related to the insulating effect of the fat layer that retards cooling of the body after death. High body temperature onset is rapid. Very active animals show rapid onset. Example is strychnine poisoning.
- 8. Organs Involved:** The degrees of the expression of post mortem changes vary from tissues to tissues. The presence of bacterial flora, enzyme secretions, and the availability of moisture and substrates influences the rate of post mortem autolysis. Pancreas-high amount of enzymes rapid changes occurs. In fibrous tissue less amount of enzyme is present slow changes occur. Retina is most sensitive and separates from choroids. Adrenals, liver and testis and abdominal organs also show rapid autolytic changes.
- 9. Others:** Like-presence of ingesta, air currents etc affects autolytic changes.

Sequence of Post Mortem Changes

1. Rigor mortis
2. Algor mortis
3. Livor mortis-Hypostatic congestion
4. PM clotting of blood
5. Imbibition of hemoglobin
6. Imbibition of bile
7. PM desquamation
8. PM Softening
9. PM Discoloration
10. PM Distention
11. PM Displacement
12. PM rupture of organ and tissue

1. Rigor Mortis: It refers to the contraction and stiffening of muscles after death. Most literatures consider the fall in the availability of adenosine triphosphate (ATP) as the possible cause of terminal muscle fiber contraction after death. Other plausible explanation includes the influx of calcium ions after cessation of the sodium pump. Classically, rigor mortis begins from 1-6 hours after death and passes off in 24-48 hours. However, several factors influence the onset of rigor mortis and these include the following:

- (a) Nutritional status of the animal.
- (b) Environmental and body temperature of the cadaver.
- (c) Cause of death.

Well-fed animals have large glycogen reserves, and may show a delay in the onset of rigor, while cachectic animals may develop rigor quickly. Animal cadavers that are exposed to hot temperatures may develop and pass off rigor mortis quickly. Those that died of septicaemic diseases may not develop rigor mortis at all. Once a part of the

animal body that has passed into rigor is moved that part will pass off rigor mortis. This has some significance in human medico-legal cases. Sequence of organs involved:

1. Heart
2. Anterior portion- head, neck and limbs.
3. Posterior portion-limbs and tail

2. Algor Mortis: Gradual cooling of the animal body after death, and is associated with a fall in ATP. Rate of cooling is affected by:

1. Ingesta
2. External temperature
3. Air currents
4. Thickness of hair coat
5. Size of animal

3. Livor Mortis: the settling of blood to the down side of the animal body. Gravitational force causes this to happen. This gravitational settling of blood and body fluids results to intense reddish colouration of the organs and tissues at the down side of the cadaver.

4. PM Clotting of Blood: Degeneration of the endothelium-releasing thromboplastin in anoxia.

Chicken Fat Clot - An old term referring to the gelatinous mass formed by separation and coagulation of plasma proteins from the component of blood. This is usually seen inside the major blood vessels and the heart. They sometimes give a cast of the ramifications of the vessels. To prevent misconceptions, the use of this term should be avoided.

Currant Jelly Clot - An old term applied to coagulated blood. The term becomes obsolete and should be avoided.

5. Haemoglobin Imbibitions: Pinkish to reddish colouration imparted to tissues due to the lysis of red blood cells. This is most evident on the surfaces of large arteries and in outer surfaces of visceral organs.

6. Bile Imbibitions: Golden yellow colouration imparted on tissues following seepage of bile. Discolouration is most evident on the surfaces of organs in contact with the gall bladder, and on duodenal mucosa.

7. PM Desquamation: The mucosa of the rumen will peel off large patches because of autolysis. The intestinal mucosa may have an accumulation of cells and mucus on the surface as a result of autolysis and these changes may be easily confused with an inflammation of intestine.

8. PM Softening: Softening of tissue or organs caused by “autolysis” with assistance from saprophytic bacteria or perhaps the normal flora of the tissue. This may be easily observed in liver and kidneys. The pancreas is very sensitive because of its enzymatic contents and softens rapidly which is accentuated by handling.

9. PM Discoloration- Discoloration of the tissues or organs results from the break-down of haemoglobin and the action of bacterial hydrogen sulphide (H_2S) on haemoglobin ($Hb(Fe) + H_2S(Bacteria) = FeS(Fe sulphide) + H_2$). The pretty shades of blue, green and purple are only due to the above said reaction or changes. Discoloration also occurs because of lysis of erythrocytes with permeation of released Hb into tissues which turn them red, this is called “imbibition”. This change is much more evident on the inner surface of large arteries as often-observed in horse and in most tissue in aborted fetuses that died some time before abortion.

10. PM Distention- Occurs largely because of fermentation of ingesta with gas production in digestive tract. The gas distends all parts from stomach down and extreme pressure may build in some organs. The stomach or diaphragm may rupture or the abdominal muscle may tear to produce hernias of various types. Extreme pressure builds in organs down from stomach results in:

- (1) Bloody foam from nostrils
- (3) Pale area in liver due to pressure
- (4) Imprintation of intestinal loops on the liver
- (5) Myocardium –large pale area due to pressure
- (6) Pale viscera due to pushing of blood to other parts
- (7) Rupture of organs

Autolysis often gives the cut surface of the myocardium a mottled semi-cooked appearance that resembles degeneration and may be confusing. Gas bubbles are often present in liver and kidneys, usually in pale areas of autolysis and putrefaction, which are not uniformly spread through the tissues. Bacteria in large numbers may be seen in autolytic tissue, especially in liver, some species such as horses seen to have large number of such organisms in blood vessels in the brain, which appear often relatively shortly after death. Post-mortem abdominal distension with push blood from the venous system in the abdomen to give pale viscera lent the hind limb muscles, lungs and neck region are very congested. There is no doubt in it that tissues in the body contain a normal bacterial flora. Flora load is such more in gastro-intestinal and respiratory tract. There may be viral flora as well since viruses are found in normal tissue cultures. The normal bacterial flora is a contributory factor of causing autolysis and putrefaction if multiplication occurs after death.

Accumulation of edema fluid- Clear or red edematous fluid may be present beneath the skin as an important post-mortem changes. It is difficult to explain that why the edema fluid accumulates? This is observed particularly in sheep. The diagnosis of black quarter and malignant edema, which result in combination of gas production, haemorrhages and oedema, may be confused with the post-mortem changes.

Differentiation of Post Mortem and Ante Mortem Changes

- Inflammatory responses to dead tissue within the same section
- Biliary epithelium (digested very quickly)
- Gut mucosa (goes quickly due to bacteria and enzymes)
- Adrenal medulla
- Nervous tissue
- Pancreas
- Presence of living and dead tissue in the same section
- Condition of RBC within vessels (tend to lyse with autolysis)

Specific Post Mortem Changes in Various Organs and Tissues: Progressive autolytic changes in organs/tissues- A number of studies have been carried out to determine the progressive autolytic changes in cell at specific time period and different temperature. Experimental evidence in rats suggests that shift in weight and water levels of body organs occur after death. At 3 hours after death there was significant loss of weight from

parts of intestine and an increase in the heaviness of lungs. The reason is obviously due to blood accumulation. Other tissues that increased in weight were diaphragm, heart, abdominal muscle, ileum, cerebellum and kidneys. The testicular weight decreased. There is progressive autolytic changes marked in liver which are characterized by clumping of chromatin and loss of differential staining. The release of lysosomal hydrolases may not contribute as much as necrosis or autolysis. The nucleolus undergoes the most rapid changes and glycogen disappears from the cytoplasm quickly, endoplasmic reticulum dilates and mitochondria swell up, become round and lose their internal structure. The most sensitive tissue to autolysis is the “retina” which becomes separated from the choroids. The seminiferous tubules in which vacuolation in the between cells occurs; and the intestine in which the epithelium over the villi falls off. The brain softens into mush but is still worth fixing since cytological and structure details are surprisingly well preserved.

Main Autolytic Changes: It can be confused with pathological lesions which are following:

Gastrointestinal Tract: Segments of the gastrointestinal tract undergo rapid post mortem changes. This is due to the presence of digestive juices, bacterial flora, and substrates needed for the growth of microorganisms. Immediately after death, contractions of the villi occur and result to sloughing of the intestinal epithelia. Grossly, a mucoid whitish sludge may be observed, and this may be stained yellow with bile. As time progresses, gaseous distensions occur. This invariably results to stripping of the mucosa, perforation and rupture. Occasionally, stagnant blood may be lysed and impart a reddish colouration to the contents. Haemoglobin imbibition may impart a reddish colour to the serosa. In ruminants, sloughing of the epithelia of the forestomach occurs.

Liver and Gall Bladder: Haemoglobin imbibition imparts a shiny reddish discolouration on the surface of the liver. The liver parenchyma close to the gall bladder may be stained yellowish or greenish because of bile imbibition. As time progresses, the liver may lose its turgidity, become soft and clay-like in colour and consistency. The presence of gas-distended bubbles on the parenchyma usually suggests a more advanced stage of post mortem decomposition. The gall bladder mucosa easily detaches after death.

Pancreas: Post mortem change occurs rapidly in the pancreas. It may be discoloured red from haemoglobin imbibition, lose its lobular pattern, becomes soft and translucent. The portion that is facing the gall bladder may show bile imbibition. In advanced stages of post mortem degradation, the pancreas turns into a sac-like structure containing red-tinged fluid. It may even disappear leaving a flimsy membranous structure.

Kidneys: The kidneys may be discoloured red from haemoglobin imbibition, and later discoloured black (pseudomelanosis). The demarcation between cortex and medulla becomes not apparent. As time progresses, the organ becomes soft and gas bubbles form at the peri-renal tissues.

Spleen and Lymph Nodes: The spleen may be discoloured gray to black, becomes soft and mushy. Advanced stages of post mortem degradation usually turn the spleen into a sac-like structure containing liquefied parenchyma that oozes when cut. The lymph nodes may be discoloured, become soft and pulpy.

Lungs and Airways: Normal lungs are pale pink in colour. After death, post mortem congestion becomes evident apart from livor mortis. Blood and fluid may ooze when the surface is cut. The airways may contain considerable quantities of froth and fluid. The organ becomes soft, collapsed, discoloured red to gray, and gas bubbles may appear on the parenchyma as time progresses.

Heart and Great Vessels: The heart becomes flabby, and may contain coagulated plasma separated from the red components of blood. The endocardial surfaces and the intima of great vessels may show reddish discolouration following haemoglobin imbibition.

Brain-The brain, being located farther from the gut show delayed post mortem changes. Vacuolation of the brain parenchyma due to seepage of cerebrospinal fluid may be observed. With advanced stages of post mortem degradation, the brain turns into a soft mushy mass. Later, the brain liquefies and will ooze when the calvarium and meninges are opened.

Muscles: Discolouration of the muscles, fascia and tendon sheath occurs after death because of seepage of myoglobin. Intense colouration of the muscles may be seen at the down side of the animal. Air pockets demarcating muscle groups and fascial planes occur in advanced stages of degradation.

Eye: The cornea of the eyes turns opaque due to absorption of the aqueous humor. The globe may either protrude or collapse. When opened, detached retina may be noted.

Body Cavities and Orifices: Surfaces of body cavities may show discolouration following haemoglobin imbibition and pseudo-melanosis. The cavities may contain red-tinged fluid and may be copious in its amount. Gaseous distensions render the carcass to appear bloated. It is not uncommon to find post mortem rectal or vaginal prolapse because of developing internal pressure. Straw-coloured fluid may be seen oozing from body opening in advanced stages of post mortem decomposition.

Prevention / Delay-PM Autolysis

- Preservation by 10% formalin
- Freezing
- Refrigeration- at 4°C-small animals
- Alternate to chilling- open the abdomen and cover with a soaked cloth –cattle

Grading of Autolysis

- Carcass condition - described on the data sheet during the initial assessment.
- Guides the decision.
 - Whether or not a necropsy should be conducted?
 - Types of samples to be collected.
 - Types of pathological tests to be done.

Example: Bacteriology and virology (for disease diagnosis)- fresh carcasses. Heavy metal, pesticide and DNA analyses- samples collected from fairly decomposed animals.

Grades

Carcass Condition	Grade/ Category
Live when first reported but subsequently died.	+
Carcass in good condition (fresh/edible)	++
Carcass fair (decomposed but organs intact)	+++
Carcass poor (advanced decomposition)	++++
Mummified carcass (skin holding bones)	+++++
Disarticulated bones (no soft tissue left)	++++++

Grade Description

Categories 1–2 (Fresh)

- Little or no bloating due to general tissue decomposition.
- Skin not sloughed.
- Internally all organs intact with material generally suitable for histopathology.

Category 3 (Moderately decomposed)

- Slight bloating.
- Some skin sloughing or stiffening of flippers.
- All internal organs including the liver show integrity, although autolysis and decomposition may render the tissue matrix unsuitable for standard histopathology.

Category 4 (Badly Decomposed)

- Usually bloated.
- Missing patches of skin.
- Internal organs, particularly the liver, show loss of integrity or complete disintegration.
- In some carcasses bloating may not be evident due to very advanced decomposition or release of gas through wounds.

Categories 5–6 (Dried Carcasses or Bones)

- Advanced to the point where little remains of the carcass other than the skeleton or hide.

Decision Based on Grades

- 1, 2 and 3 categories - necropsied in detail. If possible, these carcasses should be transported to a suitable facility for necropsy.
- 4, 5 and 6 categories - examined to the extent possible.
- Internal examination should always be conducted on intact carcasses because carcasses that appear decomposed externally can be in relatively good condition internally.

Remember

- Autolysis is self digestion by tissue enzymes present in or released into the cells after death.
- Should be differentiated from antemortem lesions by presence of inflammatory changes in surrounding organs.
- Most of Veterinary Pathologists do not do routine post mortem examinations on animals that have been dead for > 24 hours.
- Veterinary Pathologists Must be experienced enough to make distinction between autolytic changes and lesions for correct diagnosis.

Chapter 13

How to Write Post Mortem Report?

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The primary rule for the making of pathological observations and descriptions is simple one-observed facts **must be sternly separated from interpretations of these facts**. The **gross autopsy description** begins with a description of the animal which is being examined-species, breed or type, sex, age, weight, identifying marks. If the age is not known, it is estimated and the fact that it has been estimated is stated. The description of the animal also includes a statement of the nutritional condition-obese, good, moderately good, poor or inanition.

Obesity and at the other extreme, inanition are objective terms. Obesity is an abnormally great accumulation of fat in the body stores. Inanition is a state of emaciation characterised grossly by serous atrophy of adipose tissue, especially evident in the coronary grooves of the heart and the bone marrow, and some degree of atrophy of other body tissues, particularly obvious in the skeletal musculature. The other evaluations of the nutritional state are arbitrary. With experience each observer will be able to establish a standard scale. Only exceptionally will there be need to use more than a single word to describe the nutritional state (e.g. a fat animal with poorly developed muscles or an animal with serous atrophy of adipose tissue but good muscular development). Details such as the stage of oestrus, gestation, stage of lactating, dry, are also form part of the general description of the animal and recorded in the report. The general description of the body also includes a statement of the degree of rigor mortis and cadaverous changes. These facts have their major value when interpreting the results of the histological and special laboratory examinations.

Before moving on to a consideration of the changes in particular parts of the body, the pathologist looks for and records the changes involving the body as a whole e.g. dehydration, discolouration, anaemia. In making his examination, the pathologist naturally observes first whether there are deviations from the normal and then, the nature of these deviations. The trained observer goes from the general to the particular; he first observes the whole organ or lesion and then its parts. This thorough pattern is followed in the pathological description. The autopsy report is generally confined to a description of deviations from the normal. At times, however, it is necessary to state definitely that an organ or structure has a normal appearance-if, for example, there is an obvious discrepancy between the observed and the expected pattern of pathological changes or if one wants to emphasise the fact that changes are focal in an organ or limited to one member of a bilateral structure.

Macroscopical Description: Whether of an entire organ or structure or of focal pathological changes, is essentially a statement of:

- a) size
- b) shape
- c) colour
- d) consistency
- e) appearance of cut surface
- f) relation to surroundings

Description of focal lesions also requires a statement of anatomical relations and, when numerous the number (if countable) and pattern of distribution (e.g. evenly distributed, in clusters, etc.).

Size: It is an objective quality and is best indicated by actual measurements-weight, linear dimensions or sometimes volume, whichever is most appropriate to the particular structure or lesion. Words such as “enlarged”, “small”, “shrunk”, “dilated”, “contracted” have no absolute value. They represent interpretations and are used only in comparative sense or when more accurate references of size are inapplicable or inadequate to convey the over-all impression. Focal changes in an organ, when numerous, often vary in size and shape. In such cases, state the range and the modal characteristics.

Shape: It is irregular, three dimensional structures and lesions can seldom be described with pretensions to accuracy. One generally has to be contented with stating the over-all appearance-pyramidal, oval, circular, egg-shaped, stellate, irregular etc. Referring the size and shape of structures or lesions to the dimensions and form of common objects is sometimes an aid to communication, but must be practised with discretion if it is to be meaningful.

Colour: The description of **colour** offers no difficulties provided one refers only to the names of the colours in the spectrum-red, orange, yellow, green, blue and violet-together with brown, black, white and grey. Combination of any two of these adjectives will cover all the colour tints likely to be encountered (e.g. “green-blue” means more blue than green). “Light” (pale) and “dark” (deep) will suffice as modifying adjectives.

Consistency: It is referred to as soft, hard, firm or resilient with the usual modifications of slightly, moderately or very.

Fluids and Exudates: These are described as watery (serous) or viscid (mucoid) with appropriate modifying adjective. Exudates may also be pasty, crumbly or inspissated. A statement of colour is modified by “clear” or “turbid”. Material in suspension is described as such. Amount is expressed by volume if possible.

Appearance of Cut Surface of Organs: Description of the appearance of the cut surface of organs includes a statement of the appearance and distribution of changes and their relation to anatomical landmarks, as well as colour and consistency. It is useful to distinguish between “generalised” and “diffuse”, i.e. between distribution throughout the whole organ and distribution within one part of the organ as in the example, generalized pulmonary emphysema vs. diffuse emphysema in the apical lobes. The characteristics of the cut surface of focal changes whether homogeneous, special structural details, capsule etc. and again colour and consistency are described as they are manifest for most of the focal changes if they are numerous and details are given for only the important variations from the general pattern.

Relation to Surrounding Tissues and Structure: It is a description of relation to surrounding tissues and structure is given for focal changes and includes such details, as sharp or diffuse transition, compression or infiltration, haemorrhagic or hyperaemic bordering zones, and so on. The principles of describing microscopical observations are the same as those for gross autopsy descriptions facts, not interpretations, and progression from the general to the specific.

If normal or recognisable structures are present in the section, one refers to these for orientation in describing first the nature, site, and degree of the principal changes and

then the variations and special features. The results of special staining procedures or of histochemical reactions may be left to the last or may be woven into the general description if by doing so the basic features can be described more precisely and concisely. In some histological specimens, particularly tumours, the best approach is to begin by describing the general architectural pattern, then continuing with details of cytology and local variations and finally describing the relation to surrounding tissues, if any, are present in the section.

Many of the terms used in histopathological descriptions- e.g., pyknosis, necrosis, hydropic degeneration, acanthosis etc. are really interpretations. Their use, however, can be recommended for conciseness and clearness. These terms have precise meanings or connotations and if there is any doubt about their meaning or applicability, consult standard textbooks or dictionaries. Except for measurements of particular structures, quantitative evaluations in microscopical description are expressed by such terms as limited or extensive, slight, moderate or severe, and in the case of numerical quantities to few, sparse, moderate or abundant. Note that severe and extensive are not synonyms.

Conclusions: A good pathological description, like good autopsy technique, is neat and efficient. A description which is precise, concise and lucid gives confidence in the reliability of the observations and the conclusions drawn from them. Obscurity, prolixity and redundancy are the cardinal sins. Simple and grammatically correct sentences, built up of a lean and exact vocabulary, arranged locally and with fact and fancy rigorously separated are professional hallmarks. Anything else should go in the waste paper basket.

Chapter 14

Diagnosis and Management of Fractures in Animals

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Fracture is a break in the continuity of bone. Fractures are common in animals and are caused either by direct trauma like automobile accidents or by indirect trauma caused while running, jumping or falling from a height. Fracture may also be caused by violent muscle contractions or by systemic/metabolic diseases like in rickets or osteomalacia, wherein the bone becomes weak leading to fracture. Based on the communication of fracture site with the environment, it can be classified as *simple (closed)* fracture or *compound (open)* fracture. Based on the extent of fracture line, it can be classified as incomplete or complete fracture. Based on the direction of fracture line in relation to the longitudinal axis of bone, fracture can be called as: *transverse*, *oblique*, *spiral*, *comminuted* (more than two bone fragments) or *multiple* (two or more independent fractures of the same bone). Based on the location of fracture, it can be classified as: *diaphyseal*, *metaphyseal* or *epiphyseal fracture*.

Diagnosis of a bone fracture is relatively easy. Anamnesis of the case generally indicates history of trauma, like a road traffic accident or fall. An animal with a long bone fracture will have non-weight bearing lameness on the affected limb. Inflammation and swelling at the site of fracture is common, especially in places with good soft tissue covering. Crepitation of bone fragments can be easily felt at the site of fracture, especially in comminuted fractures at the lower limb region. In open fractures, fractured bone fragments may be visible through the skin wound on naked eye. Confirmative diagnosis of a fracture can be done by radiographic examination of the affected site. Generally both medio-lateral and cranio-caudal or dorso-plantar/dorso-palmar views are indicated. In recent years digital X-ray has increased the accuracy of diagnosing a fracture. Modern diagnostic methods like digital X-ray, CT scan and MRI are particularly useful, where sometimes a survey radiograph fails to diagnose a fracture, especially in fractures/luxations of axial skeleton, like vertebrae.

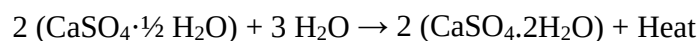
One of the goals of fracture management is to reduce the fracture fragments to near normal anatomical alignment. The reduction of fracture can be done either by closed or open method. Reduction of fracture without incising the skin or exposing the bone, through external immobilization is called *closed reduction*. It is indicated in fresh fractures in young animals, comprising only two pieces with little or no overriding of fragments. *Open reduction* is achieved by exposing the fracture fragments through an incision and is done primarily along with internal fixation. It is commonly indicated in fractures with severe comminution and/ or severe overriding of fragments.

A thorough physical examination of the animal suspected of having a fracture should be carried out prior to the decision for treatment. The patient first must be made safe from continued trauma. Temporary stabilization of limb may be performed prior to moving the animal or attempting to get the animal stand. Fractures below the level of stifle or elbow may be temporarily stabilized with splints or casts. The factors that should be considered while selecting the technique of fracture management are: species

of the animal, location and type of fracture, presence or absence of soft tissue and neuromuscular trauma, closed or open fracture environment, behavioural nature of the animal, the facilities available and the experience of the veterinary surgeon.

1. External Fixation Techniques- External fixation refers to immobilization of fractures/other skeletal abnormalities with devices applied externally, without the use of any invasive technique. Among the external fixation devices, plaster of Paris and modified Thomas splints are widely used.

A. Plaster cast- Plaster bandages consist of a cotton bandage that has been impregnated with plaster of Paris, which hardens after it has been made wet. Plaster of Paris is calcined Gypsum, ground to a fine powder by milling. When water is added, the more soluble form of calcium sulfate returns to the relatively insoluble form, and heat is produced.



Plaster cast is the most widely used external fixation technique in both small and large animals. It is indicated in immobilization of fractures below the mid-diaphysis of radius or tibia. The animal is restrained under sedation/anaesthesia. The placement of cast is facilitated by use of rope restraint. An assistant should help to maintain alignment of the limb during application, being sure to check the position of the limb in cranio-caudal and latero-medial planes. In large ruminants, the tension on the limb during casting may be achieved by placing wires through holes drilled in the hoof wall and applying tension. The talcum powder may be sprinkled over the limb and an even layer of cotton is applied around the leg in order to protect bony prominences. Over padding should be avoided because it may impair immobilization by allowing movement of fracture fragments within the cast. Splints are bandaged in place with an even pressure. Plaster of Paris bandages are soaked in warm water for a few seconds until air bubbles cease to appear. Plaster bandages are then removed from the water, squeezed and wrapped over the splints, starting at the fracture site and continuing up and down including the entire limb except the toe pads in dogs and hooves in large animals. The plaster cast is never stretched or tightened around the limb. The plaster is moulded by rubbing each section with wet hand before the plaster is set. The thickness of the cast is usually based on clinical judgement. Casts 4-5 layers thick may be adequate for small animals, 6-8 layers for calves less than 150 kg body weight, but adult cattle may require casts 12-16 layers thick. Casts used on the hind limbs must be thicker because of stress concentration by the angulation of the hock. Incorporation of splints (wooden/metal) within the cast (2 rods placed 90° to each other) can increase the strength of the cast, but is only needed in large animals. The plaster is then allowed to become dry and hard. Complete drying/hardening of the cast (attaining full strength) may take 24-72 hrs. Use of a walking bar ('U' shaped bar placed under the hoof and incorporated into the cast) will increase the distribution of loading forces into the cast and away from the distal limb.

Toes and hooves are inspected several times during the first 24 hr for any swelling, coldness or constriction. If the toe/hoof gets swollen or gets cold, pressure at the end of the cast should be relieved by removing the cast and the cast may be reapplied after swelling subsides. In open fractures, daily dressing of the skin wound should be undertaken. Cast may be maintained for up to 3-4 weeks in young dogs, 4-6 weeks in

adult dogs and calves. Fractures in adult cattle may heal within 8-10 weeks, but often require 12-16 weeks for clinical union to occur. Radiographic union of the fracture (bony union with resolution of the fracture line) is not seen for weeks to months after clinical union (defined as sufficient bridging callus to allow weight bearing without additional support to the limb) has been reached. Plaster is removed after radiographic fracture union takes place. Affected limb is massaged to promote circulation after removal of the cast. Movement is restricted till the limb regains its normal function.

B. Fiberglass cast- It is a lighter, synthetic alternative to the more traditional plaster cast. It is created by padding the extremity with cotton or waterproof padding material, followed by wrapping several layers of knitted fiberglass bandages impregnated with a water-soluble, quick-setting resin. It is lighter and more durable than plaster, so fiberglass has quickly become the preferred type of casting with many patients and medical care providers.

The fiberglass cast has many advantages over a plastic cast. The first thing most patients notice is that the cast weighs less and is more comfortable. It is made of water-activated polyurethane resin combined with bandaging materials, so this material offers greater strength and less time for setting, as well. A fiberglass cast requires less maintenance than a plaster cast and often is used after the healing process already has begun. The fiberglass bandages on the outside of the cast are waterproof, but most of the padding materials inside the cast are not. Waterproof materials have been developed in recent years to replace this inside padding with a waterproof alternative. This offers a padding material for fiberglass casts that would allow wetting. The material also is reported to cut down on odor and itching by helping to wick moisture away from the skin under the cast. This type of material combined with the fiberglass bandages provides a more breathable solution than a traditional plaster cast.

As with any type of cast, a fiberglass cast has its drawbacks. Fiberglass casts set quickly, so a less-experienced medical care provider has less time to properly wrap the injured extremity. Further, the synthetic materials leave less room for swelling. Fiberglass casts are not always appropriate for more complex fractures where the bone is out of position. Plaster is more moldable than the knitted fiberglass and resin bandages, so a more comfortable fit can sometimes be achieved with plaster. Further, plaster is smoother and less likely to snag clothing or to rub skin raw.

C. Modified Thomas splints- Modified Thomas Splint is indicated for immobilization of fractures at the distal femur, radius/ulna, distal humerus and tibia/ fibula. The splint can be prepared from aluminum rods or conduit pipes of various sizes. For fore limbs, the length of the bar should be the distance from the thigh up to the tip of the toe in an extended leg. To prepare the ring, a splint mold is used and the rod is bent around the splint mold. The use of Thomas splint and cast combination is appropriate for fractures distal to the elbow or stifle in large animals.

2. Internal Fixation Techniques- Internal fixation of fractures by open reduction has the advantage of providing good alignment of fracture fragments and also rigid fixation of bone fragments. In small animal practice, internal fixation is the preferred method for fixation of long bone fractures, especially of femur, humerus and tibia. It can be achieved

either by intramedullary fixation techniques like pinning/nailing or by extramedullary techniques like bone plating.

A. Intramedullary pinning- Intramedullary pins are the most widely and commonly used fixation devices in veterinary practice. It can be used either alone or along with other ancillary techniques like cerclage wiring. Intramedullary pins resist bending forces and maintain alignment. *Single IM pinning* is indicated in simple transverse/slight oblique diaphyseal fractures of long bones. Use of full cerclage or hemicerclage wires along with single IM pins provide greater stability against the rotational forces, especially in long oblique fractures. *Stack pinning*, i.e. the use of more than one pin (of relatively small diameter), is indicated in transverse fractures to combat torsional and rotational forces. Each pin is inserted through a separate proximal hole into the medullary cavity, preferably allowing the opposite ends of the pins to diverge within the opposite metaphysis. *Rush pinning*: Rush pins are solid curved cylindrical pins, which are used for the fixation of metaphyseal/physeal fractures, especially in young animals. Two pins are passed from opposite sides (normally medial and lateral surfaces) of the cortex in the smaller fragment, so that they are crossed just above the fracture site and then glanced off the endosteal surface of the diaphyseal bone to provide spring loaded tension. They provide rotational stability. Rush pins are also contraindicated in cases where there is longitudinal crack is present in the bone fragment. *Kuntscher nailing*: ‘Clover leaf’ or ‘V’ shaped Kuntscher nails are hollow IM nails and have the advantage of lighter in weight and can provide three point fixation. It is indicated in transverse or short oblique fractures of long bones, especially femur, humerus and tibia, where the cortex is good with no longitudinal cracks. The method is preferred in calves and large dogs, where medullary diameter is large.

B. Interlocking IM nails- These are relatively new devices used mostly in small animal practice. Interlocking Nails are a refinement of the original Kuntscher Nail. An Interlocking Nail is basically an intramedullary pin secured in position by proximal and distal transfixing screws which engage the bone to the nail to provide axial bending and torsional stability. In 1989, R. Tass Dueland, DVM (University of Wisconsin) and associates began investigations to apply this human fracture modality to veterinary orthopedics. These investigations consisted of both mechanical and clinical trials to define the nail and instrumentation requirements. This research provided encouraging clinical results and resulted in the development of the veterinary IN Systems. Now the interlocking nailing could be performed with the use of image intensifier or without (using a Jig system).

In comminuted long bone fractures, the IN System is useful due to the mechanical advantage of medullary implantation (i.e. mechanical axis) and prevention of collapse of the fracture by the interlocking effect. Ancillary cerclage wires reduce periosteal stripping and minimize disturbance to blood supply compared to plating. Because the implant is placed in the middle of the medullary cavity, it is situated along the neutral axis in the center of the bone. As such, like intramedullary pins, it counteracts bending forces very well. The screws that secure the nail to the bone counteract rotational and axial forces placed on the bone-implant construct. While Interlocking Nails provide excellent bending strength, there are at most 4 cortices of bone engaged by screws on either side of the fracture line. As such, postoperative restriction of exercise remains

critically important to a successful outcome to minimize axial and rotational loads on the screws. Screw breakage is the most common mode of failure with these implants.

- i. **Indication-** The interlocking nail is best suited for diaphyseal fractures, and is especially useful in fractures with extensive comminution of the diaphysis. Because screws are placed at either end of the nail, fractures of the metaphyseal or epiphyseal regions may not be amenable to repair with interlocking nails if there is not sufficient bone for screw placement. Veterinary Interlocking Nails are commonly placed into the femur, humerus, and tibia. Because of the curve of the canine femur, shorter nails than would ideally be used are often required, and in some cases the bone is repaired in a slightly straighter alignment than it was prior to fracture.
- ii. **Advantages-** Like plating, Interlocking Nails permit early return to limb function with the added advantage of minimal soft tissue morbidity. In human patients, nails are typically placed in a closed fashion with fluoroscopic guidance to assess fracture reduction. Without fluoroscopic guidance, the fracture is approached through as small of an approach as possible to guide the nail across the fracture and assess alignment. This limited approach preserves the soft tissue attachments to the bone fragments contributing to lower rates of postoperative infection and nonunion as compared with plating. Anatomic reconstruction of the fracture pieces is NOT performed, but anatomic realignment of the larger end portions of the fracture is achieved and maintained by the interlocking nail, facilitating secondary bone healing with callus formation.
- iii. **Insertion Technique-** A hole is created proximally and enlarged with reamers or successively larger IM pins to a size that allows passage of the nail into the bone. The largest diameter nail that will fit into the medullary cavity is chosen. Special equipment attaches the nail to drive it into the bone so that it is well seated in the bone proximally. The screws are placed through small stab incisions in the soft tissue and are directed through the bone and through the holes in the nail by utilizing a special jig that aligns the drilling site with the holes in the nail. At least one screw must be placed on either end of the nail to maintain rigid control over axial and rotational forces. Interlocking Nails typically stay in the bone forever and removal may be difficult, especially if it was well seated in the bone at placement. Bone may cover the proximal end of the nail and the screw heads over time. Such consideration should be given prior to placement of the implant system in a patient. Non-availability of proper size and shape of the nail and the cost of the instruments, however, limit their use in routine practice.

Bone Plate and Screws Fixation- Bone plate and screws fixation provide rigid and stable internal fracture fixation and early mobilization of the joints (functional recovery of the limbs). Plates can be used to function as:

- (a) *Tension band plate/compression plate*,
- (b) *Neutralization plate*, and
- (c) *Buttress plate*.

There are three basic types of plates: (a) straight plates for diaphysis, (b) special plate (like 'T' plate) for epiphysis and metaphysis, and (c) angled/hooked plates for the

proximal and distal end of long bone like femur. Straight plates are most widely used in veterinary practice. *Dynamic compression plate* (DCP) is an improvised plate over traditional round-hole plates. The special geometry of its oval screw holes has increased the potential use of the plate. Compression is achieved through eccentric placement of the screws in the oval holes of the plate.

Locking Plates- Locking plates are fracture fixation devices with threaded screw holes, which allow screws to thread to the plate and function as a fixed-angle device. These plates may have a mixture of holes that allow placement of both locking and traditional nonlocking screws (combi plates). While locking plates have been used for years in specialized research trials, they have been available for general orthopaedic applications only in the last 5 to 10 years.

The main biomechanical difference from conventional plates is the fact that the latter require compression of the plate to the bone and rely on friction at the bone-plate interface. With increasing axial loading cycles, the screws can begin to toggle, which decreases the friction force and leads to plate loosening. If this occurs prematurely, fracture instability will occur, leading to implant failure. Thus, the more difficult it is to achieve and maintain tight screw fixation (as for example, in metaphyseal and osteoporotic bone), the more difficult it is to maintain stability. This biomechanical prerequisite of conventional plates is associated with biological pitfalls due to compression of periosteal blood supply and compromise of the vascularity of the fracture. Thus, conventional plate osteosynthesis with rigid fixation (e.g., interfragmentary compression and lag screws) has been associated with a substantial complication rate, including infection, hardware failure, delayed union, and nonunion.

In contrast, locking plates follow the biomechanical principle of external fixators and do not require friction between the plate and bone. They are considered to be internal fixators from a biomechanical standpoint since the angular-stable interface between the screws and the plate allows placement of the plate without any contact to the bone. In essence, however, locking plates can be considered to be external fixators placed underneath the skin envelope, although they are more stable as a result of the shorter distance between the plate and the bone. Locking plates are substantially more expensive than conventional plates. Locking plates are more difficult to use to help achieve an adequate reduction. Particularly with specialty plates, which have only locking holes, fracture reduction must be achieved primarily, before plate fixation. Once a locking screw has been placed through the plate into bone, this particular bone segment can no longer be manipulated by insertion of additional screws or by using compression devices. Locking plates are contraindicated in simple fractures that require interfragmentary compression.

External Skeletal Fixation Techniques- External skeletal fixation refers to the stabilization of musculoskeletal injury using percutaneous fixation pins that are connected outside the body to form a rigid frame or scaffold, spanning the region of instability, e.g. transfixation pinning and casting, linear fixators and circular fixators. This type of fixation is indicated for the management of long bone fractures especially of tibia and radius, where cast immobilization is not appropriate or does not provide optimal level of fixation (fractures proximal to the distal radial or tibial physis, with soft tissue injuries and open fractures).

Advantages of ESF include: early return to function of the affected limb with excellent mechanical properties; ability to adjust the frame after bone fixation, allowing correction of rotational or angular deformities; avoidance of surgical trauma to the injured tissue; avoidance of infection associated with buried implants; ease of implant removal after fracture union; provision for transarticular fixation in the presence of severe soft tissue trauma or severe comminution of the proximal or distal end of the affected bone; preservation of joint motion and multiple applications with reusability of components.

Transfixation Pinning and Casting- Transfixation pinning and casting (TPC) may be applied either as a “hanging limb pin cast” or as external skeletal fixator. Hanging limb pin cast refers to placement of transcortical/transfixation pins through the bone proximal to the injury, followed by application of a full limb cast. The advantage of using pin-casts for ESF compared with hanging limb casts is that the fracture is more stable and the fracture fragments are not able to move within the pin-cast after the swelling within the cast resolves, and the pin-cast may not need to span adjacent joints in large animals. For management of open fractures, daily dressing of the wound may be carried out by leaving a hole in the cast (window cast) at the site of injury. However, this gives unsatisfactory access to the wound and is uncomfortable to the patient because the swelling in the limb becomes concentrated at the defect in the cast. Alternatively, metal or acrylic side bars may be used; they must be large enough to sustain the weight of the animal.

Linear and Circular Cixators- Linear fixators are rarely used in small animal practice. However, unilateral linear fixators are sometimes used along with IM pins for fixation of femoral fractures. They prevent the rotational stress and provide rigid fixation. They are also used for fixation mandibular/maxillary fractures in small animal practice. In large animal practice, the bilateral linear fixators are generally used for the fixation of tibial/radial/metatarsal or metacarpal fractures, especially for the management of compound fractures. Circular fixators were generally used in small animal practice for limb lengthening procedures and arthrodesis etc. But in large animals, their use of late has been tried in the management of long bone fractures, especially compound fractures and for fixation of fractures near the joint.

Management of animals in the immediate postoperative period is more critical, as they may feel discomfort by the presence of the external frames. They may also take some time to get accustomed; hence regular observation of the animal is required during the period. Regular antiseptic dressing of the pin/wire-skin interfaces and the open wound (in case of compound/infected fractures) is done till the healing occurs. Pin site discharge is common with ESF; hence prolonged dressing of the pin site is needed. After radiological union of fracture, the pins/wires are cut and the frame is removed.

Acrylic and Epoxy-Pin Cixation Systems- Stainless steel ESF components provide rigid fixation but are heavy. Aluminum and carbon fiber components are lighter; however there is higher cost with carbon fiber. Regardless, the size and shape of the connecting bars/rings are similar and fixation of transosseous pins is dictated by the size and location of clamps or rings. To overcome these limitations, other materials like acrylic (eg, polymethylmethacrylate) and epoxy putty have been used. Advantages of free form fixation include contouring the connecting bars to match any fracture

configuration, pin direction need not be influenced by connecting bar location, and pin diameter need not be influenced by clamp size. Free form fixators have been generally used for mandible and maxilla fractures in dogs and for fractures of small bones in birds. But in recent years, they have been found effective for management of open long bone fractures in dogs, sheep/goats, calves and foals weighing at least up to 100 kg.

Acrylic/Epoxy-pin fixators (bilateral uniplanar, bilateral multiplanar and circular) can be used for repair of open fractures/dislocations distal to the stifle and elbow joints. Fractures and dislocations are reduced and immobilized using 1.2-2.0 mm K-wires fixed at different levels (at least at 2 points in each fragment). Fixation wires in the same plane are bent and joined; and using additional wires, connecting bars/rings are constructed to make a temporary scaffold. Thoroughly mixed epoxy putty is then applied using the scaffold as guide and by incorporating the wires within the epoxy mold. In acrylic systems, the fixator components are constructed using acrylic columns with the use of flexible hollow plastic pipes, which are attached to the fixation pins. The fixation of acrylic/epoxy ESF is easy, less cumbersome, needs minimal instrumentation, economical and also provides stable fixation of fractures in animals weighing up to 100 kg, hence can be practiced by a veterinary surgeon at any remote corner in the field.



Fig. 1: *Typical sign of non-weight bearing lameness in a buffalo with fracture of metacarpus.*



Fig. 2: *Open infected fracture in a cow.*



Fig. 3: *Radiographic diagnosis of comminuted fracture of metatarsal bone.*



Fig. 4: *Fiber glass cast for fixation of radial fracture in a horse.*



Fig. 5: *Application of Modified Thomas splint for femur fracture repair in a calf.*



Fig. 6: *Rush pins used for fixation of proximal tibial fracture.*



Fig. 7: *Technique of interlocking nailing for femur fracture repair in a buffalo.*



Fig. 8: *Repair of comminuted tibial fracture using bone plate.*



Fig. 9: *Circular ring fixation for repair of radial fracture in a horse.*



Fig. 10: *Epoxy-pin fixation for repair of open metatarsal fracture in a calf.*

Chapter 15

Management of Common Surgical Affections of Livestock

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There are several affections of livestock which cannot be managed by oral/parenteral medication and demand surgical intervention. The success of surgical maneuvers depends on how early the surgeon attends it. However, certain surgical condition can be prevented by good managerial practices. Few commonly occurring surgical conditions in livestock are:

1. **Cleft Lip**- This is a common congenital anomaly occurring in pigs and calves where the upper lip has break in continuity and the dental pad/teeth are exposed and the animal faces problem in suckling which is repaired (cheiloplasty) surgically.
2. **Nasal Bleeding** (epistaxis)- It occurs when there is severe blow to the head, broken horn, high fever, entry of leech in the nostril. This can be managed by cold fomentation forehead with ice, injection or local application of styptics. A leech can be removed by pouring common salt (sodium chloride) solution.
3. **Nasal Polyps**- These are warty outgrowths due to a parasite (schistoma) normally it responds to medical management (anthiomaline) but obstinate cases needs surgical intervention (rhinotomy) for excision.
4. **Affections of Oral Cavity**-
 - (a) **Wounds** are very common foot and mouth disease and foreign body (glass, bone, wire, nail, needle, chemicals) laceration which can be successfully managed by mouth wash with 0.1% KMnO₄ solution and topical application of boroglycerine paste. However, if the wounds do not heal a thorough examination is required along with radiography for tooth fracture or deeply penetrating foreign bodies.
 - (b) **Tongue may be swollen** due to a ligature of a nylon strings or a honey cyst (ranula) or becomes thick and woody (actinobacillosis) where in the removal of foreign body/ligature or drainage/excision of cyst or application of glycerin-magnesium sulphate paste along with antibiotics may be used. At times foreign bodies like metal wires or pieces of chaff blades are founding inter dental spaces. Tongue lalooing is vice in animals and obscene either strip of tissue is removed from the ventral part of tongue or rings are pierced to avoid self suckling.
 - (c) **Teeth table** over shot or under shot or scissor bite may be found as a congenital anomaly such animals may be culled as the prehension and mastication may be interfered. Improper wearing of teeth leads to accumulation of feed in the buccal cavity it is because at every act of chewing the buccal mucosa comes in between the sharp molars of upper and lower jaw in such conditions the sharp ends are rasped by means of dental floats. In piglets the needle teeth are trimmed on second day of farrowing while in adult boars the over grown canines are either rasped or cut with help of giglie's wire saw.

- (d) **Cleft palate** is a condition where there is incomplete closure of the roof of oral cavity as a result there is through and through communication in nasal and oral cavity. It is a congenital anomaly the calf or a piglet suckles milk which comes out through nostrils. Under regional nerve block or general anesthesia the defect is repaired in two stages.
- (e) **Tumors viz.,** epulis, odontoma, adamantoma are common tumors found in oral cavity. These tumors are surgically excised. However, at times chronic, fibrosed and organized actinomycotic abscesses of the mandible are misdiagnosed tumors which invariably respond to lancing and drainage with local dressing with hydrogen peroxide and glycerin-magnesium-copper sulphate paste, parenteral streptopenicillin and oral potassium iodide 4 gm OD as an electuary with jaggery.
- (f) **Fracture of mandible** or teeth is very common due to kick, blow of iron rod or wood and pole or sudden application of brakes during transport of animal in vehicle where the chin gets dashed with the back of driver's cabin where inter dental wiring, supporting splint muzzle or acrylic external fixator can be applied.
5. **Affections of Eyes-** In ward turning of upper or out ward of lower eye lid leads to pain full and impaired vision which can be repaired under regional nerve block by z-y plasty. Pink eye condition is due to *Moraxella bovis* and can be managed by topical instillation of antibiotics. Down ward rotation of eye ball is a common condition in dairy cows which is due to paralysis of orbicularis oculi muscle which can be strengthened by anchoring it to the orbital rim. Corneal opacity occurs due to systemic diseases, trauma to the cornea by kicks, blow of tail switch, maliciously hit by stick which is treated by 5% NaCl eye drop solution. Eye worms like thelazia can be removed surgically by desensitizing the eye. Ocular squamous cell carcinoma is frequently observed in cattle and buffaloes If the eye ball have to be conserved in precious show quality specimen animals chemotherapy with levamisole or immunotherapy with BCG vaccine/ *Corynebacterium parvum* or saline-phenol extract of the partially excised tissue is of help however, if the tumor has spread in whole of the orbital cavity extirpation of eye ball is recommended under suitable analgesia.
6. **Affections of Horn-** Long horns are dangerous to the other animal housemates as well as to the human handlers but in certain show specimen breeds like khillar, kankrej, gir, Amrit Mahal or halliker the long horns are preserved and maintained for aesthetic purpose. In fighting/butting the horns break or the periople/core comes out and it bleeds profusely. The bleeding is checked by local and parenteral styptics, pain is killed by analgesics and coculet's splint is applied. In some animals the horns are sharpened, colored and polished either to hide the actual age or to look it more beautiful or the animal is exposed to sun light for longer duration or due to friction of roughness or uneven height of two animals yoked together to pull a cart or plough also due to viral affection "horn cancer" occurs levamisole or immunotherapy with BCG vaccine/*Corynebacterium parvum* or saline-phenol extract of the partially excised tissue is of help else the horn is removed surgically under proper anesthesia. To prevent this, the young calves are disbudded by chemicals (bemupbuns) or silver nitrate stick or caustic soda pebbles or electrical debudder/dehorner or red hot iron method.

7. **Affections of Ear-** Tears of earpinna are common due to entrapment of ear tags in baling wires of the fence or maliciously trimmed by quacks for the treatment of bloat, since the pinna is richly supplied by blood vessels the bleeding is profused and to be checked/controlled. Internal ear infection occur due to poor management and hygiene as crows go to eat the ear discharges and leads to chronic otorrhoea which is attempted to treat by unskilled personnel by pouring diesel on the external ear however, the structure of ear canal is like English letter “L” which can only be flushed with antibiotic added normal saline with flexible catheter.
8. **Yoke Gall-** Due to continuous mechanical irritation of rough yoke or uneven height of two animals yoked together to pull a cart or plough leads to swelling of dorsum of neck (nuchal ligament) as it becomes chronic and septic and abscessation occurs. In early stage it responds to topical application of “Sumag” but in chronic conditions it has to be excised.
9. **Affections of Trachea-** The tracheal rings collapse when strangulated due to tight neck rope or entrapped in Travis/chute which can be managed by tracheal anastomosis. In acute stage of HS, when there is inspiratory dyspnoea in such dire emergency, stab tracheostomy is performed.
10. **Choke-** The esophagus is obstructed due to consumption of placenta, rag, stone fruit or polythene rope or bag. The animal is unable to eructate gases as a result the stomach starts bloating and animal is in distress and restless. Either the obstructed material to be pushed down toward stomach by means of a probang else esophagotomy as performed under suitable anesthesia.
11. **Affections of Forelimb-**
 - (a) **Congenital contracted tendons** in calves is not uncommon (bent knee) such animal are crippled one of the recommended treatment is intravenous administrations of large doses of ox tetracycline for one week along with green bamboo splint coaptation gives promising results else supracarpal or sub carpal tenotomy of SDF tendon is performed.
 - (b) **Capped elbow/knee** is common in animals which are continuously stalled on hard floor with insufficient floor space; the skin comes in between the ground and the bony prominence. Conservative management with intralesional injection of corticosteroid is of limited help however, providing soft bedding material reduces the incidence.
 - (c) **Radial paralysis** occurs when animal is casted or remains in lateral recumbency for longer duration the radial nerve gets compressed between the ground and rib cage and the animal stand as if the elbow is dropped and stumbles while walking it is because the fetlock knuckles. The condition can be medically managed by corticosteroid, physiotherapy and neurotonics.
12. **Joint Ill-** the calves carry the infection while *in utero* or in post natal stage through the umbilical cord. The animal has an umbilical abscess and swelling of either both knees and hock or any one of these joint. The umbilical abscess to be drained when matured and flushed with antiseptics. Sufficient bedding material needs to be provided to avoid decubital (bed sores) lesions. The joint swelling can be managed by NSAID and specific antibiotics.

- 13. Hernia-** it is a condition in which the intestine or any other visceral organ or membrane from the abdominal wall below the skin. This occurs because of trauma, kick, incomplete closure of the prenatal defects, horn gore, fall over peg, dehiscence of surgical wound. Umbilical hernia occurs in calves and piglets where anxious dam over licks and pulls the a cord as a results the soft tissue and the muscles around the umbilicus get torn and due to gravity as well as abdominal pressure the contents escape under the skin. It can be conservatively managed by applying abdominal bandaged.
- 14. TRP-** A thoraco-abdominal disease complex. It develops as a consequence of perforation of reticulum by sharp foreign objects, inadvertent ingestion and swallowing of sharp pointed objects, mostly metallic. Symptoms are arched back (thoracic kyphosis and lumbar lordosis) and anxious expression reluctance to move (esp. inclined plane) and uneasy, careful gait groaning at the time of defecation, movement, lying down etc. jugular pulsation abduction of elbow splashing and frictional rubs on auscultation progressive edema of dependent parts selective right lateral recumbency recurrent tympanitis tucked up belly pain on xiphoid percussion firm dark feces in small quantity. Blood shows neutrophilic leucocytosis with shift to left (band neutrophils - 7000-10000 / cumm). Treatment consists of conservative treatment immobilization of animal administration of antimicrobials (penicillines) oral administration of magnet to immobilize the metallic objects, application of blistering ointment it causes maturation of abscess, on drainage sharp object can be retrieved back, pericardiotomy through fifth rib resection thoracotomy magnetic retrieval of metallic objects. For proper management keep animal on inclined plane (elevated in front). Prevention can be done by use of processed feed passed over magnet to remove metals and by use of reticular magnets (given at the age of 18 months or 2 years). Keep animal away from new construction sites.
- 15. Cesarean Section-** Cesarean section is the delivery of fetus at parturition by laparohysterotomy. It is indicated in fetomaternal disproportion incomplete dialation or induration of cervix irreversible uterine torsion fetal monstors faulty fetal disposition fetal emphysema uterine rupture and severe uterine haemorrhage foetal mummification. The various techniques include standing left oblique celiotomy, standing right paralumbar celiotomy, recumbent left / right paralumbar celiotomy, recumbent ventral midline celiotomy, recumbent ventral paramedian celiotomy and ventrolateral celiotomy. Post operative treatment and care Oxytocin (20-40IU) admin I/M after surgery to hasten uterine involution and to controle bleeding. Give a broad-spectrum antibiotic analgesic, anti-inflammatory drug for 3-7 days, anti histaminic glucose 5% haemostatics if necessary give laxative feeds daily dressing of a wound suture removal after 10-14 days
- 16. Vaginal Prolapse:** The condition is usually seen in mature females in the last trimester of pregnancy. Predisposing factors include increased intra-abdominal pressure associated with increased size of the pregnant uterus, intra-abdominal fat, or rumen distention superimposed upon relaxation and softening of the pelvic girdle and associated soft-tissue structures in the pelvic canal and perineum mediated by increased circulating concentrations of estrogens and relaxin during late pregnancy. Intra-abdominal pressure is increased in recumbent animals. Added to this, sheep tend

to face uphill when lying down, so that gravity assists vaginal eversion and prolapse. The organ is washed and rinsed, and the bladder is emptied if necessary. Usually, this can be achieved by elevating the prolapsus to allow straightening of the urethra; occasionally, needle puncture through the vaginal wall may be necessary. The vagina is well lubricated (glycerol provides lubrication and reduces congestion and edema by osmotic action) and replaced and then held in position until it feels warm again. Retention is achieved by insertion of a Buhner suture—a deeply buried, circumferential suture placed around the vestibulum to provide support at the point at which the initial eversion of the vaginal wall occurs.

- 17. Capped Elbow-** Capped elbow is the inflammatory swelling of the subcutaneous bursae (acquired bursitis) located over the olecranon process. Trauma from lying on poorly bedded hard floors, kicks, falls, riding the tailgate of trailers, iron shoes projecting beyond the heels, and prolonged recumbency are frequent causes. Acute early cases may respond well to applications of cold water, followed in a few days by aseptic aspiration and injection of a corticosteroid. The bursa may also be reduced in size by application of a counterirritant or by ultrasonic or radiation therapy. Older encapsulated bursae are more refractory. Surgical treatment (usually curettage and drainage) is recommended for advanced chronic cases or for those that become infected.
- 18. Contracted Flexor Tendons-** Contracted flexor tendons are probably the most prevalent abnormality of the musculoskeletal system of newborn foals and calves. An autosomal recessive gene causes this condition. In utero positioning may also affect the degree of disability. Mildly affected animals recover without treatment. In moderate cases, a splint can be applied to force the animal to bear weight on its toes. The pressure from the splint must not compromise the circulation, or the foot may undergo ischemic necrosis. Frequent manual extension of the joints and attempting to stretch the ligaments, tendons and muscles help in treating these intermediate cases. Severe cases require tenotomy of one or both flexor tendons. A plaster-of-Paris cast may also be indicated in some cases. Extreme cases may not respond to any treatment.
- 19. Hard Milker (Contracted Sphincter)-** If the constriction of the sphincter is at the terminal portion of the teats it can be incised using a stinsen's teat director. A constriction higher up in the teat canal is cut by using a teat bistoury or a concealed teat knife.
- 20. Teat Obstructions-** Acquired teat obstructions are usually the result of proliferation of granulation tissue after the occurrence of an observed or unobserved teat injury. They can range from diffuse, tightly adherent lesions to highly mobile discrete lesions that float throughout the gland cistern. Some "floaters" are caused by formation of small masses from butterfat, minerals, and tissue in mammary ducts during the dry period. These can be recognized by intermittent disruptions in milk flow. They may be removed by forced pressure downward on the teat cistern or by use of specialized instruments inserted through the teat canal. Membranous obstructions in the area of the annular fold at the base of the gland cistern are sometimes observed in heifers. Treatment of these obstructions is generally frustrating.

- 21. Udder Abscesses-** Subcutaneous abscesses of the udder (not involving the milk-producing tissue) can develop between the skin and the supporting connective tissue of the udder. Diagnosis is by needle aspiration. Abscesses usually develop secondary to wounds, advanced mastitis, infected hematomas, or severe contusions. They should be incised and drained when they are chronic and near the surface of the udder. The wound should be flushed daily with an antiseptic solution or water under pressure until healing is complete.
- 22. Teat Sphincter Inadequacy (Leakers):** High levels of intramammary pressure in high-producing dairy cows may result in milk dripping from teats in cows that are waiting to be milked. Shorter intervals or more frequent milking may be recommended when a large proportion of the herd is affected. Occasionally, cows are observed to leak milk continuously. These cows usually have sustained a severe teat injury or have an abnormal streak canal. In general, little can be done to correct this condition, and most of these cows will develop mastitis; it is recommended that persistent leakers be designated for removal from the herd.
- 23. Supernumerary Teats-**These teats are often seen in the bovine in between normal teats. Their functional capacity can only be determined after parturition of the animal. It is desirable to remove these teats for cosmetic reason and also because they interfere with the milking procedure. Supernumerary teats are removed surgically under local anaesthesia by making two elliptical incisions at the junction of teat and skin of the udder.
- 24. Teat Fistula-** An abnormal opening or a passage between the teat cistern and the teat surface through which milk flows out in lactating animals. Mostly it is acquired anomaly, traumatic in origin, and rarely congenital. It result in to leakage of the milk due to fistula and also leads to condition of mastitis. Although a teat fistula is best treated during the dry period, fresh wounds need a immediate attention. For repair of a teat fistula two elliptical incisions are given on the skin edges for debridement and undermining.
- 25. Ulcerative Thelitis-** Ulcerative thelitis is a disease affecting primarily high-yielding primiparous graded Murrah milch buffaloes and causing serious economic losses to the farmers. The disease is characterized by acute inflammation of one or more teats with subsequent thickening, narrowing or closure of teat canal leading to incomplete drainage of milk. The quality of milk appears to be normal unlike in clinical mastitis. This is followed by ulceration, focal necrosis, and sloughing off the affected teat. Healing may be delayed due to the trauma of milking and secondary bacterial infections.
- 26. Atresia Ani-** Congenital defects are observed in different parts of the body, especially last part of the digestive tract like atresia ani. This congenital anomaly has been reported in all domestic animals. It is one of the quite frequently found defects of intestine among sheep because of recessive gene. Recto-vaginal fistula is another congenital problem causing direct communication between rectum and vagina in female calves responsible for urofecal mixing and vulva if it be normal, serves as a common orifice for both digestive and urogenital tracts. This deformity is

accompanied by atresia ani and blind recta pouch ends just anterior to the approximated site of anus.

- 27. Hoof disorders-** Wall cracks are important disorders in cattle. Vertical wall cracks tend to occur most frequently in beef cattle, but they are relatively rare in dairy cattle. They tend to affect the outer hooves of the front feet and only a small percentage of lesions lead to lameness. Horizontal wall cracks are common to both beef and dairy cattle. They are indicative of physiological change and when severe enough to result in a full thickness defect in the wall, suggest complications from disease or some other stressor. Whenever wall cracks result in lameness, corrective trimming procedures and a block fitted to the healthy claw to relieve weight bearing are indicated.
- 28. Tail Gangrene-** Injuries near the end of the tail often result in dry gangrene. The tail may have been crushed or caught in something, such as in a closing door or between the shelf and the wall of the basking area. The tissue, starting at the end of the tail, begins to die, turning dark brown or black, becoming very hard and brittle, shrinking inwards, collapsing in on itself. The bony processes of the tail vertebrae are easily visualized as they create ridges under the skin of the tail. When caught in time while it is still near the end of the tail, the amputation is very quickly done. The tail must be severed in the healthy tissue. If the cut is made in the dead tissue or close to the demarcation between healthy and dead tissue, too often there is enough bacteria left in the end of the stump to continue the gangrenous process.

Chapter 16

Use of Ultrasound and Radio Diagnosis in Animal Diseases

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Modern Imaging tools have made the diagnostic approaches of diseases in Veterinary practice more precise and early, least exhaustive and non-invasive in nature. The modality included ultrasound, radiography, computed tomography (CT), magnetic resonance imaging (MRI), nuclear medicine (NM), and teleradiography. These techniques provide the clinicians with new information about the interior of the animal body that has never before been available. Applications and innovations in diagnostic aids are not abating but are in fact accelerating and, therefore, advances in image production and visualization will continue. Available diagnostic modalities are providing interdisciplinary teams of clinicians to adopt effective diagnostic, curative and preventive measures of animal diseases in a non-invasive way.

Ultrasonography (USG)- Ultrasound (U/S) means sound waves of frequencies greater than audible to human ear (greater than 20,000 Hz). Frequencies between 2 to 10 MHz are mainly used in diagnostic ultrasound. There are three main types of transducer available- linear array, convex and sector. A pulse is generated by one or more piezoelectric crystals in an ultrasound transducer. As the transducer is placed in close contact with the body surfaces through a coupling medium the ultrasound beam is absorbed, reflected and scattered. A portion of the ultrasound beam backs towards the source (echo) that is converted into visual display (Mode A, B & M). When the image displayed in B-mode scan are formed rapidly and presented in sequences, the movement of organs will be viewed in real time. In order to form these sequential images, it is necessary to sweep the ultrasound beam over the tissue by either mechanical or electronic means. The image formed on the scan screen is actually a mixture of images of different echoes depending on the area scanned.

Image interpretation- Gas, bone etc are barrier to ultrasound (acoustic shadowing) i.e. hyperechoic and appear white on screen. Ultrasound passes through fluid uninterrupted (acoustic enhancement) i.e. anechoic and appears black on screen. Images of soft tissues appear as mixed shades of gray depending upon their proportion of fat, fibrous tissue and fluid.

Advancement in ultrasound- Doppler ultrasound (Color Doppler), multi-hertz transducer, 3-dimensional, 4-dimensional ultrasound and ultrasound-guided biopsy have made the imaging modality more versatile. Percutaneous biopsy of abdominal organs like liver, kidney, spleen, prostate, abdominal mass, lesions etc. under ultrasound guidance is well documented. Endosonography refers to use of small transducers incorporated into the tip of an endoscope and inserted into a body cavity (peroral, transrectal, transvaginal) for intraluminal imaging.

Scanning approaches- A coupling medium is applied between body surface and transducer to establish an intimate contact. Transabdominal scanning commonly used in small animals is approached from ventral or lateral wall. Resolution of the image is not good in large animals because of the presence of gas and ingesta in abdomen. For

transrectal scanning, rectal transducer is used. Resolution is good and very commonly used in large animals. Transthoracic approaches are used to scan heart, major vessels, lungs and diaphragm. Sonoendoscopic probes are used to allow transluminal evaluation of hollow visceral organs. Miscellaneous approaches for tendons, udder and tests, eyes etc. are also used.

Ultrasound is now routinely used for scanning numerous organs and to study reproductive biology (viz. monitoring pregnancy, ovarian dynamics, and ovum pick-up technology), haemodynamic parameters and morphometric measurements of different organs. Ultrasonography of eyeballs, tendons and ligaments, udder and teat etc are also reported. The merits of ultrasound are its ability to characterize shape size and internal parenchymal architecture, cost effectiveness, no radiation hazards, repetitive examinations can be done, provide instant information and well tolerated by the animals. A sonologist should be aware of the common artifacts (reverberation, mirror image, comet tail) to avoid errors in image interpretation.

Radiography- Plain Radiography: More than 100 years have passed since the discovery of x-ray on 8 Nov 1895 and much advancement has been added, still conventional radiography continues to be the mainstay of diagnostic imaging in veterinary practice in most of the third world countries.

Instrumentation- X-ray machine- Portable, Mobile and Fixed x-ray machine depending on the budget, work load and type of animals to be radiographed.

Accessory equipments- Cassettes, x-ray films, hangers, markers, illuminators etc.

Dark room requirements- Film processing tanks, film processing chemicals(developer, fixer etc), safe light etc.

Protective equipments and clothing- Protective screen, lead apron, gloves, goggles, radiation detectors etc.

Principles of Radiographic Interpretation- The principle of radiography is that the animal is exposed with a beam of x-rays that are differently absorbed by the tissues, thus casting a “shadow” in the beam that can be recorded usually on a film. Bone absorbs maximum x-ray, appears white on film; muscles, viscera, fluid absorb lesser amount of x-ray, appear gray on film; fat absorbs still less x-ray, appears grayish on film; air absorbs least amount of x-ray, appears black on film. A complete clinical and physical examination of the patient to correctly locate the area suspected for pathology is always a prerequisite to the radiographic examination. Cleaning of the part to be radiographed is very essential as the dirt will obstruct the x-ray and will produce superimposing shadows thereby reducing the sharpness of the radiograph. Radiograph converts a three-dimensional subject into two-dimensional planes, so it becomes essential to take at least two views at 90° to each other. Before undertaking the task of radiographic interpretation it is essential to know the normal anatomical relationships of bones and joints, age of various epiphysis closures and developmental pattern of the bone. In case of doubt, it is advisable, to compare the radiograph with the otherwise having normal configuration. Standard signs of radiographic pathology have been classified in terms of change in size, architecture, contour, density, position and function. By noting changes in any of these, the pathological lesions are located. The best way to narrow down the differential diagnosis using radiographic interpretation for a pathological lesion is to categorize it in developmental, metabolic, traumatic, infectious, neoplastic and degenerative changes.

These changes are visualized in relation to increased soft tissue densities, periosteal proliferation, de-mineralization, decreased joint spaces, angulations and interrupted bone densities. Generally, the purpose of radiography is to confirm a clinical diagnosis, not to make a diagnosis. To make a final radiographic diagnosis, one must weigh all the evidence like species, breed, age, and sex of the animal, history, clinical signs, laboratory findings, and response to therapy. To keep all the information into account, an x-ray diagnosis is to be achieved.

Special Radiography- Use of contrast media to delineate the organ of interest from its surrounding structures. Generally barium and iodine preparations are used as positive contrast and air, carbon dioxide and oxygen are used as negative contrast.

Fluoroscopy (viewing of an x-ray image on a fluorescent screen instead of a film)- The greatest contribution of fluoroscopy is that it provides dynamic radiographic study. Many fluoroscopes are equipped with spot film device to record the fluoroscopic image. Fluoroscopy is technically simple, save time and expenses of exposing and developing x-ray film. The shortcomings of fluoroscopy are its radiation hazards and poor visibility of the images.

Image intensifier- The limitation of fluoroscopy is overcome through image intensification. X-ray beam carrying useful information passes through an image intensifier. The image produced for viewing is 1000-5000 times brighter and the image can be electronically controlled.

Subtraction technique- It is the photographic method that eliminates unwanted images and makes it easier to visualize important radiographic information on a radiograph, example cerebral angiogram where positive print of a survey radiograph is superimposed on it to mask the undesirable images of bone and make the vessels visible.

Xeroradiography- It is method of x-ray imaging in which a visible electrostatic pattern is produced on the surface of photoconductor. It is useful in soft tissue imaging. Advantages are dry procedure; enhance visualization of the border between images of different densities (edge effect), low contrast that it cannot be used for very thick parts, as very high exposure is needed.

Digital Radiography- Digital images can be electronically stored, transmitted, processed and displayed under computer control. Digital radiography based on storage phosphor technology is an imaging system in which x-ray images are correctly exposed due to storage screen's extremely wide exposure range. It offers virtually endless opportunities for qualitative and quantitative image assessment. Bone and soft tissues are clearly visible from a single exposure. It is possible that a day will come when x-ray films will no longer be used in radiography.

Computed Tomography (CT)- It produces cross sectional images of the body using x-rays, x-ray detectors and computer analysis of the information. The unique 3-D anatomical visualization free from surrounding tissue (superimposition), non-invasive nature and its ability to perform dynamic and/or quantitative tissue characterization have a tremendous impact on disease diagnosis and follow up cases. The most important role of CT for the surgeon is in its precise localization and extent of the lesion relative to surrounding tissue. Images obtained with or without the use of contrast material provide

additional information about the physical characteristic of the lesion as well as functional aspects such as vascularity.

Positron Emission Tomography (PET)- It is a new area of metabolic imaging technique of future with exciting potential. Radioisotope emits positron, combines with electron to form positronium that emits gamma rays, detected by Crystal Detector-Computer analysis and images are formed. When using PET, it is as if we are viewing the body both with a microscope and pair of binoculars at the same time i.e. looking at the cells and whole body at the same time. Brain as well as whole body can be imaged.

Nuclear Medicine (Scintigraphy)- Nuclear Medicine is a highly sensitive advance procedure in which radioisotopes are used to detect functional status of by the body systems. The principle of Nuclear Medicine is based on the use of a pharmaceutical labeled with an isotope that after entry into the blood stream gets localized in a particular tissue or organ. Thus localization of the isotope can then be detected by using detector or gamma camera (Scintillation camera) due to emission of gamma rays from the area of interest. Whole body and SPECT (Single Photon Emission Computer Tomography) procedures need the most developed instrumentation, the so called SPECT-camera, where the detector can be moved by the computer to allow imagine 3-dimentional distribution of radio pharmaceutical and a better scintigraphy and resolution of picture quality. Most commonly used radioisotope is Technetium-99 m. This isotope has the advantage of a short half-life of 6 hrs and thus animal can be discharged next day after the scan. Additionally radiation exposure to the animal and attending personnel is minimum. A computer system is usually attached to the camera for gathering data and its display and quantification. A scan appears as an image formed of dot. The interpretation is based on the appearance of increased (hot spots) decreased (cold spot) radioactivity regions. Scintigraphy has been used to detect functional status of kidney, liver, GI Tract, lung, thyroid, bone, brain etc. The limitations are high cost, safety measures, etiological non-specificity and difficulty in interpretation especially in skeletal system.

Radiation Safety: It is the responsibility of the radiologist to ensure radiation safety measures in radiology section and to educate other personnel about the potential hazards of radiation.

Magnetic Resonance Imaging (MRI): It is a highly sensitive and non-invasive technique that provides accurate and detailed anatomic images with good contrast and spatial resolution. The basic principle of MRI is that certain atomic nuclei will absorb and remit radio waves when placed in a strong magnetic field (Nuclear Magnetic Resonance- NMR). Radio waves remitted can be used to construct a diagnostic anatomic image through a computer-assisted technique termed as MRI. It has multiplanner capability and no radiation burden, and that makes MRI very attractive. All body regions in small animals and extremities and head in large animal can be imaged. The spectrum of MRI application is very wide. It has numerous musculo-skeletal applications. It is especially useful for differentiating an inflammatory process from a neoplastic mass. Contraindications to the use of MRI are presence of ferromagnetic materials in the body and pregnancy.

Teleradiology: With the advancement of information technology, it has been possible to execute e-transmission of radiological information to any corner of the globe and seek expertise opinion about it. A radiologist can also explore the facility of computer-aided radiodiagnosis from the available software and thus strengthened the diagnosis with more accuracy and consistency with minimum time.

Conclusions- Modern diagnostic imaging aids are potential armament in the hands of expert clinicians for understanding various complex biological phenomena of the animals in a non-invasive way. However, the examination takes skill and training to perform and to interpret. It is essential to produce an image of good quality that will allow the clinicians to differentiate between artifacts and real image. The clinicians must have a through understanding of the instrumentation and knowledge of anatomy and patho-physiology of the subject concerned.



Fig.a. Plain radiograph showing pulmonary metastasis.

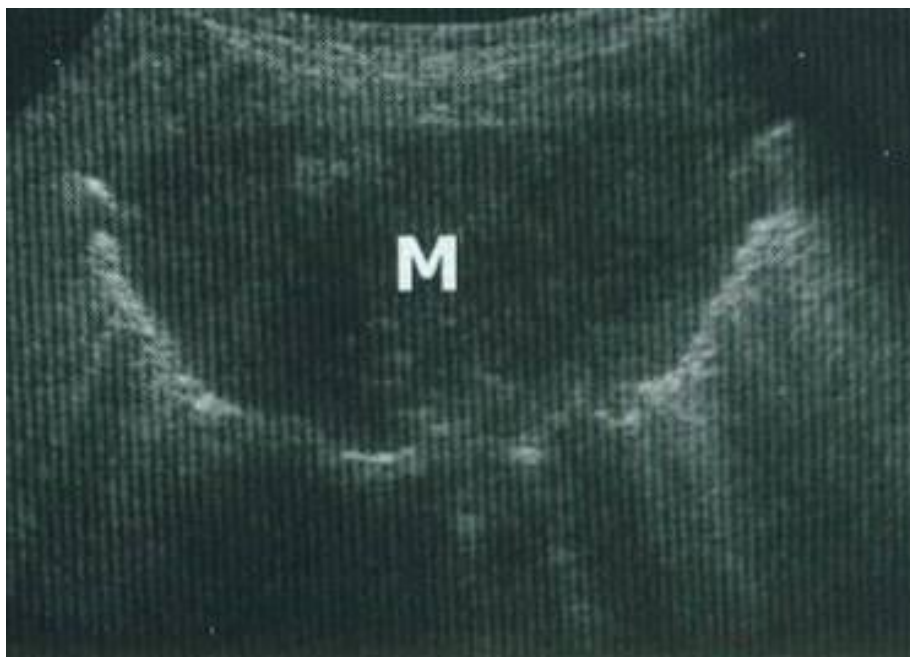


Fig. 2. Sonogram showing pulmonary metastasis.



Fig. 3. Contrast radiography of stomach and small intestine showing filling defect.

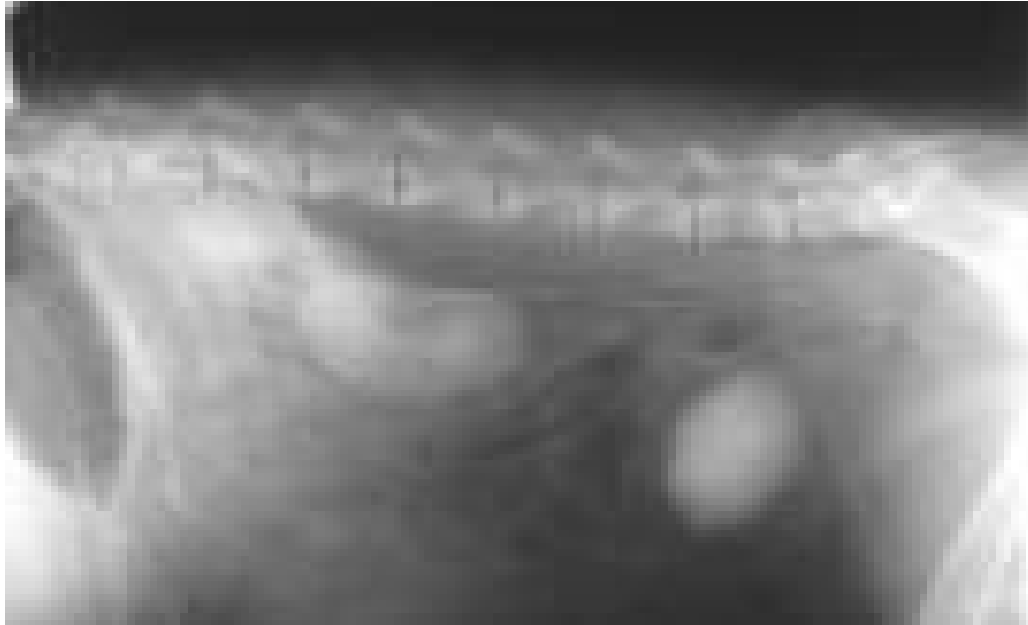


Fig. 4. Intravenous pyelography (IVP) delineating kidney, ureter and urinary bladder.

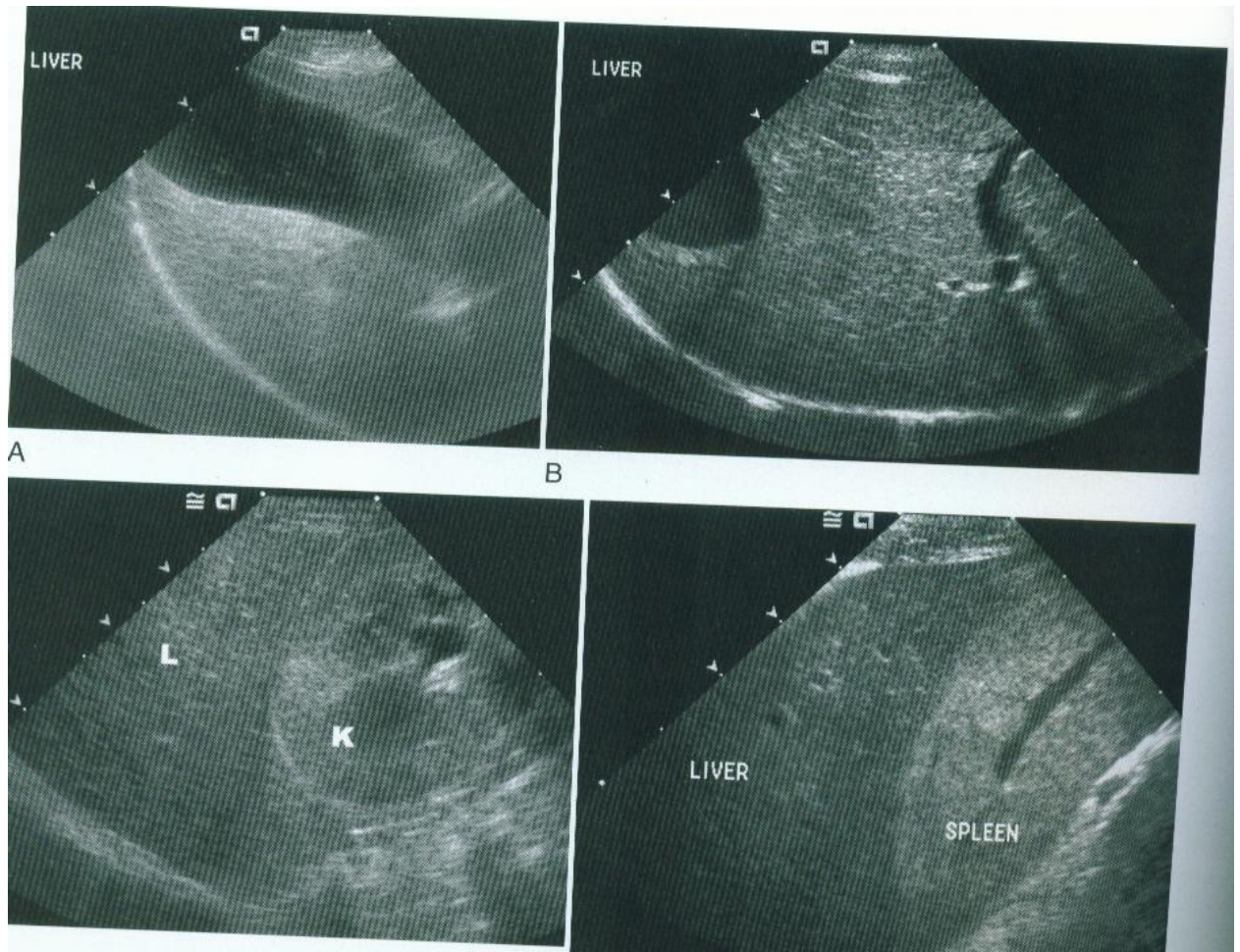


Fig. 5. Sonogram showing normal liver, kidney and spleen

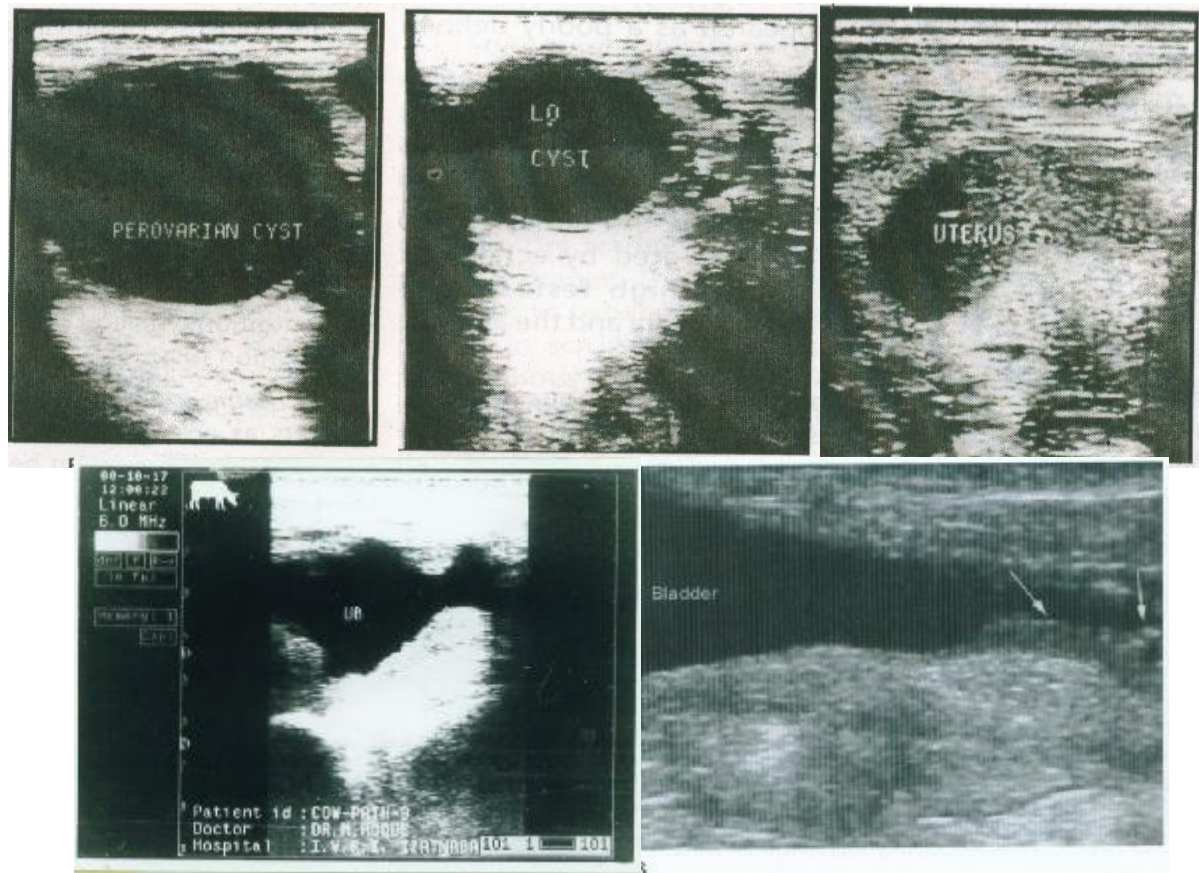


Fig. 6. Sonograms:1 &2: Ovarian cysts; 3. Pyometra ; 4&5: Soft tissue growth in the urinary bladder.

Chapter 17

Commonly Occurring Toxicities in Livestock: Diagnosis and Management

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Everything is poisonous, if dosage and route of entry into living organism are not restricted. If the animals have access to chemicals, they may ingest them out of curiosity. Man and animals always have been exposed to heavy metals in the environment, Metallic contamination of food and water probably led to the first poisoning. Animals are frequently exposed to xenobiotics such as agro-chemicals, metallic and non-metallic compounds, mycotoxins and other environmental pollutants, they may cause health hazards to man and animals. Many times heavy metals exert their toxic effects by combining with one or more reactive groups essential for normal physiological functions, Mortality in animals also causes huge economic loss to the farmers. To sustain and improve the livestock production, it is required to control and prevent the toxicosis well in advance and also to evolve line of their treatment.

Poisoning is an emergency and is to be treated immediately. Until the confirmation, treatment with antidotes can not be initiated and hence the aim is to neutralize the poison and to maintain the vital functions of body.

(I) Removal of Toxicant and Prevention of Further Absorption:

- a. In case of history of external application, wash the body surface with lukewarm water, and don't use soap and detergents.
- b. Gastric lavage is useful in small animals but should not be done in unconscious animals and after ingestion of petroleum products. Isotonic NaCl solution 10 ml/Kg is indicated for giving lavage. In rumenotomy is only satisfactory method to remove toxicants. Endotracheal intubation should be done.
- c. Emetics: Emesis in dogs, cats and swine can be induced by emetics within 2-3 hours of ingestion. Apomorphine (contraindicated in cats and pigs) - 0.04-0.07 mg/kg i.m. or s.c., Xylazine 1-2 mg/kg i.m. can be administered for inducing vomiting. Oral administration of 1% CuSO₄, 10-60 ml in dogs, 3-20 ml in cats and spoonful of Sodium chloride can also be employed. But there is a risk of aspiration pneumonia. Emetics are contraindicated in poisoning with

barbiturates, kerosin, petroleum products and in seizures/ convulsive animals.

- d. For hastening the removal from GI tract, saline purgatives can be used. Mag Sulph or Sod Sulphate- 1.0 gm / kg in dogs and cat; 100-200 gm in large animals can be administered orally. Irritant purgatives, cathartics and oil based purgatives are not used as these may cause dehydration.
- e. Neutralization of poison: Use of adsorbing agent to prevent further absorption is advocated.

Universal antidote:

Activated Vegetable Charcoal	- 10 g
MgO	- 05 g
Kaoline	- 05 g
Tannic acid	- 05 g
Water	- 200 ml

(II) Use of Specific Antidotes:

Treatment with specific antidotes can be given after confirmation of toxicity. Various antidotes available are mentioned Table.

S. No	Chemical agent (s)	Specific Antidote
1	Lead, Mercury, Arsenic	BAL
2	Iron	Deferroxamine
3	HCN	Sod. Nitrite & Sod.thiosulphate
4	Nitrate/ Nitrite	Methylen Blue
5	Copper	d-Penicillamine
6	Molybdenum	Copper
7	Lead, Zinc	CaNa ₂ -EDTA
8	Barbiturates	Bemigrade
9.	Coumarine	Vit K
10	Ethylen glycol,. methanol	Ethyl alcohol
11	Organophosphate	Oximes & atropine

(III) Supportive and Symptomatic Therapy:

Symptomatic and supportive treatment helps in counteracting the manifestations of chemical toxicity and to maintain the vital functions of the body.

- a. Analeptics in barbiturate toxicity, Anticonvulsants and pentobarbital in seizures/ convulsions, Vit K in haemorrhages, Bronchodilators and antihistaminic in anaphylactic shock, Vasopressors in hypotensive crisis are indicated.
- b. Saline/ Ringer saline are administered in cases of dehydration.
- c. Use of diuretics are advocated to enhance the urinary excretion of toxic chemical. Mannitol as 10% solution can be administered intravenously at dose of 1-2 ml/kg/24 hrs for this purpose.

The line of treatment to be adopted in various chemical toxicities occurring in livestock has been discussed as below.

A. Metal Toxicity

1. Lead

Source: Ingestion of lead-based paints, discarded crankcase oil, solid lead, asphalt, golf balls, bullets and other lead products may cause toxicity in animals. Contamination of herbage by metal processing industries and automobile exhausts is also the common source of lead toxicity in poultry, livestock and other domestic animals.

Clinical Signs

Acute toxicity: Nervous signs include excitement, bellowing, staggering, muscular spasm or tetany, convulsions and blindness.

Subacute toxicity: Digestive disorders: Such as anorexia, constipation or diarrhoea, colic, ruminal atony and voraciousness are observed.

Chronic toxicity or plumbism: Plumbism is manifested by locomotor disturbances, rigidity of joints, swaying movements, drop in milk yield, abortions, and blue-black discoloration of gums, roaring and anemia.

Treatment: Calcium-disodium EDTA should be given in large animal as slow i.v injection of 6.6% solution @ 70 mg/kg/day divided in 2-3 doses for 3-5 days along with symptomatic treatment.

2. Arsenic

Source: Consumption of arsenicals (Sodium arsenite, potassium arsenite etc.) contamination of herbage and water by arsenate sprays, arsenical dips, weedicides, rodenticides and effluents from metal smelting works. Licking of wood preserved with arsenical and dipping of unshorn sheep in arsenical dips.

Arsenicals inhibit a variety of enzymes and uncouple oxidative phosphorylation. Arsenic has special affinity for intestines, liver and kidney as they are rich in oxidative enzymes.

Clinical Signs: Hemorrhagic gastroenteritis and preserved carcass is the characteristic feature of arsenic toxicity.

Treatment: British anti lewisite (BAL) : Large animals: 2-3 mg/kg I.m. as a 5% solution in 10% benzylbenzoate solution in peanut oil. Repeat the injections at 4-6 hr intervals on first 2 or 3 days and twice a day thereafter until recovery. Small animals: 2-5 mg/kg i.m. as 10% solution in oil. Sodium thiosulfate and other symptomatic measures should also be given.

3. Mercury

Source: Accidental ingestion of seed and herbage treated with mercurial fungicides or mercurial effluents from the metallurgy and inadvertent use of mercurial ointments for skin lesions are the common source of poisoning in animals.

Clinical Signs

Acute: Nervous signs in dogs are manifested by excitement, tremors, convulsions and depression. GI disturbances in cattle, sheep and pigs include severe gastroenteritis, bloody diarrhoea, vomiting with blood and dehydration.

Chronic: Incoordination, altered gait, stumbling, head pressing, muscle spasms, pale mucosae, epistaxis, dyspnoea and fever- are commonly noted following chronic toxicity of mercury in animal.

Treatment: No effective antidote. Treatment is mainly symptomatic.

4. Selenium

Source: Toxicity occurs mainly by ingestion of forage (Astragalus, Gonopsis, Stanleya, Xylorrhiza, Aster and Astriplex spp.) rich in selenium content or grown on selenium rich soil and inadvertent use of selenium containing feed additive and shampoo.

Clinical Signs

Acute: Anorexia, colic, bloat, watery diarrhoea, pale mucosa, dyspnoea with fluid sounds from lungs are observed in acute toxicity in animals.

Sub acute: In this condition animal stumble on fixed objects or obstacle due to blindness caused by clouded cornea etc.

Chronic: Including all signs animal also shows lameness due to erosion of articular surface of long bones and separation of hoof wall below the coronary band. Hair defects are also seen.

Treatment: No effective antidote. The treatment must be aimed to expel out the unabsorbed selenium from GIT. The affected animal should be shifted to protein rich diets.

B. Non Metal Toxicity

1. Strychnine

Source: Strychnine is an alkaloid obtained from the *Strychnos nuxvomica*. Strychnine is used in the form of baits to control various pests such as squirrels, rats, wild carnivores, rabbits etc. Nux vomica is also used as a bitter principle and stomachic.

Clinical Signs: Clinical signs of strychnine poisoning occur as a result of overflow of efferent nerve impulses to the skeletal muscles and are characterized by hyperesthesia, restlessness, panting, nausea, vomiting, frequent defecation and urination, colic convulsions, prostration, stiff gait, extension of limb, opisthotonus, struggling to rise, vocalization (In dogs, cats etc.) and laboured breathing. Death occurs due to cerebral anoxia.

Treatment: Pentobarbital sodium (10-20 mg/kg, i.v. or i.p.) in small animals and chloral hydrate (5 gm/50 kg, i.v. as 6% sol.) in large animals are the drugs of choice to overcome convulsion. Diazepam, methocarbamol and xylazine are also effective in sedative doses. Stomach must be washed out with 1:250 dilution of tincture iodine solution followed by i.v. infusion of isotonic saline or glucose. Intubation of trachea may be done to facilitate respiration.

2. Nitrates Nitrites

Source: Nitrates are naturally present in soil, ground water, forages, and silages

and raw crops. Nitrate or nitrite poisoning in animals occurs after the ingestion of nitrate fertilizers (sodium nitrate, calcium nitrate, ammonium nitrate etc.), nitrate rich crops, drinking of high nitrate containing water or overdosing of sodium nitrate during the treatment of cyanide poisoning. Nitrite ions produce toxic effect by relaxing the smooth muscle of the blood vessels leading to hypotensive effect. Nitrite ions also oxidize ferrous hemoglobin to ferric from i.e. methemoglobin.

Clinical Signs: Clinical signs appear within few minutes to 4-5h depending on the nitrate content ingested and are characterized by the respiratory and GIT disturbance. They include frequent urination, vomiting, bloody diarrhea and colic. Respiratory insufficiency is characterized by dyspnoea, cyanosis, rapid and weak pulse, chocolate color blood, tremors and convulsions. Death occurs due to anoxia. Chronic toxicity exhibits loss of body weight,

decreased milk production, general weakness, abortion, still births and birth of weak calves.

Treatment: Methylene blue (1%) w/v in isotonic saline, slow i.v. is administered at the dose of 8.8 mg/kg in ruminates and 4.4 mg/kg in small animals and can be repeated with caution if required. Blood transfusion and oxygen therapy are helpful. Saline purgative to empty the GI tract, drinking of 10-15 liters of cold water to reduce conversion of nitrate to nitrite, and soothing agent for GIT (liquid paraffin) and antibiotic therapy are also helpful to animals for rapid recovery.

3. Cyanide

Source: Ingestion of cyanogenic glycoside such as Amygdalin In wild cherry and dhurra in fodder grass like sorghum, sudangrass poisoning in animals. In addition, cyanide compounds used as rodenticide, fumigants and fertilizers, are the source of poisoning in animals. Animals may also be poisoned maliciously by cyanide compounds. Cyanide radical inhibits cellular respiratory enzymes, binds with the ferric iron and has vasoconstrictor action and reduces blood flow to vital organs.

Clinical Signs: Clinical signs appear within few minutes of ingestion and may vary from mild to severe panting, gasping and violent behavior. Other prominent signs include profuse salivation, lacrimation, urination, frequent defecation, severe colic, emesis, muscle tremors, prostration. Brightness of the mucous membrane, clonic convulsions, and mydriasis. The death in acute poisoning even sometimes occurs without these clinical signs. Animals die because of respiratory failure followed by cardiac arrest.

Treatment: Sodium nitrite (1%) @ 15-25 mg/kg i.v. in combination of sodium thiosulphate (25%) @ 1.25 g/kg body weight is the antidote of choice for the treatment of cyanide poisoning. In addition, four liters of vinegar diluted in 10-15 liters of cold water is given orally to prevent hydrolysis of cyanogenic glycosides. Liquid paraffin can be administered to soothe the irritated mucous membrane of the gastro-intestinal tract.

C. Insecticides Toxicity

1. Organophosphate and Carbamate

Source: Organophosphorus and carbamate compounds are used as insecticide, pesticides, herbicides, defoliants and soil nematocides etc. In agriculture practice and to control external and internal parasites in animals. Livestock may be poisoned by improper dipping, dusting, spraying, accidental ingestion or malicious use of these insecticides. Organophosphate

insecticides irreversibly inhibit the acetylcholinesterase (AChE) enzymes whereas the carbamates inhibit the enzyme reversibly.

Clinical Signs: Characteristic signs of o.p. and carbamate poisoning in animals include profuse salivation, lacrimation, diarrhoea, coughing, tremors convulsions, incoordination, laboured breathing, dyspnoea, hyperaesthesia and CNS stimulation. Depending upon dose, death of animal occurs within few minutes to hours due to respiratory failure.

Treatment: Atropine sulphate is administered at a dose of 0.2-0.5 mg/kg and the dosage can be increased up to 1 mg/kg in severe cases. One fourth of the dose can be administered as i.v. and rest by i.m or s.c. route which can be repeated at 3-6 hrs interval as per clinical manifestation. Excess atropinization should be avoided which may worsen the condition of animal. AChE reactivator, pyridine-2 – aldoxime methiodine (2-PAM, pralidoxime) is a good adjunct to atropine sulphate in early stage of the poisoning and is administered at the dose rate of 20-50 mg/kg as a 10% solution by slow i.v. infusion. 2-PAM or any other AChE reactivator is not recommended for the treatment of carbamate poisoning and also in the later stage of organophosphate poisoning. Therapy with multivitamin, fluid (dextrose saline) electrolyte may also be recommended.

2. Organo-Chlorine or Chlorinated Hydrocarbons

Source: Chlorinated hydrocarbons are widely used for prevention and control of insects infestation around agricultural farms, house premises and on animals. Animals may be poisoned by ingestion of contaminated feed, water and air. Sprayed foliage or fodder or direct spray are the major sources of poisoning of organochlorine pesticides. These insecticides have tendency of persistence in environment and ability to accumulate in biologic food chains, and thus these residues upon consumption may cause chronic toxicity in man and animals.

These are non-specific stimulants to the central nervous system.

Clinical Signs: The clinical signs of poisoning of organochlorine insecticides are almost similar in all species, however, may vary in onset, severity and duration depending on quantity and type of the insecticide. Clinical signs appear within a few minutes to several hours or even days and are characterized by hyperaesthesia, profuse salivation, lacrimation, diarrhea; urination, piloerection, incoordination, jaw clamping, respiratory distress, laboured breathing, clonic convulsions and fever. Death may occur within minutes to few hours or even after several days. Respiratory failure is the main cause of death.

Treatment: There is no specific antidotal therapy for the treatment of organochlorine insecticide poisoning. Barbiturates or sedatives such as pentobarbitone, 10 mg/kg i/v or i/m or phenobarbitone, 25-50 mg/kg intramuscular in small animals; or chloral hydrate, 5 gm/50 kg b.wt. by oral route in large animals have beneficial effect to control convulsion, Atropine sulphate (0.1-0.2 mg/kg; s/c) can be given to check profuse salivation, lacrimation and nasal secretions. The use of combination of cholestyramine with non-absorbable oils for maximum fecal elimination of these pesticides.

3. Synthetic Pyrethroid

Source: Insecticidal properties of pyrethrum flowers (*Chrysanthemum cinerariaefolium*, *C. roseum*) are well recognized. Natural pyrethrins are now restricted and almost being replaced by synthetic pyrethroids. Most commonly used synthetic pyrethroid in India belong to alpha-cyano pyrethroid such as deltamethrin, cypermethrin, fenvalerate etc. and non alpha cyano including allethrin, permethrin and are used to check proliferation of pest and insects and to boost agriculture production and disease control programmes. They can also be used to kill ectoparasites in animals.

Clinical Signs: Pyrethroid insecticides have been classified into two groups, Type I and II, on the basis of poisoning syndrome. Type I produce tremor syndrome. Type II produced choreoathetosis and salivation (CS) syndrome. The clinical manifestations observed after oral administration in rats include ataxia, abasia and gait abnormalities, choreoathetosis, "tip-toe" walk and increased salivation, lacrimation, piloerection, tremor and convulsions. The mortalities occur within the first 3 hr and surviving animals recover within 7 days.

Neurobehavioural syndromes of Type I and Type II pyrethroids are quite different. Tremors, hyperthermia and decreased motor activity in Type I and pawing, borrowing, salivation, whole body tremor to choreoathetosis, hypothermia and lowered motor activity in Type II are evident.

Treatment: No selective antidote is available for the treatment of synthetic pyrethroid insecticide poisoning. Sedation by barbiturate such as pentobarbitone 10 mg/kg, i.v. or i.m.; phenobarbitone, 25-50 mg/kg intramuscular in small animals; or chloral hydrate, 5mg/50 kg b.w. by oral route in large animals have beneficial effect to control convulsions. Atropine sulphate (0.1-0.5 mg/kg i.m.) can be given to check profuse salivation, lacrimation and nasal secretions.

D. Mycotoxins

1. Aflatoxins

Source: These are produced by *Aspergillus* spp (*A. flavus*, *A. clavatus* and *A. parasiticus*). The main aflatoxins are named as B1, and B2. The toxins are potent carcinogenic, mutagenic, teratogenic and hepatotoxic. They produce chronic type of toxicity. Aflatoxin cause hepatotoxicity, nephro-toxicity and immuno-suppression is due to blinding to nuclear DNA and inhibition of RNA formation resulting in the impairment of synthesis of enzymes and essential proteins.

Clinical Signs: Clinical signs of aflatoxicosis include gradual reduction in feed

efficiency, milk production, weight gain, anaemia, Jaundice, haemorrhages, uraemia and abortion.

Treatment: There is no specific antidote for the treatment of aflatoxicosis. Symptomatic treatment is given. Supplementation of hepatoprotective agents, choline and aflatoxin binders are advocated.

Ergot Toxins

Source: Produced by the fungi of genus *Claviceps purpurea* and *C. paspali*. The mould contains various alkaloids which are ergotamine, ergonovine, ergocryptine, ergocornine, ergostin and ergosine. The nervous signs are produced due to interfering with neurotransmitter function of 5-HT, dopamine and norepinephrine in CNS. Ergot alkaloids cause intense vasoconstriction, damage to capillary endothelium and promotion of thrombus formation leading to stasis of blood supply in the extremities.

Clinical Signs:

Acute: Hyperexcitability of nerves i.e. spasms, tremors, incoordination, convulsions and paralysis.

Chronic: Necrosis, gangrene and sloughing of the extremities (tail, ears, feet or nose), and becoming sensitive to secondary bacterial infections.

Treatment: There is no specific antidote. However Vit E., Selenium combined therapy is useful in combating ergotoxin induced gangrene hooves in cattle and buffaloes.

Chapter 18

Diagnosis of Common Animal Bacterial Diseases

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Diagnosis of disease is the supreme importance in case of veterinary point of view as the patient remains dumb to explain the problem. So job of the clinician becomes tough that he has to rule out all the possible chances about the cause of the illness. Every disease has its own characteristic signs, lesions etc, with which one can come to a half way diagnosis. Due to the advent in the field of diagnosis lot and lot of improvements has come so that one can use the aid of the kits, equipments to accurately diagnose the disease. Though disease investigation is a science it is an art and needs special skill.

Points to Ponder During Disease Investigation:

1. Reliable and correct history
2. Careful clinical examination of sick animals
3. Accurate epidemiological data
4. Proper necropsy examination of dead animals by a qualified personnel and recording of lesions in a proper format
5. Proper collection, preservation and dispatch of samples to designated referral laboratory
6. Careful and wise interpretation
7. Specimen should be in adequate quantity.
8. Specimen must be collected before the administration of antimicrobial agents.
9. Contamination of specimen with externally present organisms or normal flora of body must be prevented.
10. Specimen should not get contaminated during storage.
11. Specimen handling should not be risky to individual
12. Apply strict aseptic techniques throughout the procedure.
13. Wash hands before and after the collection.
14. Collect the specimen at the optimum time.
15. Make certain that the specimen is representative of infectious process (e.g. sputum).
16. Collect or place the specimen aseptically in an appropriate sterile container.
17. Ensure that outside of specimen container is clean and uncontaminated.
18. Tightly close the container so that its contents do not leak during transportation.
19. Label and date the container complete the requisition form.
20. Arrange for immediate transportation of specimen to laboratory.

Haemorrhagic Septicemia

Haemorrhagic septicemia is a highly contagious bacterial disease of animals characterized by oedematous swelling in neck region, dyspnoea, pneumonia and wide spread haemorrhage in visceral organs.

Etiology:

1. *Pasteurella multocida*
2. *Mannheimia hemolytica*. They are Gram negative organisms.

Symptoms: The symptoms of Haemorrhagic Septicemia include high fever (105-107°F), oedema occur in the throat and brisket regions, which contains a gelatinous material. Ghar-ghar sound can be heard while breathing.

Diagnosis: Diagnosis can be carried out based on the symptoms, lesions seen. Samples from heart blood, liver, spleen, lungs or exudates can be taken for staining by leishman's method which reveals bipolar organisms. Heart blood is the choice of sample for isolation and identification of the organism. Animal inoculation is done by inoculating the suspected material or bacterial culture in mice and rabbits by scarification or subcutaneous route. The inoculated animal dies within 24-72 hours with haemorrhagic tracheitis.

Glanders

Glanders is an infectious disease of equines caused by *Pseudomonas* sp. characterized by ulcers in nasal passage, miliary nodules in lungs, oedema of lymph nodes, lymphangitis and lymphadenitis.

Etiology: *Pseudomonas mallei* which are a Gram negative bacteria. There are two forms of the disease. "Glanders" form: When the principal lesions are seen in the nostrils, submaxillary glands and lungs. "Farcy" form: When lesions are on the surface of limbs or body.

Symptoms: Symptoms include high fever (103-105°F), difficulty in breathing, oily pus drains from lymphatics and lymph nodes especially of hind legs. Punched out ulcers in the lungs is the pathognomonic lesion of glanders, ulcers are also seen in nasal passage.

Specimens for Diagnosis: Urine, pus, affected tissues and swab from infected tissue surfaces are suitable for *P. aeruginosa*. In case of *P. mallei*, collect tissue containing early nodules or pus from ulcers.

Diagnosis: Diagnosis can be done by isolation and identification of the agent, based on clinical signs and lesions. Animal inoculation tests can also be done.

Straus Reaction: The Straus reaction is seen in male guinea pigs inoculated intra peritoneally with infective material containing either *P. pseudomallei* or *P. mallei*. The reactions of swelling of the testes, inflammation of the tunica vaginalis and ulceration of the scrotal skin develops in 2-3 days. This test is not confirmatory one because Straus reaction is also produced by *Brucella*, Preisz-nocard bacillus, and *Actinobacillus ligniersii*.

Mallein Test: This is used to demonstrate the hypersensitivity developed after infection with *P. mallei*. Mallein is a glycoprotein extracted from the bacterium. The test can be done by three ways:

Subcutaneous Test: Swelling at the injection site and fever.

Ophthalmic Test: Instilled on conjunctival sac. Inflammatory and purulent reaction occurs within 6-12 hrs.

Intrapalpebral: Inoculate at skin of the lower eyelid. Localized, oedematous swelling and purulent conjunctivitis.

Serological Tests: Serological tests like CFT, Agglutination, IHA, and CIE are used in the diagnosis of glanders.

Anthrax

Anthrax is an acute septicemic disease of animals caused by *Bacillus* sp. and characterized by sudden death, absence of rigor mortis, tarry colour blood from natural openings and widespread haemorrhage. It is dreadful pathogen which can affect almost all mammals. It is a potential biological weapon. .

Etiology: *Bacillus anthracis*. It is a gram Positive non motile capsulated rods.

Symptoms: Clinical signs include high fever (105-107°F), haemorrhages from natural orifices, dyspnoea, oozing of tarry coloured blood and sudden mortality up to 90%

Diagnosis:

Based on Clinical Symptoms: Blood films from dead animals made by puncturing the superficial vein of the ear or in the region of the foot. Care should be taken to seal the injection site by placing cotton soaked in alcohol and ignited. The smears are heat fixed and stained by Wright's or Giemsa's stain to reveal *B. anthracis* as large blue rods with characteristic dark pink or purple coloured capsules. In case of horses and pigs since peripheral blood contains fewer organisms, smears should be made from the edematous fluid or LN's.

Cultural Examination: Swabs from blood are inoculated on to blood agar plates and incubated at 37°C for 24 hrs and examined for their typical growth. Swabs are inoculated in agar enriched with blood or serum and incubated for 6 hrs at 37°C and examined by stained smears.

Ascoli's Precipitation Test: Grind up the organ or blood of suspected animal and suspend in 5-10 parts of saline and boil for 15 minutes. Filter through filter paper and allow it to cool. Place 0.5 ml of anti-anthrax serum (1:50) in a small test tube and overlay with 0.5 ml of clear filtrate. Stand at room temperature for 15 minutes. A white ring of precipitation indicates a positive reaction.

String of Pearl's Test: *B. anthracis* produces swollen round cells in chains (string of pearls) when incubated for 3-6 hrs on tryptose agar containing 0.05 –0.5 I.U of penicillin/ml. It is considered reliable but needs experience. Cherry gamma phage can be used. It can be propagated on *B. anthracis* strain 14, which makes a suitable control. On one half of blood agar plate, streak *B. anthracis* strain 14 culture as control and on the other half streak the culture to be identified. Add drops of phage preparation (1:10 dilution) on both halves. Incubate the plates in the upright position for 8-12 hrs. Zones of clearing will be seen on control culture and on the suspected half if it is *B. anthracis*.

Animal Inoculation: Guinea pigs & mice are highly susceptible. Materials are inoculated by dermal scarification, subcutaneous or intramuscular. Death occurs in 2-3 days and the organisms will be readily identified in the blood & tissues. Apart from this laser pyrolysis, gas-liquid chromatography and mass spectrometry are used for detection of anthrax toxin productions. There are also fluorescent Ab-, ELISA- and PCR-based laboratory tests available and an immunochromatographic rapid diagnosis field test.

Prevention and Control: Post mortem should not be carried out in case of anthrax. Once opened there is always chance of spread of spores into the environment thus spreads to the human.

Black Quarter

Black Quarter is an acute disease of cattle and sheep caused by *Clostridium chauveii* and characterized by lameness, hot painful swelling in thigh muscles producing crepitating sound on pressure & accumulation of serosanguinous fluid in affected area.

Etiology: *Clostridium chauveii* Gram positive anaerobe.

Symptoms: When infection begins, the animal may develop a fever, and the affected limb can feel hot to touch. The limb usually swells significantly, and the animal can develop lameness on the affected leg. Crepitation (the sensation of air under the skin) can be noticed in many infections, as the area seems to crackle under pressure. Once clinical signs develop, the animal may only live a short while, sometimes as little as 12 hours. Occasionally, cattle will succumb to the disease without showing any symptoms, and only a post-mortem reveals the cause. During a necropsy, a diagnosis is usually made very quickly, as the affected muscle is usually mottled with black patches, which are dead tissue, killed by the toxins that the bacteria release when they infect live tissue. If viewed under a microscope, small rod-like bacteria can be seen to confirm the diagnosis.

Diagnosis: Diagnosis can be done based on history, based on symptoms like crepitate swelling particularly in the hind or fore quarter which crackles when rubbed with the fingers as a result of gas production. Smears prepared from the lesions and oedematous fluid reveal Gram positive rod. Isolation can be done from the center of the lesion, oedematous fluid and from heart blood & spleen. FAT is used to differentiate from *Cl. septicum*. Broth cultures or oedematous fluid from the lesions can be tested for toxicity and specific neutralization by antitoxin in mice or guinea pigs.

Control and Prevention: When an animal has died as a result of the blackleg disease, obey the following instructions:

1. Burn the carcass or bury it deeply with lime.
2. Burn any contaminated materials, including feces.
3. Disinfect any contaminated areas.
4. Do not conduct any biopsy on the animal.
5. Do not feed the carcass to any other animal(s).
6. Promptly contact a veterinarian or the State livestock sanitary official.

Enterotoxaemia

It is an acute disease of sheep, goat and camel caused by *Clostridium perfringens* and characterized by endocardial haemorrhage, gastroenteritis and pulpy kidneys.

Etiology:

1. *Clostridium perfringens* type D-sheep
2. *Clostridium perfringens* type A-camel

The causative agent is *Cl. perfringens* type D. However, predisposing factors also are essential; the most common of these is the ingestion of excessive amounts of feed or milk in the very young and of grain in feedlot lambs. In young lambs, the disease usually is restricted to the single lambs, because a ewe with twins seldom gives enough milk to allow enterotoxemia to develop. In the feedlot, the disease usually is seen in lambs switched rapidly to high-grain diets. As the starch intake increases, it provides a suitable medium for growth of the causative bacteria, which produce ϵ toxin. A major effect of

the toxin is to cause vascular damage, particularly of capillaries in the brain. Many sheep carry strains of *C. perfringens* type D as part of the normal microflora of the intestine and serve as the source of organisms to infect the newborn. Most such carriers have nonvaccinal antitoxin titers.

Clinical Signs: Usually, sudden deaths in the best-conditioned lambs are the first indication of enterotoxemia. In some cases, excitement, incoordination, and convulsions occur before death. Opisthotonos, circling, and pushing the head against fixed objects are common signs of CNS involvement; frequently, hyperglycemia or glycosuria is seen. Diarrhea may or may not develop. Occasionally, adult sheep are affected; they show weakness, incoordination, and convulsions and die within 24 hr. In goats, the course of disease ranges from peracute to chronic, with signs that vary from sudden death to watery diarrhea with or without blood. Acutely affected calves not found dead show mania, convulsions, blindness, and death in a few hours. Subacutely affected calves are stuporous for a few days and may recover. In goats, diarrhea and nervous signs are seen, and death occurs in several weeks. Type D enterotoxemia occasionally is seen in young horses that have overeaten.

Diagnosis: A presumptive diagnosis of enterotoxemia is based on sudden, convulsive deaths in lambs on carbohydrate-rich feed. Smears of intestinal contents reveal many short, thick gram-positive rods. Confirmation requires demonstration of ϵ toxin in the small-intestinal fluid. Fluid, not ingesta, should be collected in a sterile vial within a few hours after death and sent under refrigeration to a laboratory for toxin identification. Chloroform, added at 1 drop for each 10 mL of intestinal fluid, will stabilize any toxin present. Although immunologic tests have been developed to replace the traditional mouse assay for detection of toxin, they are less sensitive than the mouse assay. A PCR protocol for detection of the gene for ϵ toxin is effective in identifying isolates as either type B or D.

Control: The method of control depends on the age of the lambs, the frequency with which the disease appears on a particular property, and the method of husbandry. If the disease is seen consistently in young lambs on a property, ewe immunization probably is the most satisfactory method of control. Breeding ewes should be given 2 injections of type D toxoid their first year, and 1 injection 4-6 wk before lambing and each year thereafter.

Tetanus

Tetanus is a bacterial disease of animals caused by the toxins of clostridia and characterized by prolonged spasmodic contractions of muscles, stiffness and immobilization. It is also known as “*lock jaw*”.

Etiology: *Cl. tetani* is a straight, slender, Gram positive rod that characteristically produces a terminal, spherical endospores that bulges the cell giving the characteristic drumstick appearance. (The young spore may be oval rather than spherical). It occurs singly and occasionally in chains. It is non-capsulated and motile by peritrichous flagella.

Symptoms: Classical clinical signs include convulsions, immobilization, stiffness of muscles and lock jaw.

Diagnosis: Diagnosis can be done based on-

Direct Microscopy: Demonstration of characteristic drumstick spores of *C. tetani* by Gram stained smears of material from a wound (but it is not confirmative, because *C. tetanomorphum* and *C. tetanoides* also produces drumstick spores).

Isolation: Necrotic tissue from a wound or wound exudates can be heated to 80°C for 20mts and used to inoculate a blood agar plate and another blood agar plate containing stiff agar. A tube of thioglycollate medium or cooked meat broth could also be inoculated and sub cultured into blood agar. The plates are to be incubated anaerobically for 2-3 days. Growth is noticed using a hand lens as a filamentous growth spreading throughout the medium. Confirmation is done by identification of toxin. The toxin present in animal's serum or in filtrate from cooked meat broth or thioglycollate medium can be inoculated in to mice S/C or I/M-ly and identified by neutralization or protection tests using specific antitoxin.

Swine Erysipelas

It is an important disease of pigs caused by Gram-positive bacteria and characterized by arthritis, vegetative endocarditis and rhomboid shaped erythematous skin lesions. It is also known as “diamond skin disease”.

Etiology: *Erysipelothrix rhusiopathiae* (previously named *Erysipelothrix insidiosa* form S (Smooth) - form colonies and usually from acute syndromes is a Gram-positive rod, the R (rough) form colonies usually from chronic disease is a Gram-positive filament. The organism is non-motile, non-spore forming, non-acid fast, occur either in singly, in groups or in chains.

Symptoms: Symptoms include arthritis, erythema on skin, later sloughing of skin and patches on abdomen which has appearance of diamond markings.

Diagnosis: Diagnosis can be done by the Diamond shaped skin lesions are pathognomonic. Staining which reveals Gram-positive rods in acute cases and Gram-positive filaments in chronic cases. Based on cultural characters and biochemical tests diagnosis can be done. Serological tests are not applicable for diagnosis.

Listeriosis

Listeriosis is an acute infectious disease of animals caused by *Listeria* sp. and characterized by congestion of meninges and brain, presence of micro abscess in brain and lymphocytic leptomeningitis.

Etiology: *L. monocytogenes* is medium sized, Gram +ve rods, non-spore forming and non-acid fast organism.

Symptoms: It include high fever (105-107°F), circling movements, abortion and torticollis

Diagnosis: Diagnosis can be done with staining which reveal Gram +ve rods (often coccobacillary). Histopathological examination of brain tissue can often give a presumptive diagnosis of neural listeriosis. Isolation and Identification can be done by inoculation of specimens on selective media include blood agar with an antibiotic supplement or blood agar containing 0.05% potassium tellurite (inhibitory to Gram –ve). Inoculation in developing chicken embryos causes development of focal necrotic lesions on the chorio allantoic membrane (CAM).

Animal inoculation test can be carried out-Anton's test: Inoculation of live bacterial suspension into the conjunctiva of a rabbit or guinea pig only *L. monocytogenes* causes a

purulent keratoconjunctivitis within 24-36hrs of inoculation. Intra peritoneal inoculation of mice with a 24 hr broth culture. Both *L. monocytogenes* and *L. ivanovii* are pathogenic for mice. They die within in 5 days with necrotic lesions present in the liver.

Control: Poor quality silage should be avoided. Vaccination with killed vaccines, which do not induce effective cell-mediated immune response, is not protective because *L. monocytogenes* is an intracellular pathogen. Live, attenuated vaccines, which contain serovars 1/2a, 1/2b, and 4b are reported to reduce the prevalence of listeriosis in sheep.

Public Health Significance

1. *L. monocytogenes* causes meningoencephalitis, meningitis, encephalitis and uterine infection with abortion, stillbirths, granulomatosis and valvular endocarditis in humans.
2. The source could be soil, contaminated milk, cheese, meat, vegetables and human carriers.
3. Infection is frequently associated with immuno-compromised persons.

Tuberculosis

Tuberculosis is a deadly chronic disease caused by *Mycobacteria spp.* which are slender rods of varying lengths that sometimes show branching filamentous form resembling 'fungal mycelium'. Hence, the name mycobacteria, meaning fungus like bacteria. Although cytochemically Gram positive, the Mycobacteria do not take up the dyes of the Gram stain because the cell walls are rich in lipids mycolic acid. They are non-motile, non-sporing and non-capsulated.

Clinical Signs: It includes weakness, cough, skin and bone condition.

Diagnosis: Diagnosis can be made based on history, direct microscopic examination of acid fast stained slides which shows organisms are appearing as slender, often beaded, red staining rods against a blue background. Isolation and identification is a tough task. Animal inoculation can be done either by i/v or s/c or i/m route.

S.. No	Inoculated Suspected Material	<i>M. tuberculosis</i>	<i>M. bovis</i>	<i>M. avium</i>
1.	Rabbits (I/V)	+	++	++
2.	Guinea pig (S/C)	++	++	-
3.	Chicken (I/V)	-	-	++

Tuberculin test (Intra dermal, double intra dermal, ophthalmic tests in cattle and wattle test in case of poultry) which is the herd test commonly used. Specimens from live animals include aspirates from cavities, lymph nodes, biopsies, tracheobronchial lavages and centrifuged deposit from about 50 ml of milk in the case of suspected tuberculous mastitis. With dead animals, collect fresh and fixed (10% formalin) samples of lesions.

Specimens to be collected for diagnosis are live animals include aspirates from cavities, lymph nodes, biopsies, trachea-bronchial lavages and centrifuged deposit from about 50 ml of milk in the case of suspected tuberculous mastitis. With dead animals, collect fresh and fixed (10% formalin) samples of lesions.

Control and Prevention: Treatment and vaccination are inappropriate in control programmes for cattle. In many countries, tuberculin testing followed by isolation and slaughter of reactors has been implemented as the basis of national eradication schemes.

Brucellosis

Brucellosis is a highly zoonotic bacterial disease caused by several species of *Brucella* causing abortion in cattle in the last trimester of pregnancy. *B. abortus* mainly affects the cattle and *B. melitensis* affects goat and it is zoonotic. Brucellae are coccobacilli or short rods, arranged singly or in short chains. They are non-motile, non-capsulated, non-sporing and partial acid fast. They are stained by Grams, MZN and Koster's stain.

Clinical Symptoms: The main clinical symptoms include abortion which occur in the last trimester of pregnancy, hygroma in chronic cases, orchitis in males, sometimes there is mastitis.

Diagnosis: It is done based on history, signs, staining, isolation and identification. Animal inoculation can be done which is called as Straus test in which inoculated guinea pigs develop orchitis.

Serological Tests: There are a range of serological tests which are carried out in order to diagnose brucellosis. Tests include RBPT - Antigen consists of suspension of brucellae organisms stained with rose bengal and adjusted to pH 3.6. STAT- titre > 1:40 is +ve, titre < 1:20 is doubtful, ABRT/Milk ring test, Coombs test is used to detect incomplete antibodies. CFT is a recommended test for detection of brucella antibodies. Indirect ELISA can also be performed. Rivanol precipitation and Mercaptoethanol agglutination are used to detect primary IgM antibodies. PCR can also be employed to detect brucella DNA.

Leptospirosis

Leptospirosis is characterised by jaundice, which is caused by spirochetes namely *Leptospira* spp. It is also one of the zoonotic diseases and can cause a great damage to the human beings as well. Leptospire are actively motile, helically shaped, slender spirochaetes possessing large number of light and fine spirals. Characteristically, *Leptospira* have hooked ends and two periplasmic flagella (PF), also known as axial filament and endoflagella. They stain weakly or not at all with both aniline and Romanowsky stains. They may be stained with Giemsa stain. But they can be best viewed under dark field illumination microscope and can be stained by silver impregnation technique of Fontana. *Leptospira* divide by binary fission.

Clinical Signs: Characteristic signs include fever, jaundice, it can also cause mastitis and hence there is change in consistency and colour of the milk.

Diagnosis: It can be done based on direct microscopic examination, serological tests, animal inoculation and PCR

Direct Microscopy: Leptospire can be demonstrated in urine, other body fluids, and tissues by darkfield microscopy (DFM) and by FAT. Urine is centrifuged to concentrate the leptospire.

Animal Inoculation: Guinea pigs, hamsters and weaning gerbils can be inoculated i/p with 0.5 to 1 ml of neutralized urine, unclotted blood or a 10% tissue suspension in EMJH or 1% BSA. Cardiac blood is taken aseptically when a temperature rise is detected or at 5, 8, 10 and 14 days post infection.

Serology: Macroscopic agglutination test is a screening test and uses dead antigen (lack of specificity). Microscopic agglutination test (MAT) uses live leptospire as antigen and is highly sensitive and serovar specific. CFT and ELISA are also useful for detection of

leptospiral antibodies in serum. To identify the serovar, MAT, restriction endonucleases, DNA analysis and monoclonal antibodies are useful.

Campylobacteriosis

Campylobacteriosis causes greater damage to the farming community by causing abortion in case of bovines. Like brucellosis this is also an important disease in case of bovines is concerned. The main causative agent is *C. fetus*. The *Campylobacter* species are thin, curved, Gram-negative, microaerophilic, motile rods. They are motile by means of polar flagella. Small, curved or seagull-shaped rods can be demonstrated by using DCF-stained (dilute carbol fuchsin for 4 minutes) smears prepared from colonies. Wet mounts under phase contrast or darkfield microscopy reveal the characteristic curved forms with darting motility.

Diagnosis:

Based on Direct Microscopy: Fluorescent antibody staining of smears from foetal, abomasal contents, cervical mucus and preputial washings is most reliable, especially when small numbers of the bacterium are present. *C. jejuni* can be seen in wet mounts of faeces by phase contrast or dark field microscopy. The typical darting motility of corkscrew-like organism is suggestive of *Campylobacter* species.

Based on Isolation and Identification: *Campylobacter fetus* (both subspecies): cervical mucus and preputial washings can be passed through a 0.65 µm membrane filter to reduce contamination. Foetal abomasal contents and filtrates are useful for isolation and identification.

Serological Tests: The cervical mucus agglutination test for *C. fetus* subsp. *venerealis* is accurate if carried out 2-7 months post-infection. A vaginal mucus agglutination test has been found useful but the serum agglutination test is unreliable. A four-fold increase in an agglutinating antibody titre to the bacterium would suggest involvement of the organism in the diarrhoea. Fluorescent antibody-stained smears can be used to identify *C. fetus*, but the technique does not distinguish between the subtypes.

Mating Test or Virgin Heifer Test: To detect infected bull, virgin heifers are inseminated with semen and preputial washings from suspected animals and examined the heifer's vaginal mucous during the following 3-4 weeks. DNA probes are available for diagnosis.

Salmonellosis

Salmonellosis is an infectious disease of calves, lambs and piglets caused by *Salmonella* sp. and characterized by septicemia, wide spread haemorrhage on visceral organs, typhoid nodules in liver and haemorrhagic enteritis.

Etiology:

1. *Salmonella enteritidis* var. Typhimurium (S. Typhimurium)
2. *Salmonella enteritidis* var. Dublin (S. Dublin)
3. *Salmonella enteritidis* var. Choleraesuis (S. Choleraesuis)

These are gram negative, motile, non-lactose fermenter bacillus.

Symptoms: There is high rise of fever (102°F -105°F), weakness, icterus, anemia, diarrhoea, melena and dehydration noticed.

Diagnosis: Diagnosis is based on the symptoms, isolation of organism from clinical samples like faeces. Staining is a simplest way to detect the organisms.

Immunodiagnostic tests can also be done. PCR is also available for the diagnosis of Salmonellosis.

Colibacillosis

Colibacillosis is an infectious disease of new born animals caused by *E. coli* and characterized by enteritis, swelling of lymph nodes and peyer's patches, pneumonia and haemorrhage on endocardium.

Etiology: *Escherichia coli* is a gram negative, motile, lactose fermenter bacilli.

Symptoms: There is marked fever (103-106°F), diarrhoea, dehydration and nasal discharges. A condition "naval ill" usually noticed in young ones.

Diagnosis: Diagnosis is based on the symptoms, isolation of organism from clinical samples like faeces. Staining is a simplest way to detect the organism. Immunodiagnostic tests can also be done. PCR is also available for the diagnosis of causative agent.

Strangles

Strangles is an acute infectious disease of horses caused by streptococci and characterized by abscess in pharyngeal and maxillary lymph nodes, pleurisy, pericarditis, suppurative pneumonia, presence of abscess on liver, kidneys and spleen.

Etiology: *Streptococcus equi* is a gram positive cocci having chain formation.

Symptoms: There is marked fever (104°F -107°F), abscess in throat region and dyspnoea.

Diagnosis: Culture of nasal swabs, nasal washes or pus from abscesses is essential for confirming the presence of *S. equi*. However, culture may fail to detect the organism during the incubation period, in early clinical phases, and in the guttural pouch carriage in apparently normal horses following recovery from strangles. PCR, combined with culture, increases the carrier detection rate while serology is not very useful in the detection of *S. equi* infection.